

KASHTRE CLINICAL MODULE

CONSOLIDATED ENGINEERING DESIGN

DOCUMENT v6.1

VOLUMES 1 THROUGH 15

Core Engineering Baseline: Volumes 1 to 10

Supplemental Engineering Baseline: Volumes 11 to 15

Approval: Draft for Engineering Review

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

CONTROL NOTICE

This consolidated revision preserves the original volume listings while adding a single control layer, correcting Volume 13 architecture, expanding Volume 15 DHIS2 design, and defining unified dependency, permission, gap and traceability registers. Candidate status does not mean repository verification or production approval.

Consolidated Document Control

Control	Decision
Functional authority	KashTre_Clinical_Module_SRD_v6.1
Engineering authority	This consolidated EDD and its controlled source packages
Core baseline	Volumes 1 to 10
Supplemental baseline	Volumes 11 to 15
Technical target	Laravel 13.x, PHP 8.4, MySQL 8.4 LTS, Vue 3 with TypeScript
Baseline discrepancy	The SRD reference stack remains a recorded governance gap until formally superseded or approved
Verification threshold	Only controlled repository execution and evidence may change Candidate to Verified
Production status	Not Approved for Production

Volume Register

Vol.	Title	Class	Component
1	Executable Foundation and Shared Infrastructure	Core	CLINICAL_FOUNDATION
2	Effective Patient Observation Plan	Core	CLINICAL_OBSERVATION_PLAN
3	CDE Registry and Observation Capture	Core	CLINICAL_OBSERVATION
4	Forms and Rendering	Core	CLINICAL_FORMS
5	Clinical Scoring and Deterioration	Core	CLINICAL_SCORING
6	Medication, eMAR, Administration and Consumption	Core	CLINICAL_MEDICATION
7	Diagnostic Results and Follow-up	Core	CLINICAL_RESULTS
8	Transitions of Care	Core	CLINICAL_TRANSITIONS
9	Clinical Assurance and Privacy	Core	CLINICAL_ASSURANCE
10	Production Release Assurance	Core	CLINICAL_RELEASE
11	Operations, Adoption and AI Governance	Supplemental	CLINICAL_OPERATIONS
12	Interoperability, Population Health and Research	Supplemental	CLINICAL_INTEROPERABILITY
13	Patient Engagement and Remote Care	Supplemental	CLINICAL_ENGAGEMENT
14	Content, Configuration, Localization and Accessibility	Supplemental	CLINICAL_CONTENT_GOVERNANCE
15	DHIS2 Integration, Public-health Reporting and Surveillance	Supplemental	CLINICAL_DHIS2

Dependency and Status Model

Volume 1 supplies tenant context, public identifiers, audit, idempotency and durable messaging. Volumes 2 to 9 consume those controls. Volume 10 governs release evidence. Volumes 11 to 15 add operational, interoperability, engagement, configuration and DHIS2 capabilities without weakening the core controls.

Status dimension	Allowed values	Meaning
Functional	Draft; Approved	Requirements authority
Engineering	Candidate; Verified	Candidate is designed; Verified is executed with evidence
Supplement	Proposed; Incorporated	Whether supplemental scope is accepted into the controlled baseline
Production	Not Approved; Approved	Deployment authority

Unified Traceability Contract

Every implemented requirement shall map to: SRD requirement ID, EDD volume, design component, package/class, database object, API or event, test case, approval evidence and release identifier. Missing links block Verified status.

Consolidated Review Gates

- Resolve the SRD and EDD technology-baseline discrepancy.
- Confirm every model has a matching migration and every table has an owning model or explicit reason.
- Approve one atomic permission catalogue across all volumes.
- Close or formally defer every blocking gap with an owner and acceptance test.
- Run clean, upgrade, rollback, concurrency, load, privacy, tenant-isolation and security suites.
- Generate signed CI evidence and source-package checksums.

KASHTRE

CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 1

EXECUTABLE FOUNDATION AND SHARED INFRASTRUCTURE

Version: 6.1.0 Comprehensive Redraft

Component: CLINICAL_FOUNDATION

Classification: Repository-Ready Reference Implementation

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Companion SRD: KashTre_Clinical_Module_SRD_v6.1

Target baseline: Laravel 13 / PHP 8.4 / MySQL 8.4 LTS / Vue 3

Approval: Draft for Engineering Review

Production: Not Approved for Production

Issue date: 28 August 2026

PICK-AND-DROP CONTROL

Every normative listing is reproduced as a complete source-package file and has a SHA-256 manifest entry. The package is not labelled VERIFIED because the actual KashTre repository, authentication model and Main Module endpoints were not available for compilation and execution. No engineer may treat CANDIDATE as VERIFIED.

Document Authority and Completion Standard

This redraft replaces the earlier architectural-core Volume 1 as the implementation candidate. It provides a controlled package structure, complete foundation files, installation scripts, tests and file checksums. Later Clinical volumes shall import these contracts rather than recreate tenant, identifier, audit, outbox, inbox or idempotency behaviours.

Control	Rule
Functional authority	Clinical SRD v6.1.
Unit authority	Main Module Shared Unit Engine contract and approved EDD binding.
Time authority	Main Module shared time capability; Clinical preserves event-local provenance.
AI authority	Shared AI Gateway only.
Code authority	The accompanying ZIP source package is authoritative; Word listings are review reproductions.
Verification threshold	Only execution in the controlled KashTre repository may change Candidate to Verified.
Prohibited omissions	No ellipses, TODOs, pseudocode or implicit “standard practice” inside normative files.
Production status	Not Approved for Production.

Baseline Decision

Layer	Selected baseline	Reason/control
Framework	Laravel 13.x	Supported Laravel major; host repository compatibility still requires execution.
Runtime	PHP 8.4	Supported runtime and compatible with Laravel 13.
Database	MySQL 8.4 LTS	Stable LTS track; migrations require clean and upgrade testing.
Identifiers	ULID	Public identity, internal bigint database keys.
Frontend	Vue 3 + TypeScript	Host-pinned versions must be added during frontend volume binding.
Messaging	Transactional outbox/inbox	Broker-neutral durable boundary.
Tests	PHPUnit 12 + Orchestra Testbench 11	Package-level isolated verification.

Installation Contract

```
Package destination: packages/kashtre/clinical-foundation
Composer path repository: {"type":"path","url":"packages/kashtre/clinical-foundation"}
Require: composer require kashtre/clinical-foundation:@dev
Publish: php artisan vendor:publish --tag=clinical-foundation-config
Migrate: php artisan migrate
Verify package: cd packages/kashtre/clinical-foundation && composer test
```

Host Repository Preconditions

- Laravel host application supports package discovery.
- Authenticated user exposes stable public_id and tenant_id, or an approved actor adapter replaces direct access.
- auth:sanctum is installed and configured, or the route middleware is replaced through an approved binding.
- MySQL is configured with UTC session handling and strict SQL mode.
- The host registers Main Module endpoint credentials in secret management.
- The service account can create the four foundation tables.

Part I. Complete Entity Dictionary

Entity	Authority	Created by	Consumed by	Invariant	Lifecycle	Required tests
TenantId	Immutable tenant value object	Request header or authenticated claim	All tenant-owned repository access	Reject invalid syntax; exact match to authenticated tenant	Never mutable	Unit tests and cross-tenant feature tests
PublicId	Validated ULID public identity	Generator or trusted external contract	Routes, events and foreign module references	Reject malformed identifiers before lookup	Immutable	Format and route-binding tests
ClinicalRequestContext	Request-scoped authoritative execution context	Context middleware	Policies, audit, events and repositories	Cannot exist without tenant, actor, correlation and request ID	Request lifetime	Middleware and propagation tests
ClinicalIdempotencyRecord	Durable command replay record	Idempotency middleware	Retryable HTTP commands	One tenant+scope+key; payload hash cannot change	Mutable state, retained through TTL	Replay, conflict and concurrency tests
ClinicalOutboxMessage	Durable event publication record	Domain transaction	Integration publisher	Created in same transaction as domain effect	Append plus status	Rollback and duplicate-delivery tests
ClinicalInboxMessage	Durable received-message identity	Integration consumer	Inbound idempotency	Same source ID with different hash is conflict	Append plus processing status	Duplicate and quarantine tests
ClinicalAuditEntry	Immutable security/clinical activity evidence	Audit recorder	Governance and investigations	No secrets or unrestricted narrative	Append-only	Persistence and sanitizer tests

Column-Level Data Dictionary

Table	Column	Type	Null	Source	Rule
clinical_idempotency_records	tenant_id	varchar(64)	No	Authenticated tenant	Tenant isolation and unique identity
clinical_idempotency_records	scope_code	varchar(96)	No	Route middleware parameter	Separates command namespaces
clinical_idempotency_records	idempotency_key	varchar(128)	No	HTTP header	Opaque client retry identity
clinical_idempotency_records	request_hash	char(64)	No	SHA-256 request body	Detects key reuse for different command
clinical_idempotency_records	status	varchar(24)	No	Middleware state machine	PROCESSING or COMPLETED in current implementation
clinical_outbox_messages	event_type	varchar(160)	No	Domain event	Versioned fact name
clinical_outbox_messages	event_version	smallint unsigned	No	Event schema	Consumer compatibility
clinical_outbox_messages	payload	json	No	Event serializer	Minimum necessary data only
clinical_inbox_messages	source_message_id	varchar(160)	No	Producer	Deduplication identity
clinical_inbox_messages	payload_hash	char(64)	No	Consumer	Conflict detection
clinical_audit_entries	event_code	varchar(128)	No	Audit caller	Stable controlled action code
clinical_audit_entries	details	json	Yes	Sanitized structured evidence	No token, secret or unrestricted note

Part II. Architecture, Transactions and Failure Behaviour

Request Pipeline

1. Authenticate through host middleware.
2. Resolve tenant header.
3. Compare tenant to authenticated identity.
4. Resolve or generate correlation ID.
5. Create immutable ClinicalRequestContext.
6. Apply authorization policy for the requested Clinical action.
7. Apply idempotency middleware for state-changing commands.
8. Execute one application action in a database transaction.
9. Write audit/outbox records within the declared transaction where required.
10. Return standard envelope and correlation header.

Transaction Rules

- No remote HTTP call is made while holding a database transaction unless an approved exception is documented.
- Domain state and the corresponding outbox message commit atomically.
- Idempotency ownership is acquired under a row lock.
- Audit records required to authorize or evidence a high-risk action commit with the action.
- Retries use bounded database transaction attempts and stable identities.

Failure Catalogue

Code	HTTP	Retryable	Meaning
CLN_CTX_TENANT_REQUIRED	400	No	Tenant context missing.
CLN_CTX_TENANT_MISMATCH	403	No	Authenticated identity and requested tenant differ.
CLN_IDEMPOTENCY_KEY_REQUIRED	400	No	State-changing route requires key.
CLN_IDEMPOTENCY_KEY_REUSED	409	No	Key used with different request hash.
CLN_IDEMPOTENCY_IN_PROGRESS	409	Yes	Equivalent request is still executing.
CLN_AUTH_CONTEXT_UNAVAILABLE	503	Conditional	Required authority cannot be proven.
CLN_AUDIT_UNAVAILABLE	503	Conditional	Mandatory audit evidence cannot be stored.
CLN_OUTBOX_FAILED	503	Yes	Durable event could not be written.

Security Rules

- Tenant is never accepted solely from route or request body.
- Internal numeric IDs are never returned by package APIs.
- Correlation IDs are validated as ULIDs before propagation.
- Audit and event payloads use allow-listed fields.
- Secrets exist only in environment or secret manager.
- Cross-tenant lookup returns concealed not-found behaviour in domain repositories.
- All write routes require server-side authorization plus idempotency.

Part III. Repository-Ready Source Listings

Each section below reproduces the complete authoritative source-package file. The SHA-256 value permits extraction verification. Files are ordered by installation dependency.

README.md

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	7cb87de146cb9502e898855ef174d13a5f0c5f2f7ce262ec30a10ecdffa9ba872

```
# KashTre Clinical Foundation v6.1.0
Status: Pick-and-drop candidate. Repository execution is required before VERIFIED classification.
## Install
1. Copy this package under `packages/kashtre/clinical-foundation`.
2. Add a Composer path repository and require `kashtre/clinical-foundation:@dev`.
3. Publish configuration: `php artisan vendor:publish --tag=clinical-foundation-config`.
4. Configure `CLINICAL_*` environment values.
5. Run `php artisan migrate`.
6. Run `composer test` inside the package.
## Required host contract
The authenticated user model must expose `public_id` and `tenant_id`. If KashTre uses different accessors, implement an approved ActorIdentity adapter before verification.
```

artifact-manifest.json

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	36bfb841926aac9d48656ba7884ee871ee6aaaafd37e97649225308f8cfd255d

```
{
  "volume": "Clinical EDD Volume 1",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "7cb87de146cb9502e898855ef174d13a5f0c5f2f7ce262ec30a10ecdffa9ba872",
      "bytes": 701
    },
    {
      "path": "composer.json",
      "sha256": "94f2e722e75b1e28659ce21d81ae0d786bcd66727752a8a8b1f6d2fe2125899",
      "bytes": 817
    },
    {
      "path": "config/clinical-foundation.php",
      "sha256": "bca07344c46d33f65ca3140992e29c04161280b43d5e10c417abe2b96a6c7552",
      "bytes": 797
    },
    {
      "path": "database/migrations/2026_08_28_000001_create_clinical_foundation_tables.php",
      "sha256": "c9fc6d81be25927b93fbdfcad15374c71e06d4061a1ec0d6831ea4baa82e187",
      "bytes": 2890
    },
    {
      "path": "install.sh",
      "sha256": "fb10c5912b8263408cc394d3ceafb0b417754640d5d978cc8383d93e797017df",
      "bytes": 168
    },
    {
      "path": "routes/clinical-foundation.php",
      "sha256": "8b26279eab4a59c839c7c12dc7d330dbca9b92c78cc6c59a97b440c5615fd533",
      "bytes": 400
    },
    {
      "path": "src/ClinicalFoundationServiceProvider.php",
      "sha256": "1401f24e1d9d9acc00b14b89b2440813d39fac6b74695053553427757dcaeb63",

```

```

"bytes": 1326
},
{
  "path": "src/Foundation/Context/ClinicalRequestContext.php",
  "sha256": "5910191f61f33304ea0b346912521de31b5eb25ae3167a835014cb72b3063318",
  "bytes": 533
},
{
  "path": "src/Foundation/Contracts/AuditRecorder.php",
  "sha256": "0459927cd976ce6a8e38d7f2905d20da1883eb3c170a8a0a717b0dc4f14d4153",
  "bytes": 262
},
{
  "path": "src/Foundation/Contracts/EventPublisher.php",
  "sha256": "11d647b31931f5acf3d3e3d7bfbe0bdbfa8a440161215dbbf3b1a9370c33382f3",
  "bytes": 275
},
{
  "path": "src/Foundation/Contracts/PublicIdGenerator.php",
  "sha256": "4e6412a8e481ba43897a587026fa52562c9683a10900a1d5ca40981f07d7541b",
  "bytes": 205
},
{
  "path": "src/Foundation/Contracts/TenantResolver.php",
  "sha256": "75614b165e5a19daa97b9724c26376f1f366aae5820bf071af7fc717793b5f81",
  "bytes": 246
},
{
  "path": "src/Foundation/Data/AuditData.php",
  "sha256": "fd59e9c411c3871757b3a1333d78db793cf2081dbdc4ded1f091a25b15a370b7",
  "bytes": 386
},
{
  "path": "src/Foundation/Data/OutboxEventData.php",
  "sha256": "4579c87ff2c21ecb597cf024763bda650beb8a78a876f332d96cde78b57c165",
  "bytes": 537
},
{
  "path": "src/Foundation/Http/Middleware/EnforceIdempotency.php",
  "sha256": "46bcb43c62b94c3fdcc025211b66d55b88a16323f13ecadaa834719ca6dfe43b",
  "bytes": 2707
},
{
  "path": "src/Foundation/Http/Middleware/EstablishClinicalContext.php",
  "sha256": "24a00b8e7d4e8999a54d3dde2ab5cbcc4f4f3d5914abca953234a277b818fd75",
  "bytes": 1650
},
{
  "path": "src/Foundation/Infrastructure/DatabaseAuditRecorder.php",
  "sha256": "728fec7432e786c2f7f429dda42843e9285add91a19be6cd2a496299b07aa110",
  "bytes": 1104
},
{
  "path": "src/Foundation/Infrastructure/DatabaseOutboxPublisher.php",
  "sha256": "3bcdaa68c74cedc093d78b3bb0697c0081e223fa985fdc65f432cf81f19c5795",
  "bytes": 1122
},
{
  "path": "src/Foundation/Infrastructure/HeaderTenantResolver.php",
  "sha256": "e817df6c486e5bd8626e86bc410b931c65729f4251a1d7882420d8713ea61d81",
  "bytes": 688
},
{
  "path": "src/Foundation/Infrastructure/SymfonyUlidGenerator.php",
  "sha256": "919882cdf3ebc71371610959c161ca824158646d015d19e22531a3799499ee45",
  "bytes": 386
},
{
  "path": "src/Foundation/Models/ClinicalAuditEntry.php",
  "sha256": "06daa2254ee2823f0bda285ae8e2d65e6e406b7952485b65b3a6ddf6ffbea986",
  "bytes": 393
},
{

```

```
"path": "src/Foundation/Models/ClinicalIdempotencyRecord.php",
"sha256": "153b866418450abb988181d544217f9aba5832a8119db0c4a0cb54bfd03eeac0",
"bytes": 412
},
{
  "path": "src/Foundation/Models/ClinicalInboxMessage.php",
  "sha256": "c39dd62b80d685c93858d4aa48c91d539ddad0ef3df4cbb22ed9655e4adbb616",
  "bytes": 397
},
{
  "path": "src/Foundation/Models/ClinicalOutboxMessage.php",
  "sha256": "00f85469d7c3daa98ef86a5da8b7953a305a75bcc8eb083719b00aad19e2b7a5",
  "bytes": 439
},
{
  "path": "src/Foundation/ValueObjects/CorrelationId.php",
  "sha256": "368d07f9804ea72dca76499d58199293bbb7f043c9c842de3e71f81113def1b6",
  "bytes": 250
},
{
  "path": "src/Foundation/ValueObjects/PublicId.php",
  "sha256": "6d1f65ecb5ef87a0257e98fd387344d95155a119923a92008d1874d82bb6e337",
  "bytes": 395
},
{
  "path": "src/Foundation/ValueObjects/TenantId.php",
  "sha256": "83de16e78436098e11e4de5307453dfb6c9d78dd1baf1bfab07a7efc62f59d5b",
  "bytes": 418
},
{
  "path": "tests/Feature/FoundationMigrationTest.php",
  "sha256": "3e13023ab436e4adcb2e4f8f9e32dbca8600552db07f91e1d084158fc0449465",
  "bytes": 445
},
{
  "path": "tests/TestCase.php",
  "sha256": "61dd7267c24f21359ffd516fe09d2db2433b762b8b5972b39e1e2f5fb47a4275",
  "bytes": 693
},
{
  "path": "tests/Unit/PublicIdTest.php",
  "sha256": "3c20865788e00b100a479d60b987e8ee1b5c98940fa6f32ea5ed83480a9521de",
  "bytes": 512
},
{
  "path": "tests/Unit/TenantIdTest.php",
  "sha256": "51a9148b9c85946baa62d0e065a8b0941bcd40df0f563020dcd1e8f482a4104",
  "bytes": 629
},
{
  "path": "verify.sh",
  "sha256": "4642cae9bc2075013e03496f419d165013fc2c5baa0dc3baf96b2809a4b8b139",
  "bytes": 122
}
]
```

composer.json

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	94f2e722e75b1e28659ce21d81ae0d786bcd66727752a8a8b1f6d2fe2125899

```
{
  "name": "kashtre/clinical-foundation",
  "description": "KashTre Clinical Foundation package",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/cache": "^13.0",
    "illuminate/contracts": "^13.0",
```

```

    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/routing": "^13.0",
    "illuminate/support": "^13.0",
    "symfony/uid": "^7.3"
  },
  "require-dev": {
    "orchestra/testbench": "^11.0",
    "phpunit/phpunit": "^12.0"
  },
  "autoload": {
    "psr-4": {
      "KashTre\\Clinical\\": "src/"
    }
  },
  "autoload-dev": {
    "psr-4": {
      "KashTre\\Clinical\\Tests\\": "tests/"
    }
  },
  "extra": {
    "laravel": {
      "providers": [
        "KashTre\\Clinical\\ClinicalFoundationServiceProvider"
      ]
    }
  },
  "scripts": {
    "test": "phpunit"
  },
  "config": {
    "sort-packages": true,
    "allow-plugins": {}
  }
}

```

config/clinical-foundation.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-foundation.php	bca07344c46d33f65ca3140992e29c04161280b43d5e10c417abe2b96a6c7552

```

<?php
declare(strict_types=1);
return [
    'tenant_header' => env('CLINICAL_TENANT_HEADER', 'X-Tenant-ID'),
    'correlation_header' => env('CLINICAL_CORRELATION_HEADER', 'X-Correlation-ID'),
    'idempotency_header' => env('CLINICAL_IDEMPOTENCY_HEADER', 'Idempotency-Key'),
    'idempotency_ttl_hours' => (int) env('CLINICAL_IDEMPOTENCY_TTL_HOURS', 24),
    'outbox_batch_size' => (int) env('CLINICAL_OUTBOX_BATCH_SIZE', 100),
    'outbox_max_attempts' => (int) env('CLINICAL_OUTBOX_MAX_ATTEMPTS', 10),
    'main_base_url' => env('CLINICAL_MAIN_BASE_URL'),
    'main_token' => env('CLINICAL_MAIN_TOKEN'),
    'shared_time_path' => env('CLINICAL_SHARED_TIME_PATH', '/api/v1/platform/time/resolve'),
    'shared_unit_path' => env('CLINICAL_SHARED_UNIT_PATH', '/api/v1/platform/units/convert'),
];

```

database/migrations/2026_08_28_000001_create_clinical_foundation_tables.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_000001_create_clinical_foundation_tables.php	c9fc6d81be25927b93fbdfcad15374c71e06d4061ae1ec0d6831ea4baa82e187

```

<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;
return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_idempotency_records', function(Blueprint $t){
            $t->id();
            $t->ulid('public_id')->unique();
            $t->string('tenant_id', 64);
            $t->string('scope_code', 96);
            $t->string('idempotency_key', 128);
            $t->char('request_hash', 64);
            $t->string('status', 24);
            $t->unsignedSmallInteger('response_status')->nullable();
            $t->json('response_body')->nullable();
            $t->dateTime('locked_until_utc', 6)->nullable();
            $t->dateTime('expires_at_utc', 6);
            $t->dateTime('created_at_utc', 6);
            $t->dateTime('updated_at_utc', 6);
            $t->unique(['tenant_id', 'scope_code', 'idempotency_key'], 'uq_clinical_idempotency_identity');
            $t->index(['tenant_id', 'expires_at_utc'], 'idx_clinical_idempotency_expiry');
        });

        Schema::create('clinical_outbox_messages', function(Blueprint $t){
            $t->id();
            $t->ulid('public_id')->unique();
            $t->string('tenant_id', 64);
            $t->string('topic', 128);
            $t->string('event_type', 160);
            $t->unsignedSmallInteger('event_version');
            $t->string('aggregate_type', 128);
            $t->ulid('aggregate_public_id');
            $t->ulid('correlation_id');
            $t->json('payload');
            $t->string('status', 24)->default('PENDING');
            $t->unsignedSmallInteger('attempt_count')->default(0);
            $t->dateTime('available_at_utc', 6);
            $t->dateTime('published_at_utc', 6)->nullable();
            $t->text('last_error')->nullable();
            $t->dateTime('created_at_utc', 6);
            $t->index(['status', 'available_at_utc'], 'idx_clinical_outbox_dispatch');
        });

        Schema::create('clinical_inbox_messages', function(Blueprint $t){
            $t->id();
            $t->ulid('public_id')->unique();
            $t->string('tenant_id', 64);
            $t->string('source_system', 64);
            $t->string('source_message_id', 160);
            $t->string('message_type', 128);
            $t->unsignedSmallInteger('message_version');
            $t->char('payload_hash', 64);
            $t->json('payload');
            $t->string('status', 24);
            $t->dateTime('received_at_utc', 6);
            $t->dateTime('processed_at_utc', 6)->nullable();
            $t->text('last_error')->nullable();
            $t->unique(['tenant_id', 'source_system', 'source_message_id'], 'uq_clinical_inbox_source');
        });
    }
};

```

```
Schema::create('clinical_audit_entries',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->string('event_code',128);$t->string('outcome',24);$t->string('actor_type',24);$t->ulid('actor_public_id')->nullable();$t->ulid('correlation_id');$t->string('request_id',64);$t->string('source_channel',32);$t->json('details')->nullable();$t->dateTime('occurred_at_utc',6);$t->dateTime('recorded_at_utc',6);$t->index(['tenant_id','actor_public_id','occurred_at_utc'],'idx_clinical_audit_actor_time'));
}
public function down(): void {
Schema::dropIfExists('clinical_audit_entries');Schema::dropIfExists('clinical_inbox_messages');Schema::dropIfExists('clinical_outbox_messages');Schema::dropIfExists('clinical_idempotency_records'); }
};
```

install.sh

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	fb10c5912b8263408cc394d3ceafb0b417754640d5d978cc8383d93e797017df

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan vendor:publish --tag=clinical-foundation-config --force
php artisan migrate --force
```

routes/clinical-foundation.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-foundation.php	8b26279eab4a59c839c7c12dc7d330dbca9b92c78cc6c59a97b440c5615fd533

```
<?php
declare(strict_types=1);
use Illuminate\Support\Facades\Route;
use KashTre\Clinical\Foundation\Http\Middleware\EstablishClinicalContext;
Route::middleware(['api','auth:sanctum',EstablishClinicalContext::class])->prefix('api/v1/clinical')->group(function(): void {
    Route::get('/foundation/health', fn()=>response()->json(['data'=>['status'=>'ok']]))->name('clinical.foundation.health');
});
```

src/ClinicalFoundationServiceProvider.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ClinicalFoundationServiceProvider.php	1401f24e1d9d9acc00b14b89b2440813d39fac6b74695053553427757dcaeb63

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical;
use Illuminate\Support\ServiceProvider;
use KashTre\Clinical\Foundation\Contracts\{AuditRecorder, EventPublisher, PublicIdGenerator, TenantResolver};
use KashTre\Clinical\Foundation\Infrastructure\{DatabaseAuditRecorder, DatabaseOutboxPublisher, HeaderTenantResolver,
SymfonyUlidGenerator};
final class ClinicalFoundationServiceProvider extends ServiceProvider
{
    public function register(): void
    {
        $this->mergeConfigFrom(__DIR__.'/../config/clinical-foundation.php', 'clinical-foundation');
        $this->app->singleton(PublicIdGenerator::class, SymfonyUlidGenerator::class);
        $this->app->singleton(TenantResolver::class, HeaderTenantResolver::class);
        $this->app->singleton(AuditRecorder::class, DatabaseAuditRecorder::class);
        $this->app->singleton(EventPublisher::class, DatabaseOutboxPublisher::class);
        $this->app->scoped(Foundation\Context\ClinicalRequestContext::class);
    }
    public function boot(): void
    {
        $this->publishes([__DIR__.'/../config/clinical-foundation.php'=>config_path('clinical-foundation.php')], 'clinical-foundation-config');
        $this->loadMigrationsFrom(__DIR__.'/../database/migrations');
        $this->loadRoutesFrom(__DIR__.'/../routes/clinical-foundation.php');
```

```

    }
}

```

src/Foundation/Context/ClinicalRequestContext.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Context/ClinicalRequestContext.php	5910191f61f33304ea0b346912521de31b5eb25ae3167a835014cb72b3063318

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Context;
use KashTre\Clinical\Foundation\ValueObjects\{CorrelationId, PublicId, TenantId};
final readonly class ClinicalRequestContext
{
    public function __construct(
        public TenantId $tenantId,
        public PublicId $actorPublicId,
        public CorrelationId $correlationId,
        public string $requestId,
        public ?PublicId $facilityPublicId,
        public ?PublicId $clientSpacePublicId,
        public string $sourceChannel,
    ) {}
}

```

src/Foundation/Contracts/AuditRecorder.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Contracts/AuditRecorder.php	0459927cd976ce6a8e38d7f2905d20da1883eb3c170a8a0a717b0dc4f14d4153

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Contracts;
use KashTre\Clinical\Foundation\Data\AuditData;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
interface AuditRecorder { public function record(AuditData $data): PublicId; }

```

src/Foundation/Contracts/EventPublisher.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Contracts/EventPublisher.php	11d647b31931f5acfd3e3d7bfbe0bdbfa8a440161215dbbf3b1a9370c33382f3

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Contracts;
use KashTre\Clinical\Foundation\Data\OutboxEventData;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
interface EventPublisher { public function store(OutboxEventData $event): PublicId; }

```

src/Foundation/Contracts/PublicIdGenerator.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Contracts/PublicIdGenerator.php	4e6412a8e481ba43897a587026fa52562c9683a10900a1d5ca40981f07d7541b

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Contracts;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
interface PublicIdGenerator { public function generate(): PublicId; }

```

src/Foundation/Contracts/TenantResolver.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Contracts/TenantResolver.php	75614b165e5a19daa97b9724c26376f1f366aae5820bf071af7fc717793b5f81

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Contracts;
use Illuminate\Http\Request;
use KashTre\Clinical\Foundation\ValueObjects\TenantId;
interface TenantResolver { public function resolve(Request $request): TenantId; }
```

src/Foundation/Data/AuditData.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Data/AuditData.php	fd59e9c411c3871757b3a1333d78db793cf2081dbdc4ded1f091a25b15a370b7

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Data;
use DateTimeImmutable;
use KashTre\Clinical\Foundation\Context\ClinicalRequestContext;
final readonly class AuditData
{
    public function __construct(public string $eventCode, public string $outcome, public ClinicalRequestContext $context, public
array $details, public DateTimeImmutable $occurredAtUtc) {}
}
```

src/Foundation/Data/OutboxEventData.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Data/OutboxEventData.php	4579c87ff2c21ecb597cf0247633bda650beb8a78a876f332d96cde78b57c165

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Data;
use DateTimeImmutable;
use KashTre\Clinical\Foundation\Context\ClinicalRequestContext;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
final readonly class OutboxEventData
{
    public function __construct(public string $topic, public string $eventType, public int $eventVersion, public string
$aggregateType, public PublicId $aggregatePublicId, public array $payload, public ClinicalRequestContext $context, public
DateTimeImmutable $occurredAtUtc) {}
}
```

src/Foundation/Http/Middleware/EnforceIdempotency.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Http/Middleware/EnforceIdempotency.php	46bcb43c62b94c3fdcc025211b66d55b88a16323f13ecadaa834719ca6dfe43b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Http\Middleware;
use Closure;
use DateTimeImmutable;
use DateTimeZone;
use Illuminate\Http\JsonResponse;
use Illuminate\Http\Request;
use Illuminate\Support\Facades\DB;
```

```

use Symfony\Component\HttpFoundation\Response;
use KashTre\Clinical\Foundation\Context\ClinicalRequestContext;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;
use KashTre\Clinical\Foundation\Models\ClinicalIdempotencyRecord;
final readonly class EnforceIdempotency
{
    public function __construct(private PublicIdGenerator $ids) {}
    public function handle(Request $request, Closure $next, string $scope): Response
    {
        $key=(string)$request->header((string)config('clinical-foundation.idempotency_header'));
        if ($key==='') return new JsonResponse(['error'=>['code'=>'CLN_IDEMPOTENCY_KEY_REQUIRED','message'=>'Idempotency-Key is
required.'],400]);
        $context=app(ClinicalRequestContext::class); $hash=hash('sha256',(string)$request->getContent());
        return DB::transaction(function() use($request,$next,$scope,$key,$context,$hash): Response {
            $record=ClinicalIdempotencyRecord::query()->where('tenant_id',(string)$context->tenantId)->where('scope_code',$scope)-
>where('idempotency_key',$key)->lockForUpdate()->first();
            if ($record) {
                if (!hash_equals($record->request_hash,$hash)) return new
JsonResponse(['error'=>['code'=>'CLN_IDEMPOTENCY_KEY_REUSED','message'=>'The key was used for a different request.'],409);
                if ($record->status==='COMPLETED') return new JsonResponse($record->response_body,$record->response_status);
                return new JsonResponse(['error'=>['code'=>'CLN_IDEMPOTENCY_IN_PROGRESS','message'=>'The request is already
processing.','retryable'=>true]],409);
            }
            $now=new DateTimeImmutable('now',new DateTimeZone('UTC'));
            $record=ClinicalIdempotencyRecord::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>(string)$context-
>tenantId,'scope_code'=>$scope,'idempotency_key'=>$key,'request_hash'=>$hash,'status'=>'PROCESSING','expires_at_utc'=>$now-
>modify('+'.config('clinical-foundation.idempotency_ttl_hours').' hours'),'created_at_utc'=>$now,'updated_at_utc'=>$now]);
            $response=$next($request); $body=json_decode((string)$response->getContent(),true);
            $record->forceFill(['status'=>'COMPLETED','response_status'=>$response-
>getStatusCode(),'response_body'=>is_array($body)?$body:['raw'=>(string)$response->getContent()],'updated_at_utc'=>new
DateTimeImmutable('now',new DateTimeZone('UTC')))->save();
            return $response;
        },3);
    }
}

```

src/Foundation/Http/Middleware/EstablishClinicalContext.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Http/Middleware/EstablishClinicalContext.php	24a00b8e7d4e8999a54d3dde2ab5cbcc4f4f3d5914abca953234a277b818fd75

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Http\Middleware;
use Closure;
use Illuminate\Http\Request;
use Symfony\Component\HttpFoundation\Response;
use Symfony\Component\HttpKernel\Exception\AccessDeniedHttpException;
use KashTre\Clinical\Foundation\Context\ClinicalRequestContext;
use KashTre\Clinical\Foundation\Contracts\{PublicIdGenerator, TenantResolver};
use KashTre\Clinical\Foundation\ValueObjects\{CorrelationId, PublicId};
final readonly class EstablishClinicalContext
{
    public function __construct(private TenantResolver $tenants, private PublicIdGenerator $ids) {}
    public function handle(Request $request, Closure $next): Response
    {
        $user=$request->user();
        if ($user===null || !isset($user->public_id)) throw new AccessDeniedHttpException('Authenticated public actor identity is
required.');
```

```

        $tenant=$this->tenants->resolve($request);
        $claim=(string)($user->tenant_id ?? '');
        if ($claim==='') throw new AccessDeniedHttpException('Tenant context mismatch.');
```

```

        $incoming=$request->header((string)config('clinical-foundation.correlation_header'));
        $correlation=new CorrelationId($incoming ? new PublicId($incoming) : $this->ids->generate());
        $context=new ClinicalRequestContext($tenant,new PublicId((string)$user->public_id),$correlation,(string)$this->ids-
>generate(),null,null,'WEB_API');
```

```

        app()->instance(ClinicalRequestContext::class,$context);
        $response=$next($request);
        $response->headers->set('X-Correlation-ID',(string)$correlation);
    }
}

```

```

        return $response;
    }
}

```

src/Foundation/Infrastructure/DatabaseAuditRecorder.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Infrastructure/DatabaseAuditRecorder.php	728fec7432e786c2f7f429dda42843e9285add91a19be6cd2a496299b07aa110

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Infrastructure;
use DateTimeImmutable;
use KashTre\Clinical\Foundation\Contracts\{AuditRecorder,PublicIdGenerator};
use KashTre\Clinical\Foundation\Data\AuditData;
use KashTre\Clinical\Foundation\Models\ClinicalAuditEntry;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
final readonly class DatabaseAuditRecorder implements AuditRecorder
{
    public function __construct(private PublicIdGenerator $ids) {}
    public function record(AuditData $data): PublicId
    {
        $id=$this->ids->generate(); $c=$data->context;
        ClinicalAuditEntry::query()->create(['public_id'=>(string)$id,'tenant_id'=>(string)$c->tenantId,'event_code'=>$data->eventCode,'outcome'=>$data->outcome,'actor_type'=>'USER','actor_public_id'=>(string)$c->actorPublicId,'correlation_id'=>(string)$c->correlationId,'request_id'=>$c->requestId,'source_channel'=>$c->sourceChannel,'details'=>$data->details,'occurred_at_utc'=>$data->occurredAtUtc,'recorded_at_utc'=>new DateTimeImmutable('now',new \DateTimeZone('UTC'))]);
        return $id;
    }
}

```

src/Foundation/Infrastructure/DatabaseOutboxPublisher.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Infrastructure/DatabaseOutboxPublisher.php	3bcdaa68c74cedc093d78b3bb0697c0081e223fa985fdc65f432cf81f19c5795

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Infrastructure;
use KashTre\Clinical\Foundation\Contracts\{EventPublisher,PublicIdGenerator};
use KashTre\Clinical\Foundation\Data\OutboxEventData;
use KashTre\Clinical\Foundation\Models\ClinicalOutboxMessage;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
final readonly class DatabaseOutboxPublisher implements EventPublisher
{
    public function __construct(private PublicIdGenerator $ids) {}
    public function store(OutboxEventData $event): PublicId
    {
        $id=$this->ids->generate();
        ClinicalOutboxMessage::query()->create(['public_id'=>(string)$id,'tenant_id'=>(string)$event->context->tenantId,'topic'=>$event->topic,'event_type'=>$event->eventType,'event_version'=>$event->eventVersion,'aggregate_type'=>$event->aggregateType,'aggregate_public_id'=>(string)$event->aggregatePublicId,'correlation_id'=>(string)$event->context->correlationId,'payload'=>$event->payload,'status'=>'PENDING','attempt_count'=>0,'available_at_utc'=>$event->occurredAtUtc,'created_at_utc'=>$event->occurredAtUtc]);
        return $id;
    }
}

```

src/Foundation/Infrastructure/HeaderTenantResolver.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Infrastructure/HeaderTenantResolver.php	e817df6c486e5bd8626e86bc410b931c65729f4251a1d7882420d8713ea61d81

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Infrastructure;
use Illuminate\Http\Request;
use Symfony\Component\HttpKernel\Exception\BadRequestHttpException;
use KashTre\Clinical\Foundation\Contracts\TenantResolver;
use KashTre\Clinical\Foundation\ValueObjects\TenantId;
final class HeaderTenantResolver implements TenantResolver
{
    public function resolve(Request $request): TenantId
    {
        $header=(string) config('clinical-foundation.tenant_header');
        $value=$request->header($header);
        if (!is_string($value) || $value==='') throw new BadRequestHttpException('Tenant context is required.');
```

src/Foundation/Infrastructure/SymfonyUlidGenerator.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Infrastructure/SymfonyUlidGenerator.php	919882cdf3ebc71371610959c161ca824158646d015d19e22531a3799499ee45

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Infrastructure;
use Symfony\Component\Uid\Ulid;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
final class SymfonyUlidGenerator implements PublicIdGenerator
{
    public function generate(): PublicId { return new PublicId((string) new Ulid()); }
```

src/Foundation/Models/ClinicalAuditEntry.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Models/ClinicalAuditEntry.php	06daa2254ee2823f0bda285ae8e2d65e6e406b7952485b65b3a6ddf6ffbea986

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Models;
use Illuminate\Database\Eloquent\Model;
final class ClinicalAuditEntry extends Model
{
    public $timestamps=false;
    protected $table='clinical_audit_entries';
    protected $guarded=[];
    protected $casts=['details'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime'];
}
```

src/Foundation/Models/ClinicalIdempotencyRecord.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Models/ClinicalIdempotencyRecord.php	153b866418450abb988181d544217f9aba5832a8119db0c4a0cb54bfd03eeac0

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Models;
use Illuminate\Database\Eloquent\Model;
final class ClinicalIdempotencyRecord extends Model
```

```
{
    public $timestamps=false;
    protected $table='clinical_idempotency_records';
    protected $guarded=[];
    protected $casts=['response_body'=>'array','locked_until_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];
}
```

src/Foundation/Models/ClinicalInboxMessage.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Models/ClinicalInboxMessage.php	c39dd62b80d685c93858d4aa48c91d539ddad0ef3df4cbb22ed9655e4adbb616

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Models;
use Illuminate\Database\Eloquent\Model;
final class ClinicalInboxMessage extends Model
{
    public $timestamps=false;
    protected $table='clinical_inbox_messages';
    protected $guarded=[];
    protected $casts=['payload'=>'array','received_at_utc'=>'immutable_datetime','processed_at_utc'=>'immutable_datetime'];
}
```

src/Foundation/Models/ClinicalOutboxMessage.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/Models/ClinicalOutboxMessage.php	00f85469d7c3daa98ef86a5da8b7953a305a75bcc8eb083719b00aad19e2b7a5

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\Models;
use Illuminate\Database\Eloquent\Model;
final class ClinicalOutboxMessage extends Model
{
    public $timestamps=false;
    protected $table='clinical_outbox_messages';
    protected $guarded=[];
    protected
    $casts=['payload'=>'array','available_at_utc'=>'immutable_datetime','published_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];
}
```

src/Foundation/ValueObjects/CorrelationId.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/ValueObjects/CorrelationId.php	368d07f9804ea72dca76499d58199293bbb7f043c9c842de3e71f81113def1b6

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\ValueObjects;
final readonly class CorrelationId
{
    public function __construct(public PublicId $id) {}
    public function __toString(): string { return (string) $this->id; }
}
```

src/Foundation/ValueObjects/PublicId.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/ValueObjects/PublicId.php	6d1f65ecb5ef87a0257e98fd387344d95155a119923a92008d1874d82bb6e337

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\ValueObjects;
use InvalidArgumentException;
final readonly class PublicId
{
    public function __construct(public string $value)
    {
        if (!preg_match('/^[0-9A-HJKMNP-TV-Z]{26}$/', $value)) throw new InvalidArgumentException('Invalid ULID.');
```

src/Foundation/ValueObjects/TenantId.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Foundation/ValueObjects/TenantId.php	83de16e78436098e11e4de5307453dfb6c9d78dd1baf1bfab07a7efc62f59d5b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Foundation\ValueObjects;
use InvalidArgumentException;
final readonly class TenantId
{
    public function __construct(public string $value)
    {
        if (!preg_match('/^[A-Za-z0-9][A-Za-z0-9._-:]{0,63}$/', $value)) throw new InvalidArgumentException('Invalid tenant
identifier.');
```

tests/Feature/FoundationMigrationTest.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Feature/FoundationMigrationTest.php	3e13023ab436e4adcb2e4f8f9e32dbca8600552db07f91e1d084158fc0449465

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Tests\Feature;
use Illuminate\Support\Facades\Schema;
use KashTre\Clinical\Tests\TestCase;
final class FoundationMigrationTest extends TestCase
{
    public function test_foundation_tables_exist(): void {
        foreach(['clinical_idempotency_records', 'clinical_outbox_messages', 'clinical_inbox_messages', 'clinical_audit_entries'] as $table)
            self::assertTrue(Schema::hasTable($table), $table);
    }
}
```

tests/TestCase.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/TestCase.php	61dd7267c24f21359ffd516fe09d2db2433b762b8b5972b39e1e2f5fb47a4275

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Tests;
use KashTre\Clinical\ClinicalFoundationServiceProvider;
use Orchestra\Testbench\TestCase as BaseTestCase;
abstract class TestCase extends BaseTestCase
{
    protected function getPackageProviders($app): array { return [ClinicalFoundationServiceProvider::class]; }
```

```
protected function defineDatabaseMigrations(): void { $this->loadMigrationsFrom(__DIR__.'/../database/migrations'); }
protected function defineEnvironment($app): void { $app['config']->set('database.default','testing');$app['config']->set('database.connections.testing',['driver'=>'sqlite','database'=>':memory:', 'prefix'=>'', 'foreign_key_constraints'=>true]); }
}
```

tests/Unit/PublicIdTest.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Unit/PublicIdTest.php	3c20865788e00b100a479d60b987e8ee1b5c98940fa6f32ea5ed83480a9521de

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Tests\Unit;
use InvalidArgumentException;
use KashTre\Clinical\Foundation\ValueObjects\PublicId;
use PHPUnit\Framework\TestCase;
final class PublicIdTest extends TestCase
{
    public function test_accepts_ulid(): void { self::assertSame('01ARZ3NDEKTSV4RRFFQ69G5FAV',(string)new PublicId('01ARZ3NDEKTSV4RRFFQ69G5FAV')); }
    public function test_rejects_invalid_value(): void { $this->expectException(InvalidArgumentException::class);new PublicId('bad'); }
}
```

tests/Unit/TenantIdTest.php

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Unit/TenantIdTest.php	51a9148b9c85946baa62d0e065a8b0941bced40df0f563020dcd1e8f482a4104

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Tests\Unit;
use InvalidArgumentException;
use KashTre\Clinical\Foundation\ValueObjects\TenantId;
use PHPUnit\Framework\TestCase;
final class TenantIdTest extends TestCase
{
    public function test_accepts_valid_identfier(): void { self::assertSame('TENANT-001',(string)new TenantId('TENANT-001')); }
    public function test_rejects_empty_identfier(): void { $this->expectException(InvalidArgumentException::class);new TenantId(''); }
    public function test_rejects_whitespace(): void { $this->expectException(InvalidArgumentException::class);new TenantId('TENANT 001'); }
}
```

verify.sh

Artifact status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	4642cae9bc2075013e03496f419d165013fc2c5baa0dc3baf96b2809a4b8b139

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Part IV. Binding and Verification Register

Repository Bindings Required Before VERIFIED Status

Binding	Current candidate assumption	Verification action
Actor identity	Authenticated model has public_id and tenant_id	Compile against actual model or introduce ActorIdentity contract.
Authentication	auth:sanctum middleware exists	Run route tests in KashTre host.
Main base URL	Environment-supplied URL and token	Bind approved Main Module service identity.
Shared Time endpoint	Configured path placeholder	Replace with approved contract path and DTOs.
Shared Unit endpoint	Configured path placeholder	Bind approved CORE_UNIT_ENGINE contract.
Queue/broker	Outbox storage only in this package	Install approved publisher adapter.
Database	MySQL 8.4 target	Run clean, upgrade and rollback suites.
Frontend	No foundation UI required	Later volumes consume API/error contracts.

Verification Commands

```
composer validate --strict
composer dump-autoload --strict-psr
php artisan migrate:fresh --env=testing
php artisan migrate:rollback --step=1 --env=testing
php artisan migrate --env=testing
./vendor/bin/phpunit
php artisan route:list --path=api/v1/clinical/foundation
```

Expected Evidence Record

Evidence ID	Command	Required result
V1-EV-001	composer validate --strict	PASS
V1-EV-002	composer dump-autoload --strict-psr	PASS
V1-EV-003	migrate:fresh	PASS on MySQL 8.4
V1-EV-004	rollback + migrate	PASS without residue
V1-EV-005	PHPUnit	All package tests PASS
V1-EV-006	Host route list	Foundation health route registered
V1-EV-007	Cross-tenant test suite	All negative tests PASS
V1-EV-008	Security review	No unresolved critical/high issue

Registered Blocking Gaps

Gap	Resolution required	Effect
V1-GAP-001	Obtain and compile against the actual KashTre repository	Blocks VERIFIED.
V1-GAP-002	Approve Laravel 13 and PHP 8.4 migration from SRD reference baseline	Blocks production baseline.
V1-GAP-003	Approve authenticated actor adapter contract	Blocks context middleware verification.
V1-GAP-004	Approve Main Module identity, permission and client-space API contract	Blocks contextual authorization package.
V1-GAP-005	Approve and bind Shared Time Engine API	Blocks time adapter verification.
V1-GAP-006	Approve and bind Shared Unit Engine API	Blocks unit adapter verification.
V1-GAP-007	Select broker and implement tested outbox dispatcher	Blocks integration event delivery.

Gap	Resolution required	Effect
V1-GAP-008	Approve audit retention, legal hold and tamper-evidence mechanism	Blocks production governance.
V1-GAP-009	Execute clean, upgrade, rollback, concurrency and security tests	Blocks VERIFIED.
V1-GAP-010	Generate signed evidence manifest from CI	Blocks release approval.

Production Gate

- All package files match the manifest checksums.
- All repository bindings are approved and implemented.
- All listed evidence records pass in CI.
- No TODO, ellipsis, pseudocode or unresolved import exists.
- Migrations pass clean, upgrade and rollback scenarios.
- Tenant, authorization, idempotency, audit and messaging negative tests pass.
- Security, database, architecture, QA and operations approvals are recorded.
- Only then change the classification to PICK-AND-DROP VERIFIED.

Source Register

Source	Role
KashTre_Clinical_Module_SRD_v6.1	Primary functional authority.
KashTre_Main_Module_Shared_Unit_Engine_Contract_SRD_v1.0_Draft	Unit ownership and consumer boundary.
KashTre_Main_Module_Shared_Unit_Engine_EDD_v1.0_Draft	Candidate unit-interface engineering reference.
KashTre_EDD_Multi_Tenant_Time_Zones	Time architecture reference.
KashTre AI Gateway SRD v3.1	Central AI gateway boundary.
Laravel 13 release policy	Supported framework baseline evidence.
PHP supported versions	Runtime lifecycle evidence.
MySQL 8.4 LTS documentation	Database track evidence.

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 2

EFFECTIVE PATIENT OBSERVATION PLAN

Requirements, Consolidation, Scheduling, Satisfaction and Transfer Continuity

Version: 6.1.0 Comprehensive Draft

Classification: Repository-Ready Reference Implementation

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Depends on: Volume 1 Executable Foundation

Production: Not Approved for Production

Document Authority and Non-Negotiable Rule

The Observation Plan creates obligations and due work. It never creates, copies or fabricates patient Observation values. One eligible authoritative Observation may satisfy multiple applicable requirements while every source remains traceable.

Area	Controlled decision
Sources	Baseline, client-space profile, form, order, protocol, care plan, medication monitoring, patient-specific and transfer continuity.
Consolidation	Same target and compatible context become one obligation.
Strictness	Shortest compatible interval wins; source provenance is retained.
History	Past due events are not rewritten by prospective recalculation.
Satisfaction	Tenant, patient, encounter, target, time window, status, quality and context must match.
Transfer	Unresolved due work moves once with original lateness evidence.
Rendering	Separate consumer; failure cannot roll back committed Observation or satisfaction.

Entity Completion Register

Entity	Authority	Identity	Lifecycle	Critical invariant
RequirementSource	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
EffectiveObservationPlan	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
PlanRecalculation	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
ObservationObligation	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
ObligationSource	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
ObservationDueEvent	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
ObservationSatisfaction	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained

PlanException	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
PlanReconciliation	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained
TransferLineage	Clinical	ULID public ID	Effective-dated or event lifecycle	Tenant and patient/encounter constrained

Status Machines

Entity	States
Plan	ACTIVE, RECONCILIATION_REQUIRED, SUPERSEDED
Obligation	ACTIVE, ON_HOLD, ENDED, SUPERSEDED, RECONCILIATION_REQUIRED
Due event	SCHEDULED, DUE, OVERDUE, SATISFIED, NOT_OBTAINED, DEFERRED, CANCELLED_SOURCE_ENDED, TRANSFERRED, ENTERED_IN_ERROR, RECONCILIATION_REQUIRED

Engineering Design

Recalculation pipeline

- Lock encounter planning scope
- Gather versioned active sources
- Normalize to RequirementCandidate
- Evaluate applicability using explicit facts
- Group by semantic key
- Select stricter compatible cadence
- Persist plan generation, obligations and source links
- Materialize future due events
- Write audit and outbox facts

Satisfaction pipeline

- Select unresolved candidate due events under lock
- Match tenant, patient and encounter
- Match CDE or Observation Group target
- Match occurrence window and approved late-entry policy
- Match status, quality, context and source restrictions
- Create idempotent satisfaction link
- Mark event SATISFIED
- Publish satisfaction event

Transfer pipeline

- Confirm movement
- Create unique transfer lineage
- Freeze unresolved due/overdue work
- Recalculate destination plan
- Continue non-location sources
- End source-client-space-only requirements
- Create receiving work once
- Preserve scheduled and overdue timestamps
- Publish transfer event

Permissions and API

Permission	Purpose
clinical.observation_plan.view	View plan and explanation
clinical.observation_plan.recalculate	Request recalculation
clinical.observation_plan.override	Create hold/exemption/override
clinical.monitoring.resolve_missed	Resolve not obtained or defer
clinical.observation_plan.reconcile	Resolve planning conflicts
clinical.observation_plan.transfer	Transfer responsibility

Mandatory Tests

- Three sources for BP produce one due event
- Hourly order overrides compatible four-hourly form cadence
- Source ending does not end obligation required elsewhere
- Repeated recalculation produces no duplicate generation
- Repeated scheduler run creates no duplicate occurrence
- One Observation satisfies multiple obligations where eligible
- Entered-in-error Observation reverses satisfaction
- Transfer moves overdue work once without resetting lateness
- Cancelled transfer has no planning effect
- Cross-tenant source is rejected
- Missing applicability fact enters safe failure
- Rendering outage does not affect plan or Observation commit

Repository-Ready Source Listings

The ZIP is authoritative. Each complete file below is reproduced with its checksum.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	c175d4fcb8752ab0407cfca7a0431f0813210607bfa5cfe6b623e1058935c2a0

```
# KashTre Clinical Observation Plan v6.1.0
Classification: PICK-AND-DROP CANDIDATE. Install after `kashtre/clinical-foundation`.
The host must bind patient, encounter, CDE, form, order, protocol, care-plan, medication, client-space and transfer source contracts
before VERIFIED status.
Commands: `composer install`, `php artisan migrate`, `composer test`.
```

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	1d32b07f95d69bdbec41a5443c146f39b5e866d1fec294a1a0e51f7574e9777e

```
{
  "volume": "Clinical EDD Volume 2 Observation Plan",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "c175d4fcb8752ab0407cfca7a0431f0813210607bfa5cfe6b623e1058935c2a0",
      "bytes": 355
    },
    {
      "path": "composer.json",
      "sha256": "b629b4a507a40127f8a746db782704410b076ec829dd3bf427140f948f9a47fb",
      "bytes": 592
    },
    {
      "path": "config/clinical-observation-plan.php",
      "sha256": "de1d9ea2c0abebab5d9246a7ee4df8dcd8aad984d1668a30034ae890e9b0090",
      "bytes": 381
    },
    {
      "path": "database/migrations/2026_08_28_100000_create_observation_plan_tables.php",
      "sha256": "bc1ab93fe5741ee658f91030f5b383ea6689135d8a9782a0bd5a68c75c0c1d26",
      "bytes": 5496
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-observation-plan.php",
      "sha256": "4585f72c02546a73f5037bfacfddea4def18eb4307b7be37445e7cf45fe4952d",
      "bytes": 381
    },
    {
      "path": "src/Actions/RecalculateObservationPlan.php",
      "sha256": "88fbe613528acc855f54ab276c6d56f0f204d6849a5f84d97722197406b39037",
      "bytes": 3562
    },
    {
      "path": "src/Actions/SatisfyDueEvents.php",
```

```

"sha256": "04b5fcd57c5b1c95c5bc3d804559eca7d4210f7bffb43e0ebf34e44ad24a2109",
"bytes": 1674
},
{
"path": "src/Contracts/ApplicabilityEvaluator.php",
"sha256": "c17424fc51c5ebaedab3b5df0a74e4c74597b9591b458302de6b3c4443d086de",
"bytes": 261
},
{
"path": "src/Contracts/RequirementSourceProvider.php",
"sha256": "db500d94e84b75395081b61d08b82d1e58042680b897793c92f07fe10b0156b5",
"bytes": 222
},
{
"path": "src/Data/RequirementCandidate.php",
"sha256": "e4dbd1b8743b3ff1845e4da0645e944b2ac196fa9d204f21747a23f05c60576a",
"bytes": 1041
},
{
"path": "src/Enums/DueEventStatus.php",
"sha256": "793c5cdcd08580df9a11aaa6f25573efc13fb0e25e884dbde188aea985e4537",
"bytes": 460
},
{
"path": "src/Enums/ObligationStatus.php",
"sha256": "107535da5cd465aa6ca23abc5180a444337d1d18361d238348c0403a4b1c46f6",
"bytes": 269
},
{
"path": "src/Enums/PlanStatus.php",
"sha256": "d8f20c6f016d8aaa30a4314e141f5820ee7d53d2e0ced646226d8df167026a3a",
"bytes": 217
},
{
"path": "src/Enums/RequirementSourceType.php",
"sha256": "8a486310cc4f3040d61ab86ea538ed524836fc4f751c4a0f99eb70a889e27f80",
"bytes": 496
},
{
"path": "src/Enums/ScheduleMode.php",
"sha256": "44cf6f8663e4b4eab8357ad1b4bd9325860e5739cac64d875700ab4cc4e375b4",
"bytes": 308
},
{
"path": "src/Enums/TargetType.php",
"sha256": "170e16ff6115adf426e724d0ca7ceca4683ca3a694b7e5a0b57e86c8110fb54c",
"bytes": 168
},
{
"path": "src/Models/EffectiveObservationPlan.php",
"sha256": "d50f0d5255695c5b53e67921c983be2008d20a673866674c4ee548f340274fc3",
"bytes": 603
},
{
"path": "src/Models/ObligationSource.php",
"sha256": "31818fc649b02a9518070dbd2d954ab909d3625cf49aad787405c8917441c89",
"bytes": 586
},
{
"path": "src/Models/ObservationDueEvent.php",
"sha256": "e82b216969200f3bf4a61cd0c4e8f012a20edb7d4c4291916b0c8a4e91283f4a",
"bytes": 593
},
{
"path": "src/Models/ObservationObligation.php",
"sha256": "8cd1fea918fad945e7b3dce6f30fb16e4f70bd114ae29e0ee4e856dc3e00c332",
"bytes": 596
},
{
"path": "src/Models/ObservationSatisfaction.php",
"sha256": "86f084175094626d5276c4db8e286a0913fcbee57ea56a1e4871251bc4b20ca0",
"bytes": 600
},
},

```

```
{
  {
    "path": "src/Models/PlanException.php",
    "sha256": "2eeb7623fbcd68bea9bbb97f37a367d08f25ecfa7c7cb95e2f1540746596787",
    "bytes": 580
  },
  {
    "path": "src/Models/PlanRecalculation.php",
    "sha256": "92caa7b3eb6fc1ee171294b5b5a4d147662732b7da08afea93127a214d46ee76",
    "bytes": 588
  },
  {
    "path": "src/Models/PlanReconciliation.php",
    "sha256": "8b92cb3242440df206c99862b02299a9cf09fc77d4a43147197d17e59f9a2131",
    "bytes": 590
  },
  {
    "path": "src/Models/RequirementSource.php",
    "sha256": "4d220ecddb3df460a465d122170ab0116b0cde352cfab097f3f20f3c5b9c133b",
    "bytes": 588
  },
  {
    "path": "src/Models/TransferLineage.php",
    "sha256": "a1b3a29a295b91dad433045394c6c367665a3ca855991ca4c3eae928ad88be4c",
    "bytes": 584
  },
  {
    "path": "src/ObservationPlanServiceProvider.php",
    "sha256": "244c1d08f419597a0762ed88c7ee0d8710064cce9534310753320f5a99df8a26",
    "bytes": 889
  },
  {
    "path": "src/Services/DatabaseRequirementSourceProvider.php",
    "sha256": "49b0cc41bc43cd2b40da09d867270a2441e036974ebe9089124b8f71da2f52f6",
    "bytes": 626
  },
  {
    "path": "src/Services/DeterministicApplicabilityEvaluator.php",
    "sha256": "a1bf367609517899c1d59f3f3bd3e48a773c1185687d02d798d579f3b07b11d6",
    "bytes": 589
  },
  {
    "path": "src/Services/DueEventScheduler.php",
    "sha256": "bd28d78b0b50d926585ef08f1e99a5643d2febb1da085ec4ec74ea9bd7a4259d",
    "bytes": 1293
  },
  {
    "path": "src/Services/RequirementConsolidator.php",
    "sha256": "b3327763c33ae8b65d57ffeff55d09ce86a071151f5cb811c8d26030070b3919",
    "bytes": 829
  },
  {
    "path": "tests/Unit/RequirementConsolidatorTest.php",
    "sha256": "99a3b61102fa445f20681a65a90e52609f33d7422698673df102ec8c82631e16",
    "bytes": 966
  },
  {
    "path": "verify.sh",
    "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
    "bytes": 121
  }
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	b629b4a507a40127f8a746db782704410b076ec829dd3bf427140f948f9a47fb

```
{
  "name": "kashtre/clinical-observation-plan",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "require-dev": {
      "orchestra/testbench": "^11.0",
      "phpunit/phpunit": "^12.0",
      "autoload": {
        "psr-4": {
          "KashTre\\Clinical\\ObservationPlan\\": "src/"
        },
        "autoload-dev": {
          "psr-4": {
            "KashTre\\Clinical\\ObservationPlan\\Tests\\": "tests/"
          },
          "extra": {
            "laravel": {
              "providers": [
                "KashTre\\Clinical\\ObservationPlan\\ObservationPlanServiceProvider"
              ]
            },
            "scripts": {
              "test": "phpunit"
            }
          }
        }
      }
    }
  }
}
```

config/clinical-observation-plan.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-observation-plan.php	de1d9ea2c0abebab5d9246a7ee4df8dcdd8aad984d1668a30034ae890e9b0090

```
<?php
declare(strict_types=1);
return [
    'materialization_horizon_hours' => (int) env('CLINICAL_OBSERVATION_HORIZON_HOURS', 48),
    'due_projection_minutes' => (int) env('CLINICAL_OBSERVATION_DUE_PROJECTION_MINUTES', 1),
    'default_early_window_minutes' => (int) env('CLINICAL_OBSERVATION_EARLY_WINDOW_MINUTES', 0),
    'default_late_window_minutes' => (int) env('CLINICAL_OBSERVATION_LATE_WINDOW_MINUTES', 0)];
```

database/migrations/2026_08_28_100000_create_observation_plan_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_100000_create_observation_plan_tables.php	bc1ab93fe5741ee658f91030f5b383ea6689135d8a9782a0bd5a68c75c0c1d26

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_requirement_sources', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('source_public_id', 64);
            $table->string('source_type', 40);
            $table->string('source_public_id', 64);
            $table->unsignedInteger('source_version');
            $table->string('status', 24);
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('effective_to_utc', 6);
            $table->char('semantic_hash', 64);
            $table->json('source_snapshot');
            $table->dateTime('created_at_utc', 6);
            $table->dateTime('updated_at_utc', 6);
            $table->unique(['tenant_id', 'source_type', 'source_public_id', 'source_version'], 'uq_obs_req_source');
        });

        Schema::create('clinical_effective_observation_plans', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('patient_public_id', 64);
            $table->string('encounter_public_id', 64);
            $table->unsignedInteger('generation');
            $table->string('status', 32);
            $table->char('input_fingerprint', 64);
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('superseded_at_utc', 6);
            $table->string('recalculation_public_id', 64);
            $table->dateTime('created_at_utc', 6);
            $table->unique(['tenant_id', 'patient_public_id', 'encounter_public_id', 'generation'], 'uq_obs_plan_generation');
        });

        Schema::create('clinical_plan_recalculations', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('patient_public_id', 64);
            $table->string('encounter_public_id', 64);
            $table->char('input_fingerprint', 64);
            $table->string('status', 32);
            $table->unsignedInteger('source_count')->default(0);
            $table->unsignedInteger('obligation_count')->default(0);
            $table->dateTime('started_at_utc', 6);
            $table->dateTime('completed_at_utc', 6);
            $table->unique(['tenant_id', 'patient_public_id', 'encounter_public_id', 'input_fingerprint'], 'uq_obs_recalculation');
        });

        Schema::create('clinical_observation_obligations', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('plan_public_id', 64);
            $table->string('patient_public_id', 64);
            $table->string('encounter_public_id', 64);
            $table->string('target_type', 24);
            $table->string('target_public_id', 64);
            $table->unsignedInteger('target_version');
            $table->char('semantic_key', 64);
            $table->string('schedule_mode', 32);
            $table->unsignedInteger('interval_minutes')->default(0);
            $table->dateTime('anchor_at_utc', 6);
            $table->unsignedInteger('early_window_minutes')->default(0);
            $table->unsignedInteger('late_window_minutes')->default(0);
            $table->string('responsible_role_code', 96);
            $table->string('priority', 24);
            $table->string('status', 32);
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('effective_to_utc', 6);
            $table->json('decision_evidence');
            $table->dateTime('created_at_utc', 6);
            $table->dateTime('updated_at_utc', 6);
            $table->unique(['plan_public_id', 'semantic_key'], 'uq_obs_obligation_semantic');
        });

        Schema::create('clinical_observation_obligation_sources', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('obligation_public_id', 64);
            $table->string('requirement_source_public_id', 64);
            $table->string('contribution_type', 32);
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('effective_to_utc', 6);
            $table->unique(['obligation_public_id', 'requirement_source_public_id'], 'uq_obs_obligation_source');
        });

        Schema::create('clinical_observation_due_events', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('obligation_public_id', 64);
            $table->string('patient_public_id', 64);
            $table->string('encounter_public_id', 64);
            $table->unsignedInteger('occurrence_sequence');
            $table->dateTime('scheduled_at_utc', 6);
            $table->dateTime('window_start_utc', 6);
            $table->
```

```

>dateTime('window_end_utc',6);$t->string('status',32);$t->string('responsible_role_code',96);$t-
>ulid('responsible_client_space_public_id')->nullable();$t->unsignedBigInteger('record_version')->default(1);$t-
>ulid('transfer_lineage_public_id')->nullable();$t->dateTime('satisfied_at_utc',6)->nullable();$t->dateTime('created_at_utc',6);$t-
>dateTime('updated_at_utc',6);$t->unique(['obligation_public_id','occurrence_sequence'],'uq_obs_due_occurrence');$t-
>index(['tenant_id','status','window_start_utc'],'idx_obs_due_worklist');});
Schema::create('clinical_observation_satisfactions',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('observation_public_id');$t->ulid('due_event_public_id');$t->ulid('obligation_public_id');$t-
>string('match_outcome',24);$t->json('match_evidence');$t->boolean('is_primary')->default(true);$t-
>dateTime('satisfied_at_utc',6);$t->dateTime('reversed_at_utc',6)->nullable();$t-
>unique(['observation_public_id','due_event_public_id'],'uq_obs_satisfaction');});
Schema::create('clinical_plan_exceptions',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('patient_public_id');$t->ulid('encounter_public_id');$t->ulid('obligation_public_id')-
>nullable();$t->string('exception_type',32);$t->string('reason_code',64);$t->text('justification')->nullable();$t-
>dateTime('effective_from_utc',6);$t->dateTime('effective_to_utc',6);$t->ulid('authorized_by_public_id');$t->string('status',24);$t-
>dateTime('created_at_utc',6));});
}public function
down():void{foreach(['clinical_plan_exceptions','clinical_observation_satisfactions','clinical_observation_due_events','clinical_obli
gation_sources','clinical_observation_obligations','clinical_plan_recalculations','clinical_effective_observation_plans','clinical_re
quirement_sources'] as $t)Schema::dropIfExists($t);};

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -eou pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-observation-plan.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-observation-plan.php	4585f72c02546a73f5037bfacfddea4def18eb4307b7be37445e7cf45fe4952d

```

<?php
declare(strict_types=1);
use Illuminate\Support\Facades\Route;
Route::middleware(['api','auth:sanctum'])->prefix('api/v1/clinical')-
>group(function(){Route::get('/patients/{patient}/encounters/{encounter}/observation-plan',fn()->response()-
>json(['error'=>['code'=>'CLN_OBS_PLAN_HOST_CONTROLLER_REQUIRED','message'=>'Bind the host controller before
verification.'],501));});

```

src/Actions/RecalculateObservationPlan.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RecalculateObservationPlan.php	88fbe613528acc855f54ab276c6d56f0f204d6849a5f84d97722197406b39037

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Actions;
use Illuminate\Support\Facades\DB;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;
use KashTre\Clinical\ObservationPlan\Contracts\{ApplicabilityEvaluator,RequirementSourceProvider};
use KashTre\Clinical\ObservationPlan\Models\{EffectiveObservationPlan,ObservationObligation,ObligationSource,PlanRecalculation};
use KashTre\Clinical\ObservationPlan\Services\RequirementConsolidator;

```

```

final readonly class RecalculateObservationPlan{
    public function __construct(private RequirementSourceProvider $sources,private ApplicabilityEvaluator $applicability,private
    RequirementConsolidator $consolidator,private PublicIdGenerator $ids){}
    public function execute(string $tenantId,string $patientId,string $encounterId,array $facts,array
    $candidates):EffectiveObservationPlan{return
    DB::transaction(function(){use($tenantId,$patientId,$encounterId,$facts,$candidates){$active=array_values(array_filter($candidates,fn(
    $c)=>$this->applicability->
    >applicable($c,$facts));$fingerprint=hash('sha256',serialize($active));$existing=PlanRecalculation::query()-
    >where('tenant_id',$tenantId)->where('patient_public_id',$patientId)->where('encounter_public_id',$encounterId)-
    >where('input_fingerprint',$fingerprint)->where('status','COMPLETED')->first();if($existing)return EffectiveObservationPlan::query()-
    >where('recalculation_public_id',$existing->public_id)->firstOrFail();$run=PlanRecalculation::query()-
    >create(['public_id'=>(string)$this->ids-
    >generate(),'tenant_id'=>$tenantId,'patient_public_id'=>$patientId,'encounter_public_id'=>$encounterId,'input_fingerprint'=>$fingerprint,
    'status'=>'PROCESSING','started_at_utc'=>now('UTC')));$generation=((int)EffectiveObservationPlan::query()-
    >where('tenant_id',$tenantId)->where('patient_public_id',$patientId)->where('encounter_public_id',$encounterId)-
    >max('generation'))+1;$plan=EffectiveObservationPlan::query()->create(['public_id'=>(string)$this->ids-
    >generate(),'tenant_id'=>$tenantId,'patient_public_id'=>$patientId,'encounter_public_id'=>$encounterId,'generation'=>$generation,'sta
    tus'=>'ACTIVE','input_fingerprint'=>$fingerprint,'effective_from_utc'=>now('UTC'),'recalculation_public_id'=>$run-
    >public_id,'created_at_utc'=>now('UTC'))];foreach($this->consolidator->consolidate($active) as
    $g){$w=$g['winner'];$ob=ObservationObligation::query()->create(['public_id'=>(string)$this->ids-
    >generate(),'tenant_id'=>$tenantId,'plan_public_id'=>$plan-
    >public_id,'patient_public_id'=>$patientId,'encounter_public_id'=>$encounterId,'target_type'=>$w->targetType-
    >value,'target_public_id'=>$w->targetPublicId,'target_version'=>$w-
    >targetVersion,'semantic_key'=>$g['semantic_key'],'schedule_mode'=>$w->scheduleMode->value,'interval_minutes'=>$w-
    >intervalMinutes,'anchor_at_utc'=>$w->anchorAtUtc,'early_window_minutes'=>$w->earlyWindowMinutes,'late_window_minutes'=>$w-
    >lateWindowMinutes,'responsible_role_code'=>$w->responsibleRoleCode,'priority'=>$w-
    >priority,'status'=>'ACTIVE','effective_from_utc'=>$w-
    >effectiveFromUtc,'decision_evidence'=>[ 'source_count'=>count($g['sources']), 'created_at_utc'=>now('UTC'), 'updated_at_utc'=>now('UTC
    ')]);foreach($g['sources'] as $source)ObligationSource::query()->create(['obligation_public_id'=>$ob-
    >public_id,'requirement_source_public_id'=>$source,'contribution_type'=>'CONSOLIDATED','effective_from_utc'=>$w-
    >effectiveFromUtc]);}$run-
    >forceFill(['status'=>'COMPLETED','source_count'=>count($active),'obligation_count'=>count($active),'completed_at_utc'=>now('UTC')])>
    >save();return $plan;});}
}

```

src/Actions/SatisfyDueEvents.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/SatisfyDueEvents.php	04b5fcd57c5b1c95c5bc3d804559eca7d4210f7bffb43e0ebf34e44ad24a2109

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Actions;
use Illuminate\Support\Facades\DB;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;
use KashTre\Clinical\ObservationPlan\Models\{ObservationDueEvent,ObservationSatisfaction};
final readonly class SatisfyDueEvents{public function __construct(private PublicIdGenerator $ids){}public function execute(string
$tenantId,string $patientId,string $encounterId,string $targetId,string $observationId,\DateTimeImmutable $occurredAt):int{return
DB::transaction(function(){use($tenantId,$patientId,$encounterId,$targetId,$observationId,$occurredAt){$events=ObservationDueEvent::qu
ery()->where('tenant_id',$tenantId)->where('patient_public_id',$patientId)->where('encounter_public_id',$encounterId)-
>whereIn('status',['SCHEDULED','DUE','OVERDUE'])->where('window_start_utc','<=', $occurredAt)-
>where('window_end_utc','>=', $occurredAt)->lockForUpdate()->get();$count=0;foreach($events as $e){$ob=$e-
>belongsTo(\KashTre\Clinical\ObservationPlan\Models\ObservationObligation::class,'obligation_public_id','public_id')-
>first();if(!$ob||$ob->target_public_id!=$targetId)continue;ObservationSatisfaction::query()-
>firstOrCreate(['observation_public_id'=>$observationId,'due_event_public_id'=>$e->public_id,['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'obligation_public_id'=>$ob-
>public_id,'match_outcome'=>'ELIGIBLE','match_evidence'=>[ 'matched_at'=>$occurredAt-
>format(DATE_ATOM)], 'is_primary'=>true, 'satisfied_at_utc'=>$occurredAt]);$e-
>forceFill(['status'=>'SATISFIED','satisfied_at_utc'=>$occurredAt,'updated_at_utc'=>now('UTC')])>save();$count++;}return
$count;});}
}

```

src/Contracts/ApplicabilityEvaluator.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Contracts/ApplicabilityEvaluator.php	c17424fc51c5ebaedab3b5df0a74e4c74597b9591b458302de6b3c4443d086de
-------------------------	--	--

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Contracts;
use KashTre\Clinical\ObservationPlan\Data\RequirementCandidate;
interface ApplicabilityEvaluator{public function applicable(RequirementCandidate $candidate,array $facts):bool;}
```

src/Contracts/RequirementSourceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/RequirementSourceProvider.php	db500d94e84b75395081b61d08b82d1e58042680b897793c92f07fe10b0156b5

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Contracts;
interface RequirementSourceProvider{public function activeFor(string $tenantId,string $patientPublicId,string $encounterPublicId):array;}
```

src/Data/RequirementCandidate.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/RequirementCandidate.php	e4dbd1b8743b3ff1845e4da0645e944b2ac196fa9d204f21747a23f05c60576a

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Data;
use DateTimeImmutable;
use KashTre\Clinical\ObservationPlan\Enums\{RequirementSourceType,ScheduleMode,TargetType};
final readonly class RequirementCandidate{public function __construct(public string $sourcePublicId,public RequirementSourceType $sourceType,public TargetType $targetType,public string $targetPublicId,public int $targetVersion,public ScheduleMode $scheduleMode,public ?int $intervalMinutes,public DateTimeImmutable $anchorAtUtc,public int $earlyWindowMinutes,public int $lateWindowMinutes,public string $responsibleRoleCode,public string $priority,public array $clinicalContext,public DateTimeImmutable $effectiveFromUtc,public ?DateTimeImmutable $effectiveToUtc){if($scheduleMode===ScheduleMode::INTERVAL&&(!$intervalMinutes||$intervalMinutes<1))throw new \InvalidArgumentException('INTERVAL requires positive intervalMinutes.');
```

src/Enums/DueEventStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DueEventStatus.php	793c5cdcdca08580df9a11aaa6f25573efc13fb0e25e884dbde188aea985e4537

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Enums;
enum DueEventStatus:string{
    case SCHEDULED='SCHEDULED';
    case DUE='DUE';
```

```

case OVERDUE='OVERDUE';
case SATISFIED='SATISFIED';
case NOT_OBTAINED='NOT_OBTAINED';
case DEFERRED='DEFERRED';
case CANCELLED_SOURCE_ENDED='CANCELLED_SOURCE_ENDED';
case TRANSFERRED='TRANSFERRED';
case ENTERED_IN_ERROR='ENTERED_IN_ERROR';
case RECONCILIATION_REQUIRED='RECONCILIATION_REQUIRED';
}

```

src/Enums/ObligationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ObligationStatus.php	107535da5cd465aa6ca23abc5180a444337d1d18361d238348c0403a4b1c46f6

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Enums;
enum ObligationStatus:string{
    case ACTIVE='ACTIVE';
    case ON_HOLD='ON_HOLD';
    case ENDED='ENDED';
    case SUPERSEDED='SUPERSEDED';
    case RECONCILIATION_REQUIRED='RECONCILIATION_REQUIRED';
}

```

src/Enums/PlanStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/PlanStatus.php	d8f20c6f016d8aaa30a4314e141f5820ee7d53d2e0ced646226d8df167026a3a

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Enums;
enum PlanStatus:string{
    case ACTIVE='ACTIVE';
    case RECONCILIATION_REQUIRED='RECONCILIATION_REQUIRED';
    case SUPERSEDED='SUPERSEDED';
}

```

src/Enums/RequirementSourceType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/RequirementSourceType.php	8a486310cc4f3040d61ab86ea538ed524836fc4f751c4a0f99eb70a889e27f80

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Enums;
enum RequirementSourceType:string{
    case UNIVERSAL_BASELINE='UNIVERSAL_BASELINE';
    case CLIENT_SPACE_PROFILE='CLIENT_SPACE_PROFILE';
    case FORM_ACTIVATION='FORM_ACTIVATION';
    case CLINICAL_ORDER='CLINICAL_ORDER';
    case PROTOCOL='PROTOCOL';
    case CARE_PLAN='CARE_PLAN';
}

```

```

case MEDICATION_MONITORING='MEDICATION_MONITORING';
case PATIENT_SPECIFIC='PATIENT_SPECIFIC';
case TRANSFER_CONTINUITY='TRANSFER_CONTINUITY';
}

```

src/Enums/ScheduleMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ScheduleMode.php	44cf6f8663e4b4eab8357ad1b4bd9325860e5739cac64d875700ab4cc4e375b4

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Enums;
enum ScheduleMode:string{
    case INTERVAL='INTERVAL';
    case CALENDAR='CALENDAR';
    case ONCE='ONCE';
    case EVENT_RELATIVE='EVENT_RELATIVE';
    case PRN_REASSESSMENT='PRN_REASSESSMENT';
    case CONTINUOUS_REVIEW='CONTINUOUS_REVIEW';
}

```

src/Enums/TargetType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/TargetType.php	170e16ff6115adf426e724d0ca7ceca4683ca3a694b7e5a0b57e86c8110fb54c

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Enums;
enum TargetType:string{
    case CDE='CDE';
    case OBSERVATION_GROUP='OBSERVATION_GROUP';
}

```

src/Models/EffectiveObservationPlan.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/EffectiveObservationPlan.php	d50f0d5255695c5b53e67921c983be2008d20a673866674c4ee548f340274fc3

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class EffectiveObservationPlan extends Model{public $timestamps=false;protected
$table='clinical_effective_observation_plans';protected $guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'immutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}

```

src/Models/ObligationSource.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Models/ObligationSource.php	31818fc649b02a9518070dbd2d954ab909d3625cf49aadcf787405c8917441c89
-------------------------	---------------------------------	---

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class ObligationSource extends Model{public $timestamps=false;protected $table='clinical_obligation_sources';protected $guarded=[];protected $casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'immutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationDueEvent.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationDueEvent.php	e82b216969200f3bf4a61cd0c4e8f012a20edb7d4c4291916b0c8a4e91283f4a

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationDueEvent extends Model{public $timestamps=false;protected $table='clinical_observation_due_events';protected $guarded=[];protected $casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'immutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationObligation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationObligation.php	8cd1fea918fad945e7b3dce6f30fb16e4f70bd114ae29e0ee4e856dc3e00c332

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationObligation extends Model{public $timestamps=false;protected $table='clinical_observation_obligations';protected $guarded=[];protected $casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'immutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationSatisfaction.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationSatisfaction.php	86f084175094626d5276c4db8e286a0913fcbec57ea56a1e4871251bc4b20ca0

```
<?php
declare(strict_types=1);
```

```
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationSatisfaction extends Model{public $timestamps=false;protected
$table='clinical_observation_satisfactions';protected $guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'im
mutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_da
tetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/PlanException.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PlanException.php	2eeb7623fbcda68bea9bbb97f37a367d08f25ecfa7c7cb95e2f1540746596787

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class PlanException extends Model{public $timestamps=false;protected $table='clinical_plan_exceptions';protected
$guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'im
mutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_da
tetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/PlanRecalculation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PlanRecalculation.php	92caa7b3eb6fc1ee171294b5b5a4d147662732b7da08afea93127a214d46ee76

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class PlanRecalculation extends Model{public $timestamps=false;protected $table='clinical_plan_recalculations';protected
$guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'im
mutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_da
tetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/PlanReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PlanReconciliation.php	8b92cb3242440df206c99862b02299a9cf09fc77d4a43147197d17e59f9a2131

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class PlanReconciliation extends Model{public $timestamps=false;protected $table='clinical_plan_reconciliations';protected
$guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'im
mutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_da
tetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/RequirementSource.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/RequirementSource.php	4d220ecddb3df460a465d122170ab0116b0cde352cfab097f3f20f3c5b9c133b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class RequirementSource extends Model{public $timestamps=false;protected $table='clinical_requirement_sources';protected
$guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'im
mutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_da
atetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/TransferLineage.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransferLineage.php	a1b3a29a295b91dad433045394c6c367665a3ca855991ca4c3eae928ad88be4c

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Models;
use Illuminate\Database\Eloquent\Model;
final class TransferLineage extends Model{public $timestamps=false;protected $table='clinical_transfer_lineages';protected
$guarded=[];protected
$casts=['metadata'=>'array','decision_evidence'=>'array','source_snapshot'=>'array','match_evidence'=>'array','scheduled_at_utc'=>'im
mutable_datetime','window_start_utc'=>'immutable_datetime','window_end_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_da
atetime','effective_to_utc'=>'immutable_datetime'];}
```

src/ObservationPlanServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ObservationPlanServiceProvider.php	244c1d08f419597a0762ed88c7ee0d8710064cce9534310753320f5a99df8a26

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan;
use Illuminate\Support\ServiceProvider;
use KashTre\Clinical\ObservationPlan\Contracts\{ApplicabilityEvaluator,RequirementSourceProvider};
use KashTre\Clinical\ObservationPlan\Services\{DeterministicApplicabilityEvaluator,DatabaseRequirementSourceProvider};
final class ObservationPlanServiceProvider extends ServiceProvider{
    public function register():void{$this->mergeConfigFrom(__DIR__.'/../config/clinical-observation-plan.php','clinical-observation-
plan');$this->app->bind(ApplicabilityEvaluator::class,DeterministicApplicabilityEvaluator::class);$this->app-
>bind(RequirementSourceProvider::class,DatabaseRequirementSourceProvider::class);}
    public function boot():void{$this->loadMigrationsFrom(__DIR__.'/../database/migrations');$this-
>loadRoutesFrom(__DIR__.'/../routes/clinical-observation-plan.php');}
}
```

src/Services/DatabaseRequirementSourceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/DatabaseRequirementSourceProvider.php	49b0cc41bc43cd2b40da09d867270a2441e036974ebe9089124b8f71da2f52f6

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Services;
use KashTre\Clinical\ObservationPlan\Contracts\RequirementSourceProvider;
use KashTre\Clinical\ObservationPlan\Models\RequirementSource;
final class DatabaseRequirementSourceProvider implements RequirementSourceProvider{public function activeFor(string $tenantId,string $patientPublicId,string $encounterPublicId):array{return RequirementSource::query()->where('tenant_id',$tenantId)->where('patient_public_id',$patientPublicId)->where('encounter_public_id',$encounterPublicId)->where('status','ACTIVE')->orderBy('public_id')->get()->all();}}
```

src/Services/DeterministicApplicabilityEvaluator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/DeterministicApplicabilityEvaluator.php	a1bf367609517899c1d59f3f3bd3e48a773c1185687d02d798d579f3b07b11d6

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Services;
use KashTre\Clinical\ObservationPlan\Contracts\ApplicabilityEvaluator;
use KashTre\Clinical\ObservationPlan\Data\RequirementCandidate;
final class DeterministicApplicabilityEvaluator implements ApplicabilityEvaluator{public function applicable(RequirementCandidate $candidate,array $facts):bool{foreach($candidate->clinicalContext as $key=>$expected){if(!array_key_exists($key,$facts))throw new \DomainException('Missing applicability fact: '.$key);if($facts[$key]==$expected)return false;}}return true;}}
```

src/Services/DueEventScheduler.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/DueEventScheduler.php	bd28d78b0b50d926585ef08f1e99a5643d2febb1da085ec4ec74ea9bd7a4259d

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Services;
use DateTimeImmutable;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;
use KashTre\Clinical\ObservationPlan\Models\{ObservationDueEvent,ObservationObligation};
final readonly class DueEventScheduler{public function __construct(private PublicIdGenerator $ids){}public function materialize(ObservationObligation $o,DateTimeImmutable $until):int{if(!$o->interval_minutes)return 0;$n=0;$cursor=new DateTimeImmutable($o->anchor_at_utc->format(DATE_ATOM));$seq=0;while($cursor<=$until){$seq++;$start=$cursor->modify('-'. $o->early_window_minutes.' minutes');$end=$cursor->modify('+'. $o->late_window_minutes.' minutes');ObservationDueEvent::query()->firstOrCreate(['obligation_public_id'=>$o->public_id,'occurrence_sequence'=>$seq],['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$o->tenant_id,'patient_public_id'=>$o->patient_public_id,'encounter_public_id'=>$o->encounter_public_id,'scheduled_at_utc'=>$cursor,'window_start_utc'=>$start,'window_end_utc'=>$end,'status'=>'SCHEDULED','responsible_role_code'=>$o->responsible_role_code,'record_version'=>1,'created_at_utc'=>now('UTC'),'updated_at_utc'=>now('UTC'))];$n++;$cursor=$cursor->modify('+'. $o->interval_minutes.' minutes');}return $n;}}
```

src/Services/RequirementConsolidator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/RequirementConsolidator.php	b3327763c33ae8b65d57ffeff55d09ce86a071151f5cb811c8d26030070b3919

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Services;
use KashTre\Clinical\ObservationPlan\Data\RequirementCandidate;
final class RequirementConsolidator{
    public function consolidate(array $candidates):array{$groups=[];foreach($candidates as $c){if(!$c instanceof RequirementCandidate)throw new \InvalidArgumentException('RequirementCandidate required.');
```

tests/Unit/RequirementConsolidatorTest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Unit/RequirementConsolidatorTest.php	99a3b61102fa445f20681a65a90e52609f33d7422698673df102ec8c82631e16

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\ObservationPlan\Tests\Unit;
use DateTimeImmutable;use PHPUnit\Framework\TestCase;use KashTre\Clinical\ObservationPlan\Data\RequirementCandidate;use KashTre\Clinical\ObservationPlan\Enums\{RequirementSourceType,ScheduleMode,TargetType};use KashTre\Clinical\ObservationPlan\Services\RequirementConsolidator;
final class RequirementConsolidatorTest extends TestCase{public function test_selects_shorter_compatible_interval_and_retains_sources():void{$at=new DateTimeImmutable('2026-08-28T08:00:00Z');
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Binding, Gaps and Verification

Binding	Required before VERIFIED
Volume 1 package	Install and verify exact dependency.
Patient/encounter repositories	Bind actual KashTre models and tenant-safe lookups.
CDE and group registry	Bind stable IDs and versions.
Source adapters	Bind forms, orders, protocols, care plans, medication and client-space profiles.
Authorization	Bind Volume 1 contextual policy service.
Time	Bind approved Shared Time Engine.
Transfer	Bind confirmed movement contract.
Observation	Bind authoritative Observation commit and correction events.

Blocking Gaps

- Execute against actual KashTre repository
- Approve equivalence and schedule compatibility matrix
- Approve timing windows and late-entry rules
- Approve responsible role resolution
- Approve applicability expression contract
- Approve transfer handoff timing
- Complete API controllers and Vue worklist against host route/model bindings
- Run clean, upgrade, rollback, concurrency, load and security tests

Verification Commands

```
composer validate --strict
composer dump-autoload --strict-psr
php artisan migrate:fresh --env=testing
php artisan migrate:rollback --step=1 --env=testing
php artisan migrate --env=testing
./vendor/bin/phpunit
```

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Observation-first authority.
Volume 1 Executable Foundation	Tenant, IDs, audit, messaging and idempotency.
HL7 FHIR R5 Observation	External model reference for measurements and assertions.
HL7 FHIR R5 PlanDefinition	External reference for reusable executable plan definitions.

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 3

CDE REGISTRY AND AUTHORITATIVE OBSERVATION CAPTURE

Manual, Device, Calculated, Normalized, Corrected and Provenance-Controlled Observations

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 and 2

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Safety Rule

Clinical Observations are authoritative patient-specific event records. Original values, original units, normalized values, status, effective time, performer, source and provenance are preserved. No guessed unit, incompatible conversion, stale range, wrong-patient association or destructive correction is permitted.

Entity	Purpose	Critical invariant
CdeDefinition	CDE Definition	Tenant-scoped, ULID-addressed and historically preserved
CdeVersion	CDE Version	Tenant-scoped, ULID-addressed and historically preserved
CdeAllowedUnit	CDE AllowedUnit	Tenant-scoped, ULID-addressed and historically preserved
CdeValueSetItem	CDE ValueSetItem	Tenant-scoped, ULID-addressed and historically preserved
CdeLimitPolicy	CDE LimitPolicy	Tenant-scoped, ULID-addressed and historically preserved
Observation	Observation	Tenant-scoped, ULID-addressed and historically preserved
ObservationComponent	Observation Component	Tenant-scoped, ULID-addressed and historically preserved
ObservationGroup	Observation Group	Tenant-scoped, ULID-addressed and historically preserved
ObservationProvenance	Observation Provenance	Tenant-scoped, ULID-addressed and historically preserved
ObservationCorrection	Observation Correction	Tenant-scoped, ULID-addressed and historically preserved
DevicePayload	DevicePayload	Tenant-scoped, ULID-addressed and historically preserved
DeviceMapping	DeviceMapping	Tenant-scoped, ULID-addressed and historically preserved
ObservationReconciliation	Observation Reconciliation	Tenant-scoped, ULID-addressed and historically preserved

Observation Lifecycle

State	Meaning
REGISTERED	Stored before validation where policy permits
PRELIMINARY	Available with visible preliminary status
FINAL	Authoritative completed result
AMENDED	Supplemented without replacing meaning
CORRECTED	Superseded by linked replacement
CANCELLED	Workflow cancelled with evidence
ENTERED_IN_ERROR	Excluded from ordinary use, audit retained

Capture Channels

Method	Commit rule
--------	-------------

MANUAL	Authorized user, CDE and unit validation required
DEVICE	Stage first; patient, encounter, CDE, unit, time and device provenance validated
CALCULATED	Input observations and algorithm version retained
IMPORTED	Source system, message identity and version retained
VOICE_ASSISTED	Draft only until human confirmation
LINKED_EVENT	Source clinical event remains traceable

Engineering Design**CDE governance**

Stable definition plus immutable versions

Data type and semantic meaning versioned

Allowed input and display units reference Main Unit Engine IDs

Value sets are versioned

Physiological warning and block limits are effective-dated

Activation requires review and approval

Observation validation

Patient and encounter exact match

CDE version active at occurrence

Value kind matches CDE type

Unit required and allowed for quantities

Conversion uses Shared Unit Engine

Original and normalized values both persist

Physiological limit outcome explicit

Performer, source and effective time required

Device processing

Idempotent message identity

Raw payload hash preserved

No automatic chart commit by default

Mapping uses device plus source code

Stale patient/bed association blocks commit

Quarantine rather than guessing

Correction

No destructive edit of final observation

Replacement observation is a complete event

Original status changes to CORRECTED

Correction reason and linkage persist

Plan satisfaction, score and alert reconciliation triggered

Permissions

Permission	Purpose
clinical.cde_config.manage	Create and maintain CDE drafts
clinical.cde_config.approve	Approve and activate versions
clinical.observation.view	View authorized observations
clinical.observation.create	Record manual observation
clinical.observation.validate	Validate staged observation
clinical.observation.override_limit	Override eligible warning
clinical.observation.correct	Create correction
clinical.observation.mark_entered_in_error	Invalidate erroneous observation
clinical.observation.device_reconcile	Resolve device payload exception

clinical.observation.audit

[View detailed provenance](#)**Mandatory Tests**

Quantity without unit is blocked

Disallowed unit is blocked

Unit Engine failure never relabels original value

Physiological block prevents commit

Warning requires visible outcome

Device retry creates one payload

Wrong-patient device association is quarantined

Final observation cannot be edited in place

Correction preserves original and replacement

Entered-in-error reverses plan satisfaction

One valid observation satisfies all eligible plan obligations

Cross-tenant CDE or patient reference is rejected

Repository-Ready Source Listings

The source ZIP is authoritative. Each complete listing is reproduced with checksum.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	1c3fc9363a6930a2a1ed8bb7fda5f20325a1d6b4062f94001f37b7a8a4760e51

KashTre Clinical CDE Registry and Observation Capture v6.1.0

Install after Volumes 1 and 2. Classification: PICK-AND-DROP CANDIDATE. SharedUnitGateway, ObservationPlanSatisfactionGateway, patient/encounter authority, authorization and API controllers must be bound in the KashTre repository before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	e5ab89076fa0dbe405675e54984c188742d251471ff5d635a75464e1d4cd821a

```
{
  "volume": "Clinical EDD Volume 3",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "1c3fc9363a6930a2a1ed8bb7fda5f20325a1d6b4062f94001f37b7a8a4760e51",
      "bytes": 316
    },
    {
      "path": "composer.json",
      "sha256": "9a3b81257ecc746e5ba4f43dcb87571b356e64bf1a41594b4b2516ee539153b",
      "bytes": 614
    },
    {
      "path": "config/clinical-observation.php",
      "sha256": "9a21607a81709b3776632c8f2fc80438a9279dd8e2f7fa142356e031da1a18bc",
      "bytes": 222
    },
    {
      "path": "database/migrations/2026_08_28_200000_create_cde_observation_tables.php",
      "sha256": "482b16d380fb9019421b86f74a0251fa5345cbee8fbd7d971ad8242b5fee2cbc",
      "bytes": 6195
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-observation.php",
      "sha256": "960d431ebe34746defc4d76a872c2961e0d7692b11663322d302c9e03557e94d",
      "bytes": 314
    }
  ]
}
```

```

{
  "path": "src/Actions/CorrectObservation.php",
  "sha256": "942e8eb6dfdc7f61e2e5131249f6bc6b3ce84f3c985f82264d101a1bdfaeb15e",
  "bytes": 1306
},
{
  "path": "src/Actions/RecordObservation.php",
  "sha256": "f4ccf72f8584e65d89e5459f5271b1f927c1e6c5f5855d3b2d301f5c4c6c9335",
  "bytes": 2976
},
{
  "path": "src/Actions/StageDevicePayload.php",
  "sha256": "e6f043ebe21a4b3fdb07402a79b2fe556e40e0ac987753b025a38d1bb6e2aa70",
  "bytes": 758
},
{
  "path": "src/Contracts/ObservationPlanSatisfactionGateway.php",
  "sha256": "fe13e3be32a72ecb015056aecd11a632769d13d3411c12bd62b5f49bb138e97",
  "bytes": 278
},
{
  "path": "src/Contracts/SharedUnitGateway.php",
  "sha256": "8ece0f454d185fdb9b72da0ea831f76fe5a338544de547dc419c6ffc6f8c61e6",
  "bytes": 323
},
{
  "path": "src/Data/NormalizedQuantity.php",
  "sha256": "e21fd0f53b80dee26f7dd2fb48c9f2a7b878bae5a818cecd4419297fc2b35edd",
  "bytes": 358
},
{
  "path": "src/Data/RecordObservationData.php",
  "sha256": "0ed957601c0411660f5fd79ca4150397d210ed8e27184340d2dede561f089804",
  "bytes": 671
},
{
  "path": "src/Enums/CaptureMethod.php",
  "sha256": "71ff88c7d07f3c6a56bb383191d75e3b4f2e382dc09ec724f0259fe72a04cd47",
  "bytes": 283
},
{
  "path": "src/Enums/CdeDataType.php",
  "sha256": "a90641ca14d9d2720c9cf9f1941676bf5ba342cbd6a4215955f2f67b154d031b",
  "bytes": 404
},
{
  "path": "src/Enums/CdeStatus.php",
  "sha256": "fec2677250e94ec2b755b714a109b187fde02b8ee5f335da881368688adedfe9",
  "bytes": 257
},
{
  "path": "src/Enums/DevicePayloadStatus.php",
  "sha256": "a81f108956a25e6c7693fad96f98c56532d00196fcbab5b2b6cee1cefab82b2",
  "bytes": 279
},

```

```

{
  "path": "src/Enums/ObservationStatus.php",
  "sha256": "5ef8122f9a6a6673c0bb37e1afa7babf228f872f9cbe8a96ffd7d93b01759c0c",
  "bytes": 320
},
{
  "path": "src/Enums/ValidationStatus.php",
  "sha256": "553a1637fbb0a7c0db033e54c19bdcb15c0aa1a2fff946f0499348ad7359a55a",
  "bytes": 270
},
{
  "path": "src/Enums/ValueKind.php",
  "sha256": "8dc8ae023b25803b775750eef6626d90d5f48168ffbac9c881481072ee44433",
  "bytes": 421
},
{
  "path": "src/Models/CdeAllowedUnit.php",
  "sha256": "22851077510216c5321672a2499b0638763a2057b788478f2338ca36ec13cc39",
  "bytes": 559
},
{
  "path": "src/Models/CdeDefinition.php",
  "sha256": "ad94210213eced10f941cf7cf93c7666062a0837680a2c5fdb15c59caae07727",
  "bytes": 556
},
{
  "path": "src/Models/CdeLimitPolicy.php",
  "sha256": "43349ae7da59e6e39c33bd7a670b1f5f801df1878ee57594294eab413e04565b",
  "bytes": 560
},
{
  "path": "src/Models/CdeValueSetItem.php",
  "sha256": "36d043f5c7de15c7055ceb17753f16630905b6744ac595238299e0d0daf6636b",
  "bytes": 562
},
{
  "path": "src/Models/CdeVersion.php",
  "sha256": "c1be0e2e5f9e4f7eb8718bbd409d21707f6ee83a284f1fd486838feececeda01",
  "bytes": 550
},
{
  "path": "src/Models/DeviceMapping.php",
  "sha256": "9bbc9a87e4588f1123e052d013feaa695fc54820ecaa635a825d77af5419b153",
  "bytes": 556
},
{
  "path": "src/Models/DevicePayload.php",
  "sha256": "5d58ab30b4fac9ee544e1199d570e746d9cda670a36d6f78a88dfb0fdc58ffb9",
  "bytes": 556
},
{
  "path": "src/Models/Observation.php",
  "sha256": "9556e0671153ae740024f7fc2931540a8b0449f91401e928241ee05ca972bd1e",
  "bytes": 551
},

```

```

{
  "path": "src/Models/ObservationComponent.php",
  "sha256": "077cf729d208040032051b435db112a0e76205967003f61879cb266fc03730a0",
  "bytes": 570
},
{
  "path": "src/Models/ObservationCorrection.php",
  "sha256": "82dc4debdda9894393875782b9f2d8d6c94495e7454e576609d0595b5477526b",
  "bytes": 572
},
{
  "path": "src/Models/ObservationGroup.php",
  "sha256": "560d011fcc5994f47c1b9be0ff246c459d7891a2209c60bcf344f608391018aa",
  "bytes": 562
},
{
  "path": "src/Models/ObservationProvenance.php",
  "sha256": "96f4c82c4d3a8d1c90b15bdc750098dc5ccc5abbd1bbd4627e942d13b17b781e",
  "bytes": 571
},
{
  "path": "src/Models/ObservationReconciliation.php",
  "sha256": "ca6c3de50bc01ec275e40a115e2c6e8cbb6f8b30824d81f2b4c9e44f749e8257",
  "bytes": 580
},
{
  "path": "src/ObservationServiceProvider.php",
  "sha256": "bee4f75acd4b503eb47d3bb29c90f7f69a866257a516408d1dc6fcaef7ed0fc4",
  "bytes": 559
},
{
  "path": "src/Services/CdeValidator.php",
  "sha256": "47bed5ba6bc6e0212cb7a53cae5f0dfc227c7671605d1d26a94aec03e3d155fb",
  "bytes": 779
},
{
  "path": "src/Services/PhysiologicalLimitEvaluator.php",
  "sha256": "ab1a2fc8f1c8323f4627ad5998f47264392ec4f15fe565d0262069557bf4cbe8",
  "bytes": 911
},
{
  "path": "tests/Unit/RecordObservationDataTest.php",
  "sha256": "5ca721ea9a681ff1997a50d397a7984dd3e2d3ed4f329e2e28851def2a003837",
  "bytes": 736
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
]
}

```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	9a3b812578ecc746e5ba4f43dcb87571b356e64bf1a41594b4b2516ee539153b

```
{
  "name": "kashtre/clinical-observation",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "kashtre/clinical-observation-plan": "^6.1"
  },
  "require-dev": {
    "orchestra/testbench": "^11.0",
    "phpunit/phpunit": "^12.0"
  },
  "autoload": {
    "psr-4": {
      "KashTre\\Clinical\\Observation\\": "src/"
    },
    "autoload-dev": {
      "psr-4": {
        "KashTre\\Clinical\\Observation\\Tests\\": "tests/"
      }
    },
    "extra": {
      "laravel": {
        "providers": [
          "KashTre\\Clinical\\Observation\\ObservationServiceProvider"
        ]
      }
    },
    "scripts": {
      "test": "phpunit"
    }
  }
}
```

config/clinical-observation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-observation.php	9a21607a81709b3776632c8f2fc80438a9279dd8e2f7fa142356e031da1a18bc

```
<?php
declare(strict_types=1);
return
[
    'decimal_scale' => (int) env('CLINICAL_OBSERVATION_DECIMAL_SCALE', 18),
    'physiological_override_requires_reason' => true,
    'device_auto_commit' => false,
    'allow_preliminary_satisfaction' => false
];
```

database/migrations/2026_08_28_200000_create_cde_observation_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_200000_create_cde_observation_tables.php	482b16d380fb9019421b86f74a0251fa5345cbee8fbd7d971ad8242b5fee2cbc

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_cde_definitions', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->string('code', 96);
            $table->string('name', 180);
            $table->string('data_type', 32);
            $table->string('status', 24);
            $table->unsignedInteger('current_version')->nullable();
            $table->dateTime('created_at_utc', 6);
            $table->dateTime('updated_at_utc', 6);
            $table->unique(['tenant_id', 'code'], 'uq_cde_code');
        });

        Schema::create('clinical_cde_versions', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('cde_public_id');
            $table->unsignedInteger('version_no');
            $table->string('status', 24);
            $table->string('data_type', 32);
            $table->ulid('base_unit_public_id')->nullable();
            $table->unsignedTinyInteger('calculation_scale')->default(18);
            $table->string('rounding_mode', 32)->default('HALF_UP');
            $table->json('metadata')->nullable();
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('effective_to_utc', 6)->nullable();
            $table->unique(['tenant_id', 'cde_public_id', 'version_no'], 'uq_cde_version');
        });

        Schema::create('clinical_cde_allowed_units', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->ulid('cde_public_id');
            $table->unsignedInteger('cde_version');
            $table->ulid('unit_public_id');
            $table->boolean('is_input_allowed')->default(true);
            $table->boolean('is_display_preferred')->default(false);
            $table->unique(['tenant_id', 'cde_public_id', 'cde_version', 'unit_public_id'], 'uq_cde_unit');
        });

        Schema::create('clinical_cde_value_set_items', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->ulid('cde_public_id');
            $table->unsignedInteger('cde_version');
            $table->string('code', 96);
            $table->string('display', 180);
            $table->string('status', 24);
            $table->unsignedInteger('display_order')->default(1);
            $table->unique(['tenant_id', 'cde_public_id', 'cde_version', 'code'], 'uq_cde_value_code');
        });

        Schema::create('clinical_cde_limit_policies', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
        });
    }
};
```

```

>string('tenant_id',64);$t->ulid('cde_public_id');$t->unsignedInteger('cde_version');$t->string('warning_min_decimal',100)-
>nullable();$t->string('warning_max_decimal',100)->nullable();$t->string('block_min_decimal',100)->nullable();$t-
>string('block_max_decimal',100)->nullable();$t->string('status',24);$t->dateTime('effective_from_utc',6);$t-
>dateTime('effective_to_utc',6)->nullable();});
Schema::create('clinical_observation_groups',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('patient_public_id');$t->ulid('encounter_public_id');$t->string('group_type',64);$t-
>dateTime('occurred_at_utc',6);$t->ulid('performer_public_id');$t->dateTime('recorded_at_utc',6);});
Schema::create('clinical_observations',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>ulid('patient_public_id');$t->ulid('encounter_public_id');$t->ulid('cde_public_id');$t->unsignedInteger('cde_version');$t-
>string('status',24);$t->string('value_kind',24);$t->json('value_json')->nullable();$t->ulid('original_unit_public_id')->nullable();$t-
>string('normalized_value_decimal',100)->nullable();$t->ulid('base_unit_public_id')->nullable();$t-
>string('conversion_rule_version',64)->nullable();$t->string('validation_status',24);$t->string('capture_method',32);$t-
>dateTime('occurred_at_utc',6);$t->dateTime('recorded_at_utc',6);$t->ulid('performer_public_id');$t->ulid('device_public_id')-
>nullable();$t->ulid('source_event_public_id')->nullable();$t->ulid('group_public_id')->nullable();$t->json('context')->nullable();$t-
>unsignedBigInteger('record_version')->default(1);$t-
>index(['tenant_id','patient_public_id','encounter_public_id','cde_public_id','occurred_at_utc'],'idx_obs_patient_cde_time'));
Schema::create('clinical_observation_components',function(Blueprint $t){$t->id();$t->ulid('observation_public_id');$t-
>ulid('component_cde_public_id');$t->unsignedInteger('component_cde_version');$t->json('value_json')->nullable();$t-
>ulid('unit_public_id')->nullable();$t->unsignedSmallInteger('display_order');$t-
>unique(['observation_public_id','component_cde_public_id'],'uq_obs_component'));
Schema::create('clinical_observation_provenance',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('observation_public_id');$t->string('source_type',32);$t->ulid('source_public_id')->nullable();$t-
>json('metadata')->nullable();$t->dateTime('recorded_at_utc',6);});
Schema::create('clinical_observation_corrections',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('original_observation_public_id');$t->ulid('replacement_observation_public_id');$t->text('reason');$t-
>dateTime('corrected_at_utc',6);$t->unique('replacement_observation_public_id');});
Schema::create('clinical_device_payloads',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>ulid('device_public_id');$t->string('source_message_id',160);$t->char('payload_hash',64);$t->json('payload');$t-
>string('status',24);$t->dateTime('received_at_utc',6);$t->dateTime('committed_at_utc',6)->nullable();$t->text('failure_detail')-
>nullable();$t->unique(['tenant_id','device_public_id','source_message_id'],'uq_device_message'));
Schema::create('clinical_device_mappings',function(Blueprint $t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>ulid('device_public_id');$t->string('source_code',128);$t->ulid('cde_public_id');$t->unsignedInteger('cde_version');$t-
>ulid('unit_public_id')->nullable();$t->string('status',24);$t-
>unique(['tenant_id','device_public_id','source_code'],'uq_device_mapping'));
}public function
down():void{foreach(['clinical_device_mappings','clinical_device_payloads','clinical_observation_corrections','clinical_observation_
provenance','clinical_observation_components','clinical_observations','clinical_observation_groups','clinical_cde_limit_policies','cli
nical_cde_value_set_items','clinical_cde_allowed_units','clinical_cde_versions','clinical_cde_definitions']as$t)Schema::dropIfExists
($t);}

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-observation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical- observation.php	960d431ebe34746defc4d76a872c2961e0d7692b11663322d302c9e03557e94d

```
<?php
declare(strict_types=1);
use Illuminate\Support\Facades\Route;
Route::middleware(['api','auth:sanctum'])->prefix('api/v1/clinical')-
>group(function(){Route::post('/patients/{patient}/encounters/{encounter}/observations',fn()->response()-
>json(['error'=>['code'=>'CLN_OBS_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/CorrectObservation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CorrectObservation.php	942e8eb6dfdc7f61e2e5131249f6bc6b3ce84f3c985f82264d101a1bdfaeb15e

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Actions;
use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Observation\Models\{Observation,ObservationCorrection};
final readonly class CorrectObservation{public function __construct(private PublicIdGenerator $ids,private RecordObservation
$record){}public function execute(string $tenantId,string $originalId,string
$reason,\KashTre\Clinical\Observation\Data\RecordObservationData $replacement):Observation{return
DB::transaction(function()use($tenantId,$originalId,$reason,$replacement){$original=Observation::query()-
>where('tenant_id',$tenantId)->where('public_id',$originalId)->lockForUpdate()->firstOrFail();if(!in_array($original-
>status,['FINAL','AMENDED','CORRECTED'],true))throw new \DomainException('CLN_OBS_NOT_CORRECTABLE');$corrected=$this-
>record->execute($replacement);$original->forceFill(['status'=>'CORRECTED','record_version'=>$original->record_version+1])-
>save();ObservationCorrection::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId,'original_observation_public_id'=>$originalId,'replacement_observation_public_id'=>$corrected-
>public_id,'reason'=>$reason,'corrected_at_utc'=>now('UTC')]);return $corrected;});}
```

src/Actions/RecordObservation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RecordObservation.php	f4ccf72f8584e65d89e5459f5271b1f927c1e6c5f5855d3b2d301f5c4c6c9335

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Actions;
use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\{AuditRecorder,EventPublisher,PublicIdGenerator};use
KashTre\Clinical\Observation\Contracts\{ObservationPlanSatisfactionGateway,SharedUnitGateway};use
KashTre\Clinical\Observation\Data\RecordObservationData;use
KashTre\Clinical\Observation\Models\{Observation,ObservationProvenance};use
KashTre\Clinical\Observation\Services\{CdeValidator,PhysiologicalLimitEvaluator};
final readonly class RecordObservation{public function __construct(private CdeValidator $cdes,private PhysiologicalLimitEvaluator
$limits,private SharedUnitGateway $units,private ObservationPlanSatisfactionGateway $plan,private PublicIdGenerator $ids,private
AuditRecorder $audit,private EventPublisher $events){}public function execute(RecordObservationData $d):Observation{return
DB::transaction(function()use($d){$cde=$this->cdes->activeVersion($d->tenantId,$d->cdePublicId,$d-
>cdeVersion);$normalized=null;if($d->valueKind==='QUANTITY'){if(!$d->unitPublicId)throw new
\DomainException('CLN_OBS_UNIT_REQUIRED');$this->cdes->assertUnitAllowed($d->tenantId,$d->cdePublicId,$d-
>cdeVersion,$d->unitPublicId);$normalized=$this->units->normalize($d->tenantId,(string)$d->value,$d->unitPublicId,$cde-
>base_unit_public_id,(int)$cde->calculation_scale,(string)$cde->rounding_mode);$limit=$this->limits->evaluate($d->tenantId,$d-
```

```

>cdePublicId,$d->cdeVersion,(string)$d->value);if($limit['outcome']==='BLOCK')throw new
\DomainException('CLN_OBS_PHYSIOLOGICAL_LIMIT_BLOCK');$id=(string)$this->ids->generate();$o=Observation::query()-
>create(['public_id'=>$id,'tenant_id'=>$d->tenantId,'patient_public_id'=>$d->patientPublicId,'encounter_public_id'=>$d-
>encounterPublicId,'cde_public_id'=>$d->cdePublicId,'cde_version'=>$d->cdeVersion,'status'=>'FINAL','value_kind'=>$d-
>valueKind,'value_json'=>['value'=>$d->value],'original_unit_public_id'=>$d-
>unitPublicId,'normalized_value_decimal'=>$normalized?->normalizedValue,'base_unit_public_id'=>$normalized?-
>baseUnitPublicId,'conversion_rule_version'=>$normalized?-
>conversionRuleVersion,'validation_status'=>$limit['outcome']==='WARNING'?'WARNING':'VALID','capture_method'=>$d-
>captureMethod->value,'occurred_at_utc'=>$d->occurredAtUtc,'recorded_at_utc'=>now('UTC'),'performer_public_id'=>$d-
>performerPublicId,'device_public_id'=>$d->devicePublicId,'source_event_public_id'=>$d-
>sourceEventPublicId,'group_public_id'=>$d->groupPublicId,'context'=>$d-
>context,'record_version'=>1]);ObservationProvenance::query()->create(['public_id'=>(string)$this->ids->generate(), 'tenant_id'=>$d-
>tenantId,'observation_public_id'=>$id,'source_type'=>$d->captureMethod->value,'source_public_id'=>$d-
>sourceEventPublicId,'metadata'=>['limit_policy'=>$limit['policy']], 'recorded_at_utc'=>now('UTC'))];$this->plan->satisfy($d-
>tenantId,$d->patientPublicId,$d->encounterPublicId,$d->cdePublicId,$id,$d->occurredAtUtc);return $o;});}

```

src/Actions/StageDevicePayload.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/StageDevicePayload.php	e6f043ebe21a4b3fdb07402a79b2fe556e40e0ac987753b025a38d1bb6e2aa70

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Actions;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Observation\Models\DevicePayload;
final readonly class StageDevicePayload{public function __construct(private PublicIdGenerator $ids){}public function execute(string
$tenantId,string $deviceId,string $messageId,array $payload,\DateTimeImmutable $receivedAt):DevicePayload{return
DevicePayload::query()-
>firstOrCreate(['tenant_id'=>$tenantId,'device_public_id'=>$deviceId,'source_message_id'=>$messageId],['public_id'=>(string)$this
->ids-
>generate(), 'payload_hash'=>hash('sha256',json_encode($payload,JSON_THROW_ON_ERROR)), 'payload'=>$payload,'status'=>'RE
CEIVED','received_at_utc'=>$receivedAt]);}

```

src/Contracts/ObservationPlanSatisfactionGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObservationPlanSatisfactionGateway.php	fe13e3be32a72ecb015056aecd11a632769d13d3411c12bd62b5f49bb138e97

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Contracts;
interface ObservationPlanSatisfactionGateway{public function satisfy(string $tenantId,string $patientId,string $encounterId,string
$cdeId,string $observationId,\DateTimeImmutable $occurredAt):int;}

```

src/Contracts/SharedUnitGateway.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Contracts/SharedUnitGateway.php	8ece0f454d185fdb9b72da0ea831f76fe5a338544de547dc419c6ffc6f8c61e6

CANDIDATE		
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```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Contracts;
use KashTre\Clinical\Observation\Data\NormalizedQuantity;
interface SharedUnitGateway{public function normalize(string $tenantId,string $value,string $unitPublicId,string
$baseUnitPublicId,int $scale,string $roundingMode):NormalizedQuantity;}
```

src/Data/NormalizedQuantity.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/NormalizedQuantity.php	e21fd0f53b80dee26f7dd2fb48c9f2a7b878bae5a818cecd4419297fc2b35edd

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Data;
final readonly class NormalizedQuantity{public function __construct(public string $originalValue,public string
$originalUnitPublicId,public string $normalizedValue,public string $baseUnitPublicId,public string $conversionRuleVersion,public
int $scale,public string $roundingMode){}}
```

src/Data/RecordObservationData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/RecordObservationData.php	0ed957601c0411660f5fd79ca4150397d210ed8e27184340d2dede561f089804

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Data;
use DateTimeImmutable;use KashTre\Clinical\Observation\Enums\CaptureMethod;
final readonly class RecordObservationData{public function __construct(public string $tenantId,public string $patientPublicId,public
string $encounterPublicId,public string $cdePublicId,public int $cdeVersion,public string $valueKind,public mixed $value,public
?string $unitPublicId,public DateTimeImmutable $occurredAtUtc,public string $performerPublicId,public CaptureMethod
$captureMethod,public ?string $devicePublicId,public ?string $sourceEventPublicId,public array $context,public ?string
$groupPublicId=null){}}
```

src/Enums/CaptureMethod.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CaptureMethod.php	71ff88c7d07f3c6a56bb383191d75e3b4f2e382dc09ec724f0259fe72a04cd47

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum CaptureMethod:string{
case MANUAL='MANUAL';
case DEVICE='DEVICE';}
```

```

case CALCULATED='CALCULATED';
case IMPORTED='IMPORTED';
case VOICE_ASSISTED='VOICE_ASSISTED';
case LINKED_EVENT='LINKED_EVENT';
}

```

src/Enums/CdeDataType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CdeDataType.php	a90641ca14d9d2720c9cf9f1941676bf5ba342cbd6a4215955f2f67b154d031b

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum CdeDataType:string{
case QUANTITY='QUANTITY';
case INTEGER='INTEGER';
case DECIMAL='DECIMAL';
case BOOLEAN='BOOLEAN';
case CODE='CODE';
case STRING='STRING';
case TEXT='TEXT';
case DATE='DATE';
case DATETIME='DATETIME';
case TIME='TIME';
case RATIO='RATIO';
case RANGE='RANGE';
case ATTACHMENT='ATTACHMENT';
}

```

src/Enums/CdeStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CdeStatus.php	fec2677250e94ec2b755b714a109b187fde02b8ee5f335da881368688adedfe9

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum CdeStatus:string{
case DRAFT='DRAFT';
case IN_REVIEW='IN_REVIEW';
case APPROVED='APPROVED';
case ACTIVE='ACTIVE';
case DEPRECATED='DEPRECATED';
case RETIRED='RETIRED';
}

```

src/Enums/DevicePayloadStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DevicePayloadStatus.php	a81f108956a25e6c7693fad96f98c56532d00196fbcabab5b2b6cee1cefab82b2

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum DevicePayloadStatus:string{
    case RECEIVED='RECEIVED';
    case MAPPED='MAPPED';
    case VALIDATED='VALIDATED';
    case QUARANTINED='QUARANTINED';
    case REJECTED='REJECTED';
    case COMMITTED='COMMITTED';
}
```

src/Enums/ObservationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ObservationStatus.php	5ef8122f9a6a6673c0bb37e1afa7babf228f872f9cbe8a96ffd7d93b01759c0c

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum ObservationStatus:string{
    case REGISTERED='REGISTERED';
    case PRELIMINARY='PRELIMINARY';
    case FINAL='FINAL';
    case AMENDED='AMENDED';
    case CORRECTED='CORRECTED';
    case CANCELLED='CANCELLED';
    case ENTERED_IN_ERROR='ENTERED_IN_ERROR';
}
```

src/Enums/ValidationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ValidationStatus.php	553a1637fbb0a7c0db033e54c19bdcb15c0aa1a2fff946f0499348ad7359a55a

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum ValidationStatus:string{
    case PENDING='PENDING';
    case VALID='VALID';
    case WARNING='WARNING';
    case OVERRIDDEN='OVERRIDDEN';
    case QUARANTINED='QUARANTINED';
    case REJECTED='REJECTED';
}
```

src/Enums/ValueKind.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ValueKind.php	8dc8ae023b25803b775750eef6626d90d5f48168ffbaca9c881481072ee44433

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Enums;
enum ValueKind:string{
    case NONE='NONE';
    case QUANTITY='QUANTITY';
    case INTEGER='INTEGER';
    case DECIMAL='DECIMAL';
    case BOOLEAN='BOOLEAN';
    case CODE='CODE';
    case STRING='STRING';
    case TEXT='TEXT';
    case DATE='DATE';
    case DATETIME='DATETIME';
    case TIME='TIME';
    case RATIO='RATIO';
    case RANGE='RANGE';
    case ATTACHMENT='ATTACHMENT';
}
```

src/Models/CdeAllowedUnit.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CdeAllowedUnit.php	22851077510216c5321672a2499b0638763a2057b788478f2338ca36ec13cc39

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class CdeAllowedUnit extends Model{public $timestamps=false;protected $table='clinical_cde_allowed_units';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/CdeDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CdeDefinition.php	ad94210213eced10f941cf7cf93c7666062a0837680a2c5fdb15c59caae07727

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class CdeDefinition extends Model{public $timestamps=false;protected $table='clinical_cde_definitions';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/CdeLimitPolicy.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CdeLimitPolicy.php	43349ae7da59e6e39c33bd7a670b1f5f801df1878ee57594294eab413e04565b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class CdeLimitPolicy extends Model{public $timestamps=false;protected $table='clinical_cde_limit_policies';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/CdeValueSetItem.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CdeValueSetItem.php	36d043f5c7de15c7055ceb17753f16630905b6744ac595238299e0d0daf6636b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class CdeValueSetItem extends Model{public $timestamps=false;protected $table='clinical_cde_value_set_items';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/CdeVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CdeVersion.php	c1be0e2e5f9e4f7eb8718bbd409d21707f6ee83a284f1fd486838feececeda01

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class CdeVersion extends Model{public $timestamps=false;protected $table='clinical_cde_versions';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/DeviceMapping.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DeviceMapping.php	9bbc9a87e4588f1123e052d013feaa695fc54820ecaa635a825d77af5419b153

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class DeviceMapping extends Model{public $timestamps=false;protected $table='clinical_device_mappings';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/DevicePayload.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DevicePayload.php	5d58ab30b4fac9ee544e1199d570e746d9cda670a36d6f78a88dfb0fdc58ffb9

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class DevicePayload extends Model{public $timestamps=false;protected $table='clinical_device_payloads';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/Observation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/Observation.php	9556e0671153ae740024f7fc2931540a8b0449f91401e928241ee05ca972bd1e

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class Observation extends Model{public $timestamps=false;protected $table='clinical_observations';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationComponent.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDAT E	src/Models/ObservationComponent.ph p	077cf729d208040032051b435db112a0e76205967003f61879cb266fc03730 a0

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationComponent extends Model{public $timestamps=false;protected
$table='clinical_observation_components';protected $guarded=[];protected
```

```
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationCorrection.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationCorrection.php	82dc4debdda9894393875782b9f2d8d6c94495e7454e576609d0595b5477526b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationCorrection extends Model{public $timestamps=false;protected
$table='clinical_observation_corrections';protected $guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationGroup.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationGroup.php	560d011fcc5994f47c1b9be0ff246c459d7891a2209c60bcf344f608391018aa

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationGroup extends Model{public $timestamps=false;protected $table='clinical_observation_groups';protected
$guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationProvenance.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationProvenance.php	96f4c82c4d3a8d1c90b15bdc750098dc5ccc5abbd1bbd4627e942d13b17b781e

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationProvenance extends Model{public $timestamps=false;protected
$table='clinical_observation_provenance';protected $guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ObservationReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ObservationReconciliation.php	ca6c3de50bc01ec275e40a115e2c6e8cbb6f8b30824d81f2b4c9e44f749e8257

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Models;
use Illuminate\Database\Eloquent\Model;
final class ObservationReconciliation extends Model{public $timestamps=false;protected
$table='clinical_observation_reconciliations';protected $guarded=[];protected
$casts=['value_json'=>'array','context'=>'array','metadata'=>'array','provenance'=>'array','received_at_utc'=>'immutable_datetime','
occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','ef
fective_to_utc'=>'immutable_datetime'];}
```

src/ObservationServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ObservationServiceProvider.php	bee4f75acd4b503eb47d3bb29c90f7f69a866257a516408d1dc6fcaef7ed0fc4

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation;
use Illuminate\Support\ServiceProvider;
use KashTre\Clinical\Observation\Contracts\{ObservationPlanSatisfactionGateway,SharedUnitGateway};
final class ObservationServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'../config/clinical-observation.php','clinical-observation');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'../database/migrations');$this->loadRoutesFrom(__DIR__.'../routes/clinical-observation.php');}}
```

src/Services/CdeValidator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/CdeValidator.php	47bed5ba6bc6e0212cb7a53cae5f0dfc227c7671605d1d26a94aec03e3d155fb

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Services;
use KashTre\Clinical\Observation\Models\{CdeAllowedUnit,CdeVersion};
final class CdeValidator{public function activeVersion(string $tenantId,string $cdeld,int $version):CdeVersion{return
CdeVersion::query()->where('tenant_id',$tenantId)->where('cde_public_id',$cdeld)->where('version_no',$version)-
>where('status','ACTIVE')->firstOrFail();}public function assertUnitAllowed(string $tenantId,string $cdeld,int $version,string
$unitId):void{$ok=CdeAllowedUnit::query()->where('tenant_id',$tenantId)->where('cde_public_id',$cdeld)-
>where('cde_version',$version)->where('unit_public_id',$unitId)->where('is_input_allowed',true)->exists();if(!$ok)throw new
\DomainException('CLN_OBS_UNIT_NOT_ALLOWED');}}
```

src/Services/PhysiologicalLimitEvaluator.php

Status	Destination	SHA-256
PICK-AND-	src/Services/PhysiologicalLimitEvaluator.p	ab1a2fc8f1c8323f4627ad5998f47264392ec4f15fe565d0262069557bf4c

DROP CANDIDAT E	hp	be8
-----------------------	----	-----

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Services;
use KashTre\Clinical\Observation\Models\CdeLimitPolicy;
final class PhysiologicalLimitEvaluator{public function evaluate(string $tenantId,string $cdId,int $version,string
$value):array{$p=CdeLimitPolicy::query()->where('tenant_id',$tenantId)->where('cde_public_id',$cdId)-
>where('cde_version',$version)->where('status','ACTIVE')->first();if(!$p)return
['outcome'=>'VALID','policy'=>null];$v=(float)$value;if($p->block_min_decimal!==null&&$v<(float)$p->block_min_decimal||$p-
>block_max_decimal!==null&&$v>(float)$p->block_max_decimal)return ['outcome'=>'BLOCK','policy'=>$p->public_id];if($p-
>warning_min_decimal!==null&&$v<(float)$p->warning_min_decimal||$p->warning_max_decimal!==null&&$v>(float)$p-
>warning_max_decimal)return ['outcome'=>'WARNING','policy'=>$p->public_id];return ['outcome'=>'VALID','policy'=>$p-
>public_id];}}
```

tests/Unit/RecordObservationDataTest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDAT E	tests/Unit/RecordObservationDataTest.p hp	5ca721ea9a681ff1997a50d397a7984dd3e2d3ed4f329e2e28851def2a003 837

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Observation\Tests\Unit;use DateTimeImmutable;use PHPUnit\Framework\TestCase;use
KashTre\Clinical\Observation\Data\RecordObservationData;use KashTre\Clinical\Observation\Enums\CaptureMethod;
final class RecordObservationDataTest extends TestCase{public function test_preserves_value_and_context():void{$d=new
RecordObservationData('T1','01ARZ3NDEKTSV4RRFFQ69G5FAV','01ARZ3NDEKTSV4RRFFQ69G5FAW','01ARZ3NDEKTSV4RRFFQ69
G5FAX',1,'QUANTITY','37.2','01ARZ3NDEKTSV4RRFFQ69G5FAY',new DateTimeImmutable('2026-08-
28T08:00:00Z'),'01ARZ3NDEKTSV4RRFFQ69G5FAZ',CaptureMethod::MANUAL,null,null,['site'=>'oral']);self::assertSame('37.2',$d-
>value);self::assertSame('oral',$d->context['site']);}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	IDs, context, audit, events and idempotency
Volume 2	Observation Plan satisfaction and reversal
Shared Unit Engine	Normalization and conversion provenance
Patient/Encounter	Authoritative tenant-safe lookup
Authorization	Contextual Clinical policy
Device registry	Trusted device state and assignment
Frontend	Quick capture, device validation and correction workspaces

Blocking Gaps

Execute package in actual KashTre repository
 Approve initial CDE catalogue and versions
 Approve allowed units and display preferences
 Approve physiological warning/block limits
 Approve device auto-commit policies by device class
 Bind Observation Plan satisfaction and correction reversal
 Complete host controllers, requests, resources and Vue pages
 Run clean, upgrade, rollback, concurrency, security and clinical gold tests

Verification

```

composer validate --strict
composer dump-autoload --strict-psr
php artisan migrate:fresh --env=testing
php artisan migrate:rollback --step=1 --env=testing
php artisan migrate --env=testing
./vendor/bin/phpunit
  
```

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	CDE and Observation functional authority
Volume 1 Executable Foundation	Foundation contracts
Volume 2 Observation Plan	Due-work and satisfaction contract
HL7 FHIR R5 Observation	External reference for measurements and simple assertions

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 4

CLINICAL FORMS, CLIENT-SPACE PROFILES AND RENDERING ENGINE

Observation Requirements, Add Form, Continuous Population, Historical Reproduction and Print Rendering

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1, 2 and 3

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Governing Flow

Form definitions and clinical context determine required observations. Volume 2 converts those requirements into due work. Volume 3 stores each authoritative Observation once. Volume 4 renders every applicable form from those Observations and never creates an independently editable duplicate value.

Form definition + client-space/profile/patient activation

- > observation requirements
- > Effective Observation Plan
- > nurse captures Observation once
- > continuous form population
- > read-only, longitudinal and printable rendering

Entity Register

Entity	Purpose	Invariant
FormDefinition	clinical_form_definitions	Versioned, tenant-scoped and non-duplicative
FormVersion	clinical_form_versions	Versioned, tenant-scoped and non-duplicative
FormItem	clinical_form_items	Versioned, tenant-scoped and non-duplicative
FormItemRequirement	clinical_form_item_requirements	Versioned, tenant-scoped and non-duplicative
FormReferenceAsset	clinical_form_reference_assets	Versioned, tenant-scoped and non-duplicative
ClientSpaceProfile	clinical_client_space_profiles	Versioned, tenant-scoped and non-duplicative
ClientSpaceFormAssignment	clinical_client_space_form_assignments	Versioned, tenant-scoped and non-duplicative
PatientFormActivation	clinical_patient_form_activations	Versioned, tenant-scoped and non-duplicative
FormRenderSnapshot	clinical_form_render_snapshots	Versioned, tenant-scoped and non-duplicative
FormRenderBinding	clinical_form_render_bindings	Versioned, tenant-scoped and non-duplicative
FormApproval	clinical_form_approvals	Versioned, tenant-scoped and non-duplicative
FormReconciliation	clinical_form_reconciliations	Versioned, tenant-scoped and non-duplicative

Lifecycle

Object	States
Definition	DRAFT, IN_REVIEW, APPROVED, ACTIVE, DEPRECATED, RETIRED
Patient form	AVAILABLE, ACTIVE, COMPLETED, DISCONTINUED, SUPERSEDED, ENTERED_IN_ERROR
Render	QUEUED, RENDERING, READY, FAILED, STALE

Engineering Design**Form governance**

Stable code and public ID

Immutable approved versions

Unique item keys

CDE references rather than copied semantics

Separate semantic and layout changes

Reference PDF or Word is evidence, not executable form

Approval before activation

Client-space profile

Versioned facility/client-space profile

Default forms assigned by activation mode

Applicability rules explicit

Prospective changes only

Overflow location inherits governing profile

Patient-specific Add Form does not alter defaults

Continuous population

Query qualifying Observations by patient, encounter, CDE, time, status and context

One Observation may populate several forms

Source observation IDs retained per rendered item

Missing positions show pending/due/overdue/not applicable

Correction marks snapshot stale and regenerates

Rendering failure never rolls back Observation

Add Form

Filter approved eligible catalogue

Warn on equivalent active form

Create activation against exact version

Contribute requirements to Volume 2

Prepopulate existing qualifying Observations

Retain historical form after completion

Transfer

Preserve prior form activations and history

End source-space-only requirements prospectively

Activate destination defaults

Continue ordered/protocol/patient-specific forms

Never delete Labour Care Guide after transfer

Permissions

Permission	Purpose
clinical.form_definition.create	Create draft
clinical.form_definition.update	Edit draft
clinical.form_definition.review	Review version

clinical.form_definition.approve	Approve semantic and layout version
clinical.form_definition.activate	Make version usable
clinical.form_assignment.manage	Assign to client-space profile
clinical.patient_form.add	Add to one patient
clinical.patient_form.discontinue	End prospective applicability
clinical.form_render.view	View rendered form
clinical.form_render.print	Generate controlled print
clinical.form_reconcile	Resolve rendering or binding failures

Mandatory Tests

Two forms requiring BP create one due task through Volume 2

One BP Observation populates both forms

Adding form prepopulates existing valid observations

Adding form does not change client-space defaults

Unapproved form version cannot activate

Duplicate item key is rejected

Rendering failure leaves Observation intact

Corrected Observation regenerates every dependent snapshot

Transfer preserves historical Labour form

Print output identifies version, patient, encounter, timezone and generation time

Cross-tenant form cannot activate

Reference asset cannot execute active logic

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	8e613cc45ae37997b9fdd4316e3c36b686a81e4c3bf0aa5f27dfd5a9bf0ad62a

KashTre Clinical Forms and Rendering v6.1.0

Install after Volumes 1, 2 and 3. The form engine is not an Observation database. Bind ObservationQuery, ObservationPlanContributionGateway, patient/encounter authorization, object storage, print renderer and Vue interfaces before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	f0f509b4d863f136111f33226487016cd3c98fa1d578bce30616f953213e590d

```
{
  "volume": "Clinical EDD Volume 4 Forms",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "8e613cc45ae37997b9fdd4316e3c36b686a81e4c3bf0aa5f27dfd5a9bf0ad62a",
      "bytes": 293
    },
    {
      "path": "composer.json",
      "sha256": "f8635cf1635e3783c5c46fc53474d07eee42aaec5d90e3046a94dce7b98b6f92",
      "bytes": 558
    },
    {
      "path": "config/clinical-forms.php",
      "sha256": "953a902fc587d6b7c8208186a71fda89abcb47f90765a8150498281bc9d69b89",
      "bytes": 332
    },
    {
      "path": "database/migrations/2026_08_28_300000_create_clinical_form_tables.php",
      "sha256": "1685ace8c90954839696583c08fced3202188c1c3dfeecb1023ca4bccbbfe238",
      "bytes": 5462
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-forms.php",
      "sha256": "becfd7b786d643fe3de4343a22846e3615df49dc21c27c2f73aa6481dd36bafa",
      "bytes": 307
    }
  ]
}
```

```

{
  "path": "src/Actions/ActivatePatientForm.php",
  "sha256": "5bc6283241a10975f0e8f3aafea314f226a0cda9a511ac02c857135312bc4aca",
  "bytes": 2031
},
{
  "path": "src/Actions/GenerateFormSnapshot.php",
  "sha256": "b9335822ac3f5649f89c64147bc7da6960ed748dc4a60dda8c712e5a3a7dfd68",
  "bytes": 1308
},
{
  "path": "src/ClinicalFormsServiceProvider.php",
  "sha256": "02d1640d327442c1565aa211a31c0675bcd6d808d4b6d35b27cfe168096f08a6",
  "bytes": 530
},
{
  "path": "src/Contracts/ObservationPlanContributionGateway.php",
  "sha256": "24f081875a65b3f82256562ab96a8e7690b28eee8161f112ef242952b5c533ad",
  "bytes": 374
},
{
  "path": "src/Contracts/ObservationQuery.php",
  "sha256": "07f235cb95c9d24f0acced89f3c721540c819367e3159189d6f614e2114f0e36",
  "bytes": 218
},
{
  "path": "src/Data/ActivatePatientFormData.php",
  "sha256": "ac0b438a1f9d42ed9d8f4696ed84f619454e302dee952c843c085ac0f4a98a35",
  "bytes": 530
},
{
  "path": "src/Data/RenderedFormData.php",
  "sha256": "eeb41a68304058a9e97e9bac310402285425aa4bed800b2d799c50b2e07e637b",
  "bytes": 318
},
{
  "path": "src/Enums/FormActivationMode.php",
  "sha256": "fbeb4585c4824b8950834ce5408e36bf96dc369d7ccfe5da31d6db625b0031b",
  "bytes": 365
},
{
  "path": "src/Enums/FormItemType.php",
  "sha256": "0571319fd74d534c81a7df57238a6c4b803d59328c3947b9ae90cb44bfea6b3c",
  "bytes": 303
},
{
  "path": "src/Enums/FormStatus.php",
  "sha256": "70b321bc375bd44ec1f67fbc325e30b5e496f044517a4a3fcba5e3bf7195f8e8",
  "bytes": 252
},
{
  "path": "src/Enums/LayoutMode.php",
  "sha256": "5c3d308c982c709e02f9a8c0f3e25bfd0eec6b2cea89016dff9e41302a87b5b",
  "bytes": 206
},

```

```

{
  "path": "src/Enums/MatchStrategy.php",
  "sha256": "62f315d4d0536ac901f806c3c77bb45192cd7cc9921cb64f941112de6c7fe2d5",
  "bytes": 255
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{
  "path": "src/Enums/PatientFormStatus.php",
  "sha256": "530fb6280c9d8dd10247dde7cd7f02820e0169e38c67415f2d622a6103094481",
  "bytes": 293
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  "path": "src/Enums/PopulationMode.php",
  "sha256": "11cfd0371b698363c7854652e26f5b759db95e160b5b67404da89aa1b947b4df",
  "bytes": 195
},
{
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  "bytes": 215
},
{
  "path": "src/Models/ClientSpaceFormAssignment.php",
  "sha256": "7c70f1f0e929b1ec94786b7cf6da36bc720c10e113eaa6a73753e4fa2e57b3f2",
  "bytes": 629
},
{
  "path": "src/Models/ClientSpaceProfile.php",
  "sha256": "ce7ff603aa2d2878e84b4668d42a4e2abe6f734279110081889b36b66b811565",
  "bytes": 614
},
{
  "path": "src/Models/FormApproval.php",
  "sha256": "38f2eaa8a32c11b299bdfb337df06d62f7279e058b11701b07d71cbf981dea2e",
  "bytes": 601
},
{
  "path": "src/Models/FormDefinition.php",
  "sha256": "1c9b6e0c4bf5c79f5bd6a73174053e82a293f0b4461e0f1c0331bd06377e8e75",
  "bytes": 605
},
{
  "path": "src/Models/FormItem.php",
  "sha256": "feab8851faece3cd5dc89dceda5f4721c6dbb0ea09c5fe7fce0d4f5f2c2bfb3d",
  "bytes": 593
},
{
  "path": "src/Models/FormItemRequirement.php",
  "sha256": "1edc67077495499751208f80f4228bd11edbf48621e8b0ba4da9b35373ea3a03",
  "bytes": 616
},
{
  "path": "src/Models/FormReconciliation.php",
  "sha256": "6c82b7a73fa2120d664723fca5df13031074159c2fbd8b0390adafa4cbee851d",
  "bytes": 613
},

```

```
{
  "path": "src/Models/FormReferenceAsset.php",
  "sha256": "51fcbcdfb057418d5eacc5131bfa368092d7ed443a54a7e05987426ec4cf1e60",
  "bytes": 614
},
{
  "path": "src/Models/FormRenderBinding.php",
  "sha256": "9c5e87f604edaf22278cb6e34f31cca8eaf0be9ffe313a079900370094b6a23b",
  "bytes": 612
},
{
  "path": "src/Models/FormRenderSnapshot.php",
  "sha256": "cf35dc670e18d519f0a87e3030319c313600eed7dd2376e44e11568d3312a711",
  "bytes": 614
},
{
  "path": "src/Models/FormVersion.php",
  "sha256": "920a8c273a847831c9ef21a429d8f3d1d1754f9bd7d44f235d9f07b546a081d1",
  "bytes": 599
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{
  "path": "src/Models/PatientFormActivation.php",
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  "bytes": 620
},
{
  "path": "src/Services/FormDefinitionValidator.php",
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  "bytes": 752
},
{
  "path": "src/Services/FormRenderer.php",
  "sha256": "6b5a001f4237f0e951fa516155727a050b5e228c803dca182c62606f625758e0",
  "bytes": 1293
},
{
  "path": "tests/Unit/FormDefinitionValidatorTest.php",
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  "bytes": 264
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	f8635cf1635e3783c5c46fc53474d07eee42aaec5d90e3046a94dce7b98b6f92

```
{
  "name": "kashtre/clinical-forms",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "kashtre/clinical-observation-plan": "^6.1",
    "kashtre/clinical-observation": "^6.1",
    "require-dev": {
      "orchestra/testbench": "^11.0",
      "phpunit/phpunit": "^12.0",
      "autoload": {
        "psr-4": {
          "KashTre\\Clinical\\Forms\\": "src/"
        },
        "extra": {
          "laravel": {
            "providers": [
              "KashTre\\Clinical\\Forms\\ClinicalFormsServiceProvider"
            ]
          }
        },
        "scripts": {
          "test": "phpunit"
        }
      }
    }
  }
}
```

config/clinical-forms.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-forms.php	953a902fc587d6b7c8208186a71fda89abcb47f90765a8150498281bc9d69b89

```
<?php
declare(strict_types=1);
return
[
  'render_cache_seconds' => (int) env('CLINICAL_FORM_RENDER_CACHE_SECONDS', 30),
  'maximum_repeat_items' => (int) env('CLINICAL_FORM_MAX_REPEAT_ITEMS', 500),
  'reference_upload_disk' => env('CLINICAL_FORM_REFERENCE_DISK', 'private'),
  'reference_upload_max_kb' => (int) env('CLINICAL_FORM_REFERENCE_MAX_KB', 10240)
];
```

database/migrations/2026_08_28_300000_create_clinical_form_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_300000_create_clinical_form_tables.php	1685ace8c90954839696583c08fced3202188c1c3dfeecb1023ca4bccbbfe238

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_form_definitions', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->string('code', 96);
            $table->string('name', 180);
            $table->string('clinical_domain', 96);
            $table->string('status', 24);
            $table->unsignedInteger('current_version')->nullable();
            $table->dateTime('created_at_utc', 6);
            $table->dateTime('updated_at_utc', 6);
            $table->unique(['tenant_id', 'code'], 'uq_form_code');
        });

        Schema::create('clinical_form_versions', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('form_definition_public_id');
            $table->unsignedInteger('version_no');
            $table->string('status', 24);
            $table->string('population_mode', 24)->default('CONTINUOUS');
            $table->json('layout_schema');
            $table->json('configuration')->nullable();
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('effective_to_utc', 6)->nullable();
            $table->unique(['tenant_id', 'form_definition_public_id', 'version_no'], 'uq_form_version');
        });

        Schema::create('clinical_form_items', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('form_version_public_id');
            $table->string('item_key', 128);
            $table->string('section_key', 128);
            $table->string('item_type', 32);
            $table->string('label', 240);
            $table->ulid('cde_public_id')->nullable();
            $table->unsignedInteger('cde_version')->nullable();
            $table->boolean('is_required')->default(false);
            $table->boolean('is_repeatable')->default(false);
            $table->json('matching_rule')->nullable();
            $table->json('configuration')->nullable();
            $table->unsignedInteger('display_order');
            $table->unique(['form_version_public_id', 'item_key'], 'uq_form_item_key');
        });

        Schema::create('clinical_form_item_requirements', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('form_item_public_id');
            $table->string('schedule_mode', 32);
            $table->unsignedInteger('interval_minutes')->nullable();
            $table->unsignedInteger('early_window_minutes')->default(0);
            $table->unsignedInteger('late_window_minutes')->default(0);
            $table->string('responsible_role_code', 96);
            $table->json('applicability_rule')->nullable();
            $table->dateTime('effective_from_utc', 6);
            $table->dateTime('effective_to_utc', 6)->nullable();
        });

        Schema::create('clinical_form_reference_assets', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('form_version_public_id');
            $table->string('storage_disk', 64);
            $table->string('storage_path', 500);
            $table->string('mime_type', 128);
            $table->char('sha256', 64);
            $table->string('purpose', 32);
            $table->dateTime('uploaded_at_utc', 6);
        });
    }
};
```

```

Schema::create('clinical_client_space_profiles',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('facility_public_id');$t->ulid('client_space_public_id');$t->unsignedInteger('version_no');$t-
>string('status',24);$t->dateTime('effective_from_utc',6);$t->dateTime('effective_to_utc',6)->nullable();$t-
>unique(['tenant_id','client_space_public_id','version_no'],'uq_client_space_profile'));
Schema::create('clinical_client_space_form_assignments',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('client_space_profile_public_id');$t->ulid('form_definition_public_id');$t-
>ulid('form_version_public_id');$t->string('activation_mode',40);$t->json('applicability_rule')->nullable();$t-
>boolean('allow_patient_specific_discontinue')->default(true);$t->unsignedInteger('display_order')->default(1);$t-
>unique(['client_space_profile_public_id','form_definition_public_id'],'uq_space_form'));
Schema::create('clinical_patient_form_activations',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('patient_public_id');$t->ulid('encounter_public_id');$t->ulid('form_definition_public_id');$t-
>ulid('form_version_public_id');$t->string('status',24);$t->string('activation_source_type',40);$t-
>ulid('activation_source_public_id');$t->ulid('originating_client_space_public_id')->nullable();$t->ulid('activated_by_public_id');$t-
>text('activation_reason')->nullable();$t->dateTime('activated_at_utc',6);$t->dateTime('completed_at_utc',6)->nullable();$t-
>unsignedBigInteger('record_version')->default(1);$t-
>index(['tenant_id','patient_public_id','encounter_public_id','status'],'idx_patient_forms'));
Schema::create('clinical_form_render_snapshots',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('patient_form_activation_public_id');$t->ulid('form_version_public_id');$t->string('status',24);$t-
>char('content_hash',64);$t->json('rendered_payload');$t->unsignedInteger('unresolved_count')->default(0);$t-
>dateTime('generated_at_utc',6));
Schema::create('clinical_form_render_bindings',function(Blueprint$t){$t->id();$t->ulid('render_snapshot_public_id');$t-
>string('form_item_key',128);$t->ulid('observation_public_id');$t-
>unique(['render_snapshot_public_id','form_item_key','observation_public_id'],'uq_form_render_binding'));
}public function
down():void{foreach(['clinical_form_render_bindings','clinical_form_render_snapshots','clinical_patient_form_activations','clinical_c
lient_space_form_assignments','clinical_client_space_profiles','clinical_form_reference_assets','clinical_form_item_requirements','
clinical_form_items','clinical_form_versions','clinical_form_definitions']as$t)Schema::dropIfExists($t);};

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-forms.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-forms.php	becfd7b786d643fe3de4343a22846e3615df49dc21c27c2f73aa6481dd36bafa

```

<?php
declare(strict_types=1);
use Illuminate\Support\Facades\Route;
Route::middleware(['api','auth:sanctum'])->prefix('api/v1/clinical')-
>group(function(){Route::get('/patients/{patient}/encounters/{encounter}/forms',fn()->response()-
>json(['error'=>['code'=>'CLN_FORM_HOST_CONTROLLER_REQUIRED'],501)]));

```

src/Actions/ActivatePatientForm.php

Status	Destination	SHA-256
PICK-AND-	src/Actions/ActivatePatientForm.php	5bc6283241a10975f0e8f3aaf314f226a0cda9a511ac02c857135312bc4aca

DROP CANDIDATE		
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```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Actions;
use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Forms\Contracts\ObservationPlanContributionGateway;use
KashTre\Clinical\Forms\Data\ActivatePatientFormData;use
KashTre\Clinical\Forms\Models\{FormVersion, PatientFormActivation};use
KashTre\Clinical\Forms\Services\FormDefinitionValidator;
final readonly class ActivatePatientForm{public function __construct(private PublicIdGenerator $ids,private FormDefinitionValidator
$validator,private ObservationPlanContributionGateway $plan){}public function execute(ActivatePatientFormData
$d):PatientFormActivation{return DB::transaction(function()use($d){$v=FormVersion::query()->where('tenant_id',$d->tenantId)-
>where('form_definition_public_id',$d->formDefinitionPublicId)->where('version_no',$d->formVersion)->lockForUpdate()-
>firstOrFail();$this->validator->assertActivatable($v);$existing=PatientFormActivation::query()->where('tenant_id',$d->tenantId)-
>where('patient_public_id',$d->patientPublicId)->where('encounter_public_id',$d->encounterPublicId)-
>where('form_version_public_id',$v->public_id)->whereIn('status',['ACTIVE','AVAILABLE'])-
>first();if($existing)return$existing;$a=PatientFormActivation::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$d->tenantId, 'patient_public_id'=>$d->patientPublicId, 'encounter_public_id'=>$d-
>encounterPublicId, 'form_definition_public_id'=>$d->formDefinitionPublicId, 'form_version_public_id'=>$v-
>public_id, 'status'=>'ACTIVE', 'activation_source_type'=>$d->activationSourceType, 'activation_source_public_id'=>$d-
>activationSourcePublicId, 'originating_client_space_public_id'=>$d->clientSpacePublicId, 'activated_by_public_id'=>$d-
>activatedByPublicId, 'activation_reason'=>$d->reason, 'activated_at_utc'=>$d->activatedAtUtc, 'record_version'=>1]);$this->plan-
>activateFormRequirements($d->tenantId, $d->patientPublicId, $d->encounterPublicId, $a->public_id, $v->public_id);return$a;}}}
```

src/Actions/GenerateFormSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/GenerateFormSnapshot.php	b9335822ac3f5649f89c64147bc7da6960ed748dc4a60dda8c712e5a3a7dfd68

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Actions;
use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Forms\Models\{FormRenderBinding, FormRenderSnapshot, PatientFormActivation};use
KashTre\Clinical\Forms\Services\FormRenderer;
final readonly class GenerateFormSnapshot{public function __construct(private PublicIdGenerator $ids,private FormRenderer
$renderer){}public function execute(string $tenantId,string $activationId):FormRenderSnapshot{$a=PatientFormActivation::query()-
>where('tenant_id',$tenantId)->where('public_id',$activationId)->firstOrFail();$r=$this->renderer-
>render($a);$snapshot=FormRenderSnapshot::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId, 'patient_form_activation_public_id'=>$activationId, 'form_version_public_id'=>$r-
>formVersionPublicId, 'status'=>'READY', 'content_hash'=>$r->contentHash, 'rendered_payload'=>$r-
>sections, 'unresolved_count'=>count($r->unresolvedItems), 'generated_at_utc'=>$r->generatedAtUtc]);foreach($r->sections
as$section)foreach($section as$item)foreach($item['provenance']as$observationId)FormRenderBinding::query()-
>firstOrCreate(['render_snapshot_public_id'=>$snapshot-
>public_id, 'form_item_key'=>$item['key'], 'observation_public_id'=>$observationId]);return$snapshot;}}
```

src/ClinicalFormsServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ClinicalFormsServiceProvider.php	02d1640d327442c1565aa211a31c0675bcd6d808d4b6d35b27cfe168096f08a

DROP CANDIDAT E	p	6
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```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms;
use Illuminate\Support\ServiceProvider;
use KashTre\Clinical\Forms\Contracts\{ObservationQuery,ObservationPlanContributionGateway};
final class ClinicalFormsServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'../config/clinical-forms.php','clinical-forms');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'../database/migrations');$this->loadRoutesFrom(__DIR__.'../routes/clinical-forms.php');}}
```

src/Contracts/ObservationPlanContributionGateway.php

Status	Destination	SHA-256
PICK- AND- DROP CANDIDA TE	src/Contracts/ObservationPlanContributionGate way.php	24f081875a65b3f82256562ab96a8e7690b28eee8161f112ef24295 2b5c533ad

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Contracts;
interface ObservationPlanContributionGateway{public function activateFormRequirements(string $tenantId,string $patientId,string
$encounterId,string $activationId,string $formVersionId):void;public function endFormRequirements(string $tenantId,string
$activationId,\DateTimeImmutable $endedAt):void;}
```

src/Contracts/ObservationQuery.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDATE	src/Contracts/ObservationQuery.php	07f235cb95c9d24f0acced89f3c721540c819367e3159189d6f614e2114f0e36

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Contracts;
interface ObservationQuery{public function qualifying(string $tenantId,string $patientId,string $encounterId,string $cdId,array
$rule):array;}
```

src/Data/ActivatePatientFormData.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDAT E	src/Data/ActivatePatientFormData.ph p	ac0b438a1f9d42ed9d8f4696ed84f619454e302dee952c843c085ac0f4a98a3 5

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Data;
use DateTimeImmutable;
final readonly class ActivatePatientFormData{public function __construct(public string $tenantId,public string
```

```
$patientPublicId,public string $encounterPublicId,public string $formDefinitionPublicId,public int $formVersion,public string
$activationSourceType,public string $activationSourcePublicId,public ?string $clientSpacePublicId,public string
$activatedByPublicId,public DateTimeImmutable $activatedAtUtc,public ?string $reason){}}
```

src/Data/RenderedFormData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/RenderedFormData.php	eeb41a68304058a9e97e9bac310402285425aa4bed800b2d799c50b2e07e637b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Data;
final readonly class RenderedFormData{public function __construct(public string $activationPublicId,public string
$formVersionPublicId,public array $sections,public array $unresolvedItems,public string $contentHash,public string
$generatedAtUtc){}}
```

src/Enums/FormActivationMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/FormActivationMode.php	fbebc4585c4824b8950834ce5408e36bf96dc369d7ccfe5da31d6db625b0031b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum FormActivationMode:string{
    case AUTOMATIC='AUTOMATIC';
    case AVAILABLE='AVAILABLE';
    case REQUIRED='REQUIRED';
    case CONDITIONAL='CONDITIONAL';
    case ORDER_ACTIVATED='ORDER_ACTIVATED';
    case PROTOCOL_ACTIVATED='PROTOCOL_ACTIVATED';
    case CARE_PATHWAY_ACTIVATED='CARE_PATHWAY_ACTIVATED';
}
```

src/Enums/FormItemType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/FormItemType.php	0571319fd74d534c81a7df57238a6c4b803d59328c3947b9ae90cb44bfea6b3c

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum FormItemType:string{
    case SECTION='SECTION';
    case DISPLAY='DISPLAY';
    case OBSERVATION='OBSERVATION';
    case CALCULATED='CALCULATED';
    case SIGNATURE='SIGNATURE';
    case INSTRUCTION='INSTRUCTION';
}
```

```
case REFERENCE='REFERENCE';
}
```

src/Enums/FormStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/FormStatus.php	70b321bc375bd44ec1f67fbc325e30b5e496f044517a4a3fcb5e3bf7195f8e8

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum FormStatus:string{
    case DRAFT='DRAFT';
    case IN_REVIEW='IN_REVIEW';
    case APPROVED='APPROVED';
    case ACTIVE='ACTIVE';
    case DEPRECATED='DEPRECATED';
    case RETIRED='RETIRED';
}
```

src/Enums/LayoutMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/LayoutMode.php	5c3d308c982c709e02f9a8c0f3e25bfdf0eec6b2cea89016dff9e41302a87b5b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum LayoutMode:string{
    case CAPTURE='CAPTURE';
    case READ_ONLY='READ_ONLY';
    case LONGITUDINAL='LONGITUDINAL';
    case PRINT='PRINT';
}
```

src/Enums/MatchStrategy.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MatchStrategy.php	62f315d4d0536ac901f806c3c77bb45192cd7cc9921cb64f941112de6c7fe2d5

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum MatchStrategy:string{
    case EXACT_WINDOW='EXACT_WINDOW';
    case LATEST_IN_WINDOW='LATEST_IN_WINDOW';
    case CLOSEST_TO_SLOT='CLOSEST_TO_SLOT';
    case ALL_IN_PERIOD='ALL_IN_PERIOD';
}
```

src/Enums/PatientFormStatus.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Enums/PatientFormStatus.php	530fb6280c9d8dd10247dde7cd7f02820e0169e38c67415f2d622a6103094481
-------------------------	---------------------------------	--

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum PatientFormStatus:string{
    case AVAILABLE='AVAILABLE';
    case ACTIVE='ACTIVE';
    case COMPLETED='COMPLETED';
    case DISCONTINUED='DISCONTINUED';
    case SUPERSEDED='SUPERSEDED';
    case ENTERED_IN_ERROR='ENTERED_IN_ERROR';
}
```

src/Enums/PopulationMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/PopulationMode.php	11cfd0371b698363c7854652e26f5b759db95e160b5b67404da89aa1b947b4df

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum PopulationMode:string{
    case CONTINUOUS='CONTINUOUS';
    case ON_OPEN='ON_OPEN';
    case MANUAL_REFRESH='MANUAL_REFRESH';
}
```

src/Enums/RenderStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/RenderStatus.php	797999565dd1b9b06090df0143298cf1664bae1b5d7dd7f7048cd72605a5ae0e

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Enums;
enum RenderStatus:string{
    case QUEUED='QUEUED';
    case RENDERING='RENDERING';
    case READY='READY';
    case FAILED='FAILED';
    case STALE='STALE';
}
```

src/Models/ClientSpaceFormAssignment.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDAT	src/Models/ClientSpaceFormAssignment.php	7c70f1f0e929b1ec94786b7cf6da36bc720c10e113eaa6a73753e4fa2e57b3f2

E		
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```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class ClientSpaceFormAssignment extends Model{public $timestamps=false;protected
$table='clinical_client_space_form_assignments';protected $guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_dateti
me','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/ClientSpaceProfile.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ClientSpaceProfile.php	ce7ff603aa2d2878e84b4668d42a4e2abe6f734279110081889b36b66b811565

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class ClientSpaceProfile extends Model{public $timestamps=false;protected $table='clinical_client_space_profiles';protected
$guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_dateti
me','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormApproval.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormApproval.php	38f2eaa8a32c11b299bdfb337df06d62f7279e058b11701b07d71cbf981dea2e

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormApproval extends Model{public $timestamps=false;protected $table='clinical_form_approvals';protected
$guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_dateti
me','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormDefinition.php	1c9b6e0c4bf5c79f5bd6a73174053e82a293f0b4461e0f1c0331bd06377e8e75

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
```

```
use Illuminate\Database\Eloquent\Model;
final class FormDefinition extends Model{public $timestamps=false;protected $table='clinical_form_definitions';protected
$guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_dateti
me','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormItem.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormItem.php	feab8851faece3cd5dc89dceda5f4721c6dbb0ea09c5fe7fce0d4f5f2c2bfb3d

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormItem extends Model{public $timestamps=false;protected $table='clinical_form_items';protected
$guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_dateti
me','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormItemRequirement.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormItemRequirement.php	1edc67077495499751208f80f4228bd11edbf48621e8b0ba4da9b35373ea3a03

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormItemRequirement extends Model{public $timestamps=false;protected
$table='clinical_form_item_requirements';protected $guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_dateti
me','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormReconciliation.php	6c82b7a73fa2120d664723fca5df13031074159c2fbd8b0390adafa4cbee851d

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormReconciliation extends Model{public $timestamps=false;protected $table='clinical_form_reconciliations';protected
$guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'
```

```
array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormReferenceAsset.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormReferenceAsset.php	51fcbcdfb057418d5eacc5131bfa368092d7ed443a54a7e05987426ec4cf1e60

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormReferenceAsset extends Model{public $timestamps=false;protected
$table='clinical_form_reference_assets';protected $guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormRenderBinding.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormRenderBinding.php	9c5e87f604edaf22278cb6e34f31cca8eaf0be9ffe313a079900370094b6a23b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormRenderBinding extends Model{public $timestamps=false;protected
$table='clinical_form_render_bindings';protected $guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormRenderSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormRenderSnapshot.php	cf35dc670e18d519f0a87e3030319c313600eed7dd2376e44e11568d3312a711

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormRenderSnapshot extends Model{public $timestamps=false;protected
$table='clinical_form_render_snapshots';protected $guarded=[];protected
$casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/FormVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FormVersion.php	920a8c273a847831c9ef21a429d8f3d1d1754f9bd7d44f235d9f07b546a081d1

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class FormVersion extends Model{public $timestamps=false;protected $table='clinical_form_versions';protected $guarded=[];protected $casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Models/PatientFormActivation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PatientFormActivation.php	75ecd728bba5f46bd7b8f05263233b48ff7d57e4dae9b873349e307e0050a5da

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Models;
use Illuminate\Database\Eloquent\Model;
final class PatientFormActivation extends Model{public $timestamps=false;protected $table='clinical_patient_form_activations';protected $guarded=[];protected $casts=['configuration'=>'array','layout_schema'=>'array','matching_rule'=>'array','applicability_rule'=>'array','rendered_payload'=>'array','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime','activated_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','generated_at_utc'=>'immutable_datetime'];}
```

src/Services/FormDefinitionValidator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/FormDefinitionValidator.php	0a1f76647b7cf86834c40d50d02aacddd626878596c8c9342e981e3fbcf95f05

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Services;
use KashTre\Clinical\Forms\Models\{FormItem,FormVersion};
final class FormDefinitionValidator{public function assertActivatable(FormVersion $v):void{if($v->status!=='APPROVED')throw new \DomainException('CLN_FORM_VERSION_NOT_APPROVED');$items=FormItem::query()->where('form_version_public_id',$v->public_id)->orderBy('display_order')->get();if($items->isEmpty())throw new \DomainException('CLN_FORM_HAS_NO_ITEMS');$keys=[];foreach($items as $i){if(isset($keys[$i->item_key]))throw new \DomainException('CLN_FORM_DUPLICATE_ITEM_KEY');$keys[$i->item_key]=true;if($i->item_type==='OBSERVATION'&& !$i->cde_public_id)throw new \DomainException('CLN_FORM_OBSERVATION_CDE_REQUIRED');}}
```

src/Services/FormRenderer.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Services/FormRenderer.php	6b5a001f4237f0e951fa516155727a050b5e228c803dca182c62606f625758e0
-------------------------	-------------------------------	--

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Services;
use KashTre\Clinical\Forms\Contracts\ObservationQuery;use KashTre\Clinical\Forms\Data\RenderedFormData;use
KashTre\Clinical\Forms\Models\{FormItem,FormVersion,PatientFormActivation};
final readonly class FormRenderer{public function __construct(private ObservationQuery $observations){}public function
render(PatientFormActivation $a):RenderedFormData{$v=FormVersion::query()->where('public_id',$a->form_version_public_id)-
>firstOrFail();$items=FormItem::query()->where('form_version_public_id',$v->public_id)->orderBy('display_order')-
>get();$sections=[];$missing=[];foreach($items as $i){$entry=['key'=>$i->item_key,'type'=>$i->item_type,'label'=>$i-
>label,'value'=>null,'provenance'=>[]];if($i->item_type==='OBSERVATION'){ $found=$this->observations->qualifying($a-
>tenant_id,$a->patient_public_id,$a->encounter_public_id,$i->cde_public_id,$i-
>matching_rule??[]);$entry['value']=$found;$entry['provenance']=array_column($found,'public_id');if($i-
>is_required&&count($found)===0)$missing[]=$i->item_key;}$sections[$i-
>section_key][]=$entry;}$hash=hash('sha256',json_encode($sections,JSON_THROW_ON_ERROR));return new
RenderedFormData($a->public_id,$v->public_id,$sections,$missing,$hash,now('UTC')->format(DATE_ATOM));}}
```

tests/Unit/FormDefinitionValidatorTest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Unit/FormDefinitionValidatorTest.php	b4f768fbb52f9579ce163142ac01b033966f67e83cc2e625c894198a7197504c

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Forms\Tests\Unit;use PHPUnit\Framework\TestCase;
final class FormDefinitionValidatorTest extends TestCase{public function
test_placeholder_for_repository_bound_database_test():void{self::assertTrue(true);}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	Tenant, identifiers, audit, messaging and authorization
Volume 2	Form requirement contribution and ending
Volume 3	Observation query and correction events
Main Module	Facility and client-space IDs
Object storage	Private reference assets
Print service	Controlled PDF renderer
Vue	Designer, profile setup, patient Forms area, Add Form, render viewer

Blocking Gaps

Execute package in actual KashTre repository

Approve form semantic/versioning policy

Approve applicability expression language

Bind CDE registry and ObservationQuery

Bind Observation Plan contribution gateway

Implement host controllers, Form Requests, Resources and Vue pages

Select and validate controlled PDF rendering technology

Approve initial Labour Care Guide and other seed forms

Run clean, upgrade, rollback, concurrency, performance, accessibility and clinical gold tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Observation-first forms authority
Volumes 1-3	Foundation, plans and authoritative Observations
HL7 FHIR R5 Questionnaire	External form-definition reference
HL7 FHIR Structured Data Capture	External reference for prepopulation and reduced duplicate entry

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 5

CLINICAL SCORING, DETERIORATION AND ESCALATION ENGINE

Versioned Scores, Reproducible Calculations, Alerts, Acknowledgement, Action and Resolution

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 to 4

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Safety Rule

A score is a reproducible derived clinical event. Every score must retain the exact score-definition version, input Observation IDs and versions, selection rules, component points, aggregate result, algorithm version, calculation time and triggered response rules. A missing input is never silently replaced with zero unless the approved score definition explicitly requires that behaviour.

Authoritative Observations

- > score input selection
- > deterministic component scoring
- > aggregate score and calculated Observation
- > governed trigger rules
- > alert routing
- > acknowledgement
- > assessment and action
- > escalation if overdue
- > resolution with evidence

Entity Register

Entity	Purpose	Invariant
ScoreDefinition	clinical_score_definitions	Tenant-scoped, versioned and auditable
ScoreVersion	clinical_score_versions	Tenant-scoped, versioned and auditable
ScoreInputDefinition	clinical_score_input_definitions	Tenant-scoped, versioned and auditable
ScoreBand	clinical_score_bands	Tenant-scoped, versioned and auditable
ScoreRule	clinical_score_rules	Tenant-scoped, versioned and auditable
ScoreRun	clinical_score_runs	Tenant-scoped, versioned and auditable
ScoreRunInput	clinical_score_run_inputs	Tenant-scoped, versioned and auditable
ScoreRunComponent	clinical_score_run_components	Tenant-scoped, versioned and auditable
DeteriorationAlert	clinical_deterioration_alerts	Tenant-scoped, versioned and auditable
AlertDelivery	clinical_alert_deliveries	Tenant-scoped, versioned and auditable
AlertAcknowledgement	clinical_alert_acknowledgements	Tenant-scoped, versioned and auditable
AlertAssessment	clinical_alert_assessments	Tenant-scoped, versioned and auditable
AlertAction	clinical_alert_actions	Tenant-scoped, versioned and auditable
EscalationStep	clinical_escalation_steps	Tenant-scoped, versioned and auditable
EscalationOccurrence	clinical_escalation_occurrences	Tenant-scoped, versioned and auditable
ScoreReconciliation	clinical_score_reconciliations	Tenant-scoped, versioned and auditable

Lifecycle

Object	States
Score definition	DRAFT, IN_REVIEW, APPROVED, ACTIVE, DEPRECATED, RETIRED
Score run	PENDING_INPUTS, CALCULATED, INVALIDATED, RECALCULATION_REQUIRED, ENTERED_IN_ERROR
Alert	TRIGGERED, ROUTED, DELIVERED, ACKNOWLEDGED, ASSESSED, ACTION_TAKEN, ESCALATED, RESOLVED, CANCELLED, ENTERED_IN_ERROR
Escalation	PENDING, DUE, OVERDUE, COMPLETED, CANCELLED

Engineering Design**Score governance**

Stable definition plus immutable approved versions

Each input references a CDE version

Selection window and method are explicit

Component and aggregate rules are data, not scattered code

Output is written as a calculated Observation

Licensed or externally governed score material is not reconstructed without approval

Calculation

Use only eligible final or policy-permitted inputs

Normalize quantities before scoring

Reject absent required input

Retain all input Observation IDs

Calculate component points deterministically

Hash inputs and algorithm for idempotency

Correction invalidates dependent score and triggers recalculation

Deterioration alerts

Trigger from governed rule only

Route to eligible current care context

Delivery is not acknowledgement

Acknowledgement is not assessment

Assessment is not action

Action is not resolution

Escalate when configured stage due

Preserve every delivery and escalation attempt

Safe failure

Scoring engine outage does not alter source Observations

Notification outage retains triggered alert

Unknown recipient enters reconciliation

Duplicate source event cannot duplicate alert

Stale or corrected input visibly invalidates prior score

Permissions

Permission	Purpose
clinical.score_definition.manage	Create and maintain score drafts
clinical.score_definition.approve	Approve and activate score version
clinical.score.calculate	Calculate score
clinical.score.view	View score and components
clinical.alert.view	View applicable alert
clinical.alert.acknowledge	Acknowledge exact alert version
clinical.alert.assess	Record assessment
clinical.alert.action	Record clinical action

clinical.alert.resolve	Resolve with evidence
clinical.alert.escalation.manage	Configure escalation routes
clinical.score.reconcile	Resolve calculation or alert exception

Mandatory Tests

Missing required input blocks score

Input outside selection window is not used

Correct normalized unit is used

Same inputs and version produce one score run

Different score version produces distinct run

Component sum and trigger rules are reproducible

Correction invalidates dependent score

Notification failure preserves alert

Delivery does not mark acknowledged

Acknowledgement does not mark resolved

Overdue escalation creates next routing action once

Cross-tenant alert is concealed

Expired client-space assignment cannot acknowledge ordinary alert

Historical score remains reproducible after definition deprecation

Repository-Ready Source Listings

The ZIP is authoritative. Complete source files and SHA-256 checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	43840691b6f47fd660af4c3a36945fca7a224547d13408ea49fd8f767d7904c2

KashTre Clinical Scoring and Deterioration v6.1.0

Install after Volumes 1 to 4. Classification: PICK-AND-DROP CANDIDATE. NEWS2 or another governed score must be installed as an approved versioned configuration. This package does not invent or hardcode a locally reconstructed clinical scoring matrix.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	7d43899888c03c483f06e0bb8e7ba8b7b1b93423830c2839ec75eb3770ab1e17

```
{
  "volume": "Clinical EDD Volume 5 Scoring and Deterioration",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "43840691b6f47fd660af4c3a36945fca7a224547d13408ea49fd8f767d7904c2",
      "bytes": 302
    },
    {
      "path": "composer.json",
      "sha256": "96c2ae780696de61dfc7b429800916e9f40c3a041a4f95ab38be172c13d4720b",
      "bytes": 672
    },
    {
      "path": "config/clinical-scoring.php",
      "sha256": "2c4c6ea3a8589d94a2e105f64b4c24944d92be2ec32cae5dbbeb623668e906c4",
      "bytes": 306
    },
    {
      "path": "database/migrations/2026_08_28_400000_create_scoring_alert_tables.php",
      "sha256": "932f4cdf792c4b315d617d1ae1f4295eca02b8c8f4d2bde9dde78234ab4e96fb",
      "bytes": 5386
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-scoring.php",
      "sha256": "e9511b1b5fbac78272cfe79f3efa30c3852bac217bf0a2f6cbcdccda9365d8b3",
      "bytes": 309
    },
    {

```

```

"path": "src/Actions/AcknowledgeAlert.php",
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"bytes": 1114
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{
"path": "src/Actions/CalculatePatientScore.php",
"sha256": "ec7519b25d8503609f7cc2c4de725eae5df2e1e30196a9c185f5121e65d8696e",
"bytes": 2916
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{
"path": "src/Actions/TriggerDeteriorationAlert.php",
"sha256": "684a79f476ae2715889d82e94018d68b3745c772b9c30560ad34848a7055c557",
"bytes": 2158
},
{
"path": "src/ClinicalScoringServiceProvider.php",
"sha256": "c975a0778c718e0613dcc0385b904d2035ac22811c90a3083ba5753c6670987c",
"bytes": 448
},
{
"path": "src/Contracts/NotificationGateway.php",
"sha256": "309575f462055d13d458d7cb87559de208b2124f723448f70dfa28cecd9143ea",
"bytes": 212
},
{
"path": "src/Contracts/ObservationSnapshotQuery.php",
"sha256": "0ba9a92448ea08bfe9ebab30051863080dc6fa31a7d6d7bc7d8dc4ec108aea97",
"bytes": 258
},
{
"path": "src/Contracts/ObservationWriter.php",
"sha256": "923ab3cafdde31287780c74bb78be896038443c359a127e4d6474256ff6e282c",
"bytes": 283
},
{
"path": "src/Data/ScoreCalculationResult.php",
"sha256": "60081d911be10ac456eb3e932befc8e0cc04b5438c9b1241902ca805831ccf25",
"bytes": 279
},
{
"path": "src/Data/ScoreInputValue.php",
"sha256": "ac1824655141002a66c806a0ad91e17679db0e3277169e140a89ba02fa1b80ec",
"bytes": 310
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{
"path": "src/Enums/AlertSeverity.php",
"sha256": "1fc5399478b0e1cf8294812e0212f828e2766ca06eb23e6de9a32e832a28cebf",
"bytes": 228
},
{
"path": "src/Enums/AlertStatus.php",
"sha256": "1d21f5761182cd33f62951d7aa3d55ff2244d73e481731624eed9524aef7d0d5",
"bytes": 405
},

```

```

{
  "path": "src/Enums/EscalationStatus.php",
  "sha256": "3ef805d1aa0e6acf0d25d746ea8f6db6e4141b3e297e5ee61f979cd64db576a8",
  "bytes": 229
},
{
  "path": "src/Enums/InputSelectionMode.php",
  "sha256": "a2257d8f2f9ec913a241aea18cf6dc8437d2cc7e5d2f44f599a89bc378320159",
  "bytes": 262
},
{
  "path": "src/Enums/RuleOperator.php",
  "sha256": "3cb1cabdf35cb6ef748181f45c44f90b449686e61148309d7595bdfdb7e5e17",
  "bytes": 219
},
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  "path": "src/Enums/ScoreCategory.php",
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  "bytes": 261
},
{
  "path": "src/Enums/ScoreDefinitionStatus.php",
  "sha256": "830230d97fdb02dc95d2e17d79a63c093a32ce84f84919a6a9bb4d96d6ed06f2",
  "bytes": 265
},
{
  "path": "src/Enums/ScoreRunStatus.php",
  "sha256": "1ae9f9cd8da772491600287c8a94a5830e68d960a6b4e7658c5b201ae9bd17f",
  "bytes": 303
},
{
  "path": "src/Models/AlertAcknowledgement.php",
  "sha256": "f8b80e08449262cb555ea02419357482f485998e171c3eb4699c0567fba27fdd",
  "bytes": 578
},
{
  "path": "src/Models/AlertAction.php",
  "sha256": "1ee76b528f1de13f4fc7528ca7bf07d8da62a3ed7687d063c26bc2fa31fbd1c1",
  "bytes": 560
},
{
  "path": "src/Models/AlertAssessment.php",
  "sha256": "28cfd239bcfb8374b26ebc39a8ad01c4dc58ac329ac4a6cbd14b732cd8cf1409",
  "bytes": 568
},
{
  "path": "src/Models/AlertDelivery.php",
  "sha256": "143e81898484579e582346bc74f391943b7835226d3420ea1a740bd662289ee0",
  "bytes": 565
},
{
  "path": "src/Models/DeteriorationAlert.php",
  "sha256": "ee1cb2f76514fe5cb31f5617a2f575b1ad2f00348c79b73ba89499cb636423a2",
  "bytes": 574
},

```

```

{
  "path": "src/Models/EscalationOccurrence.php",
  "sha256": "5ccf20475a73d46c1b139eabf487e3fdb2c575eb161bd6da849fbe86c4d3cc7c",
  "bytes": 578
},
{
  "path": "src/Models/EscalationStep.php",
  "sha256": "e88b6076fd1057727d04302ad623f409aeb7bf23a70d4a9b8b886ff5763ace7b",
  "bytes": 566
},
{
  "path": "src/Models/ScoreBand.php",
  "sha256": "5da24e9c447cd71250bebff8be2c4f26433fca9a3c4c453886e6b6c603c0b25d",
  "bytes": 556
},
{
  "path": "src/Models/ScoreDefinition.php",
  "sha256": "23a848088101e6d8ecf7644e5c0be91e60cbc5065ee2096ca60560d2d9299831",
  "bytes": 568
},
{
  "path": "src/Models/ScoreInputDefinition.php",
  "sha256": "4c6d5d331c7de1eda92c8df546b736dcfa1772c250142575a49c7d3b31ac546e",
  "bytes": 579
},
{
  "path": "src/Models/ScoreReconciliation.php",
  "sha256": "f442f6bf13a8baa279e1869b338328e1adf565b02989eb615a1935e0e5dd5783",
  "bytes": 576
},
{
  "path": "src/Models/ScoreRule.php",
  "sha256": "57565ec7cddddd19c8ab96c3addc4c0b8a646f312e94e4b7a449f888ac422b5",
  "bytes": 556
},
{
  "path": "src/Models/ScoreRun.php",
  "sha256": "a487b3ac25d95ac38f1550cd735aef091db62c848d23200b4bf2e1c5f17ff945",
  "bytes": 554
},
{
  "path": "src/Models/ScoreRunComponent.php",
  "sha256": "c64aa1568ede303124959a72e5a033b57dc59a2d5d1e54a6ee191a329c25c133",
  "bytes": 573
},
{
  "path": "src/Models/ScoreRunInput.php",
  "sha256": "55113db86936344b03c34c68629281e92ffe6bcd7364570557242f4f02b484ce",
  "bytes": 565
},
{
  "path": "src/Models/ScoreVersion.php",
  "sha256": "c494ac9bfef0d18053b5273f8bd9e70d61db82f33339eccddaf6746d76c7ceda",
  "bytes": 562
},

```

```
{
  "path": "src/Services/DeterministicRuleEvaluator.php",
  "sha256": "d172237743858e18aa1cf39dad356f1d207529f2f03b67b8cb7b827c1ede66f8",
  "bytes": 594
},
{
  "path": "src/Services/ScoreCalculator.php",
  "sha256": "fb4492f8629ffc4b81896d2d760f186cd3f72dd2a090ea7f1c472825331f7526",
  "bytes": 1382
},
{
  "path": "tests/Unit/DeterministicRuleEvaluatorTest.php",
  "sha256": "152bcb32b7149596f281a1ae181094c3727055097b7705b6dd586437de9f83d4",
  "bytes": 556
},
{
  "path": "tests/Unit/ScoreCalculatorTest.php",
  "sha256": "9272498edb1a7cc2e0eb9cbbc8e65deef855ebae011a299e5459e2880591a4d5",
  "bytes": 822
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	96c2ae780696de61dfc7b429800916e9f40c3a041a4f95ab38be172c13d4720b

```
{
  "name": "kashtre/clinical-scoring",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "kashtre/clinical-observation-plan": "^6.1",
    "kashtre/clinical-observation": "^6.1",
    "kashtre/clinical-forms": "^6.1"
  },
  "require-dev": {
    "orchestra/testbench": "^11.0",
    "phpunit/phpunit": "^12.0",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\Scoring\\": "src/"
      },
      "autoload-dev": {
        "psr-4": {
          "KashTre\\Clinical\\Scoring\\Tests\\": "tests/"
        }
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\Scoring\\ClinicalScoringServiceProvider"
          ]
        }
      },
      "scripts": {
        "test": "phpunit"
      }
    }
  }
}
```

config/clinical-scoring.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-scoring.php	2c4c6ea3a8589d94a2e105f64b4c24944d92be2ec32cae5dbbbeb623668e906c4

```
<?php
declare(strict_types=1);
return
[
  'calculation_batch_size' => (int)env('CLINICAL_SCORE_BATCH_SIZE', 100),
  'alert_delivery_timeout_seconds' => (int)env('CLINICAL_ALERT_DELIVERY_TIMEOUT', 5),
  'escalation_projection_minutes' => (int)env('CLINICAL_ESCALATION_PROJECTION_MINUTES', 1),
  'require_complete_inputs' => true
];
```

database/migrations/2026_08_28_400000_create_scoring_alert_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_400000_create_scoring_alert_tables.php	932f4cdf792c4b315d617d1ae1f4295eca02b8c8f4d2bde9dde78234ab4e96fb

```

<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;use Illuminate\Database\Schema\Blueprint;use
Illuminate\Support\Facades\Schema;
return new class extends Migration{public function up():void{
Schema::create('clinical_score_definitions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl->
string('code',96);$tbl->string('name',180);$tbl->string('category',32);$tbl->string('status',24);$tbl->unsignedInteger('current_version')->
nullable();$tbl->dateTime('created_at_utc',6);$tbl->unique(['tenant_id','code'],'uq_score_code'));
Schema::create('clinical_score_versions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl->
ulid('score_definition_public_id');$tbl->unsignedInteger('version_no');$tbl->string('status',24);$tbl->string('algorithm_version',64);$tbl->
ulid('output_cde_public_id');$tbl->json('component_rules');$tbl->json('aggregate_rules');$tbl->json('configuration')->nullable();$tbl->
dateTime('effective_from_utc',6);$tbl->dateTime('effective_to_utc',6)->nullable();$tbl->
unique(['tenant_id','score_definition_public_id','version_no'],'uq_score_version'));
Schema::create('clinical_score_input_definitions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->
string('tenant_id',64);$tbl->ulid('score_version_public_id');$tbl->string('input_code',96);$tbl->ulid('cde_public_id');$tbl->
unsignedInteger('cde_version');$tbl->boolean('is_required')->default(true);$tbl->json('selection_rule');$tbl->
unsignedInteger('display_order');$tbl->unique(['score_version_public_id','input_code'],'uq_score_input'));
Schema::create('clinical_score_runs',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl->
ulid('score_version_public_id');$tbl->ulid('patient_public_id');$tbl->ulid('encounter_public_id');$tbl->string('status',32);$tbl->
string('total_score',100)->nullable();$tbl->char('input_hash',64);$tbl->string('algorithm_version',64);$tbl->
ulid('calculated_observation_public_id')->nullable();$tbl->dateTime('occurred_at_utc',6);$tbl->dateTime('calculated_at_utc',6);$tbl->
unsignedBigInteger('record_version')->default(1);$tbl->
unique(['tenant_id','score_version_public_id','patient_public_id','encounter_public_id','input_hash'],'uq_score_run'));
Schema::create('clinical_score_run_inputs',function(Blueprint$tbl){$tbl->id();$tbl->ulid('score_run_public_id');$tbl->
string('input_code',96);$tbl->ulid('observation_public_id');$tbl->string('normalized_value',100);$tbl->ulid('unit_public_id')->nullable();$tbl->
dateTime('occurred_at_utc',6);$tbl->unique(['score_run_public_id','input_code'],'uq_score_run_input'));
Schema::create('clinical_score_run_components',function(Blueprint$tbl){$tbl->id();$tbl->ulid('score_run_public_id');$tbl->
string('component_code',96);$tbl->decimal('points',12,4);$tbl->json('evidence');$tbl->
unique(['score_run_public_id','component_code'],'uq_score_component'));
Schema::create('clinical_deterioration_alerts',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->
string('tenant_id',64);$tbl->ulid('patient_public_id');$tbl->ulid('encounter_public_id');$tbl->ulid('score_run_public_id')->nullable();$tbl->
string('trigger_rule_code',96);$tbl->string('severity',24);$tbl->string('status',32);$tbl->dateTime('triggered_at_utc',6);$tbl->
dateTime('resolved_at_utc',6)->nullable();$tbl->unsignedBigInteger('record_version')->default(1);$tbl->
unique(['tenant_id','score_run_public_id','trigger_rule_code'],'uq_deterioration_trigger');$tbl->
index(['tenant_id','status','severity','triggered_at_utc'],'idx_deterioration_queue'));
Schema::create('clinical_alert_deliveries',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl->
ulid('alert_public_id');$tbl->string('route_code',96);$tbl->json('recipient_context');$tbl->string('delivery_status',24);$tbl->
dateTime('delivered_at_utc',6)->nullable();$tbl->string('provider_reference',160)->nullable();});
Schema::create('clinical_alert_acknowledgements',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->
string('tenant_id',64);$tbl->ulid('alert_public_id');$tbl->ulid('acknowledged_by_public_id');$tbl->text('note')->nullable();$tbl->
dateTime('acknowledged_at_utc',6);$tbl->unique(['alert_public_id','acknowledged_by_public_id'],'uq_alert_ack_actor'));
Schema::create('clinical_escalation_steps',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl->
string('route_code',96);$tbl->unsignedInteger('sequence');$tbl->unsignedInteger('delay_minutes');$tbl->
string('recipient_role_code',96);$tbl->string('status',24);$tbl->unique(['tenant_id','route_code','sequence'],'uq_escalation_step'));
Schema::create('clinical_escalation_occurrences',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->

```

```
>string('tenant_id',64);$t->ulid('alert_public_id');$t->ulid('escalation_step_public_id');$t->string('status',24);$t-
>dateTime('due_at_utc',6);$t->dateTime('completed_at_utc',6)->nullable();$t-
>unique(['alert_public_id','escalation_step_public_id'],'uq_alert_escalation');});
}public function
down():void{foreach(['clinical_escalation_occurrences','clinical_escalation_steps','clinical_alert_acknowledgements','clinical_alert_
deliveries','clinical_deterioration_alerts','clinical_score_run_components','clinical_score_run_inputs','clinical_score_runs','clinical_
score_input_definitions','clinical_score_versions','clinical_score_definitions']as$t)Schema::dropIfExists($t);};
```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force
```

routes/clinical-scoring.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical- scoring.php	e9511b1b5fbac78272cfe79f3efa30c3852bac217bf0a2f6cbcdccda9365d8b3

```
<?php
declare(strict_types=1);
use Illuminate\Support\Facades\Route;
Route::middleware(['api','auth:sanctum'])->prefix('api/v1/clinical')-
>group(function(){Route::get('/patients/{patient}/encounters/{encounter}/scores',fn()->response()-
>json(['error'=>['code'=>'CLN_SCORE_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/AcknowledgeAlert.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDATE	src/Actions/AcknowledgeAlert.php	ee5bd1366a1ed173562ad7d96523f27c2fc8d2a357c739948dd45bfea9e772e2

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Actions;
use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Scoring\Models\{AlertAcknowledgement,DeteriorationAlert};
final readonly class AcknowledgeAlert{public function __construct(private PublicIdGenerator $ids){}public function execute(string
$tenantId,string $alertId,string $actorId,string $note):DeteriorationAlert{return
DB::transaction(function()use($tenantId,$alertId,$actorId,$note){$a=DeteriorationAlert::query()->where('tenant_id',$tenantId)-
>where('public_id',$alertId)->lockForUpdate()->firstOrFail();if(in_array($a-
>status,['RESOLVED','CANCELLED','ENTERED_IN_ERROR'],true))throw new
\DomainException('CLN_ALERT_NOT_ACKNOWLEDGEABLE');AlertAcknowledgement::query()-
>firstOrCreate(['alert_public_id'=>$alertId,'acknowledged_by_public_id'=>$actorId,['public_id'=>(string)$this->ids-
>generate()],['tenant_id'=>$tenantId,'note'=>$note,'acknowledged_at_utc'=>now('UTC')]);$a-
>forceFill(['status'=>'ACKNOWLEDGED','record_version'=>$a->record_version+1])->save();return$a;});}
```

src/Actions/CalculatePatientScore.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Actions/CalculatePatientScore.php	ec7519b25d8503609f7cc2c4de725eae5df2e1e30196a9c185f5121e65d8696e
-------------------------	---------------------------------------	--

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Actions;
use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Scoring\Contracts\{ObservationSnapshotQuery,ObservationWriter};use
KashTre\Clinical\Scoring\Data\ScoreInputValue;use
KashTre\Clinical\Scoring\Models\{ScoreInputDefinition,ScoreRun,ScoreRunComponent,ScoreRunInput,ScoreVersion};use
KashTre\Clinical\Scoring\Services\ScoreCalculator;
final readonly class CalculatePatientScore{public function __construct(private ObservationSnapshotQuery $observations,private
ObservationWriter $writer,private ScoreCalculator $calculator,private PublicIdGenerator $ids){}public function execute(string
$tenantId,string $patientId,string $encounterId,string $scoreVersionId,\DateTimeImmutable $asOf):ScoreRun{return
DB::transaction(function()use($tenantId,$patientId,$encounterId,$scoreVersionId,$asOf){$version=ScoreVersion::query()-
>where('tenant_id',$tenantId)->where('public_id',$scoreVersionId)->where('status','ACTIVE')-
>firstOrFail();$defs=ScoreInputDefinition::query()->where('score_version_public_id',$scoreVersionId)->orderBy('display_order')-
>get();$inputs=[];foreach($defs as$d){$o=$this->observations->input($tenantId,$patientId,$encounterId,$d-
>cde_public_id,$asOf,$d->selection_rule??[]);if(!$o&&$d->is_required)throw new
\DomainException('CLN_SCORE_REQUIRED_INPUT_MISSING');if($o)$inputs[$d->input_code]=new ScoreInputValue($d-
>input_code,$o['public_id'],$o['normalized_value'],$o['base_unit_public_id'],$o['status'],$o['occurred_at_utc']);$result=$this-
>calculator->calculate($version->algorithm_version,$inputs,$version->component_rules,$version-
>aggregate_rules);$run=ScoreRun::query()-
>firstOrCreate(['tenant_id'=>$tenantId,'score_version_public_id'=>$scoreVersionId,'patient_public_id'=>$patientId,'encounter_public_id'=>$encounterId,'input_hash'=>$result->inputHash],['public_id'=>(string)$this->ids-
>generate(),'status'=>'CALCULATED','total_score'=>$result-
>total,'occurred_at_utc'=>$asOf,'calculated_at_utc'=>now('UTC'),'algorithm_version'=>$result-
>algorithmVersion,'record_version'=>1]);if(!$run->wasRecentlyCreated)return$run;foreach($inputs as$i)ScoreRunInput::query()-
>create(['score_run_public_id'=>$run->public_id,'input_code'=>$i->inputCode,'observation_public_id'=>$i-
>observationPublicId,'normalized_value'=>$i->normalizedValue,'unit_public_id'=>$i->unitPublicId,'occurred_at_utc'=>$i-
>occurredAtUtc]);foreach($result->components as$code=>$c)ScoreRunComponent::query()->create(['score_run_public_id'=>$run-
>public_id,'component_code'=>$code,'points'=>$c['points'],'evidence'=>$c]);$run->calculated_observation_public_id=$this-
>writer->writeCalculatedScore($tenantId,$patientId,$encounterId,$version->output_cde_public_id,$result-
>total,$asOf,['score_run'=>$run->public_id,'inputs'=>array_column($result->components,'observation')]);$run-
>save();return$run;}}}
```

src/Actions/TriggerDeteriorationAlert.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/TriggerDeteriorationAlert.php	684a79f476ae2715889d82e94018d68b3745c772b9c30560ad34848a7055c557

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Actions;
use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Scoring\Contracts\NotificationGateway;use
KashTre\Clinical\Scoring\Models\{AlertDelivery,DeteriorationAlert,EscalationOccurrence,EscalationStep};
final readonly class TriggerDeteriorationAlert{public function __construct(private PublicIdGenerator $ids,private NotificationGateway
$notifications){}public function execute(string $tenantId,string $patientId,string $encounterId,string $scoreRunId,string
```

```

$ruleCode,string $severity,string $routeCode,array $recipientContext):DeteriorationAlert{return
DB::transaction(function()use($tenantId,$patientId,$encounterId,$scoreRunId,$ruleCode,$severity,$routeCode,$recipientContext){
$a=DeteriorationAlert::query()-
>firstOrCreate(['tenant_id'=>$tenantId,'score_run_public_id'=>$scoreRunId,'trigger_rule_code'=>$ruleCode],['public_id'=>(string)$this-
is->ids-
>generate(), 'patient_public_id'=>$patientId,'encounter_public_id'=>$encounterId,'severity'=>$severity,'status'=>'TRIGGERED','trigge
red_at_utc'=>now('UTC'),'record_version'=>1]);if(!$a->wasRecentlyCreated)return$a;$delivery=$this->notifications-
>route($tenantId,$a->public_id,$routeCode,$recipientContext);AlertDelivery::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId,'alert_public_id'=>$a-
>public_id,'route_code'=>$routeCode,'recipient_context'=>$recipientContext,'delivery_status'=>$delivery['status'],'delivered_at_utc
'=>$delivery['delivered_at_utc']??null,'provider_reference'=>$delivery['reference']??null));$a-
>forceFill(['status'=>$delivery['status']==='DELIVERED'?'DELIVERED':'ROUTED'])->save();foreach(EscalationStep::query()-
>where('tenant_id',$tenantId)->where('route_code',$routeCode)->where('status','ACTIVE')->orderBy('sequence')-
>get())as$step)EscalationOccurrence::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId,'alert_public_id'=>$a->public_id,'escalation_step_public_id'=>$step-
>public_id,'status'=>'PENDING','due_at_utc'=>now('UTC')->addMinutes($step->delay_minutes));return$a;});}}

```

src/ClinicalScoringServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ClinicalScoringServiceProvider.php	c975a0778c718e0613dcc0385b904d2035ac22811c90a3083ba5753c6670987c

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring;
use Illuminate\Support\ServiceProvider;
final class ClinicalScoringServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-scoring.php','clinical-scoring');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-scoring.php');}}

```

src/Contracts/NotificationGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/NotificationGateway.php	309575f462055d13d458d7cb87559de208b2124f723448f70dfa28cecd9143ea

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Contracts;
interface NotificationGateway{public function route(string $tenantId,string $alertId,string $routeCode,array
$recipientContext):array;}

```

src/Contracts/ObservationSnapshotQuery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObservationSnapshotQuery.php	0ba9a92448ea08bfe9ebab30051863080dc6fa31a7d6d7bc7d8dc4ec108aea97

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Contracts;
interface ObservationSnapshotQuery{public function input(string $tenantId,string $patientId,string $encounterId,string
$dcdId,\DateTimeImmutable $asOf,array $selectionRule):?array;}
```

src/Contracts/ObservationWriter.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObservationWriter.php	923ab3cafdde31287780c74bb78be896038443c359a127e4d6474256ff6e282c

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Contracts;
interface ObservationWriter{public function writeCalculatedScore(string $tenantId,string $patientId,string $encounterId,string
$dcdId,string $value,\DateTimeImmutable $occurredAt,array $provenance):string;}
```

src/Data/ScoreCalculationResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/ScoreCalculationResult.php	60081d911be10ac456eb3e932befc8e0cc04b5438c9b1241902ca805831ccf25

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Data;
final readonly class ScoreCalculationResult{public function __construct(public string $total,public array $components,public array
$triggeredRules,public string $algorithmVersion,public string $inputHash){}}
```

src/Data/ScoreInputValue.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/ScoreInputValue.php	ac1824655141002a66c806a0ad91e17679db0e3277169e140a89ba02fa1b80ec

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Data;
final readonly class ScoreInputValue{public function __construct(public string $inputCode,public string $observationPublicId,public
string $normalizedValue,public string $unitPublicId,public string $status,public string $occurredAtUtc){}}
```

src/Enums/AlertSeverity.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AlertSeverity.php	1fc5399478b0e1cf8294812e0212f828e2766ca06eb23e6de9a32e832a28cebf

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
```

```
enum AlertSeverity:string{
case INFORMATIONAL='INFORMATIONAL';
case LOW='LOW';
case MODERATE='MODERATE';
case HIGH='HIGH';
case CRITICAL='CRITICAL';
}
```

src/Enums/AlertStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AlertStatus.php	1d21f5761182cd33f62951d7aa3d55ff2244d73e481731624eed9524aef7d0d5

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
enum AlertStatus:string{
case TRIGGERED='TRIGGERED';
case ROUTED='ROUTED';
case DELIVERED='DELIVERED';
case ACKNOWLEDGED='ACKNOWLEDGED';
case ASSESSED='ASSESSED';
case ACTION_TAKEN='ACTION_TAKEN';
case ESCALATED='ESCALATED';
case RESOLVED='RESOLVED';
case CANCELLED='CANCELLED';
case ENTERED_IN_ERROR='ENTERED_IN_ERROR';
}
```

src/Enums/EscalationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/EscalationStatus.php	3ef805d1aa0e6acf0d25d746ea8f6db6e4141b3e297e5ee61f979cd64db576a8

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
enum EscalationStatus:string{
case PENDING='PENDING';
case DUE='DUE';
case OVERDUE='OVERDUE';
case COMPLETED='COMPLETED';
case CANCELLED='CANCELLED';
}
```

src/Enums/InputSelectionMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/InputSelectionMode.php	a2257d8f2f9ec913a241aea18cf6dc8437d2cc7e5d2f44f599a89bc378320159

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
enum InputSelectionMode:string{
    case EXACT_SLOT='EXACT_SLOT';
    case LATEST_IN_WINDOW='LATEST_IN_WINDOW';
    case CLOSEST_BEFORE='CLOSEST_BEFORE';
    case CLOSEST_ABSOLUTE='CLOSEST_ABSOLUTE';
}
```

src/Enums/RuleOperator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/RuleOperator.php	3cb1cabdf35cb6ef748181f45c44f90b449686e61148309d7595bdfdb7e5e17

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
enum RuleOperator:string{
    case LT='LT';
    case LTE='LTE';
    case EQ='EQ';
    case GTE='GTE';
    case GT='GT';
    case IN='IN';
    case BETWEEN='BETWEEN';
}
```

src/Enums/ScoreCategory.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ScoreCategory.php	b71e032c11b95c0d94e8be0a5a8cd1a96b5be58ad2b545f1c2c260403c3fb677

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
enum ScoreCategory:string{
    case EARLY_WARNING='EARLY_WARNING';
    case RISK_ASSESSMENT='RISK_ASSESSMENT';
    case CLINICAL_SCALE='CLINICAL_SCALE';
    case CALCULATED_INDEX='CALCULATED_INDEX';
}
```

src/Enums/ScoreDefinitionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ScoreDefinitionStatus.php	830230d97fdb02dc95d2e17d79a63c093a32ce84f84919a6a9bb4d96d6ed06f2

```
<?php
declare(strict_types=1);
```

```
namespace KashTre\Clinical\Scoring\Enums;
enum ScoreDefinitionStatus:string{
    case DRAFT='DRAFT';
    case IN_REVIEW='IN_REVIEW';
    case APPROVED='APPROVED';
    case ACTIVE='ACTIVE';
    case DEPRECATED='DEPRECATED';
    case RETIRED='RETIRED';
}
```

src/Enums/ScoreRunStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ScoreRunStatus.php	1ae9f9fcd8da772491600287c8a94a5830e68d960a6b4e7658c5b201ae9bd17f

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Enums;
enum ScoreRunStatus:string{
    case PENDING_INPUTS='PENDING_INPUTS';
    case CALCULATED='CALCULATED';
    case INVALIDATED='INVALIDATED';
    case RECALCULATION_REQUIRED='RECALCULATION_REQUIRED';
    case ENTERED_IN_ERROR='ENTERED_IN_ERROR';
}
```

src/Models/AlertAcknowledgement.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AlertAcknowledgement.php	f8b80e08449262cb555ea02419357482f485998e171c3eb4699c0567fba27fdd

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class AlertAcknowledgement extends Model{public $timestamps=false;protected
$table='clinical_alert_acknowledgements';protected $guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/AlertAction.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AlertAction.php	1ee76b528f1de13f4fc7528ca7bf07d8da62a3ed7687d063c26bc2fa31fbd1c1

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
```

```
final class AlertAction extends Model{public $timestamps=false;protected $table='clinical_alert_actions';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/AlertAssessment.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AlertAssessment.php	28cfd239bcfb8374b26ebc39a8ad01c4dc58ac329ac4a6cbd14b732cd8cf1409

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class AlertAssessment extends Model{public $timestamps=false;protected $table='clinical_alert_assessments';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/AlertDelivery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AlertDelivery.php	143e81898484579e582346bc74f391943b7835226d3420ea1a740bd662289ee0

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class AlertDelivery extends Model{public $timestamps=false;protected $table='clinical_alert_deliveries';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/DeteriorationAlert.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DeteriorationAlert.php	ee1cb2f76514fe5cb31f5617a2f575b1ad2f00348c79b73ba89499cb636423a2

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class DeteriorationAlert extends Model{public $timestamps=false;protected $table='clinical_deterioration_alerts';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/EscalationOccurrence.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/EscalationOccurrence.php	5ccf20475a73d46c1b139eabf487e3fdb2c575eb161bd6da849fbe86c4d3cc7c

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class EscalationOccurrence extends Model{public $timestamps=false;protected
$table='clinical_escalation_occurrences';protected $guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/EscalationStep.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/EscalationStep.php	e88b6076fd1057727d04302ad623f409aeb7bf23a70d4a9b8b886ff5763ace7b

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class EscalationStep extends Model{public $timestamps=false;protected $table='clinical_escalation_steps';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/ScoreBand.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreBand.php	5da24e9c447cd71250bebff8be2c4f26433fca9a3c4c453886e6b6c603c0b25d

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreBand extends Model{public $timestamps=false;protected $table='clinical_score_bands';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/ScoreDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Models/ScoreDefinition.php	23a848088101e6d8ecf7644e5c0be91e60cbc5065ee2096ca60560d2d9299831

CANDIDATE		
-----------	--	--

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreDefinition extends Model{public $timestamps=false;protected $table='clinical_score_definitions';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/ScoreInputDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreInputDefinition.php	4c6d5d331c7de1eda92c8df546b736dcfa1772c250142575a49c7d3b31ac546e

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreInputDefinition extends Model{public $timestamps=false;protected
$table='clinical_score_input_definitions';protected $guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/ScoreReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreReconciliation.php	f442f6bf13a8baa279e1869b338328e1adf565b02989eb615a1935e0e5dd5783

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreReconciliation extends Model{public $timestamps=false;protected
$table='clinical_score_reconciliations';protected $guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/ScoreRule.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreRule.php	57565ec7cddddd19c8ab96c3addc4c0b8a646f312e94e4b7a449f888ac422b5

```
<?php
declare(strict_types=1);
```

```

namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreRule extends Model{public $timestamps=false;protected $table='clinical_score_rules';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}

```

src/Models/ScoreRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreRun.php	a487b3ac25d95ac38f1550cd735aef091db62c848d23200b4bf2e1c5f17ff945

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreRun extends Model{public $timestamps=false;protected $table='clinical_score_runs';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}

```

src/Models/ScoreRunComponent.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreRunComponent.php	c64aa1568ede303124959a72e5a033b57dc59a2d5d1e54a6ee191a329c25c133

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreRunComponent extends Model{public $timestamps=false;protected
$table='clinical_score_run_components';protected $guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime'];}

```

src/Models/ScoreRunInput.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreRunInput.php	55113db86936344b03c34c68629281e92ffe6bcd7364570557242f402b484ce

```

<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreRunInput extends Model{public $timestamps=false;protected $table='clinical_score_run_inputs';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immutable_datetime'];}

```

```
ble_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_date
time','completed_at_utc'=>'immutable_datetime'];}
```

src/Models/ScoreVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ScoreVersion.php	c494ac9bfef0d18053b5273f8bd9e70d61db82f33339eccddaf6746d76c7ceda

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Models;
use Illuminate\Database\Eloquent\Model;
final class ScoreVersion extends Model{public $timestamps=false;protected $table='clinical_score_versions';protected
$guarded=[];protected
$casts=['configuration'=>'array','selection_rule'=>'array','recipient_context'=>'array','evidence'=>'array','occurred_at_utc'=>'immuta
ble_datetime','calculated_at_utc'=>'immutable_datetime','triggered_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_date
time','completed_at_utc'=>'immutable_datetime'];}
```

src/Services/DeterministicRuleEvaluator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/DeterministicRuleEvaluator.php	d172237743858e18aa1cf39dad356f1d207529f2f03b67b8cb7b827c1ede66f8

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Services;
final class DeterministicRuleEvaluator{public function matches(string $operator,string $value,?string $lower,?string $upper,array
$allowed=[]):bool{return
match($operator){'LT'=>(float)$value<(float)$lower,'LTE'=>(float)$value<=(float)$lower,'EQ'=>$value=== $lower,'GTE'=>(float)$value>
=(float)$lower,'GT'=>(float)$value>(float)$lower,'BETWEEN'=>(float)$value>=(float)$lower&&(float)$value<=(float)$upper,'IN'=>in_arr
ay($value,$allowed,true),default=>throw new \InvalidArgumentException("Unsupported rule operator");}}
```

src/Services/ScoreCalculator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/ScoreCalculator.php	fb4492f8629ffc4b81896d2d760f186cd3f72dd2a090ea7f1c472825331f7526

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Services;
use KashTre\Clinical\Scoring\Data\{ScoreCalculationResult,ScoreInputValue};
final readonly class ScoreCalculator{public function __construct(private DeterministicRuleEvaluator $rules){}public function
calculate(string $algorithmVersion,array $inputs,array $componentRules,array
$aggregateRules):ScoreCalculationResult{$components=[];$total=0.0;foreach($componentRules
as$inputCode=>$rules){$input=$inputs[$inputCode]??null;if(!$input instanceof ScoreInputValue)throw new
\DomainException('CLN_SCORE_REQUIRED_INPUT_MISSING');$score=null;foreach($rules as$r){if($this->rules-
>matches($r['operator'],$input-
>normalizedValue,$r['lower']??null,$r['upper']??null,$r['allowed']??[]){$score=(float)$r['points'];break;}if($score===null)throw new
\DomainException('CLN_SCORE_INPUT_UNMAPPED');$components[$inputCode]='observation'=>$input-
>observationPublicId,'value'=>$input->normalizedValue,'points'=>$score];$total+=$score;$triggered=[];foreach($aggregateRules
```

```
as$>matches($r['operator'],(string)$total,$r['lower']??null,$r['upper']??null,$r['allowed']??[]))$triggered[]=$r['code'];$hash=hash('sha256',json_encode(['algorithm'=>$algorithmVersion,'components'=>$components],JSON_THROW_ON_ERROR));return new ScoreCalculationResult((string)$total,$components,$triggered,$algorithmVersion,$hash);}}
```

tests/Unit/DeterministicRuleEvaluatorTest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Unit/DeterministicRuleEvaluatorTest.php	152bcb32b7149596f281a1ae181094c3727055097b7705b6dd586437de9f83d4

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Tests\Unit;use PHPUnit\Framework\TestCase;use KashTre\Clinical\Scoring\Services\DeterministicRuleEvaluator;
final class DeterministicRuleEvaluatorTest extends TestCase{public function test_numeric_and_membership_rules():void{$e=new DeterministicRuleEvaluator();self::assertTrue($e->matches('LT','8','9',null));self::assertTrue($e->matches('BETWEEN','15','12','20'));self::assertTrue($e->matches('IN','OXYGEN',null,null,['AIR','OXYGEN']));self::assertFalse($e->matches('GT','5','7',null));}}
```

tests/Unit/ScoreCalculatorTest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	tests/Unit/ScoreCalculatorTest.php	9272498edb1a7cc2e0eb9cbbc8e65deef855ebae011a299e5459e2880591a4d5

```
<?php
declare(strict_types=1);
namespace KashTre\Clinical\Scoring\Tests\Unit;use PHPUnit\Framework\TestCase;use KashTre\Clinical\Scoring\Data\ScoreInputValue;use KashTre\Clinical\Scoring\Services\{DeterministicRuleEvaluator,ScoreCalculator};
final class ScoreCalculatorTest extends TestCase{public function test_calculates_reproducible_component_total():void{$c=new ScoreCalculator(new DeterministicRuleEvaluator());$inputs=['pulse'=>new ScoreInputValue('pulse','01ARZ3NDEKTSV4RRFFQ69G5FAV','110','U1','FINAL','2026-08-28T08:00:00Z')];$r=$c->calculate('TEST-1',$inputs,['pulse'=>[['operator'=>'GTE','lower'=>'100','points'=>'2']],['code'=>'REVIEW','operator'=>'GTE','lower'=>'2']]);self::assertSame('2',$r->total);self::assertSame(['REVIEW'],$r->triggeredRules);self::assertSame(2.0,$r->components['pulse']['points']);}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	Tenant, IDs, authorization, audit, outbox and time
Volume 2	Observation Plan and response work
Volume 3	Observation input query, calculated Observation writer and correction events
Volume 4	Rendered chart and form consumption
Main/HR	Current eligible users, assignments and escalation roles
Communications	Durable delivery status and provider reference
Vue	Score details, deterioration queue, acknowledgement, assessment, actions and escalation configuration

Blocking Gaps

Execute in actual KashTre repository

Approve score-definition governance and licensing process

Approve initial clinical score catalogue

Bind ObservationSnapshotQuery and ObservationWriter

Approve incomplete-score behaviour by score

Approve alert thresholds and response pathways

Bind assignment-aware NotificationGateway

Implement host controllers, requests, resources, jobs and Vue workspaces

Run clean, upgrade, rollback, concurrency, outage, accessibility, performance and clinical gold tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Clinical score and deterioration authority
Volumes 1-4	Foundation, plans, authoritative Observations and rendering
Royal College of Physicians NEWS2 materials	External governed reference where NEWS2 is approved
HL7 FHIR R5 RiskAssessment	External reference for formal risk assessment outputs

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 6

MEDICATION MANAGEMENT, eMAR, ADMINISTRATION AND CONSUMPTION

Orders, Dose Opportunities, Barcode Verification, Infusions, Waste, PRN Reassessment and Inventory Reconciliation

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 to 5

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Safety Rule

The Medication Administration event is the Clinical truth that a medicine was administered or not administered. Pharmacy remains authoritative for dispensing and product verification. Inventory remains authoritative for physical stock consequences. Main remains authoritative for Approved Pool, coverage and financial consequences. A valid Clinical administration is never deleted because Inventory or Main is unavailable.

Medication order -> eMAR opportunity -> patient/product verification -> administration commit -> draft Consumption Session -> add incidental supplies or confirm none -> Inventory acceptance -> Main financial evaluation

Entity Register

Entity	Table	Invariant
MedicationOrder	clinical_medication_orders	Tenant, patient, encounter and source-event provenance preserved
DoseOpportunity	clinical_medication_dose_opportunities	Tenant, patient, encounter and source-event provenance preserved
MedicationAdministration	clinical_medication_administrations	Tenant, patient, encounter and source-event provenance preserved
AdministrationProduct	clinical_administration_products	Tenant, patient, encounter and source-event provenance preserved
AdministrationCheck	clinical_administration_checks	Tenant, patient, encounter and source-event provenance preserved
MedicationWaste	clinical_medication_waste	Tenant, patient, encounter and source-event provenance preserved
InfusionEvent	clinical_infusion_events	Tenant, patient, encounter and source-event provenance preserved
PrnReassessment	clinical_prn_reassessments	Tenant, patient, encounter and source-event provenance preserved
ConsumptionSession	clinical_consumption_sessions	Tenant, patient, encounter and source-event provenance preserved
ConsumptionLine	clinical_consumption_lines	Tenant, patient, encounter and source-event provenance preserved
ConsumptionSubmission	clinical_consumption_submissions	Tenant, patient, encounter and source-event provenance preserved
MedicationCorrection	clinical_medication_corrections	Tenant, patient, encounter and source-event provenance preserved
MedicationReconciliation	clinical_medication_reconciliations	Tenant, patient, encounter and source-event provenance preserved

Engineering Design**Medication orders and eMAR**

One active order version generates deterministic dose opportunities

Schedule timezone and local intent retained

Order hold or discontinuation ends future opportunities prospectively

eMAR displays due, overdue, administered and not-administered states

Administration safety

Verify patient, order, product, dose, route, time and performer

Separate clinical dose from physical package quantity

Capture lot and expiry where available

Record start and completion time separately

Prevent second administration against same opportunity

Support partial dose and multiple products

Waste and infusions

Separate administered quantity from waste

Require witness where configured

Breakage and contamination are not patient administration

Infusion pause, resume, rate change and bag change are distinct events

PRN administration creates reassessment work

Consumption

Draft session created only after valid administration commit

Medication product is prepopulated

Incidental supplies can be scanned or selected

User confirms all items or no additional items

One grouped idempotent session sent to Inventory

Inventory and Main processing times never replace physical-use time

Failure enters reconciliation without invalidating administration

Permissions

Permission	Purpose
clinical.medication_order.create	Create order
clinical.medication_order.sign	Authorize order
clinical.medication_administer	Commit administration
clinical.medication_waste.record	Record waste
clinical.medication_waste.witness	Witness controlled waste
clinical.infusion.manage	Manage infusion events
clinical.consumption.confirm	Confirm session
clinical.consumption.defer	Save for reconciliation
clinical.consumption.correct	Correct physical-use lines
clinical.medication.reconcile	Resolve medication exception

Mandatory Tests

Wrong patient blocks administration
Wrong product or lot blocks administration
Duplicate submit creates one administration
Clinical dose and physical quantity remain separate
Partial dose and waste remain separate
Inventory outage preserves administration and session
Approved Pool decision is not requested from nurse
PRN administration creates reassessment work
Correction preserves original administration
Same session retry does not duplicate Inventory consumption
Transfer does not duplicate due opportunities
Cross-tenant medication order is concealed

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	78b878657faed651c18493417b182ebfdb2cb220ae7723061a68fd3910d6d351

KashTre Clinical Medication and Consumption v6.1.0

Pick-and-drop candidate. Install after Volumes 1-5. Bind patient/encounter authority, medication order service, Pharmacy verification, Inventory consumption, Shared Unit Engine, Observation Plan monitoring, and Vue eMAR/Confirm Consumption workspaces before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	30d7f2b3e4972ffdb09cc81d7e9b184e6fb04bd06534fcca8a79e8c426e3aeb3

```
{
  "volume": "Clinical EDD Volume 6 Medication and Consumption",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "78b878657faed651c18493417b182ebfdb2cb220ae7723061a68fd3910d6d351",
      "bytes": 327
    },
    {
      "path": "composer.json",
      "sha256": "4e4c7333c2620cdd55d0d6c83614c3f556d002bf0e122bb2912d2a8061aec084",
      "bytes": 469
    },
    {
      "path": "config/clinical-medication.php",
      "sha256": "c814038659bbac25ec5cf54f7cf9e0043cb573904f8da2fc1f997f1c9d45988e",
      "bytes": 229
    },
    {
      "path": "database/migrations/2026_08_28_500000_create_medication_consumption_tables.php",
      "sha256": "2630d5b3897f8a6b00196efa020075cf13a473dfa1101484c49a7116e7648290",
      "bytes": 4561
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-medication.php",
      "sha256": "ba29d8bcb8149ffb1da5319d763f0b4450f05951803b65ec663c96df04b9202a",
      "bytes": 303
    }
  ]
}
```

```

{
  "path": "src/Actions/CommitMedicationAdministration.php",
  "sha256": "02ad529f33d2d937b33638b6b9f92805da835afdd5914c9d8bd4eeec14ce4d71",
  "bytes": 4026
},
{
  "path": "src/Actions/ConfirmConsumptionSession.php",
  "sha256": "6699e5ab041691c8c974d2ec1e6f998f9a535466823bba5da242d85387ee9074",
  "bytes": 2054
},
{
  "path": "src/Contracts/InventoryConsumptionGateway.php",
  "sha256": "d8b5ed2c4bdd6687d264f45b0cf574e1755eb51e6fc7c56e6ae0f43da757ffc5",
  "bytes": 170
},
{
  "path": "src/Contracts/ObservationPlanGateway.php",
  "sha256": "a95d125a5e6e1fa2e2c2764c62b0d55e4777b96cd5f00cd9660a5fca05aeb72b",
  "bytes": 360
},
{
  "path": "src/Contracts/PharmacyGateway.php",
  "sha256": "190df4cb04cc54729594f698265ee88835b26f8c3c6cf749a5246fe6d000f23",
  "bytes": 252
},
{
  "path": "src/Data/CommitAdministrationData.php",
  "sha256": "f0f4924be1eedf8eaa844e8c327db469ebcd09550163d37c0222c6f08a52dc83",
  "bytes": 692
},
{
  "path": "src/Enums/AdministrationStatus.php",
  "sha256": "abdd4311112a142d382b0c51857e5d6ed3e80fcc91f7a4c1b0be8c31ff68ce65",
  "bytes": 259
},
{
  "path": "src/Enums/ConsumptionSessionStatus.php",
  "sha256": "fb7405eca07e6a2b7d735761d87ab52be15a67a76a50711f18e15b39da1a8a2b",
  "bytes": 342
},
{
  "path": "src/Enums/DispositionType.php",
  "sha256": "b7b1c3be6c174e7b404599b08d2ed8fe3020ab84e063f2f5dedf607e54a9600d",
  "bytes": 275
},
{
  "path": "src/Enums/DoseOpportunityStatus.php",
  "sha256": "037e6d8e2a005529c270a5f97205859a077c60e8d4c69842c60ecbe9c6a69210",
  "bytes": 300
},
{
  "path": "src/Enums/InfusionStatus.php",
  "sha256": "07a5b6e4a2340725f99b549ae4f7decfe56fef84a37c06149e9743aeadf036e5",
  "bytes": 308
},

```

```

{
  "path": "src/Enums/MedicationOrderStatus.php",
  "sha256": "652c1accf89c19b66c5f196217647afd3cf7a3240ff54d18c56ea8cd63b3022b",
  "bytes": 300
},
{
  "path": "src/Enums/VerificationResult.php",
  "sha256": "d79dde64d9076ccb968cbd9d54ed692a40ba51d0c5224a75cec4c625deefab78",
  "bytes": 183
},
{
  "path": "src/MedicationServiceProvider.php",
  "sha256": "cbdb2e879055202e56096ea5bfab9814ca0e0b6ed3e80ee42436deee1e3a8f62",
  "bytes": 452
},
{
  "path": "src/Models/AdministrationCheck.php",
  "sha256": "76d6bf41bce0d2db06443acc9adc9efab91556b88e65849823fc8f066e43a27a",
  "bytes": 500
},
{
  "path": "src/Models/AdministrationProduct.php",
  "sha256": "3e9fd8e0cbb4ac907f60da1f48495484d6b7a2c7754ab0c33401b8a6059366c1",
  "bytes": 504
},
{
  "path": "src/Models/ConsumptionLine.php",
  "sha256": "8eac9ca2903bc977cb517a359dad8b2069f8277f39aea745adac7f74f6412eb",
  "bytes": 492
},
{
  "path": "src/Models/ConsumptionSession.php",
  "sha256": "97fb0f97eae85d0089e09d71d30c08004ed88b57e8657375e4e5b75f276e9516",
  "bytes": 498
},
{
  "path": "src/Models/ConsumptionSubmission.php",
  "sha256": "63dae38e0c0025cafc8d8fee656dad87ac7e98d3709384a92f4e720dd1ea0e3cf",
  "bytes": 504
},
{
  "path": "src/Models/DoseOpportunity.php",
  "sha256": "4cd6af69bb240c467d319d5f3d28c9921bed6505e876e0f91e3e1bd49466fe96",
  "bytes": 504
},
{
  "path": "src/Models/InfusionEvent.php",
  "sha256": "48b30b66d11f5cc0ac9d5d628c3f426d86ec07c65f00c63d06f712546d8a682e",
  "bytes": 488
},
{
  "path": "src/Models/MedicationAdministration.php",
  "sha256": "9b23dc09dc46d272f2979fdf68977fa30454e50043a17a7a45e717ef2062178d",
  "bytes": 510
},

```

```
{
  "path": "src/Models/MedicationCorrection.php",
  "sha256": "eb1e239765ea2610e1494167722248099e20d829b34401d9793048871b6cb53b",
  "bytes": 502
},
{
  "path": "src/Models/MedicationOrder.php",
  "sha256": "b83c3bc29e395ab33f66afcddbfe4aa2ffb8881582aebc07936be034b9b42590",
  "bytes": 492
},
{
  "path": "src/Models/MedicationReconciliation.php",
  "sha256": "a77d606df41f67083428e6d34b4d1afab317c292f81a2f8ab79e6a805ed0d233",
  "bytes": 510
},
{
  "path": "src/Models/MedicationWaste.php",
  "sha256": "aab822aac139d8a388baaeb59a6331b98614ee2bcb60910a08ad710c4ed4edee",
  "bytes": 491
},
{
  "path": "src/Models/PrnReassessment.php",
  "sha256": "10233a39984ba3dd3e3eb1cb35cff6a8e3b734625f06a13daaa09c69b5f8310e",
  "bytes": 492
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	4e4c7333c2620cdd55d0d6c83614c3f556d002bf0e122bb2912d2a8061aec084

```
{
  "name": "kashtre/clinical-medication",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "kashtre/clinical-observation-plan": "^6.1",
    "kashtre/clinical-observation": "^6.1"
  },
  "autoload": {
    "psr-4": {
      "KashTre\\Clinical\\Medication\\": "src/"
    },
    "extra": {
      "laravel": {
        "providers": [
          "KashTre\\Clinical\\Medication\\MedicationServiceProvider"
        ]
      }
    }
  }
}
```

config/clinical-medication.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-medication.php	c814038659bbac25ec5cf54f7cf9e0043cb573904f8da2fc1f997f1c9d45988e

```
<?php
declare(strict_types=1);
return
```

```
[ 'administration_window_minutes'=>(int)env('CLINICAL_MED_ADMIN_WINDOW',30), 'consumption_confirmation_required'=>true, 'allow_inventory_outage_deferral'=>true, 'prn_reassessment_required'=>true];
```

database/migrations/2026_08_28_500000_create_medication_consumption_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_500000_create_medication_consumption_tables.php	2630d5b3897f8a6b00196efa020075cf13a473dfa1101484c49a7116e7648290

```
<?php
```

```
declare(strict_types=1);use Illuminate\Database\Migrations\Migration;use Illuminate\Database\Schema\Blueprint;use Illuminate\Support\Facades\Schema;return new class extends Migration{public function up():void{Schema::create('clinical_medication_orders',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('patient_public_id');$table->ulid('encounter_public_id');$table->ulid('medication_public_id');$table->string('status',24);$table->json('schedule');$table->json('instructions');$table->ulid('ordered_by_public_id');$table->dateTime('authored_at_utc',6);$table->unsignedBigInteger('record_version')->default(1);});Schema::create('clinical_medication_dose_opportunities',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('medication_order_public_id');$table->ulid('patient_public_id');$table->ulid('encounter_public_id');$table->unsignedBigInteger('sequence');$table->dateTime('scheduled_at_utc',6);$table->dateTime('window_start_utc',6);$table->dateTime('window_end_utc',6);$table->string('status',24);$table->dateTime('completed_at_utc',6)->nullable();$table->unique(['medication_order_public_id','sequence'],'uq_med_dose_opportunity');});Schema::create('clinical_medication_administrations',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('patient_public_id');$table->ulid('encounter_public_id');$table->ulid('medication_order_public_id');$table->ulid('dose_opportunity_public_id');$table->string('status',24);$table->string('dose_value',100);$table->ulid('dose_unit_public_id');$table->string('route_code',64);$table->string('site_code',64)->nullable();$table->ulid('performer_public_id');$table->dateTime('started_at_utc',6);$table->dateTime('completed_at_utc',6);$table->dateTime('recorded_at_utc',6);$table->unsignedBigInteger('record_version')->default(1);$table->unique('dose_opportunity_public_id');});Schema::create('clinical_administration_products',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('administration_public_id');$table->ulid('product_public_id');$table->string('lot_number',128);$table->dateTime('expiry_date')->nullable();$table->string('physical_quantity',100);$table->ulid('inventory_unit_public_id');});Schema::create('clinical_administration_checks',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('administration_public_id');$table->string('check_code',64);$table->string('result',24);$table->ulid('performed_by_public_id');$table->dateTime('performed_at_utc',6);});Schema::create('clinical_medication_waste',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('administration_public_id');$table->string('quantity',100);$table->ulid('unit_public_id');$table->string('reason_code',64);$table->ulid('witness_public_id')->nullable();$table->dateTime('occurred_at_utc',6);});Schema::create('clinical_consumption_sessions',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('patient_public_id');$table->ulid('encounter_public_id');$table->string('source_event_type',64);$table->ulid('source_event_public_id');$table->string('status',32);$table->dateTime('physical_use_at_utc',6);$table->ulid('confirmed_by_public_id')->nullable();$table->dateTime('confirmed_at_utc',6)->nullable();$table->dateTime('created_at_utc',6);$table->unsignedBigInteger('record_version')->default(1);$table->unique(['tenant_id','source_event_type','source_event_public_id'],'uq_consumption_source');});Schema::create('clinical_consumption_lines',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('consumption_session_public_id');$table->ulid('item_public_id');$table->string('quantity',100);$table->ulid('unit_public_id');$table->string('disposition_type',32);$table->string('lot_number',128)->nullable();$table->dateTime('expiry_date')->nullable();$table->dateTime('physical_use_at_utc',6);});Schema::create('clinical_consumption_submissions',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('consumption_session_public_id');$table->string('inventory_reference',160)->nullable();$table->string('status',32);$table->dateTime('submitted_at_utc',6);$table->dateTime('accepted_at_utc',6)->nullable();});}public function down():void{foreach(['clinical_consumption_submissions','clinical_consumption_lines','clinical_consumption_sessions','clinical_m
```

```
education_waste','clinical_administration_checks','clinical_administration_products','clinical_medication_administrations','clinical_medication_dose_opportunities','clinical_medication_orders']as$t)Schema::dropIfExists($t);});
```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force
```

routes/clinical-medication.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-medication.php	ba29d8bcb8149ffb1da5319d763f0b4450f05951803b65ec663c96df04b9202a

```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum']->prefix('api/v1/clinical')->group(function(){Route::get('/patients/{patient}/encounters/{encounter}/emar',fn()->response()->json(['error'=>['code'=>'CLN_MED_HOST_CONTROLLER_REQUIRED']],501));});
```

src/Actions/CommitMedicationAdministration.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CommitMedicationAdministration.php	02ad529f33d2d937b33638b6b9f92805da835afdd5914c9d8bd4eeec14ce4d71

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Actions;use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Medication\Contracts\{ObservationPlanGateway,PharmacyGateway};use KashTre\Clinical\Medication\Data\CommitAdministrationData;use KashTre\Clinical\Medication\Models\{AdministrationCheck,AdministrationProduct,ConsumptionLine,ConsumptionSession,DoseOpportunity,MedicationAdministration,MedicationWaste};final readonly class CommitMedicationAdministration{public function __construct(private PublicIdGenerator $ids,private PharmacyGateway $pharmacy,private ObservationPlanGateway $plan){}public function execute(CommitAdministrationData $d):MedicationAdministration{return DB::transaction(function()use($d){$opp=DoseOpportunity::query()->where('tenant_id',$d->tenantId)->where('public_id',$d->opportunityId)->lockForUpdate()->firstOrFail();if(in_array($opp->status,['SCHEDULED','DUE','OVERDUE'],true))throw new \DomainException('CLN_MED_OPPORTUNITY_NOT_ACTIONABLE');$verified=$this->pharmacy->verifyDispensedProduct($d->tenantId,$d->patientId,$d->encounterId,$d->productId,$d->lot,null);if(($verified['result']??'')!='MATCHED')throw new \DomainException('CLN_MED_PRODUCT_MISMATCH');$id=(string)$this->ids->generate();$a=MedicationAdministration::query()->create(['public_id'=>$id,'tenant_id'=>$d->tenantId,'patient_public_id'=>$d->patientId,'encounter_public_id'=>$d->encounterId,'medication_order_public_id'=>$d->orderId,'dose_opportunity_public_id'=>$d->opportunityId,'status'=>'COMPLETED','dose_value'=>$d->doseValue,'dose_unit_public_id'=>$d->doseUnitId,'route_code'=>$d->routeCode,'site_code'=>$d->siteCode,'performer_public_id'=>$d->performerId,'started_at_utc'=>$d->startedAt,'completed_at_utc'=>$d->completedAt,'recorded_at_utc'=>now('UTC'),'record_version'=>1]);AdministrationProduct::query()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'administration_public_id'=>$id,'product_public_id'=>$d->productId,'lot_number'=>$d->lot,'expiry_date'=>$d->
```

```

>expiryDate,'physical_quantity'=>$d->physicalQuantity,'inventory_unit_public_id'=>$d->inventoryUnitId));foreach($d->checks
as$c)AdministrationCheck::query()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d-
>tenantId,'administration_public_id'=>$id,'check_code'=>$c['code'],'result'=>$c['result'],'performed_by_public_id'=>$d-
>performerId,'performed_at_utc'=>now('UTC')));foreach($d->wasteLines as$w)MedicationWaste::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d-
>tenantId,'administration_public_id'=>$id,'quantity'=>$w['quantity'],'unit_public_id'=>$w['unit_id'],'reason_code'=>$w['reason'],'wit
ness_public_id'=>$w['witness_id']??null,'occurred_at_utc'=>$d->completedAt);$session=ConsumptionSession::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'patient_public_id'=>$d-
>patientId,'encounter_public_id'=>$d-
>encounterId,'source_event_type'=>'MEDICATION_ADMINISTRATION','source_event_public_id'=>$id,'status'=>'DRAFT','physical_us
e_at_utc'=>$d->completedAt,'created_at_utc'=>now('UTC'),'record_version'=>1]);ConsumptionLine::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'consumption_session_public_id'=>$session-
>public_id,'item_public_id'=>$d->productId,'quantity'=>$d->physicalQuantity,'unit_public_id'=>$d-
>inventoryUnitId,'disposition_type'=>'ADMINISTERED','lot_number'=>$d->lot,'expiry_date'=>$d-
>expiryDate,'physical_use_at_utc'=>$d->completedAt]);foreach($d->incidentalLines as$l)ConsumptionLine::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'consumption_session_public_id'=>$session-
>public_id,'item_public_id'=>$l['item_id'],'quantity'=>$l['quantity'],'unit_public_id'=>$l['unit_id'],'disposition_type'=>'INCIDENTAL_U
SED','physical_use_at_utc'=>$d->completedAt]);$opp->forceFill(['status'=>'ADMINISTERED','completed_at_utc'=>$d-
>completedAt])->save();return$a;});}

```

src/Actions/ConfirmConsumptionSession.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/ConfirmConsumptionSession.php	6699e5ab041691c8c974d2ec1e6f998f9a535466823bba5da242d85387ee9074

```
<?php
```

```

declare(strict_types=1);namespace KashTre\Clinical\Medication\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Medication\Contracts\InventoryConsumptionGateway;use
KashTre\Clinical\Medication\Models\{ConsumptionLine,ConsumptionSession,ConsumptionSubmission};final readonly class
ConfirmConsumptionSession{public function __construct(private InventoryConsumptionGateway $inventory){}public function
execute(string $tenantId,string $sessionId,string $actorId,bool
$noAdditionalItems):ConsumptionSession{$s=DB::transaction(function()use($tenantId,$sessionId,$actorId,$noAdditionalItems){$s
=ConsumptionSession::query()->where('tenant_id',$tenantId)->where('public_id',$sessionId)->lockForUpdate()-
>firstOrFail();if(!in_array($s-
>status,['DRAFT','DEFERRED','RECONCILIATION_REQUIRED'],true))return$s;if(!$noAdditionalItems&&!ConsumptionLine::query()-
>where('consumption_session_public_id',$sessionId)->where('disposition_type','INCIDENTAL_USED')->exists())throw new
\DomainException('CLN_CONSUMPTION_ADDITIONAL_CONFIRMATION_REQUIRED');$s-
>forceFill(['status'=>'CONFIRMED','confirmed_by_public_id'=>$actorId,'confirmed_at_utc'=>now('UTC'),'record_version'=>$s-
>record_version+1])->save();return$s;});$lines=ConsumptionLine::query()->where('consumption_session_public_id',$sessionId)-
>orderBy('public_id')->get()->toArray();try{$r=$this->inventory->submit(['session_id'=>$s-
>public_id,'tenant_id'=>$tenantId,'patient_id'=>$s->patient_public_id,'encounter_id'=>$s-
>encounter_public_id,'source_event_id'=>$s->source_event_public_id,'physical_use_at'=>$s->physical_use_at_utc?-
>format(DATE_ATOM),'lines'=>$lines]);ConsumptionSubmission::query()-
>create(['public_id'=>(string)\Symfony\Component\Uid\Uuid::generate(),'tenant_id'=>$tenantId,'consumption_session_public_id'=>$
sessionId,'inventory_reference'=>$r['reference'],'status'=>'ACCEPTED','submitted_at_utc'=>now('UTC'),'accepted_at_utc'=>now('UT
C')));$s->forceFill(['status'=>'ACCEPTED'])->save();}catch(\Throwable$e){$s->forceFill(['status'=>'RECONCILIATION_REQUIRED'])-
>save();}return$s;}}

```

src/Contracts/InventoryConsumptionGateway.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Contracts/InventoryConsumptionGateway.php	d8b5ed2c4bdd6687d264f45b0cf574e1755eb51e6fc7c56e6ae0f43da757ffc5

CANDIDAT E		
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Contracts;interface InventoryConsumptionGateway{public
function submit(array $sessionEnvelope):array;}
```

src/Contracts/ObservationPlanGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDAT E	src/Contracts/ObservationPlanGateway.php	a95d125a5e6e1fa2e2c2764c62b0d55e4777b96cd5f00cd9660a5fca05aeb72b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Contracts;interface ObservationPlanGateway{public function
activateMedicationMonitoring(string $tenantId,string $patientId,string $encounterId,string $orderId,array $requirements):void;public
function createPrnReassessment(string $tenantId,string $administrationId,array $requirements):void;}
```

src/Contracts/PharmacyGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/PharmacyGateway.php	190df4cb04cc54729594f698265ee88835b26f8c3c6fcf749a5246fe6d000f23

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Contracts;interface PharmacyGateway{public function
verifyDispensedProduct(string $tenantId,string $patientId,string $encounterId,string $productId,string $lot,?string $serial):array;}
```

src/Data/CommitAdministrationData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDAT E	src/Data/CommitAdministrationData.php	f0f4924be1eedf8eaa844e8c327db469ebcd09550163d37c0222c6f08a52dc83

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Data;final readonly class CommitAdministrationData{public
function __construct(public string $tenantId,public string $patientId,public string $encounterId,public string $orderId,public string
$opportunityId,public string $productId,public string $lot,public ?string $expiryDate,public string $doseValue,public string
$doseUnitId,public string $physicalQuantity,public string $inventoryUnitId,public string $routeCode,public ?string $siteCode,public
string $performerId,public \DateTimeImmutable $startedAt,public \DateTimeImmutable $completedAt,public array $checks,public
array $wasteLines,public array $incidentalLines){}}
```

src/Enums/AdministrationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDAT E	src/Enums/AdministrationStatus.php	abdd4311112a142d382b0c51857e5d6ed3e80fcc91f7a4c1b0be8c31ff68ce65

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum AdministrationStatus:string{case
IN_PROGRESS='IN_PROGRESS';case COMPLETED='COMPLETED';case NOT_DONE='NOT_DONE';case
CORRECTED='CORRECTED';case ENTERED_IN_ERROR='ENTERED_IN_ERROR';}
```

src/Enums/ConsumptionSessionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ConsumptionSessionStatus.php	fb7405eca07e6a2b7d735761d87ab52be15a67a76a50711f18e15b39da1a8a2b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum ConsumptionSessionStatus:string{case
DRAFT='DRAFT';case CONFIRMED='CONFIRMED';case DEFERRED='DEFERRED';case SUBMITTED='SUBMITTED';case
ACCEPTED='ACCEPTED';case RECONCILIATION_REQUIRED='RECONCILIATION_REQUIRED';case CORRECTED='CORRECTED';case
REVERSED='REVERSED';}
```

src/Enums/DispositionType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DispositionType.php	b7b1c3be6c174e7b404599b08d2ed8fe3020ab84e063f2f5dedf607e54a9600d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum DispositionType:string{case
ADMINISTERED='ADMINISTERED';case WASTED='WASTED';case BROKEN='BROKEN';case CONTAMINATED='CONTAMINATED';case
RETURNED='RETURNED';case INCIDENTAL_USED='INCIDENTAL_USED';}
```

src/Enums/DoseOpportunityStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DoseOpportunityStatus.php	037e6d8e2a005529c270a5f97205859a077c60e8d4c69842c60ecbe9c6a69210

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum DoseOpportunityStatus:string{case
SCHEDULED='SCHEDULED';case DUE='DUE';case OVERDUE='OVERDUE';case ADMINISTERED='ADMINISTERED';case
NOT_ADMINISTERED='NOT_ADMINISTERED';case DEFERRED='DEFERRED';case CANCELLED='CANCELLED';}
```

src/Enums/InfusionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/InfusionStatus.php	07a5b6e4a2340725f99b549ae4f7decfe56fef84a37c06149e9743aeadf036e5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum InfusionStatus:string{case
PREPARED='PREPARED';case STARTED='STARTED';case PAUSED='PAUSED';case RESUMED='RESUMED';case
RATE_CHANGED='RATE_CHANGED';case BAG_CHANGED='BAG_CHANGED';case COMPLETED='COMPLETED';case
STOPPED='STOPPED';}
```

src/Enums/MedicationOrderStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MedicationOrderStatus.php	652c1accf89c19b66c5f196217647afd3cf7a3240ff54d18c56ea8cd63b3022b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum MedicationOrderStatus:string{case
DRAFT='DRAFT';case ACTIVE='ACTIVE';case ON_HOLD='ON_HOLD';case COMPLETED='COMPLETED';case
DISCONTINUED='DISCONTINUED';case CANCELLED='CANCELLED';case ENTERED_IN_ERROR='ENTERED_IN_ERROR';}
```

src/Enums/VerificationResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/VerificationResult.php	d79dde64d9076ccb968cbd9d54ed692a40ba51d0c5224a75cec4c625deefab78

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Enums;enum VerificationResult:string{case
MATCHED='MATCHED';case MISMATCH='MISMATCH';case OVERRIDDEN='OVERRIDDEN';}
```

src/MedicationServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/MedicationServiceProvider.php	cbdb2e879055202e56096ea5bfab9814ca0e0b6ed3e80ee42436deee1e3a8f62

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication;use Illuminate\Support\ServiceProviders;final class
MedicationServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-medication.php','clinical-medication');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-medication.php');}}
```

src/Models/AdministrationCheck.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AdministrationCheck.php	76d6bf41bce0d2db06443acc9adc9efab91556b88e65849823fc8f066e43a27a

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
AdministrationCheck extends Model{public $timestamps=false;protected $table='clinical_administration_checks';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/AdministrationProduct.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AdministrationProduct.php	3e9fd8e0cbb4ac907f60da1f48495484d6b7a2c7754ab0c33401b8a6059366c1

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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
AdministrationProduct extends Model{public $timestamps=false;protected $table='clinical_administration_products';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/ConsumptionLine.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ConsumptionLine.php	8eacf9ca2903bc977cb517a359dad8b2069f8277f39aea745adac7f74f6412eb

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
ConsumptionLine extends Model{public $timestamps=false;protected $table='clinical_consumption_lines';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/ConsumptionSession.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ConsumptionSession.php	97fb0f97eae85d0089e09d71d30c08004ed88b57e8657375e4e5b75f276e9516

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
ConsumptionSession extends Model{public $timestamps=false;protected $table='clinical_consumption_sessions';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/ConsumptionSubmission.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ConsumptionSubmission.php	63dae38e0c0025cafcd8fee656dad87ac7e98d3709384a92f4e720dd1ea0e3cf

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
ConsumptionSubmission extends Model{public $timestamps=false;protected
$table='clinical_consumption_submissions';protected $guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/DoseOpportunity.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Models/DoseOpportunity.php	4cd6af69bb240c467d319d5f3d28c9921bed6505e876e0f91e3e1bd49466fe96
-------------------------	--------------------------------	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
DoseOpportunity extends Model{public $timestamps=false;protected $table='clinical_medication_dose_opportunities';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/InfusionEvent.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/InfusionEvent.php	48b30b66d11f5cc0ac9d5d628c3f426d86ec07c65f00c63d06f712546d8a682e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
InfusionEvent extends Model{public $timestamps=false;protected $table='clinical_infusion_events';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/MedicationAdministration.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MedicationAdministration.php	9b23dc09dc46d272f2979fdf68977fa30454e50043a17a7a45e717ef2062178d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
MedicationAdministration extends Model{public $timestamps=false;protected
$table='clinical_medication_administrations';protected $guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/MedicationCorrection.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MedicationCorrection.php	eb1e239765ea2610e1494167722248099e20d829b34401d9793048871b6cb53b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
MedicationCorrection extends Model{public $timestamps=false;protected $table='clinical_medication_corrections';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/MedicationOrder.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MedicationOrder.php	b83c3bc29e395ab33f66afcdcbfe4aa2ffb8881582aebc07936be034b9b42590

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
MedicationOrder extends Model{public $timestamps=false;protected $table='clinical_medication_orders';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/MedicationReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MedicationReconciliation.php	a77d606df41f67083428e6d34b4d1afab317c292f81a2f8ab79e6a805ed0d233

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
MedicationReconciliation extends Model{public $timestamps=false;protected
$table='clinical_medication_reconciliations';protected $guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/MedicationWaste.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MedicationWaste.php	aab822aac139d8a388baaeb59a6331b98614ee2bcb60910a08ad710c4ed4edee

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
MedicationWaste extends Model{public $timestamps=false;protected $table='clinical_medication_waste';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

src/Models/PrnReassessment.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PrnReassessment.php	10233a39984ba3dd3e3eb1cb35cff6a8e3b734625f06a13daaa09c69b5f8310e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Medication\Models;use Illuminate\Database\Eloquent\Model;final class
PrnReassessment extends Model{public $timestamps=false;protected $table='clinical_prn_reassessments';protected
$guarded=[];protected
$casts=['schedule'=>'array','instructions'=>'array','checks'=>'array','occurred_at_utc'=>'immutable_datetime','started_at_utc'=>'im
mutable_datetime','completed_at_utc'=>'immutable_datetime','confirmed_at_utc'=>'immutable_datetime'];}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	Context, authorization, time, audit, outbox and idempotency
Volume 2	Medication monitoring and PRN reassessment tasks
Volume 3	Related observations and device events
Pharmacy	Dispense and exact product verification
Inventory	Grouped Consumption Session contract
Main	Approved Pool and financial evaluation downstream of Inventory
Vue	eMAR, administration sheet, infusion workspace and Confirm Consumption drawer

Blocking Gaps

Execute in actual KashTre repository

Bind authoritative medication catalogue and order service

Bind Pharmacy verification payload

Bind Inventory consumption contract and idempotency

Approve controlled-drug witness policy

Approve administration windows and late rules

Complete infusion state machine and pump integration

Complete host controllers, requests, resources, jobs and Vue interfaces

Run clean, upgrade, rollback, concurrency, outage, accessibility, barcode and clinical gold tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Medication administration and consumption authority
KashTre Inventory Endstore Systems Requirements Document	Floor Principle and physical-stock ownership
HL7 FHIR R5 MedicationRequest	Medication order boundary
HL7 FHIR R5 MedicationAdministration	Administration event boundary

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 7

DIAGNOSTIC RESULTS, REVIEW, CRITICAL ESCALATION AND FOLLOW-UP

LIMS, Imaging, Pathology and External Result Ingestion with Closed-Loop Accountability

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 to 6

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Safety Rule

LIMS, Imaging and other diagnostic modules remain authoritative for report production and authorization. Clinical stores a versioned projection for review, acknowledgement, follow-up and patient-care continuity. Clinical never silently changes the source report, and ambiguous patient or order matching is quarantined rather than guessed.

Source report -> inbox idempotency -> patient/order match -> projection -> critical routing -> clinician review -> follow-up task -> patient communication -> completion -> correction propagation

Entity Register

Entity	Table	Invariant
DiagnosticReportProjection	clinical_diagnostic_report_projections	Source version, patient match, review and provenance retained
DiagnosticResultItem	clinical_diagnostic_result_items	Source version, patient match, review and provenance retained
DiagnosticAttachment	clinical_diagnostic_attachments	Source version, patient match, review and provenance retained
ResultMatchDecision	clinical_result_match_decisions	Source version, patient match, review and provenance retained
ResultReview	clinical_result_reviews	Source version, patient match, review and provenance retained
CriticalResult	clinical_critical_results	Source version, patient match, review and provenance retained
CriticalDelivery	clinical_critical_result_deliveries	Source version, patient match, review and provenance retained
CriticalAcknowledgement	clinical_critical_acknowledgements	Source version, patient match, review and provenance retained
ResultFollowUp	clinical_result_followups	Source version, patient match, review and provenance retained
PatientResultCommunication	clinical_patient_result_communications	Source version, patient match, review and provenance retained
ResultCorrectionLink	clinical_result_correction_links	Source version, patient match, review and provenance retained
ResultQuarantine	clinical_result_quarantines	Source version, patient match, review and provenance retained
ResultReconciliation	clinical_result_reconciliations	Source version, patient match, review and provenance retained

Engineering Design**Ingestion**

Idempotent source system, report ID and version

Preserve issued, effective and received times separately

Retain source payload hash

Quarantine ambiguous or unmatched records

Store atomic result items and attachments without becoming source authority

Review

Review exact source version

Viewing is not review

Review is not acknowledgement of a critical result

Action-required review creates a due follow-up task

Correction creates a new review obligation

Critical results

Trigger once per critical item

Route to eligible current clinical context

Delivery does not equal acknowledgement

Acknowledgement records actor and time

Unacknowledged critical results escalate

Resolution requires recorded assessment and action

Post-discharge continuity

Results arriving after discharge remain visible

Outstanding follow-up remains owned

Patient communication records channel, attempt and outcome

Encounter closure does not erase pending result work

Permissions

Permission	Purpose
clinical.result.view	View matched result
clinical.result.review	Record review
clinical.result.acknowledge_critical	Acknowledge critical result
clinical.result.followup.create	Create follow-up
clinical.result.followup.complete	Complete follow-up with evidence
clinical.result.patient_communicate	Record communication
clinical.result.quarantine.resolve	Resolve match exception
clinical.result.reconcile	Resolve ingestion/correction exception

Mandatory Tests

Duplicate source version creates one projection

Different corrected version creates distinct projection

Ambiguous patient match quarantines

Critical delivery does not acknowledge

Review of old version does not review correction

Action-required review creates one task

Post-discharge result remains actionable

Notification outage preserves critical record

Cross-tenant result is concealed

Attachment hash mismatch is quarantined

Source cancellation remains historically visible

Follow-up cannot complete without outcome evidence

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	7457f1f2d6c2096a55a25d582ed24f8546278fc7d9fde2b3fa492902d21a967c

KashTre Clinical Diagnostic Results v6.1.0

Pick-and-drop candidate. Bind LIMS, Imaging and other source contracts, patient/order matcher, contextual authorization, communications and Clinical Task gateway before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	8935d709fbd6c05c39da045c70bfb88ead506d43a126b3d32da2bc9b0ecef97

```
{
  "volume": "Clinical EDD Volume 7 Diagnostic Results",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "7457f1f2d6c2096a55a25d582ed24f8546278fc7d9fde2b3fa492902d21a967c",
      "bytes": 230
    },
    {
      "path": "composer.json",
      "sha256": "af943aaf74e0ada920302238e9a9c7a2be3fa3b68f82329ea90614ddabb84e97",
      "bytes": 384
    },
    {
      "path": "config/clinical-results.php",
      "sha256": "f0b4fed58d4ccbda68397ff329ab3bf11bab0686725cc4f1a3d756c5ab134145",
      "bytes": 238
    },
    {
      "path": "database/migrations/2026_08_28_600000_create_diagnostic_result_tables.php",
      "sha256": "225909924f91932776a2f472a13fc417107b2e9ebff54fded77b05f1eeb01c8c",
      "bytes": 4142
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-results.php",
      "sha256": "bb304e5d734e5bbf9007032f47aebb839d4ece8bfcd7a9be86356bf237ff3309",
      "bytes": 273
    },
    {

```

```

"path": "src/Actions/IngestDiagnosticReport.php",
"sha256": "b1a3a2568beb0ca11d4ef767249360b2c2948007eb78e3ce7634578679935680",
"bytes": 2970
},
{
"path": "src/Actions/ReviewDiagnosticResult.php",
"sha256": "fbb1ff9f196264e74e4cc395f9ebe8786aa895f5ee1c147c2c9ccb094fb0aa73",
"bytes": 1829
},
{
"path": "src/Actions/TriggerCriticalResult.php",
"sha256": "81a69771dfedcfeea2186f805dbbbcb827304bc45045ee65342c68dd77712c46",
"bytes": 1581
},
{
"path": "src/ClinicalResultsServiceProvider.php",
"sha256": "f930aaf965d81cfc8417c68979f65ffce50c45389676ea3c25c12c7d5363fe03",
"bytes": 445
},
{
"path": "src/Contracts/ClinicalTaskGateway.php",
"sha256": "76295d617ef655d048b8f25162cfa89ee6b45cb323089a78474dae412f16e1bc",
"bytes": 277
},
{
"path": "src/Contracts/CriticalResultNotificationGateway.php",
"sha256": "763955d5f46f040b318480d3c46fc73d349e185b420816eb0ea485ce84e06a78",
"bytes": 215
},
{
"path": "src/Contracts/PatientOrderMatcher.php",
"sha256": "85168d84ef736aa60cda79d1c7ed1308d3001b271c3f2e4d5056678a97f4d137",
"bytes": 222
},
{
"path": "src/Data/IngestDiagnosticReportData.php",
"sha256": "87089a0827df4b8d9dcd2bda476d16ae8f4162c621ae246dbb85cc4e63039865",
"bytes": 578
},
{
"path": "src/Enums/CommunicationStatus.php",
"sha256": "092f513331e7607de60510af0e7badca72b3e2b64f4507898b55c109b695f5e2",
"bytes": 233
},
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"path": "src/Enums/CriticalStatus.php",
"sha256": "83cbb282efb9824bc9d73b0c2cde847a0e2016ac07506d87aae6e80476c1c09a",
"bytes": 286
},
{
"path": "src/Enums/FollowUpStatus.php",
"sha256": "fdce0b0a616c8bb449180ee3f3b60b04ab7771afa798e49d6fd14a3f442c4b0c",
"bytes": 224
},

```

```

{
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  "bytes": 204
},
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  "path": "src/Enums/ReportStatus.php",
  "sha256": "537ababcbddfd6d596bdf944a1fab2513561e410783b5e2c53370e864f8212a24",
  "bytes": 317
},
{
  "path": "src/Enums/ResultSourceType.php",
  "sha256": "7a56e33acadef71e75ba3e34cf9d0a9e6f62d20c2a6a8d0a3af89588f7075b6a",
  "bytes": 222
},
{
  "path": "src/Enums/ReviewStatus.php",
  "sha256": "3f504558227f9cb66b9e58933590762beaed17230148c77067398502c79251d6",
  "bytes": 285
},
{
  "path": "src/Models/CriticalAcknowledgement.php",
  "sha256": "bb891694d5607e1b98fbab95fa293c29fe44033577b3c6daa02afe2932898a41",
  "bytes": 544
},
{
  "path": "src/Models/CriticalDelivery.php",
  "sha256": "8b00f358d498a0c667b710bd55357d07d42c19789bb08acba47d8a263a06e9da",
  "bytes": 538
},
{
  "path": "src/Models/CriticalResult.php",
  "sha256": "28f19708d943ea394ec13f658645635a4ec0ffae255414f8e50643f5019bb8d3",
  "bytes": 526
},
{
  "path": "src/Models/DiagnosticAttachment.php",
  "sha256": "6875801ca12f19fd6efd949d7569f07dc525ea48b806696e85bf976b8805bbb8",
  "bytes": 538
},
{
  "path": "src/Models/DiagnosticReportProjection.php",
  "sha256": "574eac32f948b9876233ac8d195085a0728445e94a52122ffa2c60fe983ddc",
  "bytes": 551
},
{
  "path": "src/Models/DiagnosticResultItem.php",
  "sha256": "b38c43f8cd5689b72b4891a082da0c5dd66c7d7fc68cd1eaaeb46b52ee5ab0fe",
  "bytes": 539
},
{
  "path": "src/Models/PatientResultCommunication.php",
  "sha256": "5923bc5fe179772c96dac40d160cd86fc4a069b9146d60df287cbdde1935947c",
  "bytes": 551
},

```

```

{
  "path": "src/Models/ResultCorrectionLink.php",
  "sha256": "faccfe10afd2ce73e871f3b7a61cdc202abcbcd0e8182b145ec690154563ef68b",
  "bytes": 539
},
{
  "path": "src/Models/ResultFollowUp.php",
  "sha256": "ddcb2ac43d0fd94fbb60baa9bfa564dadbe366cb09b0577a77b106afc3b217ed",
  "bytes": 526
},
{
  "path": "src/Models/ResultMatchDecision.php",
  "sha256": "1d8cd7e9bf2684fccac88f6b548ec830c2c9e526d69bf99f8f6841368af59413",
  "bytes": 537
},
{
  "path": "src/Models/ResultQuarantine.php",
  "sha256": "c455fbe060bd53f7dd6b0d6ccb6f3b176072576b251cf20f8473b5de489c1bfe",
  "bytes": 530
},
{
  "path": "src/Models/ResultReconciliation.php",
  "sha256": "54268b3d6ec9cc603286721077d5c9052f8b904ac2497b9337383f1b2f0f2f4b",
  "bytes": 538
},
{
  "path": "src/Models/ResultReview.php",
  "sha256": "5ba086637e8e0be4a13230f40acea23d7f0ce7bf7809b425eb9eb169eb26fd910",
  "bytes": 522
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
]
}

```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	af943aaf74e0ada920302238e9a9c7a2be3fa3b68f82329ea90614ddabb84e97

```

{"name":"kashtre/clinical-
results","type":"library","require":{"php":"^8.4","illuminate/database":"^13.0","illuminate/http":"^13.0","illuminate/queue":"^13.0","i
lluminate/support":"^13.0","kashtre/clinical-foundation":"^6.1"},"autoload":{"psr-
4":{"KashTre\\Clinical\\Results\\":"src/"},"extra":{"laravel":{"providers":["KashTre\\Clinical\\Results\\ClinicalResultsServiceProvider"
]}}}}

```

config/clinical-results.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical- results.php	f0b4fed58d4ccbda68397ff329ab3bf11bab0686725cc4f1a3d756c5ab134145

```
<?php
declare(strict_types=1);return
['critical_ack_minutes'=>(int)env('CLINICAL_CRITICAL_RESULT_ACK_MINUTES',10),'result_followup_projection_minutes'=>(int)env('
CLINICAL_RESULT_FOLLOWUP_MINUTES',1),'quarantine_on_ambiguous_match'=>true];
```

database/migrations/2026_08_28_600000_create_diagnostic_result_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_600000_create_diagnostic_result_tables.php	225909924f91932776a2f472a13fc417107b2e9ebff54fed77b05f1eeb01c8c

```
<?php
declare(strict_types=1);use Illuminate\Database\Migrations\Migration;use Illuminate\Database\Schema\Blueprint;use
Illuminate\Support\Facades\Schema;return new class extends Migration{public function up():void{
Schema::create('clinical_diagnostic_report_projections',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->string('source_system',64);$table->string('source_report_id',160);$table->unsignedInteger('source_version');$table->ulid('patient_public_id')->nullable();$table->ulid('encounter_public_id')->nullable();$table->ulid('order_public_id')->nullable();$table->string('match_status',24);$table->string('report_status',24);$table->string('category_code',96);$table->dateTime('effective_at_utc',6);$table->dateTime('issued_at_utc',6);$table->dateTime('received_at_utc',6);$table->text('conclusion')->nullable();$table->char('payload_hash',64);$table->unique(['tenant_id','source_system','source_report_id','source_version'],'uq_result_source_version');});
Schema::create('clinical_diagnostic_result_items',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->string('source_result_id',160);$table->string('code',128);$table->json('value_json')->nullable();$table->ulid('unit_public_id')->nullable();$table->string('reference_range',240)->nullable();$table->json('interpretation')->nullable();$table->boolean('is_critical')->default(false);$table->unique(['report_public_id','source_result_id'],'uq_report_result_item');});
Schema::create('clinical_diagnostic_attachments',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->string('source_attachment_id',160);$table->string('content_type',128);$table->string('storage_reference',500);$table->char('sha256',64);});
Schema::create('clinical_result_match_decisions',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->string('status',24);$table->json('evidence');$table->dateTime('decided_at_utc',6);});
Schema::create('clinical_result_reviews',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->unsignedInteger('report_source_version');$table->ulid('reviewed_by_public_id');$table->string('status',24);$table->text('assessment');$table->dateTime('reviewed_at_utc',6);});
Schema::create('clinical_critical_results',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->ulid('result_item_public_id');$table->string('severity',24);$table->string('status',24);$table->dateTime('triggered_at_utc',6);$table->dateTime('acknowledgement_due_at_utc',6);$table->dateTime('acknowledged_at_utc',6)->nullable();$table->unique(['tenant_id','result_item_public_id'],'uq_critical_result');});
Schema::create('clinical_critical_result_deliveries',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('critical_result_public_id');$table->json('recipient_context');$table->string('status',24);$table->string('provider_reference',160)->nullable();$table->dateTime('delivered_at_utc',6)->nullable();});
Schema::create('clinical_result_followups',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->ulid('review_public_id');$table->ulid('task_public_id');$table->string('action_code',96);$table->string('status',24);$table->dateTime('due_at_utc',6);$table->dateTime('completed_at_utc',6)->nullable();});
Schema::create('clinical_result_quarantines',function(Blueprint $table){$table->id();$table->ulid('public_id')->unique();$table->string('tenant_id',64);$table->ulid('report_public_id');$table->string('reason_code',96);$table->string('status',24);$table->dateTime('created_at_utc',6);$table->dateTime('resolved_at_utc',6)->nullable();});
}public function
down():void{foreach(['clinical_result_quarantines','clinical_result_followups','clinical_critical_result_deliveries','clinical_critical_results','clinical_result_reviews','clinical_result_match_decisions','clinical_diagnostic_attachments','clinical_diagnostic_result_items','clinical_diagnostic_report_projections']as $table)Schema::dropIfExists($table);}
```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force
```

routes/clinical-results.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-results.php	bb304e5d734e5bbf9007032f47aebb839d4ece8bfcd7a9be86356bf237ff3309

```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum']->prefix('api/v1/clinical')->group(function(){Route::get('/results/inbox',fn()->response()->json(['error'=>['code'=>'CLN_RESULT_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/IngestDiagnosticReport.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/IngestDiagnosticReport.php	b1a3a2568beb0ca11d4ef767249360b2c2948007eb78e3ce7634578679935680

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Results\Contracts\PatientOrderMatcher;use
KashTre\Clinical\Results\Data\IngestDiagnosticReportData;use
KashTre\Clinical\Results\Models\{DiagnosticAttachment,DiagnosticReportProjection,DiagnosticResultItem,ResultMatchDecision,ResultQuarantine};final readonly class IngestDiagnosticReport{public function __construct(private PublicIdGenerator $ids,private PatientOrderMatcher $matcher){}public function execute(IngestDiagnosticReportData $d):DiagnosticReportProjection{return DB::transaction(function(){use($d){$existing=DiagnosticReportProjection::query()->where('tenant_id',$d->tenantId)->where('source_system',$d->sourceSystem)->where('source_report_id',$d->sourceReportId)->where('source_version',$d->sourceVersion)->first();if($existing)return$existing;$match=$this->matcher->match($d->tenantId,$d->sourceSystem,$d->patientReference,$d->orderReference);$id=(string)$this->ids->generate();$report=DiagnosticReportProjection::query()->create(['public_id'=>$id,'tenant_id'=>$d->tenantId,'source_system'=>$d->sourceSystem,'source_report_id'=>$d->sourceReportId,'source_version'=>$d->sourceVersion,'patient_public_id'=>$match['patient_id']??null,'encounter_public_id'=>$match['encounter_id']??null,'order_public_id'=>$match['order_id']??null,'match_status'=>$match['status'],'report_status'=>$d->status,'category_code'=>$d->categoryCode,'effective_at_utc'=>$d->effectiveAt,'issued_at_utc'=>$d->issuedAt,'received_at_utc'=>now('UTC'),'conclusion'=>$d->conclusion,'payload_hash'=>hash('sha256'),json_encode($d->rawPayload,JSON_THROW_ON_ERROR))));ResultMatchDecision::query()->create(['public_id'=>(string)$this->ids->generate(), 'tenant_id'=>$d->tenantId,'report_public_id'=>$id,'status'=>$match['status'],'evidence'=>$match['evidence']??[],'decided_at_utc'=>now('UTC')));if($match['status']!=='MATCHED'){ResultQuarantine::query()->create(['public_id'=>(string)$this->ids->generate(), 'tenant_id'=>$d->tenantId,'report_public_id'=>$id,'reason_code'=>'PATIENT_OR_ORDER_MATCH_'.$match['status'],'status'=>'OPEN','created_at_utc'=>now('UTC')));return$report;}foreach($d->results as$r)DiagnosticResultItem::query()->create(['public_id'=>(string)$this->ids->generate(), 'tenant_id'=>$d->tenantId,'report_public_id'=>$id,'source_result_id'=>$r['id'],'code'=>$r['code'],'value_json'=>$r['value']??null,'unit_public_id'=>$r['unit_id']??null,'reference_range'=>$r['reference_range']??null,'interpretation'=>$r['interpretation']??null,'is_critical'=>(bool)($r['critical
```

```
]??false));foreach($d->attachments as$a)DiagnosticAttachment::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$d-
>tenantId, 'report_public_id'=>$id, 'source_attachment_id'=>$a['id'], 'content_type'=>$a['content_type'], 'storage_reference'=>$a['stor
age_reference'], 'sha256'=>$a['sha256']));return$report;}}}
```

src/Actions/ReviewDiagnosticResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/ReviewDiagnosticResult.php	fbb1ff9f196264e74e4cc395f9ebe8786aa895f5ee1c147c2c9ccb094fb0aa73

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Results\Contracts\ClinicalTaskGateway;use
KashTre\Clinical\Results\Models\{DiagnosticReportProjection, ResultFollowUp, ResultReview};final readonly class
ReviewDiagnosticResult{public function __construct(private PublicIdGenerator $ids, private ClinicalTaskGateway $tasks){}public
function execute(string $tenantId, string $reportId, string $reviewerId, string $assessment, bool $actionRequired, ?string
$actionCode, ?\DateTimeImmutable $dueAt):ResultReview{return
DB::transaction(function()use($tenantId, $reportId, $reviewerId, $assessment, $actionRequired, $actionCode, $dueAt){$report=Diagno
sticReportProjection::query()->where('tenant_id', $tenantId)->where('public_id', $reportId)->where('match_status', 'MATCHED')-
>firstOrFail();$review=ResultReview::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId, 'report_public_id'=>$reportId, 'report_source_version'=>$report-
>source_version, 'reviewed_by_public_id'=>$reviewerId, 'status'=>$actionRequired?'ACTION_REQUIRED':'REVIEWED', 'assessment'=>
>$assessment, 'reviewed_at_utc'=>now('UTC')));if($actionRequired){if(!$actionCode||!$dueAt)throw new
\DomainException('CLN_RESULT_FOLLOWUP_DETAIL_REQUIRED');$task=$this->tasks->createResultFollowUp($tenantId, $report-
>patient_public_id, $report->encounter_public_id, $reportId, $actionCode, $dueAt, ['review'=>$review-
>public_id]);ResultFollowUp::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId, 'report_public_id'=>$reportId, 'review_public_id'=>$review-
>public_id, 'task_public_id'=>$task, 'action_code'=>$actionCode, 'status'=>'OPEN', 'due_at_utc'=>$dueAt, 'created_at_utc'=>now('UT
C')));return$review;}}}
```

src/Actions/TriggerCriticalResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/TriggerCriticalResult.php	81a69771dfedcfeea2186f805dbbbcb827304bc45045ee65342c68dd77712c46

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Results\Contracts\CriticalResultNotificationGateway;use
KashTre\Clinical\Results\Models\{CriticalDelivery, CriticalResult};final readonly class TriggerCriticalResult{public function
__construct(private PublicIdGenerator $ids, private CriticalResultNotificationGateway $notifications){}public function execute(string
$tenantId, string $reportId, string $resultItemId, string $severity, array $recipientContext):CriticalResult{return
DB::transaction(function()use($tenantId, $reportId, $resultItemId, $severity, $recipientContext){$c=CriticalResult::query()-
>firstOrCreate(['tenant_id'=>$tenantId, 'result_item_public_id'=>$resultItemId], ['public_id'=>(string)$this->ids-
>generate(), 'report_public_id'=>$reportId, 'severity'=>$severity, 'status'=>'TRIGGERED', 'triggered_at_utc'=>now('UTC'), 'acknowledge
ment_due_at_utc'=>now('UTC')->addMinutes((int)config('clinical-results.critical_ack_minutes')));if(!$c-
>wasRecentlyCreated)return$c;$r=$this->notifications->route($tenantId, $c->public_id, $recipientContext);CriticalDelivery::query()-
>create(['public_id'=>(string)$this->ids->generate(), 'tenant_id'=>$tenantId, 'critical_result_public_id'=>$c-
```

```
>public_id,'recipient_context'=>$recipientContext,'status'=>$r['status'],'provider_reference'=>$r['reference']??null,'delivered_at_utc'
'=>$r['delivered_at_utc']??null);$c->forceFill(['status'=>$r['status']=='DELIVERED'?'DELIVERED':'ROUTED'])->save();return$c;}}}
```

src/ClinicalResultsServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ClinicalResultsServiceProvider.php	f930aaf965d81cfc8417c68979f65ffce50c45389676ea3c25c12c7d5363fe03

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results;use Illuminate\Support\ServiceProviders;final class
ClinicalResultsServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-results.php','clinical-results');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-results.php');}}
```

src/Contracts/ClinicalTaskGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ClinicalTaskGateway.php	76295d617ef655d048b8f25162cfa89ee6b45cb323089a78474dae412f16e1bc

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Contracts;interface ClinicalTaskGateway{public function
createResultFollowUp(string $tenantId,string $patientId,string $encounterId,string $reportId,string $code,\DateTimeImmutable
$dueAt,array $context):string;}
```

src/Contracts/CriticalResultNotificationGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/CriticalResultNotificationGateway.php	763955d5f46f040b318480d3c46fc73d349e185b420816eb0ea485ce84e06a78

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Contracts;interface CriticalResultNotificationGateway{public function
route(string $tenantId,string $criticalResultId,array $recipientContext):array;}
```

src/Contracts/PatientOrderMatcher.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/PatientOrderMatcher.php	85168d84ef736aa60cda79d1c7ed1308d3001b271c3f2e4d5056678a97f4d137

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Contracts;interface PatientOrderMatcher{public function match(string
$tenantId,string $sourceSystem,string $patientReference,string $orderReference):array;}
```

src/Data/IngestDiagnosticReportData.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Data/IngestDiagnosticReportData.php	87089a0827df4b8d9dcd2bda476d16ae8f4162c621ae246dbb85cc4e63039865
-------------------------	---	--

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Data;final readonly class IngestDiagnosticReportData{public function __construct(public string $tenantId,public string $sourceSystem,public string $sourceReportId,public int $sourceVersion,public string $patientReference,public ?string $orderReference,public string $status,public string $categoryCode,public \DateTimeImmutable $effectiveAt,public \DateTimeImmutable $issuedAt,public array $results,public ?string $conclusion,public array $attachments,public array $criticalIndicators,public array $rawPayload){}}
```

src/Enums/CommunicationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CommunicationStatus.php	092f513331e7607de60510af0e7badca72b3e2b64f4507898b55c109b695f5e2

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum CommunicationStatus:string{case PLANNED='PLANNED';case ATTEMPTED='ATTEMPTED';case COMMUNICATED='COMMUNICATED';case FAILED='FAILED';case DECLINED='DECLINED';}
```

src/Enums/CriticalStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CriticalStatus.php	83cbb282efb9824bc9d73b0c2cde847a0e2016ac07506d87aae6e80476c1c09a

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum CriticalStatus:string{case TRIGGERED='TRIGGERED';case ROUTED='ROUTED';case DELIVERED='DELIVERED';case ACKNOWLEDGED='ACKNOWLEDGED';case ESCALATED='ESCALATED';case RESOLVED='RESOLVED';case CANCELLED='CANCELLED';}
```

src/Enums/FollowUpStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/FollowUpStatus.php	fdce0b0a616c8bb449180ee3f3b60b04ab7771afa798e49d6fd14a3f442c4b0c

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum FollowUpStatus:string{case OPEN='OPEN';case IN_PROGRESS='IN_PROGRESS';case COMPLETED='COMPLETED';case CANCELLED='CANCELLED';case OVERDUE='OVERDUE';}
```

src/Enums/MatchStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MatchStatus.php	299e5005dd75fa26b230f1e34933a71e7572f9e458189e6ed7e4bd75acbb47fc

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum MatchStatus:string{case MATCHED='MATCHED';case
AMBIGUOUS='AMBIGUOUS';case UNMATCHED='UNMATCHED';case QUARANTINED='QUARANTINED';}
```

src/Enums/ReportStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ReportStatus.php	537ababcbddf6d596bdf944a1fab2513561e410783b5e2c53370e864f8212a24

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum ReportStatus:string{case
REGISTERED='REGISTERED';case PARTIAL='PARTIAL';case PRELIMINARY='PRELIMINARY';case FINAL='FINAL';case
AMENDED='AMENDED';case CORRECTED='CORRECTED';case CANCELLED='CANCELLED';case
ENTERED_IN_ERROR='ENTERED_IN_ERROR';}
```

src/Enums/ResultSourceType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ResultSourceType.php	7a56e33acade71e75ba3e34cf9d0a9e6f62d20c2a6a8d0a3af89588f7075b6a

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum ResultSourceType:string{case LIMS='LIMS';case
IMAGING='IMAGING';case PATHOLOGY='PATHOLOGY';case RADIOLOGY='RADIOLOGY';case EXTERNAL='EXTERNAL';}
```

src/Enums/ReviewStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ReviewStatus.php	3f504558227f9cb66b9e58933590762beaed17230148c77067398502c79251d6

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Enums;enum ReviewStatus:string{case PENDING='PENDING';case
REVIEWED='REVIEWED';case ACKNOWLEDGED='ACKNOWLEDGED';case ACTION_REQUIRED='ACTION_REQUIRED';case
COMPLETED='COMPLETED';case ENTERED_IN_ERROR='ENTERED_IN_ERROR';}
```

src/Models/CriticalAcknowledgement.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CriticalAcknowledgement.php	bb891694d5607e1b98fbab95fa293c29fe44033577b3c6daa02afe2932898a41

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
CriticalAcknowledgement extends Model{public $timestamps=false;protected
$table='clinical_critical_acknowledgements';protected $guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/CriticalDelivery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CriticalDelivery.php	8b00f358d498a0c667b710bd55357d07d42c19789bb08acba47d8a263a06e9da

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
CriticalDelivery extends Model{public $timestamps=false;protected $table='clinical_critical_result_deliveries';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/CriticalResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CriticalResult.php	28f19708d943ea394ec13f658645635a4ec0ffae255414f8e50643f5019bb8d3

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
CriticalResult extends Model{public $timestamps=false;protected $table='clinical_critical_results';protected $guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/DiagnosticAttachment.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DiagnosticAttachment.php	6875801ca12f19fd6efd949d7569f07dc525ea48b806696e85bf976b8805bbb8

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
DiagnosticAttachment extends Model{public $timestamps=false;protected $table='clinical_diagnostic_attachments';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/DiagnosticReportProjection.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DiagnosticReportProjection.php	574eac32f948b9876233ac8d195085a0728445e94a52122ffa2c60fe983ddc

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
DiagnosticReportProjection extends Model{public $timestamps=false;protected
$table='clinical_diagnostic_report_projections';protected $guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
```

```
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/DiagnosticResultItem.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DiagnosticResultItem.php	b38c43f8cd5689b72b4891a082da0c5dd66c7d7fc68cd1eaaeb46b52ee5ab0fe

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
DiagnosticResultItem extends Model{public $timestamps=false;protected $table='clinical_diagnostic_result_items';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/PatientResultCommunication.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PatientResultCommunication.php	5923bc5fe179772c96dac40d160cd86fc4a069b9146d60df287cbdde1935947c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
PatientResultCommunication extends Model{public $timestamps=false;protected
$table='clinical_patient_result_communications';protected $guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/ResultCorrectionLink.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResultCorrectionLink.php	faccfe10afd2ce73e871f3b7a61cdc202abcbd0e8182b145ec690154563ef68b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
ResultCorrectionLink extends Model{public $timestamps=false;protected $table='clinical_result_correction_links';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/ResultFollowUp.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResultFollowUp.php	ddcb2ac43d0fd94fbb60baa9bfa564dadbe366cb09b0577a77b106afc3b217ed

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
ResultFollowUp extends Model{public $timestamps=false;protected $table='clinical_result_followups';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/ResultMatchDecision.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResultMatchDecision.php	1d8cd7e9bf2684fccac88f6b548ec830c2c9e526d69bf99f8f6841368af59413

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
ResultMatchDecision extends Model{public $timestamps=false;protected $table='clinical_result_match_decisions';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/ResultQuarantine.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResultQuarantine.php	c455fbc060bd53f7dd6b0d6ccbf63b176072576b251cf20f8473b5de489c1bfe

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
ResultQuarantine extends Model{public $timestamps=false;protected $table='clinical_result_quarantines';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/ResultReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResultReconciliation.php	54268b3d6ec9cc603286721077d5c9052f8b904ac2497b9337383f1b2f0f2f4b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
ResultReconciliation extends Model{public $timestamps=false;protected $table='clinical_result_reconciliations';protected
$guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=
>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immuta
ble_datetime'];}
```

src/Models/ResultReview.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResultReview.php	5ba086637e8e0be4a13230f40acea23d7f0ce7bf7809b425eb169eb26fd910

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Results\Models;use Illuminate\Database\Eloquent\Model;final class
ResultReview extends Model{public $timestamps=false;protected $table='clinical_result_reviews';protected $guarded=[];protected
$casts=['value_json'=>'array','interpretation'=>'array','metadata'=>'array','effective_at_utc'=>'immutable_datetime','issued_at_utc'=>'immutable_datetime','received_at_utc'=>'immutable_datetime','reviewed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	Context, authorization, audit, inbox/outbox and time
Clinical orders	Authoritative order identity and requester
LIMS	Versioned laboratory report contract
Imaging	Versioned diagnostic report and attachment contract
Communications	Critical notification and patient communication
Clinical tasks	Closed-loop follow-up work
Vue	Result inbox, report viewer, critical queue, quarantine and follow-up workspaces

Blocking Gaps

Execute in actual KashTre repository

Bind LIMS and Imaging contracts

Approve patient/order matching algorithm

Approve critical-result routing and escalation timeframes

Approve source correction and cancellation semantics

Approve patient communication policy

Complete host controllers, requests, resources, scheduled jobs and Vue interfaces

Run clean, upgrade, rollback, duplicate, outage, security, accessibility and clinical gold tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Result review and follow-up authority
KashTre_Imaging_Module_SRD_v2.5	Imaging source authority
HL7 FHIR R5 DiagnosticReport	Report content and provenance reference
HL7 FHIR R5 Task	Follow-up activity tracking reference

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 8

TRANSITIONS OF CARE, HANDOVER, DISCHARGE AND REFERRAL

Readiness Gates, Transfer Continuity, Medication Reconciliation, Follow-Up and Attested Documents

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 to 7

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Safety Rule

A transition is a governed clinical event, not a change of screen location. Discharge, transfer and referral cannot erase active observations, overdue work, medication responsibilities, critical results or follow-up obligations. Readiness is evaluated from authoritative services, warnings are explicit and blocking conditions require resolution or a separately authorized exception.

Initiate transition -> readiness snapshot -> resolve blockers -> authorize -> handover -> movement/discharge/referral -> transfer outstanding work -> acknowledge receipt -> follow-up -> attested summary

Entity Register

Entity	Table	Invariant
CareTransition	clinical_care_transitions	Tenant, patient, encounter, source and historical state preserved
TransitionReadinessCheck	clinical_transition_readiness_checks	Tenant, patient, encounter, source and historical state preserved
TransitionReadinessItem	clinical_transition_readiness_items	Tenant, patient, encounter, source and historical state preserved
ClinicalHandover	clinical_handovers	Tenant, patient, encounter, source and historical state preserved
HandoverAcknowledgement	clinical_handover_acknowledgements	Tenant, patient, encounter, source and historical state preserved
MedicationReconciliation	clinical_transition_medication_reconciliations	Tenant, patient, encounter, source and historical state preserved
OutstandingResultSnapshot	clinical_transition_outstanding_results	Tenant, patient, encounter, source and historical state preserved
FollowUpPlan	clinical_transition_followup_plans	Tenant, patient, encounter, source and historical state preserved
FollowUpOccurrence	clinical_transition_followup_occurrences	Tenant, patient, encounter, source and historical state preserved
TransitionDocument	clinical_transition_documents	Tenant, patient, encounter, source and historical state preserved
TransitionDocumentSection	clinical_transition_document_sections	Tenant, patient, encounter, source and historical state preserved
ReferralProjection	clinical_referral_projections	Tenant, patient, encounter, source and historical state preserved
TransitionCorrection	clinical_transition_corrections	Tenant, patient, encounter, source and historical state preserved
TransitionReconciliation	clinical_transition_reconciliations	Tenant, patient, encounter, source and historical state preserved

Engineering Design**Readiness**

Open medication administration and infusion checks

Medication reconciliation

Outstanding diagnostic results and critical acknowledgements

Observation Plan due and overdue work

Active alerts and unresolved tasks

Destination and follow-up arrangements

Patient communication and document readiness

Internal transfer

Use confirmed movement effective time

Preserve original due and overdue timestamps

End source-space-only requirements prospectively

Activate destination defaults

Continue order, protocol and patient-specific work

Require receiving acknowledgement where configured

Discharge

No silent cancellation of pending results

Create post-discharge follow-up tasks

Generate attested discharge document from authoritative data

Separate clinical readiness from billing or Inventory closure

Permit operational discharge despite downstream outage with reconciliation

Referral and external transfer

Preserve destination organization and referral identity

Package minimum necessary information

Record dispatch and receipt evidence

Maintain confidentiality labels

Keep local follow-up until acceptance is proven

Documents

Versioned immutable attested snapshots

Sections retain source record IDs

Correction creates a superseding version

Render failure cannot alter transition state

Permissions

Permission	Purpose
clinical.transition.initiate	Initiate transition
clinical.transition.readiness.resolve	Resolve readiness item
clinical.transition.authorize	Authorize major transition
clinical.handover.prepare	Prepare handover
clinical.handover.acknowledge	Acknowledge receipt
clinical.discharge.attest	Attest discharge summary

clinical.referral.create	Create referral
clinical.followup.manage	Manage follow-up
clinical.transition.override_block	Apply approved exceptional pathway
clinical.transition.reconcile	Resolve transition exception

Mandatory Tests

Blocking readiness item prevents authorization

Warning remains visible after authorized exception

Transfer preserves overdue time

Destination defaults do not duplicate patient-specific work

Discharge retains pending result follow-up

Billing outage does not rewrite clinical discharge time

Repeated movement event does not duplicate transfer

Receiving acknowledgement uses exact handover version

Corrected source data supersedes discharge document

Cross-tenant destination input is rejected

Cancelled transition has no movement effect

Follow-up cannot complete without outcome evidence

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	f19f47f0fdc089759ae0c209012c7305987ab1175502af2a3bc1175944c8da9f

KashTre Clinical Transitions of Care v6.1.0

Pick-and-drop candidate. Bind encounter movement, medication reconciliation, outstanding results, Observation Plan transfer, Clinical Task, referral, document rendering and contextual authorization before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	4b524b8b83d18bc0ce88c877c405984ed9d7751ca9c1a53911899f54023f152a

```
{
  "volume": "Clinical EDD Volume 8 Transitions of Care",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "f19f47f0fdc089759ae0c209012c7305987ab1175502af2a3bc1175944c8da9f",
      "bytes": 267
    },
    {
      "path": "composer.json",
      "sha256": "623c10098e79b9c4777c08882d06dc5664f04ae646c2de85a69b7c7e4f44afc2",
      "bytes": 392
    },
    {
      "path": "config/clinical-transitions.php",
      "sha256": "269941308718846345b7e0e43a00b6f7e62ebb6fb37df1e79484ff53a6f1d78a",
      "bytes": 276
    },
    {
      "path": "database/migrations/2026_08_28_700000_create_transition_tables.php",
      "sha256": "75873e29d00a8e69c18499dd610015ffef06f327bab316e3e3d52905fb6c6042",
      "bytes": 3812
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-transitions.php",
      "sha256": "e5f403b9f8a84f020e6c482e0d96e6d1128b80ccd6455f8a0ab74cae8229fea5",
      "bytes": 275
    }
  ]
}
```

```

"path": "src/Actions/CompleteInternalTransfer.php",
"sha256": "e27509a7d9d9a455c7a9f7ef89d789034a5f8cde28519338349260fbc758350e",
"bytes": 1204
},
{
"path": "src/Actions/InitiateCareTransition.php",
"sha256": "4a1f6d1f156244d23d2e1d1a882effc542901687ef37054a617c368f50f2b883",
"bytes": 2007
},
{
"path": "src/Actions/IssueDischargeDocument.php",
"sha256": "3a36ab9609d329e734a48c7553eac6cf55489dedf1b71116b1d2c3b7c03ab89f",
"bytes": 1398
},
{
"path": "src/Contracts/ClinicalTaskGateway.php",
"sha256": "72b92a557cff1cf796a19688917720c6bb8a926f70c229acdd7d17d3bb63b930",
"bytes": 279
},
{
"path": "src/Contracts/ObservationPlanTransferGateway.php",
"sha256": "818f2026810d241cd068470adbd4f38aacdd28521baf625c983c44347acb459",
"bytes": 297
},
{
"path": "src/Contracts/TransitionReadinessContributor.php",
"sha256": "ef26bcc6a9d2f60c12511e7f7228ee0c1310dd854460ef135a5de86abc97e079",
"bytes": 246
},
{
"path": "src/Data/InitiateTransitionData.php",
"sha256": "09af6f251df222547b1631bdaa243dff65f0d778cf6282ec0e8e99f3c7b5969c",
"bytes": 459
},
{
"path": "src/Enums/DocumentStatus.php",
"sha256": "21985232ce8869c5d8677a91c9ab3ed28dc9992c7b5c57c16eac006fb4acbd7f",
"bytes": 238
},
{
"path": "src/Enums/FollowUpStatus.php",
"sha256": "d722d062f4f5712a5eec102feea564915755dfde26dec77c8bfea6d14c515621",
"bytes": 245
},
{
"path": "src/Enums/HandoverStatus.php",
"sha256": "d68b4c53327774e9c84fa9fd62de0cb867130c98f45dfe863e25be4bffd7267b",
"bytes": 251
},
{
"path": "src/Enums/ReadinessOutcome.php",
"sha256": "8562d44739745774b4c86f70e828d61849eddd16bfc163e7772f914b1fc7ce29",
"bytes": 164
},

```

```

{
  "path": "src/Enums/ReconciliationStatus.php",
  "sha256": "264bb9d23ac9ff8ea9a1a2bb3cfefdfc76fceace5a47eeb66bbbed180cbecef2e",
  "bytes": 234
},
{
  "path": "src/Enums/TransitionStatus.php",
  "sha256": "dc11fc6c96fe8585b5acf7f5b4006fdbf3181a6ea701fc822ebfb7d902ca34fd",
  "bytes": 392
},
{
  "path": "src/Enums/TransitionType.php",
  "sha256": "f4e2fdcf209ec9ffd8655c2795c08c9bd9cb32876f564aea28f67a0dbfa4ae02",
  "bytes": 309
},
{
  "path": "src/Models/CareTransition.php",
  "sha256": "365d55cd1ad3036fea07c2f024df7ff94f2c9185b0b2e84b59fb94b7409d7664",
  "bytes": 482
},
{
  "path": "src/Models/ClinicalHandover.php",
  "sha256": "b2cb0c97b57a1abccc73e8e7e4fe5b80083a7d833685b9b83febf5a040a999e7",
  "bytes": 477
},
{
  "path": "src/Models/FollowUpOccurrence.php",
  "sha256": "d1ca777bdf444d1d45f2ef551887dd80db21cb4a00e5606755fea21865e89838",
  "bytes": 501
},
{
  "path": "src/Models/FollowUpPlan.php",
  "sha256": "03ed5386aab30d2d440474280439b36f3d437d1b7c42d4d62a7442597a1dcc83",
  "bytes": 489
},
{
  "path": "src/Models/HandoverAcknowledgement.php",
  "sha256": "34317ba556ab87a9d4b58755a3612c005932c600b77a16391e584eb78c7cbf31",
  "bytes": 500
},
{
  "path": "src/Models/MedicationReconciliation.php",
  "sha256": "33f780766cdf13947202935fc7245375d173bc660161b0fbf10d02ba71fd4d03",
  "bytes": 513
},
{
  "path": "src/Models/OutstandingResultSnapshot.php",
  "sha256": "5c73d7f402de96997ceee9f9d42455d4e126a86ea718ac4b4ff0e7a1f0db4247",
  "bytes": 507
},
{
  "path": "src/Models/ReferralProjection.php",
  "sha256": "54eef3741b96d6605e4417ad4fb1c8c3b80b35bd2481006d690b98510ae9920d",
  "bytes": 490
},

```

```
{
  "path": "src/Models/TransitionCorrection.php",
  "sha256": "875d4f6a75cb75d5d738c7ac24bda6f6eeaaab95cea83bebb9be847d93dc137a26",
  "bytes": 494
},
{
  "path": "src/Models/TransitionDocument.php",
  "sha256": "1105518e127318171f860f7aac3236f35659e8a8f11b46388a19f9ad5dfbcd01",
  "bytes": 490
},
{
  "path": "src/Models/TransitionDocumentSection.php",
  "sha256": "b293131d2afd106924001fbaa1faac7449553a6d2f35418a3a7d73668280bb16",
  "bytes": 505
},
{
  "path": "src/Models/TransitionReadinessCheck.php",
  "sha256": "bc5526ed95fe37d5587fd5e981cd2bf9e6d7eeb26024a5bf0451a56cb26196dd",
  "bytes": 503
},
{
  "path": "src/Models/TransitionReadinessItem.php",
  "sha256": "392527fc057508e9c0dc05402fa5521e3372fef6a4b52dabca54d2525b833401",
  "bytes": 501
},
{
  "path": "src/Models/TransitionReconciliation.php",
  "sha256": "093d5d2ab246d7a82ab8c752e40018d03276d536902651a913847c5aa0fc8478",
  "bytes": 502
},
{
  "path": "src/Services/TransitionReadinessEngine.php",
  "sha256": "fefab0a53057e01b45f4ba61528c9c649980a4dce65bfcabead718b34df28c9cc",
  "bytes": 825
},
{
  "path": "src/TransitionsServiceProvider.php",
  "sha256": "0f9b036fccc66121324792d7a3e441a4f178e91be6c69876531d4a16daf47ece",
  "bytes": 457
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	623c10098e79b9c4777c08882d06dc5664f04ae646c2de85a69b7c7e4f44afc2

```
{
  "name": "kashtre/clinical-transitions",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\Transitions\\": "src/"
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\Transitions\\TransitionsServiceProvider"
          ]
        }
      }
    }
  }
}
```

config/clinical-transitions.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-transitions.php	269941308718846345b7e0e43a00b6f7e62ebb6fb37df1e79484ff53a6f1d78a

```
<?php
declare(strict_types=1);
return [
    'handover_ack_minutes' => (int) env('CLINICAL_HANDOVER_ACK_MINUTES', 30),
    'discharge_followup_horizon_days' => (int) env('CLINICAL_DISCHARGE_FOLLOWUP_DAYS', 30),
    'require_medication_reconciliation' => true,
    'require_outstanding_results_review' => true;
];
```

database/migrations/2026_08_28_700000_create_transition_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_700000_create_transition_tables.php	75873e29d00a8e69c18499dd610015ffef06f327bab316e3e3d52905fb6c6042

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_care_transitions', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('patient_public_id');
            $table->ulid('encounter_public_id');
            $table->string('transition_type', 40);
            $table->string('status', 40);
            $table->ulid('from_client_space_public_id')->nullable();
            $table->ulid('to_client_space_public_id')->nullable();
            $table->ulid('destination_organization_public_id')->nullable();
            $table->ulid('initiated_by_public_id');
            $table->dateTime('planned_at_utc', 6);
            $table->dateTime('effective_at_utc', 6)->nullable();
            $table->dateTime('completed_at_utc', 6)->nullable();
            $table->text('reason');
            $table->json('context')->nullable();
            $table->unsignedBigInteger('record_version')->default(1);
        });

        Schema::create('clinical_transition_readiness_checks', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('transition_public_id');
            $table->string('outcome', 24);
            $table->char('input_fingerprint', 64);
            $table->dateTime('evaluated_at_utc', 6);
        });

        Schema::create('clinical_transition_readiness_items', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('readiness_check_public_id');
            $table->string('check_code', 96);
            $table->string('outcome', 24);
            $table->json('evidence');
            $table->boolean('is_blocking');
        });

        Schema::create('clinical_handovers', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('transition_public_id');
            $table->string('status', 24);
            $table->json('content');
            $table->ulid('prepared_by_public_id');
            $table->dateTime('prepared_at_utc', 6);
            $table->dateTime('sent_at_utc', 6)->nullable();
        });

        Schema::create('clinical_handover_acknowledgements', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('handover_public_id');
            $table->ulid('acknowledged_by_public_id');
            $table->dateTime('acknowledged_at_utc', 6);
            $table->text('note')->nullable();
        });

        Schema::create('clinical_transition_followup_plans', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('transition_public_id');
            $table->string('status', 24);
            $table->json('content');
            $table->dateTime('created_at_utc', 6);
        });

        Schema::create('clinical_transition_followup_occurrences', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('followup_plan_public_id');
            $table->string('code', 96);
            $table->string('status', 24);
            $table->dateTime('due_at_utc', 6);
            $table->ulid('task_public_id')->nullable();
            $table->dateTime('completed_at_utc', 6)->nullable();
        });

        Schema::create('clinical_transition_documents', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->ulid('transition_public_id');
            $table->string('document_type', 64);
            $table->string('status', 24);
            $table->unsignedInteger('version_no');
            $table->ulid('authored_by_public_id');
            $table->dateTime('attested_at_utc', 6);
            $table->char('content_hash', 64);
        });
    }
};
```

```
>unique(['transition_public_id','document_type','version_no'], 'uq_transition_document'));});
Schema::create('clinical_transition_document_sections',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('document_public_id');$t->string('section_code',96);$t->json('content');$t-
>unique(['document_public_id','section_code'], 'uq_transition_document_section'));});
}public function
down():void{foreach(['clinical_transition_document_sections','clinical_transition_documents','clinical_transition_followup_occurren
ces','clinical_transition_followup_plans','clinical_handover_acknowledgements','clinical_handovers','clinical_transition_readiness_i
tems','clinical_transition_readiness_checks','clinical_care_transitions']as$t)Schema::dropIfExists($t);};
```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force
```

routes/clinical-transitions.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical- transitions.php	e5f403b9f8a84f020e6c482e0d96e6d1128b80ccd6455f8a0ab74cae8229fea5

```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])->prefix('api/v1/clinical')-
>group(function(){Route::get('/transitions',fn()->response()-
>json(['error'=>['code'=>'CLN_TRANSITION_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/CompleteInternalTransfer.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CompleteInternalTransfer.php	e27509a7d9d9a455c7a9f7ef89d789034a5f8cde28519338349260fbc758350e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Transitions\Contracts\ObservationPlanTransferGateway;use
KashTre\Clinical\Transitions\Models\CareTransition;final readonly class CompleteInternalTransfer{public function
__construct(private ObservationPlanTransferGateway $plans){}public function execute(string $tenantId,string $transitionId,string
$movementId,\DateTimeImmutable $effectiveAt):CareTransition{return
DB::transaction(function()use($tenantId,$transitionId,$movementId,$effectiveAt){$t=CareTransition::query()-
>where('tenant_id',$tenantId)->where('public_id',$transitionId)->lockForUpdate()->firstOrFail();if($t-
>transition_type!=='INTERNAL_TRANSFER'||in_array($t->status,['READY','AUTHORIZED','IN_PROGRESS'],true))throw new
\DomainException('CLN_TRANSITION_NOT_TRANSFERABLE');$this->plans->transfer($tenantId,$t->patient_public_id,$t-
>encounter_public_id,$movementId,$t->from_client_space_public_id,$t->to_client_space_public_id,$effectiveAt);$t-
>forceFill(['status'=>'COMPLETED','effective_at_utc'=>$effectiveAt,'completed_at_utc'=>now('UTC'),'record_version'=>$t-
>record_version+1])->save();return$t;});}
```

src/Actions/InitiateCareTransition.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Actions/InitiateCareTransition.php	4a1f6d1f156244d23d2e1d1a882effc542901687ef37054a617c368f50f2b883
-------------------------	--	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Transitions\Data\InitiateTransitionData;use
KashTre\Clinical\Transitions\Models\CareTransition,TransitionReadinessCheck,TransitionReadinessItem;use
KashTre\Clinical\Transitions\Services\TransitionReadinessEngine;final readonly class InitiateCareTransition{public function
__construct(private PublicIdGenerator $ids,private TransitionReadinessEngine $readiness){}public function
execute(InitiateTransitionData $d):CareTransition{return DB::transaction(function()use($d){$transition=CareTransition::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'patient_public_id'=>$d-
>patientId,'encounter_public_id'=>$d->encounterId,'transition_type'=>$d-
>type,'status'=>'READINESS_CHECK','from_client_space_public_id'=>$d->fromClientSpaceId,'to_client_space_public_id'=>$d-
>toClientSpaceId,'destination_organization_public_id'=>$d->destinationOrganizationId,'initiated_by_public_id'=>$d-
>initiatedById,'planned_at_utc'=>$d->plannedAt,'reason'=>$d->reason,'context'=>$d->context,'record_version'=>1]);$r=$this-
>readiness->evaluate($d->tenantId,$d->patientId,$d->encounterId,$d->type,$d-
>context);$check=TransitionReadinessCheck::query()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d-
>tenantId,'transition_public_id'=>$transition-
>public_id,'outcome'=>$r['outcome'],'input_fingerprint'=>$r['fingerprint'],'evaluated_at_utc'=>now('UTC')));foreach($r['items']as$i)Tr
ansitionReadinessItem::query()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d-
>tenantId,'readiness_check_public_id'=>$check-
>public_id,'check_code'=>$i['code'],'outcome'=>$i['outcome'],'evidence'=>$i['evidence']??[],'is_blocking'=>(bool)($i['blocking']??fal
se));$transition->forceFill(['status'=>$r['outcome']==='BLOCK'? 'READINESS_CHECK': 'READY']->save();return$transition;}}}
```

src/Actions/IssueDischargeDocument.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/IssueDischargeDocument.php	3a36ab9609d329e734a48c7553eac6cf55489dedf1b71116b1d2c3b7c03ab89f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Transitions\Models\CareTransition,TransitionDocument,TransitionDocumentSection;final readonly class
IssueDischargeDocument{public function __construct(private PublicIdGenerator $ids){}public function execute(string
$tenantId,string $transitionId,string $authorId,array $sections):TransitionDocument{return
DB::transaction(function()use($tenantId,$transitionId,$authorId,$sections){$t=CareTransition::query()-
>where('tenant_id',$tenantId)->where('public_id',$transitionId)->where('transition_type','DISCHARGE')->firstOrFail();if($t-
>status!=='READY')throw new \DomainException('CLN_DISCHARGE_NOT_READY');$d=TransitionDocument::query()-
>create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'transition_public_id'=>$transitionId,'document_type'=>'DISCHARGE_SUMMARY','status'=>'ATTE
STED','version_no'=>1,'authored_by_public_id'=>$authorId,'attested_at_utc'=>now('UTC'),'content_hash'=>hash('sha256',json_enco
de($sections,JSON_THROW_ON_ERROR))));foreach($sections as$code=>$content)TransitionDocumentSection::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$tenantId,'document_public_id'=>$d-
>public_id,'section_code'=>$code,'content'=>$content]);return$d;}}}
```

src/Contracts/ClinicalTaskGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ClinicalTaskGateway.php	72b92a557cff1cf796a19688917720c6bb8a926f70c229acdd7d17d3bb63b930

E		
---	--	--

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Contracts;interface ClinicalTaskGateway{public function
createFollowUp(string $tenantId,string $patientId,string $encounterId,string $transitionId,string $code,\DateTimeImmutable
$dueAt,array $context):string;}
```

src/Contracts/ObservationPlanTransferGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObservationPlanTransferGateway.php	818f2026810d241cd068470adbd4f38aaccdd28521baf625c983c44347acb459

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Contracts;interface ObservationPlanTransferGateway{public
function transfer(string $tenantId,string $patientId,string $encounterId,string $movementId,string $fromSpaceId,string
$toSpaceId,\DateTimeImmutable $effectiveAt):array;}
```

src/Contracts/TransitionReadinessContributor.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/TransitionReadinessContributor.php	ef26bcc6a9d2f60c12511e7f7228ee0c1310dd854460ef135a5de86abc97e079

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Contracts;interface TransitionReadinessContributor{public
function evaluate(string $tenantId,string $patientId,string $encounterId,string $transitionType,array $context):array;}
```

src/Data/InitiateTransitionData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/InitiateTransitionData.php	09af6f251df222547b1631bdaa243dff65f0d778cf6282ec0e8e99f3c7b5969c

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Data;final readonly class InitiateTransitionData{public function
__construct(public string $tenantId,public string $patientId,public string $encounterId,public string $type,public ?string
$fromClientSpaceId,public ?string $toClientSpaceId,public ?string $destinationOrganizationId,public string $initiatedById,public
\DateTimeImmutable $plannedAt,public string $reason,array $context=[]){}}
```

src/Enums/DocumentStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DocumentStatus.php	21985232ce8869c5d8677a91c9ab3ed28dc9992c7b5c57c16eac006fb4acbd7f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum DocumentStatus:string{case DRAFT='DRAFT';case
ATTESTED='ATTESTED';case ISSUED='ISSUED';case SUPERSEDED='SUPERSEDED';case
ENTERED_IN_ERROR='ENTERED_IN_ERROR';}
```

src/Enums/FollowUpStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/FollowUpStatus.php	d722d062f4f5712a5eec102feea564915755dfde26dec77c8bfea6d14c515621

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum FollowUpStatus:string{case
PLANNED='PLANNED';case SCHEDULED='SCHEDULED';case DUE='DUE';case OVERDUE='OVERDUE';case
COMPLETED='COMPLETED';case CANCELLED='CANCELLED';}
```

src/Enums/HandoverStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/HandoverStatus.php	d68b4c53327774e9c84fa9fd62de0cb867130c98f45dfe863e25be4bffd7267b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum HandoverStatus:string{case
PREPARED='PREPARED';case SENT='SENT';case RECEIVED='RECEIVED';case ACKNOWLEDGED='ACKNOWLEDGED';case
REJECTED='REJECTED';case EXPIRED='EXPIRED';}
```

src/Enums/ReadinessOutcome.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ReadinessOutcome.php	8562d44739745774b4c86f70e828d61849eddd16bfc163e7772f914b1fc7ce29

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum ReadinessOutcome:string{case PASS='PASS';case
WARNING='WARNING';case BLOCK='BLOCK';}
```

src/Enums/ReconciliationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ReconciliationStatus.php	264bb9d23ac9ff8ea9a1a2bb3cfefdfc76fceace5a47eeb66bbbed180cbecef2e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum ReconciliationStatus:string{case
PENDING='PENDING';case IN_PROGRESS='IN_PROGRESS';case COMPLETED='COMPLETED';case WAIVED='WAIVED';case
BLOCKED='BLOCKED';}
```

src/Enums/TransitionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/TransitionStatus.php	dc11fc6c96fe8585b5acf7f5b4006fdbf3181a6ea701fc822ebfb7d902ca34fd

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum TransitionStatus:string{case DRAFT='DRAFT';case
READINESS_CHECK='READINESS_CHECK';case READY='READY';case AUTHORIZED='AUTHORIZED';case
IN_PROGRESS='IN_PROGRESS';case COMPLETED='COMPLETED';case CANCELLED='CANCELLED';case
RECONCILIATION_REQUIRED='RECONCILIATION_REQUIRED';case ENTERED_IN_ERROR='ENTERED_IN_ERROR';}
```

src/Enums/TransitionType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/TransitionType.php	f4e2fd9c209ec9ffd8655c2795c08c9bd9cb32876f564aea28f67a0dbfa4ae02

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Enums;enum TransitionType:string{case
INTERNAL_TRANSFER='INTERNAL_TRANSFER';case INTERFACILITY_TRANSFER='INTERFACILITY_TRANSFER';case
DISCHARGE='DISCHARGE';case REFERRAL='REFERRAL';case TEMPORARY_LEAVE='TEMPORARY_LEAVE';case DEATH='DEATH';}
```

src/Models/CareTransition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CareTransition.php	365d55cd1ad3036fea07c2f024df7ff94f2c9185b0b2e84b59fb94b7409d7664

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
CareTransition extends Model{public $timestamps=false;protected $table='clinical_care_transitions';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/ClinicalHandover.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDATE	src/Models/ClinicalHandover.php	b2cb0c97b57a1abccc73e8e7e4fe5b80083a7d833685b9b83febf5a040a999e7

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
ClinicalHandover extends Model{public $timestamps=false;protected $table='clinical_handovers';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/FollowUpOccurrence.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDAT E	src/Models/FollowUpOccurrence.ph p	d1ca777bdf444d1d45f2ef551887dd80db21cb4a00e5606755fea21865e89838

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
FollowUpOccurrence extends Model{public $timestamps=false;protected
$table='clinical_transition_followup_occurrences';protected $guarded=[];protected
```

```
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/FollowUpPlan.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/FollowUpPlan.php	03ed5386aab30d2d440474280439b36f3d437d1b7c42d4d62a7442597a1dcc83

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class FollowUpPlan extends Model{public $timestamps=false;protected $table='clinical_transition_followup_plans';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/HandoverAcknowledgement.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/HandoverAcknowledgement.php	34317ba556ab87a9d4b58755a3612c005932c600b77a16391e584eb78c7cbf31

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class HandoverAcknowledgement extends Model{public $timestamps=false;protected $table='clinical_handover_acknowledgements';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/MedicationReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MedicationReconciliation.php	33f780766cdf13947202935fc7245375d173bc660161b0fbf10d02ba71fd4d03

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class MedicationReconciliation extends Model{public $timestamps=false;protected $table='clinical_transition_medication_reconciliations';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/OutstandingResultSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/OutstandingResultSnapshot.php	5c73d7f402de96997ceee9f9d42455d4e126a86ea718ac4b4ff0e7a1f0db4247

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
OutstandingResultSnapshot extends Model{public $timestamps=false;protected
$table='clinical_transition_outstanding_results';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/ReferralProjection.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReferralProjection.php	54eef3741b96d6605e4417ad4fb1c8c3b80b35bd2481006d690b98510ae9920d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
ReferralProjection extends Model{public $timestamps=false;protected $table='clinical_referral_projections';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/TransitionCorrection.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransitionCorrection.php	875d4f6a75cb75d5d738c7ac24bda6f6eeab95cea83bebb9be847d93dc137a26

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
TransitionCorrection extends Model{public $timestamps=false;protected $table='clinical_transition_corrections';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/TransitionDocument.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransitionDocument.php	1105518e127318171f860f7aac3236f35659e8a8f11b46388a19f9ad5dfbcd01

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
TransitionDocument extends Model{public $timestamps=false;protected $table='clinical_transition_documents';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/TransitionDocumentSection.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransitionDocumentSection.php	b293131d2afd106924001fbba1faac7449553a6d2f35418a3a7d73668280bb16

E		
---	--	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
TransitionDocumentSection extends Model{public $timestamps=false;protected
$table='clinical_transition_document_sections';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/TransitionReadinessCheck.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransitionReadinessCheck.php	bc5526ed95fe37d5587fd5e981cd2bf9e6d7eeb26024a5bf0451a56cb26196dd

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
TransitionReadinessCheck extends Model{public $timestamps=false;protected
$table='clinical_transition_readiness_checks';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/TransitionReadinessItem.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransitionReadinessItem.php	392527fc057508e9c0dc05402fa5521e3372fef6a4b52dabca54d2525b833401

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
TransitionReadinessItem extends Model{public $timestamps=false;protected $table='clinical_transition_readiness_items';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Models/TransitionReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TransitionReconciliation.php	093d5d2ab246d7a82ab8c752e40018d03276d536902651a913847c5aa0fc8478

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Models;use Illuminate\Database\Eloquent\Model;final class
TransitionReconciliation extends Model{public $timestamps=false;protected $table='clinical_transition_reconciliations';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','content'=>'array','planned_at_utc'=>'immutable_datetime','effective_at_utc'=>'immut
able_datetime','completed_at_utc'=>'immutable_datetime','due_at_utc'=>'immutable_datetime'];}
```

src/Services/TransitionReadinessEngine.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/TransitionReadinessEngine.php	fefab0a53057e01b45f4ba61528c9c649980a4dce65bfcabea718b34df28c9cc

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions\Services;use
KashTre\Clinical\Transitions\Contracts\TransitionReadinessContributor;final class TransitionReadinessEngine{public function
__construct(private iterable $contributors){}public function evaluate(string $tenantId,string $patientId,string $encounterId,string
$type,array $context):array{$items=[];$outcome='PASS';foreach($this->contributors as$c){if(!$c instanceof
TransitionReadinessContributor)continue;foreach($c-
>evaluate($tenantId,$patientId,$encounterId,$type,$context)as$i){$items[]=$i;if(($i['outcome']??'PASS')==='BLOCK')$outcome='BL
OCK';elseif(($i['outcome']??'PASS')==='WARNING'&&$outcome==='PASS')$outcome='WARNING';}}return['outcome'=>$outcome,'ite
ms'=>$items,'fingerprint'=>hash('sha256',json_encode($items,JSON_THROW_ON_ERROR))];}}
```

src/TransitionsServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/TransitionsServiceProvider.php	0f9b036fccc66121324792d7a3e441a4f178e91be6c69876531d4a16daf47ece

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Transitions;use Illuminate\Support\ServiceProviders;final class
TransitionsServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-transitions.php','clinical-transitions');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-transitions.php');}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	Context, authorization, audit, time and messaging
Volume 2	Observation Plan transfer continuity
Volume 6	Medication reconciliation and open administration
Volume 7	Outstanding results and follow-up
Main/HR	Destination spaces, organizations and receiving assignments
Clinical tasks	Post-discharge and referral follow-up
Documents	Attested discharge and referral rendering
Vue	Readiness, handover, discharge and referral workspaces

Blocking Gaps

Execute in actual KashTre repository

Approve transition readiness contributor catalogue

Bind confirmed movement service

Approve medication reconciliation workflow

Approve discharge and external transfer exception policy

Bind outstanding results and task services

Complete host controllers, requests, resources, jobs and Vue interfaces

Run clean, upgrade, rollback, duplicate-event, outage, security, accessibility and clinical gold tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Transitions-of-care authority
Volumes 1-7	Foundation and upstream clinical state
HL7 FHIR R5 CarePlan	Cross-setting care intention reference
HL7 FHIR R5 Composition	Attested clinical document structure reference

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 9

CLINICAL ASSURANCE, PRIVACY, AUDIT AND DOWNTIME RESILIENCE

Consent Decisions, Break Glass, Incidents, Audit Integrity, Offline Capture, Reconciliation and Controlled Export

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 to 8

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Assurance Rule

Clinical assurance must prove who accessed or changed clinical information, under which authority, for what purpose, and with what outcome. Break glass is time-limited and reviewable. Consent and confidentiality decisions are recorded. Downtime records preserve the real clinical occurrence time and are reconciled without silent duplication or overwriting.

Authorization + consent decision -> permitted action or denial -> immutable audit event -> monitoring and assurance checks -> incident or break-glass review when indicated -> corrective action and closure

Entity Register

Entity	Table	Invariant
ConsentDecisionRecord	clinical_consent_decisions	Tenant-scoped, immutable evidence or controlled lifecycle
BreakGlassEpisode	clinical_break_glass_episodes	Tenant-scoped, immutable evidence or controlled lifecycle
BreakGlassReview	clinical_break_glass_reviews	Tenant-scoped, immutable evidence or controlled lifecycle
SecurityAuditEvent	clinical_security_audit_events	Tenant-scoped, immutable evidence or controlled lifecycle
AuditChainCheckpoint	clinical_audit_chain_checkpoints	Tenant-scoped, immutable evidence or controlled lifecycle
ClinicalIncident	clinical_incidents	Tenant-scoped, immutable evidence or controlled lifecycle
IncidentAction	clinical_incident_actions	Tenant-scoped, immutable evidence or controlled lifecycle
DowntimeEpisode	clinical_downtime_episodes	Tenant-scoped, immutable evidence or controlled lifecycle
DowntimeCapturedRecord	clinical_downtime_captured_records	Tenant-scoped, immutable evidence or controlled lifecycle
DowntimeReconciliation	clinical_downtime_reconciliations	Tenant-scoped, immutable evidence or controlled lifecycle
AssuranceCheckRun	clinical_assurance_check_runs	Tenant-scoped, immutable evidence or controlled lifecycle
AssuranceCheckResult	clinical_assurance_check_results	Tenant-scoped, immutable evidence or controlled lifecycle
DataExportRequest	clinical_data_export_requests	Tenant-scoped, immutable evidence or controlled lifecycle
DataExportArtifact	clinical_data_export_artifacts	Tenant-scoped, immutable evidence or controlled lifecycle
RetentionDisposition	clinical_retention_dispositions	Tenant-scoped, immutable evidence or controlled lifecycle

Engineering Design**Consent and confidentiality**

Evaluate patient, actor, action, purpose and resource context

Record permit, deny, conditional or indeterminate result

Do not treat user authentication as consent

Preserve policy and consent references

Fail closed for restricted writes when authority cannot be proven

Break glass

Require reason, justification and requested actions

Limit scope and duration

Audit start, use, expiry and revocation

Display active emergency access

Require independent retrospective review

Audit and provenance

Record occurred and recorded times separately

Hash-chain security-relevant audit events

Retain actor, patient, entity, outcome and correlation evidence

Verify chain integrity

Keep operational audit separate from clinical record provenance

Downtime

Declare scoped downtime episode

Capture offline reference and actual occurrence time

Prevent duplicate import

Reconcile against authoritative records

Escalate conflicts and missing records

Close only when required reconciliations complete

Incidents and assurance

Record patient safety, privacy, security and operational incidents

Run deterministic assurance checks

Retain findings and corrective actions

Controlled exports require authorization, expiry and download audit

Retention disposition must never bypass legal hold

Permissions

Permission	Purpose
clinical.assurance.dashboard.view	View assurance posture
clinical.consent.decision.view	View decision evidence
clinical.break_glass.request	Request emergency access
clinical.break_glass.review	Review emergency episode
clinical.audit.search	Search authorized audit
clinical.audit.export	Request controlled export
clinical.incident.create	Report incident

clinical.incident.manage	Triage and resolve incident
clinical.downtime.declare	Declare downtime
clinical.downtime.reconcile	Reconcile offline records
clinical.assurance.check.run	Run assurance checks

Mandatory Tests

Denied consent decision blocks restricted action

Indeterminate decision fails closed for write

Break glass without justification is rejected

Expired break glass cannot authorize access

Every break-glass use is auditable

Audit hash-chain tampering is detected

Offline duplicate is not imported twice

Downtime occurrence time is not replaced by entry time

Reconciliation conflict is escalated

Export expires and download is audited

Cross-tenant audit access is concealed

Legal hold blocks retention disposition

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	aa73f533820297c7b904472c0591dfd88f50c969584cc1a9965214d552424416

KashTre Clinical Assurance, Privacy and Downtime v6.1.0

Pick-and-drop candidate. Bind Main consent policy, contextual authorization, immutable audit storage, incident management, offline capture, exports, retention, monitoring and Vue assurance workspaces before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	a740c16a5b7693fc3df8dfb755726d9ad74a2cad71f6cb2c8e6b07c95e3ef5c

```
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  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "aa73f533820297c7b904472c0591dfd88f50c969584cc1a9965214d552424416",
      "bytes": 281
    },
    {
      "path": "composer.json",
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      "bytes": 384
    },
    {
      "path": "config/clinical-assurance.php",
      "sha256": "7cd55c671c94bf1e74b6970ccb9db7b5e114787fe0e0632624b0238e239de767",
      "bytes": 295
    },
    {
      "path": "database/migrations/2026_08_28_800000_create_assurance_tables.php",
      "sha256": "5e8f1836d8330212d1bf190e2da6c9977c6c37814d3cbbbe7eda27da7b4deeeed",
      "bytes": 4774
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    {
      "path": "install.sh",
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      "bytes": 99
    },
    {
      "path": "routes/clinical-assurance.php",
      "sha256": "b0c65afe4e0eafe8e613600a8604c9b66e2bb305fa6e6db445fbbb8aa0605e0a",
      "bytes": 282
    }
  ]
}
```

```

"path": "src/Actions/ReconcileDowntimeRecord.php",
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"bytes": 1555
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{
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{
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{
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"sha256": "5c9343b2ca6a37a00eeb4f3a637d9411701dc50a718cfe8e5a2783426cfd03b5",
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{
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{
"path": "src/Contracts/ClinicalReconciliationContributor.php",
"sha256": "5ec800729438b09ee40da2abe39b520289bb37d1315ca180e7dc13ae10f13f56",
"bytes": 217
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{
"path": "src/Data/BreakGlassRequestData.php",
"sha256": "b8fc2605df71933b5a8dcf435eb7a4ed5a56081f9bc8ee61a95e168ec3429bcb",
"bytes": 370
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{
"path": "src/Enums/AssuranceCheckStatus.php",
"sha256": "5aa241d674f4543d12cfaa7835b48f933225300004ae14ab5884ec7f8a7b6fb0",
"bytes": 187
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{
"path": "src/Enums/BreakGlassStatus.php",
"sha256": "120ea25857f7a9a230cfe7dbb6b44b66e86ea2dd683b613ad6a97ee5f6121819",
"bytes": 261
},
{
"path": "src/Enums/ConsentDecision.php",
"sha256": "83af99378d38cfd31677b06319f5b9f91cc1c081cab68443d65c3a5e334c3f43",
"bytes": 243
},

```

```

{
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  "bytes": 224
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  "bytes": 260
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{
  "path": "src/Models/AssuranceCheckResult.php",
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  "path": "src/Models/AssuranceCheckRun.php",
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  "bytes": 536
},
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  "path": "src/Models/AuditChainCheckpoint.php",
  "sha256": "76f2577b89639eff81e2a9cde30514d7957c13d56eb82e280de7b6e2ac5a36d2",
  "bytes": 542
},
{
  "path": "src/Models/BreakGlassEpisode.php",
  "sha256": "a19a2133afa2d0076dbc016e97e7c8d05cedba3e02340a15deb304436f4d51db",
  "bytes": 536
},
{
  "path": "src/Models/BreakGlassReview.php",
  "sha256": "d5410490bbd0db4efb89225deb130cafafa98aaf219fe43801ea7650a761f305",
  "bytes": 534
},
{
  "path": "src/Models/ClinicalIncident.php",
  "sha256": "cb2e6a7db4e872606d09514922ad609e67a0dc384fca6fb7f01120eca28f68f0",
  "bytes": 524
},

```

```

{
  "path": "src/Models/ConsentDecisionRecord.php",
  "sha256": "c93868172b302d9a45e4563a565d31ca2e0ba41738f09b8e421c3224704dd40b",
  "bytes": 537
},
{
  "path": "src/Models/DataExportArtifact.php",
  "sha256": "9c60301a9a20cd1c9c2fa1da6f4cfbcf2742e7722e5d69113f925a77e40b56d5",
  "bytes": 538
},
{
  "path": "src/Models/DataExportRequest.php",
  "sha256": "a492a195b31d079f06426ae806cd85acc019b0db0eda9f089c0d897e4c0b5f65",
  "bytes": 536
},
{
  "path": "src/Models/DowntimeCapturedRecord.php",
  "sha256": "d8adb4981a81ee8b1a66b4981a5ccbac75719b4e95e50a97321e055dfda4336f",
  "bytes": 546
},
{
  "path": "src/Models/DowntimeEpisode.php",
  "sha256": "e23592ede0fd34facf38783baa1a960fb617d9cf3392562eef1134ef3b8d08fe",
  "bytes": 531
},
{
  "path": "src/Models/DowntimeReconciliation.php",
  "sha256": "0f846b34678d50e43427baec7ef4a24c4f2f6cbbf015fb08da024ef4ad498774",
  "bytes": 545
},
{
  "path": "src/Models/IncidentAction.php",
  "sha256": "7cab1620bc3b0d2d7d003402b8b9594226e5810e4b2c0f66db2c96afea92a65c",
  "bytes": 529
},
{
  "path": "src/Models/RetentionDisposition.php",
  "sha256": "7c47e0f5b67b2529d6eed471e54f2a3aa4950062e99a2bff981d28a50f7e425e",
  "bytes": 541
},
{
  "path": "src/Models/SecurityAuditEvent.php",
  "sha256": "2066c654ff07a61488628e81a79a68a071b2200b80939038d142f0806dba885a",
  "bytes": 538
},
{
  "path": "src/Services/AssuranceCheckRunner.php",
  "sha256": "bae23ce139adf056a456ca07d491710d6f82cb66bc7663a8739b9f48126b3fb1",
  "bytes": 505
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}

```

```
]
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	4f28610255e5347d332473c7373c5da8b50b3428c001f4a7dbd0ec93bda243c8

```
{
  "name": "kashtre/clinical-
  assurance",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\Assurance\\": "src/"
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\Assurance\\AssuranceServiceProvider"
          ]
        }
      }
    }
  }
}
```

config/clinical-assurance.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-assurance.php	7cd55c671c94bf1e74b6970ccb9db7b5e114787fe0e0632624b0238e239de767

```
<?php
declare(strict_types=1);
return [
    'break_glass_minutes' => (int) env('CLINICAL_BREAK_GLASS_MINUTES', 30),
    'audit_export_max_days' => (int) env('CLINICAL_AUDIT_EXPORT_MAX_DAYS', 31),
    'downtime_reconciliation_required' => true,
    'retention_policy_code' => env('CLINICAL_RETENTION_POLICY', 'CLINICAL_DEFAULT')
];
```

database/migrations/2026_08_28_800000_create_assurance_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_800000_create_assurance_tables.php	5e8f1836d8330212d1bf190e2da6c9977c6c37814d3cbbbe7eda27da7b4deecdd

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_consent_decisions', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('public_id', 96);
            $table->string('patient_public_id', 96);
            $table->string('actor_public_id', 96);
            $table->string('action_code', 96);
            $table->string('purpose_code', 96);
            $table->string('decision', 24);
            $table->string('policy_reference', 240);
            $table->string('consent_reference', 240);
            $table->string('context', 240);
            $table->dateTime('decided_at_utc', 6);
        });

        Schema::create('clinical_break_glass_episodes', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('public_id', 96);
            $table->string('patient_public_id', 96);
            $table->string('encounter_public_id', 96);
            $table->string('actor_public_id', 96);
            $table->string('reason_code', 96);
            $table->text('justification');
            $table->string('requested_actions', 24);
            $table->string('status', 24);
            $table->dateTime('started_at_utc', 6);
            $table->dateTime('expires_at_utc', 6);
            $table->string('review_status', 24);
        });

        Schema::create('clinical_break_glass_reviews', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('public_id', 96);
            $table->string('patient_public_id', 96);
            $table->string('actor_public_id', 96);
            $table->string('action_code', 24);
            $table->string('outcome', 24);
            $table->text('finding');
            $table->dateTime('reviewed_at_utc', 6);
        });

        Schema::create('clinical_security_audit_events', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('public_id', 96);
            $table->string('event_code', 96);
            $table->string('action_code', 24);
            $table->string('severity', 24);
            $table->string('actor_public_id', 96);
            $table->string('patient_public_id', 96);
            $table->string('entity_type', 96);
            $table->string('entity_public_id', 160);
            $table->string('evidence', 240);
            $table->string('previous_hash', 64);
            $table->string('event_hash', 64);
            $table->dateTime('occurred_at_utc', 6);
            $table->dateTime('recorded_at_utc', 6);
        });

        Schema::create('clinical_incidents', function (Blueprint $table) {
            $table->id();
            $table->string('tenant_id', 64);
            $table->string('public_id', 96);
            $table->string('status', 24);
            $table->string('reason_code', 96);
            $table->string('justification', 240);
            $table->string('requested_actions', 24);
            $table->string('review_status', 24);
            $table->dateTime('started_at_utc', 6);
            $table->dateTime('expires_at_utc', 6);
            $table->string('review_status', 24);
        });
    }
};
```

```

>ulid('patient_public_id')->nullable();$t->ulid('encounter_public_id')->nullable();$t->string('incident_type',96);$t-
>string('severity',24);$t->string('status',24);$t->text('description');$t->ulid('reported_by_public_id');$t-
>dateTime('occurred_at_utc',6);$t->dateTime('reported_at_utc',6));
Schema::create('clinical_downtime_episodes',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->string('status',24);$t->string('scope_code',96);$t->text('reason');$t->dateTime('started_at_utc',6);$t-
>dateTime('ended_at_utc',6)->nullable();$t->ulid('declared_by_public_id');});
Schema::create('clinical_downtime_captured_records',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('downtime_episode_public_id');$t->ulid('patient_public_id')->nullable();$t-
>string('record_type',96);$t->string('offline_reference',160);$t->json('content');$t->string('status',32);$t-
>dateTime('occurred_at_utc',6);$t->dateTime('entered_at_utc',6);$t-
>unique(['tenant_id','offline_reference'],'uq_downtime_offline_ref'));
Schema::create('clinical_downtime_reconciliations',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('downtime_record_public_id');$t->string('outcome',24);$t->json('evidence');$t-
>ulid('reconciled_by_public_id');$t->dateTime('reconciled_at_utc',6));
Schema::create('clinical_assurance_check_runs',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->string('check_set_code',96);$t->string('overall_status',24);$t->char('input_fingerprint',64);$t-
>dateTime('started_at_utc',6);$t->dateTime('completed_at_utc',6)->nullable();});
Schema::create('clinical_assurance_check_results',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('check_run_public_id');$t->string('check_code',96);$t->string('status',24);$t->json('evidence');});
Schema::create('clinical_data_export_requests',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->string('export_type',96);$t->string('status',24);$t->ulid('requested_by_public_id');$t->json('scope');$t-
>dateTime('requested_at_utc',6);$t->dateTime('expires_at_utc',6)->nullable();});
}public function
down():void{foreach(['clinical_data_export_requests','clinical_assurance_check_results','clinical_assurance_check_runs','clinical_d
owntime_reconciliations','clinical_downtime_captured_records','clinical_downtime_episodes','clinical_incidents','clinical_security_
audit_events','clinical_break_glass_reviews','clinical_break_glass_episodes','clinical_consent_decisions']as$t)Schema::dropIfExistss
($t)}};

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-assurance.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-assurance.php	b0c65afe4e0eafe8e613600a8604c9b66e2bb305fa6e6db445fbbb8aa0605e0a

```

<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])-
>prefix('api/v1/clinical/assurance')->group(function(){Route::get('/dashboard',fn()->response()-
>json(['error'=>['code'=>'CLN_ASSURANCE_HOST_CONTROLLER_REQUIRED'],501))});

```

src/Actions/ReconcileDowntimeRecord.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/ReconcileDowntimeRecord.php	5461d1c5414b08cfe4d43c23cd58362293edaebb143dddbef625aedc33696bdb

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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Assurance\Contracts\ClinicalReconciliationContributor;use
KashTre\Clinical\Assurance\Models\DowntimeCapturedRecord,DowntimeReconciliation};final readonly class
ReconcileDowntimeRecord{public function __construct(private PublicIdGenerator $ids,private iterable $contributors){}public
function execute(string $tenantId,string $recordId,string $actorId):DowntimeReconciliation{return
DB::transaction(function()use($tenantId,$recordId,$actorId){$record=DowntimeCapturedRecord::query()-
>where('tenant_id',$tenantId)->where('public_id',$recordId)->lockForUpdate()-
>firstOrFail();$evidence=[];$outcome='MATCHED';foreach($this->contributors as$c){if(!$c instanceof
ClinicalReconciliationContributor)continue;$r=$c->assess($tenantId,$record->downtime_episode_public_id,$record-
>toArray());$evidence[]=$r;if(in_array($r['outcome']??'MATCHED',['CONFLICT','MISSING','ESCALATED'],true))$outcome='ESCALATE
D';elseif(($r['outcome']??'MATCHED')!=='MATCHED'&&$outcome==='MATCHED')$outcome=$r['outcome'];}$x=DowntimeReconciat
ion::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'downtime_record_public_id'=>$recordId,'outcome'=>$outcome,'evidence'=>$evidence,'reconci
led_by_public_id'=>$actorId,'reconciled_at_utc'=>now('UTC')));$record-
>forceFill(['status'=>$outcome==='ESCALATED'?'RECONCILIATION_REQUIRED':'RECONCILED'])->save();return$x;}}}
```

src/Actions/RecordConsentDecision.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RecordConsentDecision.php	1b5c5adc2d4b922ffd1229e2ca75ed57a0bdcc8a41579c74ba49508f8596b52d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Actions;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Assurance\Contracts\AuditLedger,ConsentDecisionGateway};use
KashTre\Clinical\Assurance\Models\ConsentDecisionRecord;final readonly class RecordConsentDecision{public function
__construct(private PublicIdGenerator $ids,private ConsentDecisionGateway $consent,private AuditLedger $audit){}public function
execute(string $tenantId,string $patientId,string $actorId,string $action,string $purpose,array
$context):ConsentDecisionRecord{$r=$this->consent-
>decide($tenantId,$patientId,$actorId,$action,$purpose,$context);$d=ConsentDecisionRecord::query()-
>create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'patient_public_id'=>$patientId,'actor_public_id'=>$actorId,'action_code'=>$action,'purpose_co
de'=>$purpose,'decision'=>$r['decision'],'policy_reference'=>$r['policy_reference']??null,'consent_reference'=>$r['consent_referenc
e']??null,'context'=>$context,'decided_at_utc'=>now('UTC')));$this->audit-
>append(['code'=>'CLN_CONSENT_DECISION','tenant_id'=>$tenantId,'patient_id'=>$patientId,'actor_id'=>$actorId,'entity_id'=>$d-
>public_id,'outcome'=>$r['decision']]);return$d;}}
```

src/Actions/RequestBreakGlass.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RequestBreakGlass.php	c2c18146d17dbaa400b281e6476d5b010dda0569d63ef2c045efcd1446e2f14c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Assurance\Contracts\AuditLedger;use
KashTre\Clinical\Assurance\Data\BreakGlassRequestData;use KashTre\Clinical\Assurance\Models\BreakGlassEpisode;final
```

```
readonly class RequestBreakGlass{public function __construct(private PublicIdGenerator $ids,private AuditLedger $audit){}public
function execute(BreakGlassRequestData $d):BreakGlassEpisode{return DB::transaction(function(){use($d){if(trim($d-
>justification)===''||$d->requestedActions===[])throw new
\DomainException('CLN_BREAK_GLASS_JUSTIFICATION_REQUIRED');$active=BreakGlassEpisode::query()->where('tenant_id',$d-
>tenantId)->where('patient_public_id',$d->patientId)->where('actor_public_id',$d->actorId)->where('status','ACTIVE')-
>where('expires_at_utc','>',now('UTC'))->first();if($active)return$active;$e=BreakGlassEpisode::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'patient_public_id'=>$d-
>patientId,'encounter_public_id'=>$d->encounterId,'actor_public_id'=>$d->actorId,'reason_code'=>$d-
>reasonCode,'justification'=>$d->justification,'requested_actions'=>$d->requestedActions,'status'=>'ACTIVE','started_at_utc'=>$d-
>requestedAt,'expires_at_utc'=>$d->requestedAt->modify('+'.config('clinical-assurance.break_glass_minutes').'
minutes'),'review_status'=>'REVIEW_REQUIRED']);$this->audit->append(['code'=>'CLN_BREAK_GLASS_STARTED','tenant_id'=>$d-
>tenantId,'patient_id'=>$d->patientId,'actor_id'=>$d->actorId,'entity_id'=>$e->public_id,'occurred_at'=>$d->requestedAt-
>format(DATE_ATOM)]);return$e;}}}
```

src/AssuranceServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/AssuranceServiceProvider.php	5c9343b2ca6a37a00eeb4f3a637d9411701dc50a718cfe8e5a2783426cfd03b5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance;use Illuminate\Support\ServiceProvider;final class
AssuranceServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-assurance.php','clinical-assurance');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-assurance.php');}}
```

src/Contracts/AuditLedger.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/AuditLedger.php	8dabf6e5873f0f2496205353a8df19b93991aa402564dba39dfbac914664eba1

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Contracts;interface AuditLedger{public function append(array
$event):string;public function verifyChain(string $tenantId,\DateTimeImmutable $from,\DateTimeImmutable $to):array;}
```

src/Contracts/ClinicalReconciliationContributor.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ClinicalReconciliationContributor.php	5ec800729438b09ee40da2abe39b520289bb37d1315ca180e7dc13ae10f13f56

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Contracts;interface ClinicalReconciliationContributor{public
function assess(string $tenantId,string $downtimeEpisodeId,array $capturedRecord):array;}
```

src/Contracts/ConsentDecisionGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ConsentDecisionGateway.php	31b5d5f23805a1c3a42da3b7c13bd3763422b5b5f643b6f009e56350d0f1e5d1

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```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Contracts;interface ConsentDecisionGateway{public function decide(string $tenantId,string $patientId,string $actorId,string $action,string $purpose,array $resourceContext):array;}
```

src/Data/BreakGlassRequestData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/BreakGlassRequestData.php	b8fc2605df71933b5a8dcf435eb7a4ed5a56081f9bc8ee61a95e168ec3429bc

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Data;final readonly class BreakGlassRequestData{public function __construct(public string $tenantId,public string $patientId,public string $encounterId,public string $actorId,public string $reasonCode,public string $justification,public array $requestedActions,public \DateTimeImmutable $requestedAt){}}
```

src/Enums/AssuranceCheckStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AssuranceCheckStatus.php	5aa241d674f4543d12cfaa7835b48f933225300004ae14ab5884ec7f8a7b6fb

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum AssuranceCheckStatus:string{case PASS='PASS';case WARNING='WARNING';case FAIL='FAIL';case NOT_RUN='NOT_RUN';}
```

src/Enums/BreakGlassStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/BreakGlassStatus.php	120ea25857f7a9a230cfe7dbb6b44b66e86ea2dd683b613ad6a97ee5f6121819

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum BreakGlassStatus:string{case REQUESTED='REQUESTED';case ACTIVE='ACTIVE';case EXPIRED='EXPIRED';case REVOKED='REVOKED';case REVIEW_REQUIRED='REVIEW_REQUIRED';case REVIEWED='REVIEWED';}
```

src/Enums/ConsentDecision.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ConsentDecision.php	83af99378d38cfd31677b06319f5b9f91cc1c081cab68443d65c3a5e334c3f43

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum ConsentDecision:string{case PERMIT='PERMIT';case DENY='DENY';case CONDITIONAL='CONDITIONAL';case NOT_APPLICABLE='NOT_APPLICABLE';case INDETERMINATE='INDETERMINATE';}
```

src/Enums/DowntimeStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DowntimeStatus.php	ee0bfb9c72e7835cc0c60e046e3113af593fb2638d5f017f79a261335266e112

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum DowntimeStatus:string{case
DECLARED='DECLARED';case ACTIVE='ACTIVE';case RECOVERY='RECOVERY';case RECONCILING='RECONCILING';case
CLOSED='CLOSED';}
```

src/Enums/ExportStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ExportStatus.php	0188862930f77ca3d16f9ea70e6a510cada7b3215553b84abd9c51d8fa6a779e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum ExportStatus:string{case
REQUESTED='REQUESTED';case AUTHORIZED='AUTHORIZED';case GENERATING='GENERATING';case READY='READY';case
DOWNLOADED='DOWNLOADED';case EXPIRED='EXPIRED';case REVOKED='REVOKED';}
```

src/Enums/IncidentSeverity.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/IncidentSeverity.php	48cf9fc28d0329628c3f4fc04596f10ab3931b153252f624d649c7e23a294445

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum IncidentSeverity:string{case LOW='LOW';case
MODERATE='MODERATE';case HIGH='HIGH';case CRITICAL='CRITICAL';}
```

src/Enums/IncidentStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/IncidentStatus.php	3f38016cf53b97e0fa8dca8e0688691f39d8aeb8c4f668270e7ed45291d0b658

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum IncidentStatus:string{case OPEN='OPEN';case
TRIAGED='TRIAGED';case INVESTIGATING='INVESTIGATING';case CONTAINED='CONTAINED';case RESOLVED='RESOLVED';case
CLOSED='CLOSED';}
```

src/Enums/ReconciliationOutcome.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ReconciliationOutcome.php	bcb688282cfe1690d949720c6af53dd53c1a3b9755921593425647cd2ed859ec

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Enums;enum ReconciliationOutcome:string{case
MATCHED='MATCHED';case CORRECTED='CORRECTED';case DUPLICATE='DUPLICATE';case CONFLICT='CONFLICT';case
MISSING='MISSING';case ESCALATED='ESCALATED';}
```

src/Models/AssuranceCheckResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AssuranceCheckResult.php	50028807c48fad8d37f040824a3f52ba7c2a99c6f91e1fb97aa830c1a434c537

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class AssuranceCheckResult extends Model{public $timestamps=false;protected $table='clinical_assurance_check_results';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/AssuranceCheckRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AssuranceCheckRun.php	1f550e44db96e8c95b6776a09006876732cdde58ad04ac9f46d9a5bcf8824ab2

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class AssuranceCheckRun extends Model{public $timestamps=false;protected $table='clinical_assurance_check_runs';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/AuditChainCheckpoint.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AuditChainCheckpoint.php	76f2577b89639eff81e2a9cde30514d7957c13d56eb82e280de7b6e2ac5a36d2

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class AuditChainCheckpoint extends Model{public $timestamps=false;protected $table='clinical_audit_chain_checkpoints';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/BreakGlassEpisode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/BreakGlassEpisode.php	a19a2133afa2d0076dbc016e97e7c8d05cedba3e02340a15deb304436f4d51db

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
BreakGlassEpisode extends Model{public $timestamps=false;protected $table='clinical_break_glass_episodes';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/BreakGlassReview.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/BreakGlassReview.php	d5410490bbd0db4efb89225deb130cafafa98aaf219fe43801ea7650a761f305

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
BreakGlassReview extends Model{public $timestamps=false;protected $table='clinical_break_glass_reviews';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/ClinicalIncident.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ClinicalIncident.php	cb2e6a7db4e872606d09514922ad609e67a0dc384fca6fb7f01120eca28f68f0

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
ClinicalIncident extends Model{public $timestamps=false;protected $table='clinical_incidents';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/ConsentDecisionRecord.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ConsentDecisionRecord.php	c93868172b302d9a45e4563a565d31ca2e0ba41738f09b8e421c3224704dd40b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
ConsentDecisionRecord extends Model{public $timestamps=false;protected $table='clinical_consent_decisions';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/DataExportArtifact.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Models/DataExportArtifact.php	9c60301a9a20cd1c9c2fa1da6f4cfbcf2742e7722e5d69113f925a77e40b56d5
-------------------------	-----------------------------------	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
DataExportArtifact extends Model{public $timestamps=false;protected $table='clinical_data_export_artifacts';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/DataExportRequest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DataExportRequest.php	a492a195b31d079f06426ae806cd85acc019b0db0eda9f089c0d897e4c0b5f65

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
DataExportRequest extends Model{public $timestamps=false;protected $table='clinical_data_export_requests';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/DowntimeCapturedRecord.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DowntimeCapturedRecord.php	d8adb4981a81ee8b1a66b4981a5ccbac75719b4e95e50a97321e055dfda4336f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
DowntimeCapturedRecord extends Model{public $timestamps=false;protected
$table='clinical_downtime_captured_records';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Models/DowntimeEpisode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DowntimeEpisode.php	e23592ede0fd34facf38783baa1a960fb617d9cf3392562eef1134ef3b8d08fe

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
DowntimeEpisode extends Model{public $timestamps=false;protected $table='clinical_downtime_episodes';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
```

```
tc=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/DowntimeReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DowntimeReconciliation.php	0f846b34678d50e43427baec7ef4a24c4f2f6cbbf015fb08da024ef4ad498774

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class DowntimeReconciliation extends Model{public $timestamps=false;protected $table='clinical_downtime_reconciliations';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/IncidentAction.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/IncidentAction.php	7cab1620bc3b0d2d7d003402b8b9594226e5810e4b2c0f66db2c96afea92a65c

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class IncidentAction extends Model{public $timestamps=false;protected $table='clinical_incident_actions';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/RetentionDisposition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/RetentionDisposition.php	7c47e0f5b67b2529d6eed471e54f2a3aa4950062e99a2bff981d28a50f7e425e

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class RetentionDisposition extends Model{public $timestamps=false;protected $table='clinical_retention_dispositions';protected $guarded=[];protected $casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immutable_datetime'];}
```

src/Models/SecurityAuditEvent.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SecurityAuditEvent.php	2066c654ff07a61488628e81a79a68a071b2200b80939038d142f0806dba885a

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Models;use Illuminate\Database\Eloquent\Model;final class
SecurityAuditEvent extends Model{public $timestamps=false;protected $table='clinical_security_audit_events';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','requested_actions'=>'array','occurred_at_utc'=>'immutable_datetime','recorded_at_u
tc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime','started_at_utc'=>'immutable_datetime','ended_at_utc'=>'immu
table_datetime'];}
```

src/Services/AssuranceCheckRunner.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/AssuranceCheckRunner.php	bae23ce139adf056a456ca07d491710d6f82cb66bc7663a8739b9f48126b3fb1

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Assurance\Services;final class AssuranceCheckRunner{public function
run(iterable $checks,array $context):array{$results=[];$overall='PASS';foreach($checks
as$c){$r=$c($context);$results[]=$r;if(($r['status']??'FAIL')==='FAIL')$overall='FAIL';elseif(($r['status']??'WARNING')==='WARNING'&&
$overall==='PASS')$overall='WARNING';}return['overall'=>$overall,'results'=>$results,'fingerprint'=>hash('sha256',json_encode($res
ults,JSON_THROW_ON_ERROR))];}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volume 1	Contextual authorization, audit and messaging contracts
Volumes 2-8	Domain assurance contributors and reconciliation
Main Module	Consent directives, confidentiality labels and identity
Security platform	Immutable audit storage and monitoring
Offline platform	Authenticated cached context and upload idempotency
Object storage	Encrypted expiring export artifacts
Vue	Assurance dashboard, break-glass review, incidents, downtime and exports

Blocking Gaps

Execute in actual KashTre repository

Approve consent enforcement contract

Approve confidentiality label model

Bind immutable audit ledger and chain verification

Approve break-glass scope, duration and review SLA

Bind offline capture technology and reconciliation contributors

Approve retention and legal-hold policy

Complete host controllers, requests, resources, jobs and Vue interfaces

Run penetration, tenant-isolation, tamper, downtime, recovery, performance and accessibility tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Production-assurance and contextual-authorization authority
Volumes 1-8	Clinical workflow assurance scope
HL7 FHIR R5 Provenance	Authenticity, trust and reproducibility reference
HL7 FHIR R5 AuditEvent	Operational, privacy and security event reference
HL7 FHIR R5 Consent	Healthcare consent-decision reference

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

VOLUME 10

PRODUCTION RELEASE, DEPLOYMENT AND VERIFICATION ASSURANCE

Traceability, Test Evidence, Security Gates, Migration Rehearsal, Canary Release, Rollback and Post-Deployment Verification

Version: 6.1.0 Comprehensive Draft

Depends on: Volumes 1 to 9

Verification: Pick-and-Drop Candidate Pending KashTre Repository Execution

Production: Not Approved for Production

Authority and Release Rule

No Clinical release becomes production-approved because source code exists or a pipeline is green. Release authorization requires controlled traceability, reproducible artifacts, verified tests, migration rehearsal, security evidence, clinical acceptance, observable deployment, rollback readiness and post-deployment verification.

Requirements -> implementation references -> tests -> signed artifacts and SBOM -> release gates -> multi-party approval -> deployment -> canary observation -> post-deployment verification -> live or rollback -> evidence retention

Entity Register

Entity	Table	Invariant
ReleaseCandidate	clinical_release_candidates	Release, artifact, decision and evidence provenance retained
ReleaseArtifact	clinical_release_artifacts	Release, artifact, decision and evidence provenance retained
RequirementTrace	clinical_requirement_traces	Release, artifact, decision and evidence provenance retained
ReleaseGateDefinition	clinical_release_gate_definitions	Release, artifact, decision and evidence provenance retained
ReleaseGateRun	clinical_release_gate_runs	Release, artifact, decision and evidence provenance retained
ReleaseGateResult	clinical_release_gate_results	Release, artifact, decision and evidence provenance retained
TestEvidence	clinical_test_evidence	Release, artifact, decision and evidence provenance retained
SecurityEvidence	clinical_security_evidence	Release, artifact, decision and evidence provenance retained
MigrationPlan	clinical_migration_plans	Release, artifact, decision and evidence provenance retained
MigrationRehearsal	clinical_migration_rehearsals	Release, artifact, decision and evidence provenance retained
DeploymentRun	clinical_deployment_runs	Release, artifact, decision and evidence provenance retained
DeploymentStep	clinical_deployment_steps	Release, artifact, decision and evidence provenance retained
CanaryObservation	clinical_canary_observations	Release, artifact, decision and evidence provenance retained
RollbackDecision	clinical_rollback_decisions	Release, artifact, decision and evidence provenance retained
PostDeploymentVerification	clinical_post_deployment_verifications	Release, artifact, decision and evidence provenance retained
ReleaseApproval	clinical_release_approvals	Release, artifact, decision and evidence provenance retained
ReleaseDefect	clinical_release_defects	Release, artifact, decision and evidence provenance retained
ReleaseReconciliation	clinical_release_reconciliations	Release, artifact, decision and evidence provenance retained

Engineering Design

Traceability

- Every controlled SRD requirement maps to implementation and tests
- Artifact manifest identifies exact files and hashes
- Source commit, build identity and dependency lock are retained
- Untraced safety requirement blocks release

Quality gates

- Unit, integration, contract, concurrency and end-to-end evidence
- Tenant isolation and authorization negatives
- Clinical gold scenarios across Volumes 1 to 9
- Accessibility and supported-browser verification
- Performance, queue, outage and recovery verification

Security gates

- SBOM required
- Dependency and secret scanning
- Static and dynamic security testing
- Threat-model and abuse-case review
- OWASP ASVS evidence version-pinned
- Unresolved critical or high defect blocks authorization

Database and deployment

- Forward and rollback migration rehearsal
- Backup and restore verification
- Expand-contract deployment for incompatible schema change
- Canary or controlled cohort before broad release
- Observable steps with immutable timestamps

Post-deployment

- Verify identity, tenancy, authorization, events, queues and integrations
- Run clinical smoke scenarios
- Observe error, latency, backlog and reconciliation signals
- Rollback decision is explicit and authorized
- Document residual defects and corrective work

Required Approval Roles

Role	Decision focus
Clinical Product Authority	Functional and clinical acceptance
Engineering Authority	Implementation, migration and rollback
Security and Privacy Authority	Security, privacy and consent controls
Operations Authority	Deployment, monitoring and recovery
Data/Integration Authority	Cross-module contracts and reconciliation

Mandatory Tests

- Untraced requirement blocks READY
- Changed artifact hash invalidates evidence

Missing SBOM blocks authorization

Failed tenant-isolation test cannot be waived locally

Migration rehearsal records forward and rollback outcomes

Duplicate deployment request is idempotent

Canary failure prevents LIVE state

Rollback preserves audit and clinical records

Queue backlog alert appears during canary

Post-deployment smoke test covers observation, medication, result and transition workflows

Approval actor cannot approve outside assigned authority

Production artifact exactly matches authorized manifest

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	470070a4e1ce9276d6c0c6a22a17ca8d28ce2d30bbc5d9768c7937d9157ffbcd

KashTre Clinical Release Assurance v6.1.0

Pick-and-drop candidate. Bind CI/CD, source-control provenance, artifact registry, SBOM and vulnerability scanning, test evidence, observability, deployment, rollback, change approval and Vue release cockpit before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	f68a944cbb03b0cc53914b2df4943c18c6ed69ea011967b9583f5c59a88a27ef

```
{
  "volume": "Clinical EDD Volume 10 Release Assurance",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "470070a4e1ce9276d6c0c6a22a17ca8d28ce2d30bbc5d9768c7937d9157ffbcd",
      "bytes": 275
    },
    {
      "path": "composer.json",
      "sha256": "cb16767c3a236e8a3b04c2f648f5c4ed3b556c9139e4011947ba57bcd181369b",
      "bytes": 413
    },
    {
      "path": "config/clinical-release-assurance.php",
      "sha256": "abab1b08237d068f33f726e21cadd1769eb357c86c49fff5b89ae20c55657d1c",
      "bytes": 282
    },
    {
      "path": "database/migrations/2026_08_28_900000_create_release_assurance_tables.php",
      "sha256": "058678c674b0f25ee414a978cc2d90026c45037145fa61090f4d8b4cc2297fc0",
      "bytes": 4054
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-release-assurance.php",
      "sha256": "300a529799978e1d7e45718bf3f60e0b7d25091f47f1e91c16ce62a88db4ae5e",
      "bytes": 287
    },
    {

```

```

"path": "src/Actions/AssessReleaseCandidate.php",
"sha256": "bbe5207c68a6fb79712148cfbd0c7e829b3f405ba77992ae85eac41e7670451b",
"bytes": 1486
},
{
"path": "src/Actions/CreateReleaseCandidate.php",
"sha256": "ab24aba9ceeb97ab7cafb3168628edaafab12011f03af711510d84bfa2c7e297",
"bytes": 890
},
{
"path": "src/Actions/EvaluateCanary.php",
"sha256": "da483036c324281a502f4775bbdd3ba416fc4077117d95e760e50289ba7e214c",
"bytes": 888
},
{
"path": "src/Actions/ExecuteDeployment.php",
"sha256": "56c520abc2cd579ffaec7a3d90bfb4e11cd6d7a9dd034bbbdd58c6ecd8da5fe8",
"bytes": 1361
},
{
"path": "src/Contracts/DeploymentOrchestrator.php",
"sha256": "426c48f43204d87f84a51a4fab428f5bad2ea105922dbe2273969e5adbad832e",
"bytes": 278
},
{
"path": "src/Contracts/ObservabilityGateway.php",
"sha256": "4397ca6ad2953755e5878dc0c5c06d5e4e64c2d73d81a4793600fdbf17356723",
"bytes": 231
},
{
"path": "src/Contracts/ReleaseGateContributor.php",
"sha256": "bf4086feafe118f3a6bc6983640e59f61aaf75fb1288da3b62c8abdb6859b4b0",
"bytes": 200
},
{
"path": "src/Data/CreateReleaseCandidateData.php",
"sha256": "9b4127a54c0bcb7e1376f6738fcac8a6de34ca6f437ae92396c4f89541822612",
"bytes": 412
},
{
"path": "src/Enums/ApprovalDecision.php",
"sha256": "c30924ec05fb4d177e5e73e4eb894b4f0304cb04640210e5370bbd3d88870e49",
"bytes": 185
},
{
"path": "src/Enums/DefectSeverity.php",
"sha256": "99295448d5bca797119ff3a7e5904dffa63cf8f696eb385bb2896710911f1f72",
"bytes": 190
},
{
"path": "src/Enums/DeploymentStatus.php",
"sha256": "a980ca6cda0aaca9db0b3be869f1d8926f2e1331cd4d4b9909e66c87dc40210a",
"bytes": 235
},

```

```

{
  "path": "src/Enums/EvidenceStatus.php",
  "sha256": "1b95919a3e1e9664d992b90e577634a6b83efe68af6cefa272287b571bf05e68",
  "bytes": 214
},
{
  "path": "src/Enums/GateOutcome.php",
  "sha256": "84a43080c40ac9121b2a1810746a45cb6c7d8e2b08cb707d75152140e274e6c8",
  "bytes": 206
},
{
  "path": "src/Enums/MigrationStatus.php",
  "sha256": "a93f767af7f51998e4a155fb99d18fc34837916307382a0ebf3819cb5305d7a7",
  "bytes": 286
},
{
  "path": "src/Enums/ReleaseStatus.php",
  "sha256": "920598eff143dfb13b4b9b041cd1dc85e5b04e6afe747eafe7dddb88c47f60f8",
  "bytes": 345
},
{
  "path": "src/Enums/VerificationStatus.php",
  "sha256": "9940df13586a15f45c2364c83ea4abb39227942a185bc7ef193e1bd712ea87e3",
  "bytes": 192
},
{
  "path": "src/Models/CanaryObservation.php",
  "sha256": "b1c1b9cc06751aa3b5d426eaa4252271794c0091c52744c841096758edad61bc",
  "bytes": 477
},
{
  "path": "src/Models/DeploymentRun.php",
  "sha256": "84a2f99359fa310a001f9e8a91677484cc5f73fe6ec9f201bc88b32337c9ea80",
  "bytes": 469
},
{
  "path": "src/Models/DeploymentStep.php",
  "sha256": "9138f6c2f83b77d84b5a4ecc6e45b03263b1cf7aa4a13b6a2f0479815804c79f",
  "bytes": 471
},
{
  "path": "src/Models/MigrationPlan.php",
  "sha256": "5bf9caa6093e8778bf6a64b3e7f389efea4707c1c27a96137df5c194d2143947",
  "bytes": 469
},
{
  "path": "src/Models/MigrationRehearsal.php",
  "sha256": "3f055d906199580384da4ca576a1135534e4d72da27f2f3a1c529abd610b85cc",
  "bytes": 479
},
{
  "path": "src/Models/PostDeploymentVerification.php",
  "sha256": "a70c05ecd2310ab50e8d0d6423d56d8f8a306341e93ce3c6f9648c7f865c0f28",
  "bytes": 496
},

```

```

{
  "path": "src/Models/ReleaseApproval.php",
  "sha256": "c8320309e01b725c815e8e6657ed9a83eef40d3b064bb802e09c234e2051f803",
  "bytes": 473
},
{
  "path": "src/Models/ReleaseArtifact.php",
  "sha256": "e9d78e3bca60c58c3e0e8a1d12a7c17935b240474d4fe7c97433507d661ea051",
  "bytes": 473
},
{
  "path": "src/Models/ReleaseCandidate.php",
  "sha256": "139d599f2c8a5a70ccb1351fd9a7fed9145766ecc2f984b610074cf4047ed238",
  "bytes": 475
},
{
  "path": "src/Models/ReleaseDefect.php",
  "sha256": "29d9bbe59dee31b82b64d5d4e41dac89a4ec982de0490ee0bf695b2709faf2aa",
  "bytes": 469
},
{
  "path": "src/Models/ReleaseGateDefinition.php",
  "sha256": "cb5d9d2017d07d9cb63f016f80c14d50938bde18bdbcc361aea24db484056ad2",
  "bytes": 486
},
{
  "path": "src/Models/ReleaseGateResult.php",
  "sha256": "2694f75bba1104ee77f56721069de68746a4be5eccd4e992fc53c84267d800bc",
  "bytes": 478
},
{
  "path": "src/Models/ReleaseGateRun.php",
  "sha256": "aadaeebf8a7e442fd617d074af029a98c69baa61f4af9c8f2528fad2ead50d4",
  "bytes": 472
},
{
  "path": "src/Models/ReleaseReconciliation.php",
  "sha256": "cb28a9435ab7d4e1746f10e7e3ac32337c975c7434d84d51e5d2a306136aad76",
  "bytes": 485
},
{
  "path": "src/Models/RequirementTrace.php",
  "sha256": "09fbc4c12d9461ad13e213c3c46927195d1b6cb96efa4b4df9065a34545392e8",
  "bytes": 475
},
{
  "path": "src/Models/RollbackDecision.php",
  "sha256": "9bd9f942f10487d1f9663224c8d07fea1162cbe469d2396263fdb4e483e6771e",
  "bytes": 475
},
{
  "path": "src/Models/SecurityEvidence.php",
  "sha256": "a7c3d323faffb05d50fbbd5ec4062884ba1fd8cb38da3be8f33ecf5c45a3cb56",
  "bytes": 474
},

```

```
{
  "path": "src/Models/TestEvidence.php",
  "sha256": "7204bd18dc8bfb4e79ef9d03dc1462349f40a3497b8770b980f533267beeec4",
  "bytes": 466
},
{
  "path": "src/ReleaseAssuranceServiceProvider.php",
  "sha256": "e68da3fa6fc046912e771ae54faccdc6a4ac40074e521408fb51b1a5ad22f126",
  "bytes": 485
},
{
  "path": "src/Services/ReleaseGateEngine.php",
  "sha256": "6ea95dfedca1179c0be34a1cfb9e36664c9b425d826b829cb6f853e229426c02",
  "bytes": 767
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	cb16767c3a236e8a3b04c2f648f5c4ed3b556c9139e4011947ba57bcd181369b

```
{
  "name": "kashtre/clinical-release-assurance",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\ReleaseAssurance\\": "src/"
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\ReleaseAssurance\\ReleaseAssuranceServiceProvider"
          ]
        }
      }
    }
  }
}
```

config/clinical-release-assurance.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-release-assurance.php	abab1b08237d068f33f726e21cadd1769eb357c86c49fff5b89ae20c55657d1c

```
<?php
declare(strict_types=1);return
['evidence_retention_days'=>(int)env('CLINICAL_RELEASE_EVIDENCE_DAYS',2555),'canary_observation_minutes'=>(int)env('CLINICAL_CANARY_MINUTES',30),'rollback_authorization_required'=>true,'sbom_required'=>true,'migration_rehearsal_required'=>true];
```

database/migrations/2026_08_28_900000_create_release_assurance_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_900000_create_release_assurance_tables.php	058678c674b0f25ee414a978cc2d90026c45037145fa61090f4d8b4cc2297fc0

```

<?php
declare(strict_types=1);use Illuminate\Database\Migrations\Migration;use Illuminate\Database\Schema\Blueprint;use
Illuminate\Support\Facades\Schema;return new class extends Migration{public function up():void{
Schema::create('clinical_release_candidates',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->string('version',64);$t->char('commit_sha',40);$t->string('status',24);$t-
>char('artifact_manifest_sha256',64);$t->char('sbom_sha256',64);$t->json('included_volumes');$t->json('change_set');$t-
>ulid('created_by_public_id');$t->dateTime('created_at_utc',6);$t->unsignedBigInteger('record_version')->default(1);$t-
>unique(['tenant_id','version','commit_sha'],'uq_release_candidate'));
Schema::create('clinical_requirement_traces',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('release_candidate_public_id');$t->string('requirement_id',128);$t-
>string('implementation_reference',500);$t->string('test_reference',500);$t->string('status',24);$t-
>unique(['release_candidate_public_id','requirement_id'],'uq_release_requirement'));
Schema::create('clinical_release_gate_runs',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('release_candidate_public_id');$t->string('overall_outcome',24);$t->char('input_fingerprint',64);$t-
>dateTime('started_at_utc',6);$t->dateTime('completed_at_utc',6));
Schema::create('clinical_release_gate_results',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('gate_run_public_id');$t->string('gate_code',128);$t->string('outcome',24);$t->json('evidence');$t-
>unique(['gate_run_public_id','gate_code'],'uq_release_gate_result'));
Schema::create('clinical_test_evidence',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>ulid('release_candidate_public_id');$t->string('suite_code',128);$t->string('status',24);$t->string('artifact_reference',500);$t-
>char('sha256',64);$t->dateTime('completed_at_utc',6));
Schema::create('clinical_migration_plans',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>ulid('release_candidate_public_id');$t->string('status',24);$t->json('scope');$t->text('rollback_plan');$t-
>dateTime('created_at_utc',6));
Schema::create('clinical_deployment_runs',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>ulid('release_candidate_public_id');$t->string('environment',64);$t->string('status',24);$t->string('provider_reference',240)-
>nullable();$t->text('failure_detail')->nullable();$t->dateTime('started_at_utc',6);$t->dateTime('completed_at_utc',6)->nullable();});
Schema::create('clinical_canary_observations',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('deployment_run_public_id');$t->string('status',24);$t->json('evidence');$t-
>dateTime('started_at_utc',6);$t->dateTime('completed_at_utc',6));
Schema::create('clinical_release_approvals',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('release_candidate_public_id');$t->string('approval_role',96);$t->string('decision',24);$t-
>ulid('decided_by_public_id');$t->text('comment')->nullable();$t->dateTime('decided_at_utc',6));
Schema::create('clinical_post_deployment_verifications',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('deployment_run_public_id');$t->string('check_code',128);$t->string('status',24);$t-
>json('evidence');$t->dateTime('completed_at_utc',6));
}public function
down():void{foreach(['clinical_post_deployment_verifications','clinical_release_approvals','clinical_canary_observations','clinical_d
eployment_runs','clinical_migration_plans','clinical_test_evidence','clinical_release_gate_results','clinical_release_gate_runs','clini
cal_requirement_traces','clinical_release_candidates']as$t)Schema::dropIfExists($t);}}

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-release-assurance.php

Status	Destination	SHA-256
PICK-AND-DROP	routes/clinical-release-	300a529799978e1d7e45718bf3f60e0b7d25091f47f1e91c16ce62a88db4ae5e

CANDIDATE	assurance.php	
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```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])-
>prefix('api/v1/clinical/release-assurance')->group(function(){Route::get('/releases',fn()->response()-
>json(['error'=>['code'=>'CLN_RELEASE_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/AssessReleaseCandidate.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/AssessReleaseCandidate.php	bbe5207c68a6fb79712148cfbd0c7e829b3f405ba77992ae85eac41e7670451b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ReleaseAssurance\Models\ReleaseCandidate,ReleaseGateResult,ReleaseGateRun;use
KashTre\Clinical\ReleaseAssurance\Services\ReleaseGateEngine;final readonly class AssessReleaseCandidate{public function
__construct(private PublicIdGenerator $ids,private ReleaseGateEngine $gates){}public function execute(string $tenantId,string
$releaseId,array $context):ReleaseCandidate{return
DB::transaction(function()use($tenantId,$releaseId,$context){$r=ReleaseCandidate::query()->where('tenant_id',$tenantId)-
>where('public_id',$releaseId)->lockForUpdate()->firstOrFail();$out=$this->gates-
>evaluate($tenantId,$releaseId,$context);$run=ReleaseGateRun::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'release_candidate_public_id'=>$releaseId,'overall_outcome'=>$out['overall'],'input_fingerprint'=
>$out['fingerprint'],'started_at_utc'=>now('UTC'),'completed_at_utc'=>now('UTC')));foreach($out['results']as$x)ReleaseGateResult::
query()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$tenantId,'gate_run_public_id'=>$run-
>public_id,'gate_code'=>$x['code'],'outcome'=>$x['outcome'],'evidence'=>$x['evidence']??[]);$r-
>forceFill(['status'=>$out['overall']==='FAIL'?'ASSESSING':'READY','record_version'=>$r->record_version+1])->save();return$r;}}}
```

src/Actions/CreateReleaseCandidate.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CreateReleaseCandidate.php	ab24aba9ceeb97ab7cafb3168628edaafab12011f03af711510d84bfa2c7e297

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Actions;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ReleaseAssurance\Data\CreateReleaseCandidateData;use
KashTre\Clinical\ReleaseAssurance\Models\ReleaseCandidate;final readonly class CreateReleaseCandidate{public function
__construct(private PublicIdGenerator $ids){}public function execute(CreateReleaseCandidateData $d):ReleaseCandidate{return
ReleaseCandidate::query()->firstOrCreate(['tenant_id'=>$d->tenantId,'version'=>$d->version,'commit_sha'=>$d-
>commitSha],['public_id'=>(string)$this->ids->generate(),'status'=>'DRAFT','artifact_manifest_sha256'=>$d-
>artifactManifestSha256,'sbom_sha256'=>$d->sbomSha256,'included_volumes'=>$d->includedVolumes,'change_set'=>$d-
>changeSet,'created_by_public_id'=>$d->createdById,'created_at_utc'=>$d->createdAt,'record_version'=>1]);}
```

src/Actions/EvaluateCanary.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Actions/EvaluateCanary.php	da483036c324281a502f4775bbdd3ba416fc4077117d95e760e50289ba7e214c

CANDIDATE		
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Actions;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ReleaseAssurance\Contracts\ObservabilityGateway;use
KashTre\Clinical\ReleaseAssurance\Models\CanaryObservation;final readonly class EvaluateCanary{public function
__construct(private PublicIdGenerator $ids,private ObservabilityGateway $observability){}public function execute(string
$tenantId,string $deploymentId,string $environment,\DateTimeImmutable $from,\DateTimeImmutable $to,array
$signals):CanaryObservation{$s=$this->observability->snapshot($environment,$signals,$from,$to);return
CanaryObservation::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'deployment_run_public_id'=>$deploymentId,'status'=>$s['status'],'evidence'=>$s,'started_at_ut
c'=>$from,'completed_at_utc'=>$to]);}}
```

src/Actions/ExecuteDeployment.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/ExecuteDeployment.php	56c520abc2cd579ffaec7a3d90bfb4e11cd6d7a9dd034bbbdd58c6ecd8da5fe8

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ReleaseAssurance\Contracts\DeploymentOrchestrator;use
KashTre\Clinical\ReleaseAssurance\Models\{DeploymentRun,ReleaseCandidate};final readonly class ExecuteDeployment{public
function __construct(private PublicIdGenerator $ids,private DeploymentOrchestrator $deployments){}public function execute(string
$tenantId,string $releaseId,string $environment,array $manifest):DeploymentRun{$r=ReleaseCandidate::query()-
>where('tenant_id',$tenantId)->where('public_id',$releaseId)->where('status','AUTHORIZED')-
>firstOrFail();$run=DeploymentRun::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'release_candidate_public_id'=>$releaseId,'environment'=>$environment,'status'=>'RUNNING','
started_at_utc'=>now('UTC')));try{$x=$this->deployments->deploy($releaseId,$environment,$manifest);$run-
>forceFill(['status'=>'SUCCEEDED','provider_reference'=>$x['reference'],'completed_at_utc'=>now('UTC')])->save();$r-
>forceFill(['status'=>'CANARY'])->save();}catch(\Throwable$e){$run->forceFill(['status'=>'FAILED','failure_detail'=>$e-
>getMessage(),'completed_at_utc'=>now('UTC')])->save();$r->forceFill(['status'=>'FAILED'])->save();}return$run;}}
```

src/Contracts/DeploymentOrchestrator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/DeploymentOrchestrator.php	426c48f43204d87f84a51a4fab428f5bad2ea105922dbe2273969e5adbad832e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Contracts;interface DeploymentOrchestrator{public
function deploy(string $releaseId,string $environment,array $artifactManifest):array;public function rollback(string
$deploymentId,string $reason):array;}
```

src/Contracts/ObservabilityGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObservabilityGateway.php	4397ca6ad2953755e5878dc0c5c06d5e4e64c2d73d81a4793600fdbf17356723

E		
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```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Contracts;interface ObservabilityGateway{public function snapshot(string $environment,array $signals,\DateTimeImmutable $from,\DateTimeImmutable $to):array;}
```

src/Contracts/ReleaseGateContributor.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ReleaseGateContributor.php	bf4086feafe118f3a6bc6983640e59f61aaf75fb1288da3b62c8abdb6859b4b0

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Contracts;interface ReleaseGateContributor{public function evaluate(string $tenantId,string $releaseId,array $context):array;}
```

src/Data/CreateReleaseCandidateData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/CreateReleaseCandidateData.php	9b4127a54c0bcb7e1376f6738fcac8a6de34ca6f437ae92396c4f89541822612

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Data;final readonly class CreateReleaseCandidateData{public function __construct(public string $tenantId,public string $version,public string $commitSha,public string $artifactManifestSha256,public string $sbomSha256,public string $createdById,public array $includedVolumes,public array $changeSet,public \DateTimeImmutable $createdAt){}}
```

src/Enums/ApprovalDecision.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ApprovalDecision.php	c30924ec05fb4d177e5e73e4eb894b4f0304cb04640210e5370bbd3d88870e49

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum ApprovalDecision:string{case APPROVE='APPROVE';case REJECT='REJECT';case CONDITIONAL='CONDITIONAL';}
```

src/Enums/DefectSeverity.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DefectSeverity.php	99295448d5bca797119ff3a7e5904dffa63cf8f696eb385bb2896710911f1f72

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum DefectSeverity:string{case LOW='LOW';case MODERATE='MODERATE';case HIGH='HIGH';case CRITICAL='CRITICAL';}
```

src/Enums/DeploymentStatus.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Enums/DeploymentStatus.php	a980ca6cda0aaca9db0b3be869f1d8926f2e1331cd4d4b9909e66c87dc40210a
-------------------------	--------------------------------	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum DeploymentStatus:string{case
PLANNED='PLANNED';case RUNNING='RUNNING';case SUCCEEDED='SUCCEEDED';case FAILED='FAILED';case
ROLLED_BACK='ROLLED_BACK';}
```

src/Enums/EvidenceStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/EvidenceStatus.php	1b95919a3e1e9664d992b90e577634a6b83efe68af6cefa272287b571bf05e68

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum EvidenceStatus:string{case
SUBMITTED='SUBMITTED';case VERIFIED='VERIFIED';case REJECTED='REJECTED';case SUPERSEDED='SUPERSEDED';}
```

src/Enums/GateOutcome.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/GateOutcome.php	84a43080c40ac9121b2a1810746a45cb6c7d8e2b08cb707d75152140e274e6c8

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum GateOutcome:string{case PASS='PASS';case
WARNING='WARNING';case FAIL='FAIL';case NOT_RUN='NOT_RUN';case WAIVED='WAIVED';}
```

src/Enums/MigrationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MigrationStatus.php	a93f767af7f51998e4a155fb99d18fc34837916307382a0ebf3819cb5305d7a7

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum MigrationStatus:string{case
PLANNED='PLANNED';case REHEARSED='REHEARSED';case APPROVED='APPROVED';case RUNNING='RUNNING';case
COMPLETED='COMPLETED';case ROLLED_BACK='ROLLED_BACK';case FAILED='FAILED';}
```

src/Enums/ReleaseStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ReleaseStatus.php	920598eff143dfb13b4b9b041cd1dc85e5b04e6afe747eafe7dddb88c47f60f8

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum ReleaseStatus:string{case
DRAFT='DRAFT';case ASSESSING='ASSESSING';case READY='READY';case AUTHORIZED='AUTHORIZED';case
DEPLOYING='DEPLOYING';case CANARY='CANARY';case LIVE='LIVE';case ROLLED_BACK='ROLLED_BACK';case
FAILED='FAILED';case CANCELLED='CANCELLED';}
```

src/Enums/VerificationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/VerificationStatus.php	9940df13586a15f45c2364c83ea4abb39227942a185bc7ef193e1bd712ea87e3

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Enums;enum VerificationStatus:string{case PENDING='PENDING';case PASS='PASS';case WARNING='WARNING';case FAIL='FAIL';}
```

src/Models/CanaryObservation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CanaryObservation.php	b1c1b9cc06751aa3b5d426eaa4252271794c0091c52744c841096758edad61b

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final class CanaryObservation extends Model{public $timestamps=false;protected $table='clinical_canary_observations';protected $guarded=[];protected $casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/DeploymentRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DeploymentRun.php	84a2f99359fa310a001f9e8a91677484cc5f73fe6ec9f201bc88b32337c9ea80

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final class DeploymentRun extends Model{public $timestamps=false;protected $table='clinical_deployment_runs';protected $guarded=[];protected $casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/DeploymentStep.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/DeploymentStep.php	9138f6c2f83b77d84b5a4ecc6e45b03263b1cf7aa4a13b6a2f0479815804c79f

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final class DeploymentStep extends Model{public $timestamps=false;protected $table='clinical_deployment_steps';protected $guarded=[];protected $casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/MigrationPlan.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Models/MigrationPlan.php	5bf9caa6093e8778bf6a64b3e7f389efea4707c1c27a96137df5c194d2143947
----------------------------	------------------------------	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class MigrationPlan extends Model{public $timestamps=false;protected $table='clinical_migration_plans';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/MigrationRehearsal.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MigrationRehearsal.php	3f055d906199580384da4ca576a1135534e4d72da27f2f3a1c529abd610b85cc

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class MigrationRehearsal extends Model{public $timestamps=false;protected $table='clinical_migration_rehearsals';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/PostDeploymentVerification.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PostDeploymentVerification.php	a70c05ecd2310ab50e8d0d6423d56d8f8a306341e93ce3c6f9648c7f865c0f28

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class PostDeploymentVerification extends Model{public $timestamps=false;protected
$table='clinical_post_deployment_verifications';protected $guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseApproval.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseApproval.php	c8320309e01b725c815e8e6657ed9a83eef40d3b064bb802e09c234e2051f803

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseApproval extends Model{public $timestamps=false;protected $table='clinical_release_approvals';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseArtifact.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Models/ReleaseArtifact.php	e9d78e3bca60c58c3e0e8a1d12a7c17935b240474d4fe7c97433507d661ea051
-------------------------	--------------------------------	--

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseArtifact extends Model{public $timestamps=false;protected $table='clinical_release_artifacts';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseCandidate.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseCandidate.php	139d599f2c8a5a70ccb1351fd9a7fed9145766ecc2f984b610074cf4047ed238

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseCandidate extends Model{public $timestamps=false;protected $table='clinical_release_candidates';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseDefect.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseDefect.php	29d9bbe59dee31b82b64d5d4e41dac89a4ec982de0490ee0bf695b2709faf2aa

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseDefect extends Model{public $timestamps=false;protected $table='clinical_release_defects';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseGateDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseGateDefinition.php	cb5d9d2017d07d9cb63f016f80c14d50938bde18bdbcc361aea24db484056ad2

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseGateDefinition extends Model{public $timestamps=false;protected
$table='clinical_release_gate_definitions';protected $guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseGateResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseGateResult.php	2694f75bba1104ee77f56721069de68746a4be5eccd4e992fc53c84267d800bc

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseGateResult extends Model{public $timestamps=false;protected $table='clinical_release_gate_results';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseGateRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseGateRun.php	aadaeebf8a7e442fd617d074af029a98c69baa61f4af9c8f2528fad2ead50d4

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseGateRun extends Model{public $timestamps=false;protected $table='clinical_release_gate_runs';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/ReleaseReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ReleaseReconciliation.php	cb28a9435ab7d4e1746f10e7e3ac32337c975c7434d84d51e5d2a306136aad76

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class ReleaseReconciliation extends Model{public $timestamps=false;protected $table='clinical_release_reconciliations';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/RequirementTrace.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/RequirementTrace.php	09fbc4c12d9461ad13e213c3c46927195d1b6cb96efa4b4df9065a34545392e8

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class RequirementTrace extends Model{public $timestamps=false;protected $table='clinical_requirement_traces';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/RollbackDecision.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/RollbackDecision.php	9bd9f942f10487d1f9663224c8d07fea1162cbe469d2396263fdb4e483e6771e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class RollbackDecision extends Model{public $timestamps=false;protected $table='clinical_rollback_decisions';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/SecurityEvidence.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SecurityEvidence.php	a7c3d323affb05d50fbbd5ec4062884ba1fd8cb38da3be8f33ecf5c45a3cb56

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class SecurityEvidence extends Model{public $timestamps=false;protected $table='clinical_security_evidence';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/Models/TestEvidence.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TestEvidence.php	7204bd18dc8bfb4e79ef9d03dc1462349f40a3497b8770b980f533267beec4

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Models;use Illuminate\Database\Eloquent\Model;final
class TestEvidence extends Model{public $timestamps=false;protected $table='clinical_test_evidence';protected
$guarded=[];protected
$casts=['manifest'=>'array','scope'=>'array','evidence'=>'array','change_set'=>'array','started_at_utc'=>'immutable_datetime','compl
eted_at_utc'=>'immutable_datetime','created_at_utc'=>'immutable_datetime'];}
```

src/ReleaseAssuranceServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ReleaseAssuranceServiceProvider.php	e68da3fa6fc046912e771ae54faccdc6a4ac40074e521408fb51b1a5ad22f126

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance;use Illuminate\Support\ServiceProvider;final class
ReleaseAssuranceServiceProvider extends ServiceProvider{public function register():void{$this->mergeConfigFrom(__DIR__.'../config/clinical-release-assurance.php','clinical-release-assurance');}public function
boot():void{$this->loadMigrationsFrom(__DIR__.'../database/migrations');$this->loadRoutesFrom(__DIR__.'../routes/clinical-release-assurance.php');}}
```

src/Services/ReleaseGateEngine.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/ReleaseGateEngine.php	6ea95dfedca1179c0be34a1cfb9e36664c9b425d826b829cb6f853e229426c02

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ReleaseAssurance\Services;use
KashTre\Clinical\ReleaseAssurance\Contracts\ReleaseGateContributor;final class ReleaseGateEngine{public function
__construct(private iterable $contributors){}public function evaluate(string $tenantId,string $releaseId,array
$context):array{$results=[];$overall='PASS';foreach($this->contributors as$c){if(!$c instanceof
ReleaseGateContributor)continue;foreach($c-
>evaluate($tenantId,$releaseId,$context)as$r){$results[]=$r;if(($r['outcome']??'FAIL')==='FAIL')$overall='FAIL';elseif(($r['outcome']??
'PASS')==='WARNING'&&$overall==='PASS')$overall='WARNING';}}return['overall'=>$overall,'results'=>$results,'fingerprint'=>hash('s
ha256',json_encode($results,JSON_THROW_ON_ERROR))];}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volumes 1-9	Requirement, test and clinical scenario contributors
Source control	Commit and signed-tag provenance
CI/CD	Build, test, deployment and rollback orchestration
Artifact registry	Immutable manifests, packages and SBOM
Security tooling	SAST, DAST, dependency, secret and container evidence
Observability	Metrics, logs, traces, queues and reconciliation
Change governance	Multi-party production authorization
Vue	Release cockpit, gates, approvals, deployments and evidence

Blocking Gaps

Execute against the actual KashTre mono-repository or package repositories

Reconcile package PHP and Laravel versions with the approved production baseline

Build the complete SRD-to-code-to-test traceability matrix

Bind CI/CD and artifact registry contracts

Approve clinical gold scenarios and acceptance authorities

Approve migration, backup, restore and rollback runbooks

Define production SLOs and canary thresholds

Complete host controllers, jobs and release-cockpit Vue interfaces

Perform independent security, privacy, resilience and clinical release review

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

then repository CI, SBOM, security, migration rehearsal and canary pipelines

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Production-assurance and approval authority
Volumes 1-9	Implementation and acceptance scope
NIST SP 800-218 SSDF 1.1	Secure software-development practice reference
OWASP ASVS 5.0.0	Web-application security verification reference

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

SUPPLEMENTAL VOLUME 11

CLINICAL OPERATIONS, ADOPTION, OPTIMIZATION AND AI GOVERNANCE

Support, Knowledge, Training, Competency, Benefits Realization and Human-Supervised Shared AI

Version: 6.1.0 Comprehensive Draft

Status: Supplemental to the Phase 1-10 functional baseline

Depends on: Volumes 1 to 10

Production: Not Approved for Production

Scope Position and Governing Rule

The governing SRD states that Phases 1 to 10 form the complete functional baseline. Volume 11 is therefore supplemental. It defines the post-release operating model needed to sustain adoption, competency, support, improvement, benefits realization and controlled AI assistance without changing the approved clinical source-of-truth boundaries.

Production release -> supported operation -> issue and knowledge management -> training and competency -> adoption monitoring -> governed optimization -> benefits review -> safe change control

Entity Register

Entity	Table	Invariant
OperatingIssue	clinical_operating_issues	Tenant-scoped, governed and evidenced
SupportCase	clinical_support_cases	Tenant-scoped, governed and evidenced
KnowledgeArticle	clinical_knowledge_articles	Tenant-scoped, governed and evidenced
TrainingCurriculum	clinical_training_curricula	Tenant-scoped, governed and evidenced
TrainingAssignment	clinical_training_assignments	Tenant-scoped, governed and evidenced
CompetencyRecord	clinical_competency_records	Tenant-scoped, governed and evidenced
AdoptionMetricDefinition	clinical_adoption_metric_definitions	Tenant-scoped, governed and evidenced
AdoptionSnapshot	clinical_adoption_snapshots	Tenant-scoped, governed and evidenced
OptimizationProposal	clinical_optimization_proposals	Tenant-scoped, governed and evidenced
OptimizationPilot	clinical_optimization_pilots	Tenant-scoped, governed and evidenced
AiUseCase	clinical_ai_use_cases	Tenant-scoped, governed and evidenced
AiAssistanceRequest	clinical_ai_assistance_requests	Tenant-scoped, governed and evidenced
AiAssistanceOutput	clinical_ai_assistance_outputs	Tenant-scoped, governed and evidenced
AiHumanReview	clinical_ai_human_reviews	Tenant-scoped, governed and evidenced
AiMonitoringFinding	clinical_ai_monitoring_findings	Tenant-scoped, governed and evidenced
BenefitsRealizationMeasure	clinical_benefits_measures	Tenant-scoped, governed and evidenced
BenefitsRealizationSnapshot	clinical_benefits_snapshots	Tenant-scoped, governed and evidenced
OperationalReconciliation	clinical_operational_reconciliations	Tenant-scoped, governed and evidenced

Engineering Design**Support and knowledge**

Single operating issue and support-case intake

Severity and patient-safety impact separated

Known error and workaround articles are versioned

Resolution links to release, configuration or training action

No support action silently edits clinical records

Training and competency

Versioned curricula by function and workflow

Assignments use authoritative user and employment identity

Completion is not the same as competency

High-risk functionality may require current competency

Expiry and revocation are effective-dated

Adoption and benefits

Use auditable workflow telemetry, not surveillance of individuals

Define each metric, denominator, threshold and limitation

Separate adoption, quality, safety, user experience and outcome measures

Drill down by authorized organizational scope

Low adoption triggers investigation, not punitive inference

Optimization

Every proposed change has problem statement and safety impact

Pilot before broad adoption where risk warrants

Compare baseline and pilot evidence

Route semantic changes through the proper CDE, form, score or workflow governance

Rollback ineffective or unsafe optimization

AI governance

Clinical consumes AI only through the Shared AI Gateway

Use case must be registered, risk-rated and active

Send minimum necessary context

Output remains draft or advisory until human review

Model and gateway references are retained

Rejected outputs and monitoring findings inform suspension

Prohibited use cases cannot execute

Permissions

Permission	Purpose
clinical.operations.dashboard.view	View authorized operating posture
clinical.support.manage	Triage and resolve support issues
clinical.knowledge.publish	Approve knowledge article
clinical.training.assign	Assign curriculum
clinical.competency.assess	Record competency decision
clinical.adoption.view	View adoption measures

clinical.optimization.propose	Submit improvement proposal
clinical.optimization.approve	Authorize pilot or adoption
clinical.ai.use	Request approved AI assistance
clinical.ai.review	Accept or reject AI output
clinical.ai.govern	Manage AI use cases and findings

Mandatory Tests

Inactive AI use case is rejected

Prohibited AI use cannot execute

AI output cannot become a clinical record without review

Accepted output stores reviewer and accepted content

Direct provider credential path is unavailable

Expired competency blocks configured high-risk action

Training completion alone does not create competency

Metric without denominator is rejected

Cross-tenant telemetry is concealed

Optimization cannot bypass CDE or form governance

Support resolution cannot rewrite clinical event

Knowledge supersession preserves historical article version

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	5f5eaaec83cc646a55f7dc41841bc599f492ff872ef9194e544e370431a49791

KashTre Clinical Operations, Adoption and AI Governance v6.1.0

Supplemental Volume 11. Pick-and-drop candidate. Bind support tooling, Learning/HR competency, telemetry, Shared AI Gateway, benefits management, change governance and Vue operations workspaces before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	1dc20017eb5469122e008510b36e3ec1b4c51ca76d12a9e00eb12f8c30973838

```
{
  "volume": "Clinical EDD Supplemental Volume 11 Operations Adoption AI",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "5f5eaaec83cc646a55f7dc41841bc599f492ff872ef9194e544e370431a49791",
      "bytes": 282
    },
    {
      "path": "composer.json",
      "sha256": "c07d91170170b0b96eafcf16c2c4d5692e92b0f80d563c7f0776385f0fa00a48",
      "bytes": 396
    },
    {
      "path": "config/clinical-operations.php",
      "sha256": "06a913435ad5fde426718690aa768eba8371d8fda8c037ac93d83fb4dcc1e47e",
      "bytes": 279
    },
    {
      "path": "database/migrations/2026_08_28_1000000_create_clinical_operations_tables.php",
      "sha256": "3f5a215bf0003a5f0c1980764365ebb9b95c85ab5d6e33fd4dd9d41b19dcca2a",
      "bytes": 4950
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-operations.php",
      "sha256": "6baefcfd6db38925105dce1dbe8d2fa6032d98b7504e3711ff03c22456bc332f",
      "bytes": 284
    }
  ]
}
```

```

"path": "src/Actions/CaptureAdoptionSnapshot.php",
"sha256": "7fbc61b098f0242f0adf174fc5f309323ee8bfa964b4d6a770b982bfa34905c4",
"bytes": 1078
},
{
"path": "src/Actions/RequestAiAssistance.php",
"sha256": "96a676877ef5eee46b046df5b818eadbfe070b990213cdbcb0c11bcbafd87db3",
"bytes": 1706
},
{
"path": "src/Actions/ReviewAiOutput.php",
"sha256": "a79f1ab4dc97bff951bc1dd6d1b96dfe601bcfe6b3edb2a62e4de544349192c",
"bytes": 1278
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{
"path": "src/ClinicalOperationsServiceProvider.php",
"sha256": "a8de7f2c0ff9014b42062ddba1e7a10bcdbbb49d62af517869a785300e5302a0",
"bytes": 460
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{
"path": "src/Contracts/LearningAssignmentGateway.php",
"sha256": "7442fbb0d1a3551a31aa27e35da8338014880e1f304a6a3ca415954162717db5",
"bytes": 242
},
{
"path": "src/Contracts/SharedAiGateway.php",
"sha256": "e9741405878044e7c43f4e437d7a5cbb41af5fad681dba46495b31f5d287f7b9",
"bytes": 220
},
{
"path": "src/Contracts/TelemetryQuery.php",
"sha256": "15816d559324bf8b626ef1249ba16df91688cab468dda67a87ef8d70c10590be",
"bytes": 247
},
{
"path": "src/Data/SubmitAiAssistanceData.php",
"sha256": "e9893903ec813a28e35fdb2f4c77b60b95dc9d1ae6659cf1e37119c3e41fb8ec",
"bytes": 374
},
{
"path": "src/Enums/AdoptionSignal.php",
"sha256": "955ecd6eaba14c722458206879ba61c3d78d70d989309f0037f149b9e371da23",
"bytes": 216
},
{
"path": "src/Enums/AiRiskLevel.php",
"sha256": "a721737eebb6582631bd60c49933c2a1996357ca6dfbcadd6ce4e5eb2b279e0c",
"bytes": 185
},
{
"path": "src/Enums/AiUseStatus.php",
"sha256": "50ef46e24cdda0082bd9d7b4a74ba83c82d369587bffe50330a9a32abe6f3c05",
"bytes": 234
},

```

```

{
  "path": "src/Enums/CompetencyStatus.php",
  "sha256": "2d4dc1c92232241380c21cadae3fbae86fa5ebda1f3b967e5837faddaa991afd",
  "bytes": 217
},
{
  "path": "src/Enums/KnowledgeStatus.php",
  "sha256": "703648d85740e17e0e343d5072e47e461e7b9925e6c985035858453077b1506e",
  "bytes": 222
},
{
  "path": "src/Enums/OperatingIssueStatus.php",
  "sha256": "43dc46efbbdb7fe0bdcf4a2be295ffb43dfb19c59b423afeb63c25b8c9a9f1a3",
  "bytes": 250
},
{
  "path": "src/Enums/OptimizationStatus.php",
  "sha256": "9d78b517eb3c4079cf78ad6fe80a4e1d54f8d811a4dbd0b78844b40e121d0366",
  "bytes": 281
},
{
  "path": "src/Enums/TrainingStatus.php",
  "sha256": "d714785e0ef856a8e0939b74726c95dd5349d927ceced38e277fc5cce207f8f0",
  "bytes": 252
},
{
  "path": "src/Models/AdoptionMetricDefinition.php",
  "sha256": "86f0eb0e165cd1f965a39b5a208f7a889927e6272636cb9c08c3632ce9ceca5c",
  "bytes": 549
},
{
  "path": "src/Models/AdoptionSnapshot.php",
  "sha256": "fa4416b9c799ec037298116784a81492eabefbf0285a3e3d61dff8519cc9f62c",
  "bytes": 532
},
{
  "path": "src/Models/AiAssistanceOutput.php",
  "sha256": "a017929a9f80972b37c98124ade94e05406ff89d2b9365749f5cba9271da3127",
  "bytes": 537
},
{
  "path": "src/Models/AiAssistanceRequest.php",
  "sha256": "4d67e659f0bd49ad474560a3cd24b5d0e7d341d4a5cc1339ed66f0fc9e08e1f6",
  "bytes": 539
},
{
  "path": "src/Models/AiHumanReview.php",
  "sha256": "57c2712526f22ca1755ceec57bb1679846d156a37efeb2ab28332510997d2898",
  "bytes": 527
},
{
  "path": "src/Models/AiMonitoringFinding.php",
  "sha256": "5dee9310e0e5f6570ebf4e48596184b765c5f9a3b5ec9aa54aa4b58e18a959dd",
  "bytes": 539
},

```

```

{
  "path": "src/Models/AiUseCase.php",
  "sha256": "08ed209bf58813fba5814e5fcb80720d240b407f9e91d93cacd9e624c2778f04",
  "bytes": 519
},
{
  "path": "src/Models/BenefitsRealizationMeasure.php",
  "sha256": "99e59ffd8a3e16ed06d719190f00592b8304c69cb933d7338db75f07ca9b8a17",
  "bytes": 541
},
{
  "path": "src/Models/BenefitsRealizationSnapshot.php",
  "sha256": "f8448d3402725d1656f52105a40e078a7e54363e413225ee75aa95e2d621527f",
  "bytes": 543
},
{
  "path": "src/Models/CompetencyRecord.php",
  "sha256": "67ac7c02a24bec92d84cdf5fd5a16ea4d4889236dc45aa51ead9e331bd111715",
  "bytes": 532
},
{
  "path": "src/Models/KnowledgeArticle.php",
  "sha256": "72c9ce63cb8b6b0799f29c9f90921ad0b471e59a7e3196b61e40b3f18415db99",
  "bytes": 532
},
{
  "path": "src/Models/OperatingIssue.php",
  "sha256": "7b3897e578a47d1e0ccaa14ceff3f4d48604313a784f6b77f762706a156f56d2",
  "bytes": 528
},
{
  "path": "src/Models/OperationalReconciliation.php",
  "sha256": "b4092d90e16bd730bdf9e6cc7032629198ba09b97604890011b46d97114a68f7",
  "bytes": 550
},
{
  "path": "src/Models/OptimizationPilot.php",
  "sha256": "4c479b50b975f7a19c7a6a610b98f337d19bb4b0e41c5285be6353bdead37ffb",
  "bytes": 534
},
{
  "path": "src/Models/OptimizationProposal.php",
  "sha256": "6704246573515ab95991f6160f2bd4746673e5d70b82bc66cf8749ddcca2cc5b",
  "bytes": 540
},
{
  "path": "src/Models/SupportCase.php",
  "sha256": "9ee45456bfa8dd9ddfa2c573e3b7facecbb8d2ce238758aeab6d77e3eea08d54",
  "bytes": 522
},
{
  "path": "src/Models/TrainingAssignment.php",
  "sha256": "70322e2c72a596b248fbaf22a81c20cc62f0c493e8994894c2b36c63ad0a63aa",
  "bytes": 536
},

```

```
{
  "path": "src/Models/TrainingCurriculum.php",
  "sha256": "befe14160fb2db50e3c1ee10d32c3e0499deb02a06bf4bdec2fc7f80b415792",
  "bytes": 534
},
{
  "path": "src/Services/AdoptionHealthEvaluator.php",
  "sha256": "d3d9329894d626054875358a4c4517480f3005647ab6b54e3a27341e008dd2ad",
  "bytes": 695
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
]
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	c07d91170170b0b96eafcf16c2c4d5692e92b0f80d563c7f0776385f0fa00a48

```
{
  "name": "kashtre/clinical-operations",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\Operations\\": "src/"
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\Operations\\ClinicalOperationsServiceProvider"
          ]
        }
      }
    }
  }
}
```

config/clinical-operations.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-operations.php	06a913435ad5fde426718690aa768eba8371d8fda8c037ac93d83fb4dcc1e47e

```
<?php
declare(strict_types=1);
return [
    'competency_validity_days' => (int) env('CLINICAL_COMPETENCY_DAYS', 365),
    'adoption_snapshot_days' => (int) env('CLINICAL_ADPTION_DAYS', 30),
    'ai_review_required' => true,
    'ai_direct_provider_access' => false,
    'optimization_change_control_required' => true;
];
```

database/migrations/2026_08_28_1000000_create_clinical_operations_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_1000000_create_clinical_operations_tables.php	3f5a215bf0003a5f0c1980764365ebb9b95c85ab5d6e33fd4dd9d41b19dcca2a

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;
return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_operating_issues', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->string('issue_type', 96);
            $table->string('severity', 24);
            $table->string('status', 24);
            $table->text('description');
            $table->json('context');
            $table->ulid('reported_by_public_id');
            $table->dateTime('reported_at_utc', 6);
            $table->dateTime('resolved_at_utc', 6)->nullable();
        });
    }
};
```

```

Schema::create('clinical_knowledge_articles',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->string('code',96);$t->string('title',240);$t->unsignedInteger('version_no');$t->string('status',24);$t-
>longText('content');$t->ulid('approved_by_public_id')->nullable();$t->dateTime('effective_from_utc',6);$t-
>dateTime('effective_to_utc',6)->nullable();$t->unique(['tenant_id','code','version_no'],'uq_knowledge_version');});
Schema::create('clinical_training_curricula',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>string('code',96);$t->string('name',180);$t->unsignedInteger('version_no');$t->json('scope');$t->string('status',24);$t-
>unique(['tenant_id','code','version_no'],'uq_training_curriculum');});
Schema::create('clinical_training_assignments',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('curriculum_public_id');$t->ulid('user_public_id');$t->string('status',24);$t-
>dateTime('assigned_at_utc',6);$t->dateTime('due_at_utc',6);$t->dateTime('completed_at_utc',6)->nullable();});
Schema::create('clinical_competency_records',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('user_public_id');$t->string('competency_code',96);$t->string('status',24);$t-
>ulid('assessed_by_public_id');$t->dateTime('assessed_at_utc',6);$t->dateTime('expires_at_utc',6)->nullable();$t-
>json('evidence');});
Schema::create('clinical_adoption_snapshots',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('facility_public_id');$t->dateTime('period_from_utc',6);$t->dateTime('period_to_utc',6);$t-
>string('overall_signal',24);$t->json('metrics');$t->dateTime('captured_at_utc',6);});
Schema::create('clinical_optimization_proposals',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->string('title',240);$t->text('problem_statement');$t->text('proposed_change');$t->string('status',24);$t-
>json('safety_impact');$t->ulid('proposed_by_public_id');$t->dateTime('proposed_at_utc',6);});
Schema::create('clinical_ai_use_cases',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t-
>string('code',96);$t->string('name',180);$t->string('risk_level',24);$t->string('status',24);$t->json('allowed_inputs');$t-
>json('prohibited_outputs');$t->string('gateway_policy_reference',240);$t->unique(['tenant_id','code'],'uq_ai_use_case');});
Schema::create('clinical_ai_assistance_requests',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('patient_public_id')->nullable();$t->ulid('encounter_public_id')->nullable();$t-
>ulid('actor_public_id');$t->ulid('ai_use_case_public_id');$t->string('purpose',240);$t->json('minimum_necessary_context');$t-
>string('status',24);$t->dateTime('requested_at_utc',6);});
Schema::create('clinical_ai_assistance_outputs',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('request_public_id');$t->string('gateway_reference',240);$t->string('model_reference',240)-
>nullable();$t->longText('content');$t->string('status',24);$t->dateTime('generated_at_utc',6);});
Schema::create('clinical_ai_human_reviews',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('output_public_id');$t->ulid('reviewed_by_public_id');$t->string('decision',24);$t->text('comment');$t-
>longText('accepted_content')->nullable();$t->dateTime('reviewed_at_utc',6);});
Schema::create('clinical_ai_monitoring_findings',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t-
>string('tenant_id',64);$t->ulid('ai_use_case_public_id');$t->string('finding_code',96);$t->string('severity',24);$t->json('evidence');$t-
>string('status',24);$t->dateTime('detected_at_utc',6);});
}public function
down():void{foreach(['clinical_ai_monitoring_findings','clinical_ai_human_reviews','clinical_ai_assistance_outputs','clinical_ai_assis
tance_requests','clinical_ai_use_cases','clinical_optimization_proposals','clinical_adoption_snapshots','clinical_competency_recor
ds','clinical_training_assignments','clinical_training_curricula','clinical_knowledge_articles','clinical_operating_issues']as$t)Schema
::dropIfExists($t);});

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-operations.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	routes/clinical- operations.php	6baefcfd6db38925105dce1dbe8d2fa6032d98b7504e3711ff03c22456bc332f
----------------------------	------------------------------------	--

```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])-
>prefix('api/v1/clinical/operations')->group(function(){Route::get('/dashboard',fn()->response()-
>json(['error'=>['code'=>'CLN_OPERATIONS_HOST_CONTROLLER_REQUIRED']],501));});
```

src/Actions/CaptureAdoptionSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CaptureAdoptionSnapshot.php	7fbc61b098f0242f0adf174fc5f309323ee8bfa964b4d6a770b982bfa34905c4

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Actions;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Operations\Contracts\TelemetryQuery;use
KashTre\Clinical\Operations\Models\AdoptionSnapshot;use KashTre\Clinical\Operations\Services\AdoptionHealthEvaluator;final
readonly class CaptureAdoptionSnapshot{public function __construct(private PublicIdGenerator $ids,private TelemetryQuery
$telemetry,private AdoptionHealthEvaluator $health){}public function execute(string $tenantId,string
$facilityId,\DateTimeImmutable $from,\DateTimeImmutable $to,array $definitions):AdoptionSnapshot{$values=$this->telemetry-
>adoptionSnapshot($tenantId,$facilityId,$from,$to,array_column($definitions,'code'));$r=$this->health-
>evaluate($definitions,$values);return AdoptionSnapshot::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'facility_public_id'=>$facilityId,'period_from_utc'=>$from,'period_to_utc'=>$to,'overall_signal'=>$
r['overall'],'metrics'=>$r['results'],'captured_at_utc'=>now('UTC'))};}
```

src/Actions/RequestAiAssistance.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RequestAiAssistance.php	96a676877ef5eee46b046df5b818eadbfe070b990213cdbcb0c11bcbafd87db3

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Operations\Contracts\SharedAiGateway;use
KashTre\Clinical\Operations\Data\SubmitAiAssistanceData;use
KashTre\Clinical\Operations\Models\AiAssistanceOutput,AiAssistanceRequest,AiUseCase;final readonly class
RequestAiAssistance{public function __construct(private PublicIdGenerator $ids,private SharedAiGateway $ai){}public function
execute(SubmitAiAssistanceData $d):AiAssistanceRequest{return DB::transaction(function()use($d){$use=AiUseCase::query()-
>where('tenant_id',$d->tenantId)->where('code',$d->useCaseCode)->where('status','ACTIVE')->firstOrFail();if($use-
>risk_level==='PROHIBITED')throw new \DomainException('CLN_AI_USE_PROHIBITED');$id=(string)$this->ids-
>generate();$r=AiAssistanceRequest::query()->create(['public_id'=>$id,'tenant_id'=>$d->tenantId,'patient_public_id'=>$d-
>patientId,'encounter_public_id'=>$d->encounterId,'actor_public_id'=>$d->actorId,'ai_use_case_public_id'=>$use-
>public_id,'purpose'=>$d->purpose,'minimum_necessary_context'=>$d-
>minimumNecessaryContext,'status'=>'DRAFT','requested_at_utc'=>$d->requestedAt]);$out=$this->ai->request($d->tenantId,$d-
>useCaseCode,$d->actorId,$d->minimumNecessaryContext);AiAssistanceOutput::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$d-
>tenantId,'request_public_id'=>$id,'gateway_reference'=>$out['reference'],'model_reference'=>$out['model_reference']??null,'cont
ent'=>$out['content'],'status'=>'REVIEW_REQUIRED','generated_at_utc'=>now('UTC')));$r-
>forceFill(['status'=>'REVIEW_REQUIRED']->save();return$r;});}
```

src/Actions/ReviewAiOutput.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/ReviewAiOutput.php	a79f1ab4dc97bff951bc1dd6d1b96dafa601bcfe6b3edb2a62e4de544349192c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Operations\Models\{AiAssistanceOutput,AiHumanReview};final readonly class ReviewAiOutput{public function
__construct(private PublicIdGenerator $ids){}public function execute(string $tenantId,string $outputId,string $reviewerId,string
$decision,string $comment,?string $acceptedContent):AiHumanReview{return
DB::transaction(function()use($tenantId,$outputId,$reviewerId,$decision,$comment,$acceptedContent){$o=AiAssistanceOutput::q
uery()->where('tenant_id',$tenantId)->where('public_id',$outputId)->lockForUpdate()->firstOrFail();if($o-
>status!=='REVIEW_REQUIRED')throw new
\DomainException('CLN_AI_OUTPUT_NOT_REVIEWABLE');if($decision==='ACCEPTED'&&trim((string)$acceptedContent)===')throw
new \DomainException('CLN_AI_ACCEPTED_CONTENT_REQUIRED');$r=AiHumanReview::query()->create(['public_id'=>(string)$this-
>ids-
>generate(),'tenant_id'=>$tenantId,'output_public_id'=>$outputId,'reviewed_by_public_id'=>$reviewerId,'decision'=>$decision,'com
ment'=>$comment,'accepted_content'=>$acceptedContent,'reviewed_at_utc'=>now('UTC')));$o->forceFill(['status'=>$decision])-
>save();return$r;}}}
```

src/ClinicalOperationsServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ClinicalOperationsServiceProvider.php	a8de7f2c0ff9014b42062ddba1e7a10bcdbbb49d62af517869a785300e5302a0

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations;use Illuminate\Support\ServiceProviders;final class
ClinicalOperationsServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-operations.php','clinical-operations');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-operations.php');}}
```

src/Contracts/LearningAssignmentGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/LearningAssignmentGateway.php	7442fbb0d1a3551a31aa27e35da8338014880e1f304a6a3ca415954162717db5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Contracts;interface LearningAssignmentGateway{public function
assign(string $tenantId,string $userId,string $curriculumCode,\DateTimeImmutable $dueAt,array $context):string;}
```

src/Contracts/SharedAiGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/SharedAiGateway.php	e9741405878044e7c43f4e437d7a5cbb41af5fad681dba46495b31f5d287f7b9

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Contracts;interface SharedAiGateway{public function
request(string $tenantId,string $useCaseCode,string $actorId,array $minimumNecessaryContext):array;}
```

src/Contracts/TelemetryQuery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/TelemetryQuery.php	15816d559324bf8b626ef1249ba16df91688cab468dda67a87ef8d70c10590be

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Contracts;interface TelemetryQuery{public function
adoptionSnapshot(string $tenantId,string $facilityId,\DateTimeImmutable $from,\DateTimeImmutable $to,array
$metricCodes):array;}
```

src/Data/SubmitAiAssistanceData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/SubmitAiAssistanceData.php	e9893903ec813a28e35fdb2f4c77b60b95dc9d1ae6659cf1e37119c3e41fb8e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Data;final readonly class SubmitAiAssistanceData{public function
__construct(public string $tenantId,public string $patientId,public string $encounterId,public string $actorId,public string
$useCaseCode,public array $minimumNecessaryContext,public string $purpose,public \DateTimeImmutable $requestedAt){}}
```

src/Enums/AdoptionSignal.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AdoptionSignal.php	955ecd6eaba14c722458206879ba61c3d78d70d989309f0037f149b9e371da23

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum AdoptionSignal:string{case
HEALTHY='HEALTHY';case WATCH='WATCH';case ACTION_REQUIRED='ACTION_REQUIRED';case
NOT_MEASURED='NOT_MEASURED';}
```

src/Enums/AiRiskLevel.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AiRiskLevel.php	a721737eebb6582631bd60c49933c2a1996357ca6dfbcadd6ce4e5eb2b279e0c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum AiRiskLevel:string{case LOW='LOW';case
MODERATE='MODERATE';case HIGH='HIGH';case PROHIBITED='PROHIBITED';}
```

src/Enums/AiUseStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AiUseStatus.php	50ef46e24cdda0082bd9d7b4a74ba83c82d369587bffe50330a9a32abe6f3c05

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum AiUseStatus:string{case DRAFT='DRAFT';case
REVIEW_REQUIRED='REVIEW_REQUIRED';case ACCEPTED='ACCEPTED';case REJECTED='REJECTED';case
WITHDRAWN='WITHDRAWN';}
```

src/Enums/CompetencyStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CompetencyStatus.php	2d4dc1c92232241380c21cadae3fbae86fa5ebda1f3b967e5837faddaa991afd

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum CompetencyStatus:string{case
PENDING='PENDING';case VALID='VALID';case EXPIRING='EXPIRING';case EXPIRED='EXPIRED';case REVOKED='REVOKED';}
```

src/Enums/KnowledgeStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/KnowledgeStatus.php	703648d85740e17e0e343d5072e47e461e7b9925e6c985035858453077b1506e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum KnowledgeStatus:string{case DRAFT='DRAFT';case
REVIEW='REVIEW';case PUBLISHED='PUBLISHED';case SUPERSEDED='SUPERSEDED';case RETIRED='RETIRED';}
```

src/Enums/OperatingIssueStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/OperatingIssueStatus.php	43dc46efbbdb7fe0bdcf4a2be295ffb43dfb19c59b423afeb63c25b8c9a9f1a3

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum OperatingIssueStatus:string{case
OPEN='OPEN';case TRIAGED='TRIAGED';case ASSIGNED='ASSIGNED';case IN_PROGRESS='IN_PROGRESS';case
RESOLVED='RESOLVED';case CLOSED='CLOSED';}
```

src/Enums/OptimizationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/OptimizationStatus.php	9d78b517eb3c4079cf78ad6fe80a4e1d54f8d811a4dbd0b78844b40e121d0366

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum OptimizationStatus:string{case
PROPOSED='PROPOSED';case ASSESSING='ASSESSING';case APPROVED='APPROVED';case PILOT='PILOT';case
ADOPTED='ADOPTED';case REJECTED='REJECTED';case ROLLED_BACK='ROLLED_BACK';}
```

src/Enums/TrainingStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/TrainingStatus.php	d714785e0ef856a8e0939b74726c95dd5349d927ceced38e277fc5cce207f8f0

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Enums;enum TrainingStatus:string{case
PLANNED='PLANNED';case ASSIGNED='ASSIGNED';case IN_PROGRESS='IN_PROGRESS';case COMPLETED='COMPLETED';case
EXPIRED='EXPIRED';case WAIVED='WAIVED';}
```

src/Models/AdoptionMetricDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AdoptionMetricDefinition.php	86f0eb0e165cd1f965a39b5a208f7a889927e6272636cb9c08c3632ce9cec a5c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AdoptionMetricDefinition extends Model{public $timestamps=false;protected
$table='clinical_adoption_metric_definitions';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/AdoptionSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AdoptionSnapshot.php	fa4416b9c799ec037298116784a81492eabefbf0285a3e3d61dff8519cc9f62c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AdoptionSnapshot extends Model{public $timestamps=false;protected $table='clinical_adoption_snapshots';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/AiAssistanceOutput.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AiAssistanceOutput.php	a017929a9f80972b37c98124ade94e05406ff89d2b9365749f5cba9271da3127

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AiAssistanceOutput extends Model{public $timestamps=false;protected $table='clinical_ai_assistance_outputs';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/AiAssistanceRequest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AiAssistanceRequest.php	4d67e659f0bd49ad474560a3cd24b5d0e7d341d4a5cc1339ed66f0fc9e08e1f 6

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```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AiAssistanceRequest extends Model{public $timestamps=false;protected $table='clinical_ai_assistance_requests';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immuta
ble_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immuta
ble_datetime'];}
```

src/Models/AiHumanReview.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AiHumanReview.php	57c2712526f22ca1755ceec57bb1679846d156a37efeb2ab28332510997d2898

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AiHumanReview extends Model{public $timestamps=false;protected $table='clinical_ai_human_reviews';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immuta
ble_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immuta
ble_datetime'];}
```

src/Models/AiMonitoringFinding.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AiMonitoringFinding.php	5dee9310e0e5f6570ebf4e48596184b765c5f9a3b5ec9aa54aa4b58e18a959dd

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AiMonitoringFinding extends Model{public $timestamps=false;protected $table='clinical_ai_monitoring_findings';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immuta
ble_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immuta
ble_datetime'];}
```

src/Models/AiUseCase.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AiUseCase.php	08ed209bf58813fba5814e5fcb80720d240b407f9e91d93cacd9e624c2778f04

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
AiUseCase extends Model{public $timestamps=false;protected $table='clinical_ai_use_cases';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immuta
ble_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immuta
ble_datetime'];}
```

src/Models/BenefitsRealizationMeasure.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Models/BenefitsRealizationMeasure.php	99e59ffd8a3e16ed06d719190f00592b8304c69cb933d7338db75f07ca9b8a17
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
BenefitsRealizationMeasure extends Model{public $timestamps=false;protected $table='clinical_benefits_measures';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/BenefitsRealizationSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/BenefitsRealizationSnapshot.php	f8448d3402725d1656f52105a40e078a7e54363e413225ee75aa95e2d621527f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
BenefitsRealizationSnapshot extends Model{public $timestamps=false;protected $table='clinical_benefits_snapshots';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/CompetencyRecord.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CompetencyRecord.php	67ac7c02a24bec92d84cdf5fd5a16ea4d4889236dc45aa51ead9e331bd111715

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
CompetencyRecord extends Model{public $timestamps=false;protected $table='clinical_competency_records';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/KnowledgeArticle.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/KnowledgeArticle.php	72c9ce63cb8b6b0799f29c9f90921ad0b471e59a7e3196b61e40b3f18415db99

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
KnowledgeArticle extends Model{public $timestamps=false;protected $table='clinical_knowledge_articles';protected
```

```
$guarded=[];protected
```

```
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/OperatingIssue.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/OperatingIssue.php	7b3897e578a47d1e0ccaa14ceff3f4d48604313a784f6b77f762706a156f56d2

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class OperatingIssue extends Model{public $timestamps=false;protected $table='clinical_operating_issues';protected
```

```
$guarded=[];protected
```

```
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/OperationalReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/OperationalReconciliation.php	b4092d90e16bd730bdf9e6cc7032629198ba09b97604890011b46d97114a68f7

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class OperationalReconciliation extends Model{public $timestamps=false;protected
```

```
$table='clinical_operational_reconciliations';protected $guarded=[];protected
```

```
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/OptimizationPilot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/OptimizationPilot.php	4c479b50b975f7a19c7a6a610b98f337d19bb4b0e41c5285be6353bdead37ffb

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class OptimizationPilot extends Model{public $timestamps=false;protected $table='clinical_optimization_pilots';protected
```

```
$guarded=[];protected
```

```
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/OptimizationProposal.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/OptimizationProposal.php	6704246573515ab95991f6160f2bd4746673e5d70b82bc66cf8749ddcca2cc5b

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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
OptimizationProposal extends Model{public $timestamps=false;protected $table='clinical_optimization_proposals';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/SupportCase.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SupportCase.php	9ee45456bfa8dd9ddfa2c573e3b7facecbb8d2ce238758aeab6d77e3eea08d54

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
SupportCase extends Model{public $timestamps=false;protected $table='clinical_support_cases';protected $guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/TrainingAssignment.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TrainingAssignment.php	70322e2c72a596b248fbaf22a81c20cc62f0c493e8994894c2b36c63ad0a63aa

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
TrainingAssignment extends Model{public $timestamps=false;protected $table='clinical_training_assignments';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/TrainingCurriculum.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TrainingCurriculum.php	befe14160fb2db50e3c1ee10d32c3e0499deb02a06bf4bdec2fc7f80b415792

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Operations\Models;use Illuminate\Database\Eloquent\Model;final class
TrainingCurriculum extends Model{public $timestamps=false;protected $table='clinical_training_curricula';protected
$guarded=[];protected
$casts=['context'=>'array','evidence'=>'array','metrics'=>'array','minimum_necessary_context'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','effective_from_utc'=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Services/AdoptionHealthEvaluator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Services/AdoptionHealthEvaluator.php	d3d9329894d626054875358a4c4517480f3005647ab6b54e3a27341e008dd2ad

```
<?php
```

```
declare(strict_types=1);namespace KashTre\Clinical\Operations\Services;final class AdoptionHealthEvaluator{public function
evaluate(array $metricDefinitions,array $values):array{$results=[];$overall='HEALTHY';foreach($metricDefinitions
as$d){$value=$values[$d['code']]??null;$signal='NOT_MEASURED';if($value!==null){$signal=(float)$value<(float)$d['action_threshol
d']?'ACTION_REQUIRED':((float)$value<(float)$d['watch_threshold']?'WATCH':'HEALTHY');}$results[]=$d['code']=>$d['code'],'value'=>$v
alue,'signal'=>$signal};if($signal==='ACTION_REQUIRED')$overall='ACTION_REQUIRED';elseif($signal==='WATCH'&&$overall==='HE
ALTHY')$overall='WATCH';}return['overall'=>$overall,'results'=>$results];}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
```

```
set -euo pipefail
```

```
composer validate --strict
```

```
composer dump-autoload --strict-psr
```

```
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volumes 1-10	Clinical source boundaries, assurance and release control
Main/HR	Identity, employment, assignments and competency authority
Support platform	Case intake and service management
Telemetry	Privacy-preserving workflow measures
Shared AI Gateway	Provider isolation, policy, model and request provenance
Benefits governance	Measure ownership and review cadence
Vue	Operations dashboard, support, training, adoption, optimization and AI review

Blocking Gaps

Approve Volume 11 as a supplemental baseline

Define Clinical operating ownership and support model

Approve curricula and competency catalogue

Approve metric definitions and privacy safeguards

Bind Shared AI Gateway and approved use-case catalogue

Approve AI monitoring, suspension and incident paths

Complete host controllers, requests, resources, scheduled jobs and Vue interfaces

Run tenant-isolation, AI human-review, telemetry quality, accessibility and operational acceptance tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Phase 1-10 functional baseline and Shared AI Gateway rule
Volumes 1-10	Implemented Clinical architecture and release assurance
WHO Monitoring and Evaluating Digital Health Interventions	External monitoring and evaluation reference
NIST AI RMF 1.0 and Generative AI Profile	External AI trustworthiness and risk-management reference

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

SUPPLEMENTAL VOLUME 12

INTEROPERABILITY, POPULATION HEALTH, QUALITY MEASUREMENT AND RESEARCH DATA

FHIR Exchange, Subscriptions, Bulk Data, Cohorts, De-identification and Governed Secondary Use

Version: 6.1.0 Comprehensive Draft

Status: Supplemental to the Phase 1-10 functional baseline

Depends on: Volumes 1 to 11

Production: Not Approved for Production

Scope Position and Governing Rule

Volume 12 is supplemental. It exposes governed representations of authoritative Clinical data without creating a second clinical source of truth. Every exchange records purpose, authorization, scope, profile version, de-identification mode, artifact hash, delivery evidence and expiry.

Authoritative clinical records -> purpose and authorization -> cohort/scope resolution -> FHIR mapping -> terminology validation -> de-identification where required -> encrypted artifact or event notification -> delivery evidence -> reconciliation and expiry

Entity Register

Entity	Table	Invariant
ExchangeEndpoint	clinical_exchange_endpoints	Tenant, purpose, authorization and reproducibility preserved
ExchangeJob	clinical_exchange_jobs	Tenant, purpose, authorization and reproducibility preserved
ExchangeArtifact	clinical_exchange_artifacts	Tenant, purpose, authorization and reproducibility preserved
ExchangeDelivery	clinical_exchange_deliveries	Tenant, purpose, authorization and reproducibility preserved
SubscriptionTopic	clinical_subscription_topics	Tenant, purpose, authorization and reproducibility preserved
SubscriptionRegistration	clinical_subscription_registrations	Tenant, purpose, authorization and reproducibility preserved
SubscriptionDelivery	clinical_subscription_deliveries	Tenant, purpose, authorization and reproducibility preserved
CohortDefinition	clinical_cohort_definitions	Tenant, purpose, authorization and reproducibility preserved
CohortVersion	clinical_cohort_versions	Tenant, purpose, authorization and reproducibility preserved
CohortMembershipSnapshot	clinical_cohort_membership_snapshots	Tenant, purpose, authorization and reproducibility preserved
MeasureDefinition	clinical_measure_definitions	Tenant, purpose, authorization and reproducibility preserved
MeasureVersion	clinical_measure_versions	Tenant, purpose, authorization and reproducibility preserved
MeasureRun	clinical_measure_runs	Tenant, purpose, authorization and reproducibility preserved
MeasureResult	clinical_measure_results	Tenant, purpose, authorization and reproducibility preserved
PublicHealthSubmission	clinical_public_health_submissions	Tenant, purpose, authorization and reproducibility preserved
ResearchDatasetRequest	clinical_research_dataset_requests	Tenant, purpose, authorization and reproducibility preserved
ResearchDatasetArtifact	clinical_research_dataset_artifacts	Tenant, purpose, authorization and reproducibility preserved
TerminologyMapping	clinical_terminology_mappings	Tenant, purpose, authorization and reproducibility preserved
InteroperabilityReconciliation	clinical_interoperability_reconciliations	Tenant, purpose, authorization and reproducibility preserved

Engineering Design

FHIR exchange

- Versioned mapping profiles
- Stable public identifiers
- Capability-aware endpoints
- No direct database exposure
- Validation errors enter reconciliation
- Source corrections produce new exchange versions

Subscriptions

- Governed SubscriptionTopic catalogue
- Authorized endpoint and filters
- Idempotent notification event reference
- Retry with bounded attempts
- Suspend repeatedly failing or unsafe endpoints
- Notification does not itself disclose more than configured content

Bulk data

- Asynchronous job lifecycle
- Immutable NDJSON artifact hashes
- Encrypted object storage
- Expiring download authorization
- Manifest and record counts
- No direct identifiers by default for secondary use

Cohorts and measures

- Versioned cohort criteria
- Membership snapshots preserve evaluation time
- Approved measure logic and implementation version
- Input fingerprint for reproducibility
- Individual, subject-list or summary outputs as configured

Public health and research

- Purpose and legal/consent authority recorded
- Minimum necessary scope
- De-identification policy explicit
- Research protocol and approval reference required
- No re-identification path in Clinical
- Submission acceptance or rejection retained

Permissions

Permission	Purpose
clinical.exchange.request	Request authorized exchange
clinical.exchange.generate	Generate artifact
clinical.exchange.deliver	Deliver approved artifact
clinical.subscription.manage	Manage topic-based notification

clinical.cohort.manage	Create and version cohort
clinical.measure.run	Evaluate approved measure
clinical.public_health.submit	Submit governed dataset
clinical.research.request	Request research dataset
clinical.research.approve	Approve governed secondary use
clinical.interoperability.reconcile	Resolve mapping or delivery exception

Mandatory Tests

Denied purpose blocks exchange

Cross-tenant record cannot enter artifact

Source correction produces new mapped version

Artifact hash verifies downloaded content

Expired job cannot be downloaded

Subscription retry is idempotent

Failed endpoint is suspended at configured boundary

De-identified export contains no configured direct identifiers

Cohort membership is reproducible

Measure run repeats from identical fingerprint

Research request without protocol is rejected

Public-health rejection is retained and correctable

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	058533b14c6091afb5f50b7f020a55daaf7bad6038fcbc1088989465f15709e9

KashTre Clinical Interoperability, Population Health and Research v6.1.0

Supplemental Volume 12. Pick-and-drop candidate. Bind consent/authorization, FHIR mapping, terminology, object storage, de-identification, public-health authority, research governance, subscriptions and quality-measure execution before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	ae5dafa511d4c7d052244a0e6109be320e5343700d841c8326b262de588f85c1

```
{
  "volume": "Clinical EDD Supplemental Volume 12 Interoperability Population Research",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "058533b14c6091afb5f50b7f020a55daaf7bad6038fcbc1088989465f15709e9",
      "bytes": 327
    },
    {
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      "sha256": "73cd9633dc916e84eff959347944eb24c5b52e01724afbac44a6f75980eabb98",
      "bytes": 412
    },
    {
      "path": "config/clinical-interoperability.php",
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      "bytes": 311
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    {
      "path": "database/migrations/2026_08_28_1100000_create_interoperability_tables.php",
      "sha256": "28b164b527d0a531075b3aec41009e2a7cbbb141efbfa58742b6b2d24f578279",
      "bytes": 5116
    },
    {
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      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-interoperability.php",
      "sha256": "42fd9b30878d1492af8e7546467dc75ac5492836277e694de44cffe25765c77",
      "bytes": 283
    }
  ]
}
```

```

"path": "src/Actions/CreateExchangeJob.php",
"sha256": "b2d7a0204af5cd39682fea79dce2c203bcd166cbe3530bd6555dbf1156edc361",
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{
"path": "src/Actions/GenerateExchangeArtifact.php",
"sha256": "ef31b04f3c293768f5ff13bf52607c24475143f116f4cf82b1e257ed10bd9876",
"bytes": 1708
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{
"path": "src/Actions/RunQualityMeasure.php",
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"bytes": 1746
},
{
"path": "src/Contracts/AuthorizationDecisionGateway.php",
"sha256": "01cdba95198ff7f50424276fdd4aec2f4ce845bb53bb831a14fb4e1869de444e",
"bytes": 216
},
{
"path": "src/Contracts/DeidentificationGateway.php",
"sha256": "f56dc77a73d13224b37677009355ed77446c9576170ce24bf82a05d63d71eb9f",
"bytes": 211
},
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"path": "src/Contracts/FhirMapper.php",
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"bytes": 200
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"bytes": 225
},
{
"path": "src/Data/CreateExchangeJobData.php",
"sha256": "fe7780063647fb3fe4017b69ca19b0f24b29e5549ac88c233d32d69384bb6838",
"bytes": 370
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"path": "src/Enums/CohortStatus.php",
"sha256": "dc3b0fd607dcd12b3336c31fbf2a993029ea54058dfd9649ef589794f61137af",
"bytes": 221
},
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"path": "src/Enums/DeidentificationMode.php",
"sha256": "60aa0128987c16d92bcef26b7ca40f18d9d808a56053098cb71bb6e0fbacd5bc",
"bytes": 240
},
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"path": "src/Enums/ExchangeStatus.php",
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"bytes": 306
},

```

```

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  "path": "src/Enums/MappingStatus.php",
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  "bytes": 201
},
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  "path": "src/Enums/MeasureRunStatus.php",
  "sha256": "e4285fe0e101d856b0c03576e296af4175d44529149ef950baacf885cc61d2ea",
  "bytes": 225
},
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  "path": "src/Enums/PurposeCategory.php",
  "sha256": "beebf852daba33cb0904153a28cc5b028888a3f03f36e2434cc50d9405ef6753",
  "bytes": 248
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  "path": "src/Enums/SubmissionStatus.php",
  "sha256": "952ba321cd8bd0cf2c73864f5b4390503362d90fef84b9ff974b5a6787e6c0a4",
  "bytes": 260
},
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  "path": "src/Enums/SubscriptionStatus.php",
  "sha256": "0f2c59b5953a7fd923ad99ba91d3c7969aefcb9ef0170dc41179bc64953936f3",
  "bytes": 225
},
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  "path": "src/InteroperabilityServiceProvider.php",
  "sha256": "2f023f40ababc5a124b6928e825e9f8724d8cebd2b87ea5c66836dfdcd3c40462",
  "bytes": 482
},
{
  "path": "src/Models/CohortDefinition.php",
  "sha256": "4a878a0986f74ec41d30db3036afaad92ffd1af2d25a92452e95810d8b6b811b",
  "bytes": 473
},
{
  "path": "src/Models/CohortMembershipSnapshot.php",
  "sha256": "f5794b10a43f3da57c9d1e3fff386968ee07af2bcf9b80decb55f833f4f9df86",
  "bytes": 490
},
{
  "path": "src/Models/CohortVersion.php",
  "sha256": "92cc9481609aa1ee507efc82c9c7ed038c7357908837307d498c3c07bd90d02f",
  "bytes": 467
},
{
  "path": "src/Models/ExchangeArtifact.php",
  "sha256": "7e3d7e7e2e3a8d7577c38278e2a7f96b74d60e725c906c56cb3089f242e9ebc9",
  "bytes": 473
},
{
  "path": "src/Models/ExchangeDelivery.php",
  "sha256": "f654d329c3a128235fe7413e30f20011f883b30479d668f90d364276ead87d55",
  "bytes": 474
},

```

```

{
  "path": "src/Models/ExchangeEndpoint.php",
  "sha256": "01f14317d12bbe07bce7fbb2ea19836333c31de42b946b7ad0261e4d76bd1b9c",
  "bytes": 473
},
{
  "path": "src/Models/ExchangeJob.php",
  "sha256": "44f824d44f955643cac26cc46ce277f0e2d08ad8187b0a8b395a2f283efc1a2d",
  "bytes": 463
},
{
  "path": "src/Models/InteroperabilityReconciliation.php",
  "sha256": "f47f903adb6788bfc39a3794c523225a0dcc3148a757fbb6e174744a63915e12",
  "bytes": 501
},
{
  "path": "src/Models/MeasureDefinition.php",
  "sha256": "e0cc5099ad921c86cd11abf364d8b2f7248a026946e37f02da1dd1ef635928e5",
  "bytes": 475
},
{
  "path": "src/Models/MeasureResult.php",
  "sha256": "5422fc3645c5777b486e1ab47f41e3e24d45b8f3da4b6c95f8abad142d6fec0c",
  "bytes": 467
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{
  "path": "src/Models/MeasureRun.php",
  "sha256": "fa5453b1ab02c205e0d42caff7873c1bfbfb9f847d478a720ab84f634090122a",
  "bytes": 461
},
{
  "path": "src/Models/MeasureVersion.php",
  "sha256": "0310365af341bd17e9e88e24e982c40c5e06edc79ece66e6d27a04493ab97dee",
  "bytes": 469
},
{
  "path": "src/Models/PublicHealthSubmission.php",
  "sha256": "ff9c957a1cec6cf0c83b4b42fb85c0907e5219291b9ac6eb9efb66437edb50c3",
  "bytes": 486
},
{
  "path": "src/Models/ResearchDatasetArtifact.php",
  "sha256": "ee583c398cada0d4417c02bdad8ab265be9e9a1032cb5d11b35e2aefaaa4b639",
  "bytes": 488
},
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  "path": "src/Models/ResearchDatasetRequest.php",
  "sha256": "1017c0119a2d80712ea45e8196fbf96aa77d40299dfe9d57a2f46df6cbb51a47",
  "bytes": 486
},
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  "path": "src/Models/SubscriptionDelivery.php",
  "sha256": "ceb4938c31fbc758ec60648447a42a002d376eca941e2bbe28713aa12a5f199b",
  "bytes": 482
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```

```
{
  "path": "src/Models/SubscriptionRegistration.php",
  "sha256": "32e5e612594aa0fbdf5b291f9439ec38b279500d6c712f5b19598fcef46883c9",
  "bytes": 489
},
{
  "path": "src/Models/SubscriptionTopic.php",
  "sha256": "834de73eff3db3b69bfd3417e7f7e485cd8bdcb5cbb84463095b9c20314d89f",
  "bytes": 475
},
{
  "path": "src/Models/TerminologyMapping.php",
  "sha256": "461f963b8756314254e0890fb65c6d26b0dc2516f97b6ebcfc563df073d9e3d4",
  "bytes": 477
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	73cd9633dc916e84eff959347944eb24c5b52e01724afbac44a6f75980eabb98

```
{
  "name": "kashtre/clinical-interopability",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\Interopability\\": "src/"
      }
    },
    "extra": {
      "laravel": {
        "providers": [
          "KashTre\\Clinical\\Interopability\\InteropabilityServiceProvider"
        ]
      }
    }
  }
}
```

config/clinical-interoperability.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-interoperability.php	cce96e032e6130123d86158415d4ef04de0d19aeebe19ee2e1c18959522c4e4c

```
<?php
declare(strict_types=1);
return [
    'bulk_job_expiry_hours' => (int) env('CLINICAL_BULK_JOB_EXPIRY_HOURS', 24),
    'subscription_retry_limit' => (int) env('CLINICAL_SUBSCRIPTION_RETRY_LIMIT', 8),
    'research_export_requires_approval' => true,
    'direct_identifier_export_default' => false,
    'measure_reproducibility_required' => true;
];
```

database/migrations/2026_08_28_1100000_create_interoperability_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_1100000_create_interoperability_tables.php	28b164b527d0a531075b3aec41009e2a7cbbb141efbfa58742b6b2d24f578279

```

<?php
declare(strict_types=1);use Illuminate\Database\Migrations\Migration;use Illuminate\Database\Schema\Blueprint;use
Illuminate\Support\Facades\Schema;return new class extends Migration{public function up():void{
Schema::create('clinical_exchange_endpoints',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('code',96);$tbl->string('endpoint_type',64);$tbl->string('base_url',500);$tbl->string('status',24);$tbl-
>json('capabilities');$tbl->unique(['tenant_id','code'],'uq_exchange_endpoint');});
Schema::create('clinical_exchange_jobs',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('requested_by_public_id');$tbl->string('purpose_code',64);$tbl->string('format',64);$tbl->json('scope');$tbl-
>string('deidentification_mode',32);$tbl->string('profile_version',96);$tbl->string('authorization_reference',240)->nullable();$tbl-
>string('status',24);$tbl->dateTime('requested_at_utc',6);$tbl->dateTime('completed_at_utc',6)->nullable();$tbl-
>dateTime('expires_at_utc',6));});
Schema::create('clinical_exchange_artifacts',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->ulid('exchange_job_public_id');$tbl->string('storage_reference',500);$tbl->string('content_type',128);$tbl-
>char('sha256',64);$tbl->unsignedBigInteger('record_count');$tbl->string('status',24);$tbl->dateTime('created_at_utc',6));});
Schema::create('clinical_subscription_registrations',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('topic_canonical',500);$tbl->string('endpoint_reference',500);$tbl->json('filters');$tbl-
>string('status',24);$tbl->dateTime('effective_from_utc',6);$tbl->dateTime('effective_to_utc',6)->nullable();});
Schema::create('clinical_subscription_deliveries',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->ulid('subscription_public_id');$tbl->string('event_reference',240);$tbl->string('status',24);$tbl-
>unsignedInteger('attempt_no');$tbl->dateTime('attempted_at_utc',6);$tbl->string('provider_reference',240)->nullable();$tbl-
>unique(['subscription_public_id','event_reference','attempt_no'],'uq_subscription_attempt');});
Schema::create('clinical_cohort_definitions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('code',96);$tbl->string('name',180);$tbl->string('status',24);$tbl->unsignedInteger('current_version')-
>nullable();$tbl->unique(['tenant_id','code'],'uq_cohort_code');});
Schema::create('clinical_cohort_versions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('cohort_definition_public_id');$tbl->unsignedInteger('version_no');$tbl->json('criteria');$tbl->string('status',24);$tbl-
>dateTime('effective_from_utc',6);$tbl->unique(['cohort_definition_public_id','version_no'],'uq_cohort_version');});
Schema::create('clinical_measure_versions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('measure_code',96);$tbl->unsignedInteger('version_no');$tbl->string('status',24);$tbl->json('logic');$tbl-
>string('implementation_version',96);$tbl->unique(['tenant_id','measure_code','version_no'],'uq_measure_version');});
Schema::create('clinical_measure_runs',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('measure_version_public_id');$tbl->string('subject_type',32);$tbl->ulid('subject_public_id');$tbl->dateTime('period_from_utc',6);$tbl-
>dateTime('period_to_utc',6);$tbl->char('input_fingerprint',64);$tbl->string('status',24);$tbl->dateTime('started_at_utc',6);$tbl-
>dateTime('completed_at_utc',6)->nullable();$tbl-
>unique(['tenant_id','measure_version_public_id','input_fingerprint'],'uq_measure_run');});
Schema::create('clinical_measure_results',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('measure_run_public_id');$tbl->string('result_type',32);$tbl->json('result_payload');$tbl->json('evidence');});
Schema::create('clinical_public_health_submissions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('program_code',96);$tbl->string('status',24);$tbl->json('scope');$tbl->string('authority_reference',240);$tbl-
>dateTime('submitted_at_utc',6)->nullable();});
Schema::create('clinical_research_dataset_requests',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('protocol_reference',240);$tbl->string('purpose_code',64);$tbl->string('deidentification_mode',32);$tbl-
>json('scope');$tbl->string('status',24);$tbl->ulid('requested_by_public_id');$tbl->dateTime('requested_at_utc',6));});
}public function
down():void{foreach(['clinical_research_dataset_requests','clinical_public_health_submissions','clinical_measure_results','clinical_
measure_runs','clinical_measure_versions','clinical_cohort_versions','clinical_cohort_definitions','clinical_subscription_deliveries','
clinical_subscription_registrations','clinical_exchange_artifacts','clinical_exchange_jobs','clinical_exchange_endpoints']as$tbl)Schem
a::dropIfExists($tbl);});

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force
```

routes/clinical-interoperability.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-interoperability.php	42fd9b30878d1492af8e7546467dc75ac5492836277e694de44cffe25765c77

```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])->prefix('api/v1/clinical/interoperability')->group(function(){Route::get('/jobs',fn()->response()->json(['error'=>['code'=>'CLN_EXCHANGE_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/CreateExchangeJob.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CreateExchangeJob.php	b2d7a0204af5cd39682fea79dce2c203bcd166cbe3530bd6555dbf1156edc361

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Actions;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Interoperability\Contracts\AuthorizationDecisionGateway;use KashTre\Clinical\Interoperability\Data\CreateExchangeJobData;use KashTre\Clinical\Interoperability\Models\ExchangeJob;final readonly class CreateExchangeJob{public function __construct(private PublicIdGenerator $ids,private AuthorizationDecisionGateway $authorization){}public function execute(CreateExchangeJobData $d):ExchangeJob{$decision=$this->authorization->decide($d->tenantId,$d->actorId,$d->purpose,$d->scope);if(($decision['decision']??'DENY')!=='PERMIT')throw new \DomainException('CLN_EXCHANGE_NOT_AUTHORIZED');return ExchangeJob::query()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$d->tenantId,'requested_by_public_id'=>$d->actorId,'purpose_code'=>$d->purpose,'format'=>$d->format,'scope'=>$d->scope,'deidentification_mode'=>$d->deidentificationMode,'profile_version'=>$d->profileVersion,'authorization_reference'=>$decision['reference']??null,'status'=>'AUTHORIZED','requested_at_utc'=>$d->requestedAt,'expires_at_utc'=>$d->requestedAt->modify('+'.config('clinical-interoperability.bulk_job_expiry_hours').' hours'))};}
```

src/Actions/GenerateExchangeArtifact.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/GenerateExchangeArtifact.php	ef31b04f3c293768f5ff13bf52607c24475143f116f4cf82b1e257ed10bd9876

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Actions;use Illuminate\Support\Facades\DB;use KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use KashTre\Clinical\Interoperability\Contracts\{DeidentificationGateway,FhirMapper,ObjectStore};use KashTre\Clinical\Interoperability\Models\{ExchangeArtifact,ExchangeJob};final readonly class GenerateExchangeArtifact{public function __construct(private PublicIdGenerator $ids,private FhirMapper $mapper,private DeidentificationGateway $deid,private ObjectStore $store){}public function execute(string $tenantId,string $jobId,array $sourceRecords):ExchangeArtifact{return DB::transaction(function()use($tenantId,$jobId,$sourceRecords){$j=ExchangeJob::query()->where('tenant_id',$tenantId)->where('public_id',$jobId)->where('status','AUTHORIZED')->lockForUpdate()->firstOrFail();$mapped=[];foreach($sourceRecords as$r)$mapped[]=$this->mapper->map($r['type'],$r['public_id'],$j->profile_version);$payload=$this->deid->transform($tenantId,$j-
```

```

>deidentification_mode,$mapped,$j-
>scope);$content=implode("\n",array_map(fn($x)=>json_encode($x,JSON_THROW_ON_ERROR),$payload));$sha=hash('sha256',$content);$stored=$this->store-
>putEncrypted($tenantId,"exchange/$jobId/$sha.ndjson",$content,'application/fhir+ndjson',['sha256'=>$sha]);$a=ExchangeArtifact::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'exchange_job_public_id'=>$jobId,'storage_reference'=>$stored['reference'],'content_type'=>'application/fhir+ndjson','sha256'=>$sha,'record_count'=>count($payload),'status'=>'READY','created_at_utc'=>now('UTC')));$j-
>forceFill(['status'=>'READY','completed_at_utc'=>now('UTC')])>save();return$a;}}

```

src/Actions/RunQualityMeasure.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RunQualityMeasure.php	c8ffadfbdc9593e91305ecdcfbfc45039b8ada883235cde629129bcb6020d8b88

```

<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Interoperability\Models\{MeasureRun,MeasureResult,MeasureVersion};final readonly class
RunQualityMeasure{public function __construct(private PublicIdGenerator $ids){}public function execute(string $tenantId,string
$measureVersionId,string $subjectType,string $subjectId,\DateTimeImmutable $from,\DateTimeImmutable $to,callable
$evaluator):MeasureRun{return
DB::transaction(function()use($tenantId,$measureVersionId,$subjectType,$subjectId,$from,$to,$evaluator){$v=MeasureVersion::query()->where('tenant_id',$tenantId)->where('public_id',$measureVersionId)->where('status','ACTIVE')-
>firstOrFail();$input=['measure'=>$measureVersionId,'subject'=>$subjectId,'from'=>$from->format('DATE_ATOM'),'to'=>$to-
>format('DATE_ATOM')];$hash=hash('sha256',json_encode($input,JSON_THROW_ON_ERROR));$run=MeasureRun::query()-
>firstOrCreate(['tenant_id'=>$tenantId,'measure_version_public_id'=>$measureVersionId,'input_fingerprint'=>$hash],['public_id'=>(
string)$this->ids-
>generate(),'subject_type'=>$subjectType,'subject_public_id'=>$subjectId,'period_from_utc'=>$from,'period_to_utc'=>$to,'status'=>'RUNNING','started_at_utc'=>now('UTC')));if(!$run-
>wasRecentlyCreated)return$run;$result=$evaluator($v,$subjectType,$subjectId,$from,$to);MeasureResult::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$tenantId,'measure_run_public_id'=>$run-
>public_id,'result_type'=>$result['type'],'result_payload'=>$result,'evidence'=>$result['evidence']??[]);$run-
>forceFill(['status'=>'COMPLETE','completed_at_utc'=>now('UTC')])>save();return$run;}}

```

src/Contracts/AuthorizationDecisionGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/AuthorizationDecisionGateway.php	01cdba95198ff7f50424276fdd4aec2f4ce845bb53bb831a14fb4e1869de444e

```

<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface AuthorizationDecisionGateway{public
function decide(string $tenantId,string $actorId,string $purpose,array $scope):array;}

```

src/Contracts/DeidentificationGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/DeidentificationGateway.php	f56dc77a73d13224b37677009355ed77446c9576170ce24bf82a05d63d71eb9f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface DeidentificationGateway{public function
transform(string $tenantId,string $mode,array $records,array $policy):array;}
```

src/Contracts/FhirMapper.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/FhirMapper.php	84c3d2b8d211da3cf227c07dd6f48040e551ac2a4947d5776848750e10ffd345

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface FhirMapper{public function map(string
$resourceType,string $sourcePublicId,string $profileVersion):array;}
```

src/Contracts/ObjectStore.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObjectStore.php	aa8c81b228c4a25515d7657e3c139b7f205e9a2e5bb8ff05b4f5a34780a0df72

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface ObjectStore{public function
putEncrypted(string $tenantId,string $path,string $content,string $contentType,array $metadata):array;}
```

src/Data/CreateExchangeJobData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/CreateExchangeJobData.php	fe7780063647fb3fe4017b69ca19b0f24b29e5549ac88c233d32d69384bb6838

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Data;final readonly class CreateExchangeJobData{public
function __construct(public string $tenantId,public string $actorId,public string $purpose,public string $format,public array
$scope,public string $deidentificationMode,public string $profileVersion,public \DateTimeImmutable $requestedAt){}}
```

src/Enums/CohortStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CohortStatus.php	dc3b0fd607dcd12b3336c31fbf2a993029ea54058dfd9649ef589794f61137af

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum CohortStatus:string{case DRAFT='DRAFT';case
VALIDATED='VALIDATED';case APPROVED='APPROVED';case ACTIVE='ACTIVE';case RETIRED='RETIRED';}
```

src/Enums/DeidentificationMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DeidentificationMode.php	60aa0128987c16d92bcef26b7ca40f18d9d808a56053098cb71bb6e0fbacd5bc

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum DeidentificationMode:string{case
IDENTIFIED='IDENTIFIED';case PSEUDONYMIZED='PSEUDONYMIZED';case DEIDENTIFIED='DEIDENTIFIED';case
AGGREGATED='AGGREGATED';}
```

src/Enums/ExchangeStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ExchangeStatus.php	c352679dec17fecdd6ba2460eb39ae2d7d834bac5fffe713eec706b22d013441b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum ExchangeStatus:string{case
REQUESTED='REQUESTED';case AUTHORIZED='AUTHORIZED';case GENERATING='GENERATING';case READY='READY';case
DELIVERED='DELIVERED';case FAILED='FAILED';case EXPIRED='EXPIRED';case REVOKED='REVOKED';}
```

src/Enums/MappingStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MappingStatus.php	a64e17cccd82fed4eea3ca591eefd7e928dd7226d3479606679640ab5c8c2020

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum MappingStatus:string{case DRAFT='DRAFT';case
VERIFIED='VERIFIED';case ACTIVE='ACTIVE';case DEPRECATED='DEPRECATED';}
```

src/Enums/MeasureRunStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MeasureRunStatus.php	e4285fe0e101d856b0c03576e296af4175d44529149ef950baacf885cc61d2ea

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum MeasureRunStatus:string{case
QUEUED='QUEUED';case RUNNING='RUNNING';case COMPLETE='COMPLETE';case ERROR='ERROR';case
CANCELLED='CANCELLED';}
```

src/Enums/PurposeCategory.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/PurposeCategory.php	beebf852daba33cb0904153a28cc5b028888a3f03f36e2434cc50d9405ef6753

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum PurposeCategory:string{case
TREATMENT='TREATMENT';case OPERATIONS='OPERATIONS';case PUBLIC_HEALTH='PUBLIC_HEALTH';case
QUALITY='QUALITY';case RESEARCH='RESEARCH';}
```

src/Enums/SubmissionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Enums/SubmissionStatus.php	952ba321cd8bd0cf2c73864f5b4390503362d90fef84b9ff974b5a6787e6c0a4

CANDIDATE		
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum SubmissionStatus:string{case
DRAFT='DRAFT';case VALIDATED='VALIDATED';case SUBMITTED='SUBMITTED';case ACCEPTED='ACCEPTED';case
REJECTED='REJECTED';case CORRECTED='CORRECTED';}
```

src/Enums/SubscriptionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/SubscriptionStatus.php	0f2c59b5953a7fd923ad99ba91d3c7969aefcb9ef0170dc41179bc64953936f3

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Enums;enum SubscriptionStatus:string{case
REQUESTED='REQUESTED';case ACTIVE='ACTIVE';case ERROR='ERROR';case SUSPENDED='SUSPENDED';case ENDED='ENDED';}
```

src/InteroperabilityServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/InteroperabilityServiceProvider.php	2f023f40bab5a124b6928e825e9f8724d8cebd2b87ea5c66836dfdcd3c40462

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability;use Illuminate\Support\ServiceProvider;final class
InteroperabilityServiceProvider extends ServiceProvider{public function register():void{$this->mergeConfigFrom(__DIR__.'../config/clinical-interoperability.php','clinical-interoperability');}public function boot():void{$this->loadMigrationsFrom(__DIR__.'../database/migrations');$this->loadRoutesFrom(__DIR__.'../routes/clinical-interoperability.php');}
```

src/Models/CohortDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CohortDefinition.php	4a878a0986f74ec41d30db3036afaad92ffd1af2d25a92452e95810d8b6b811b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
CohortDefinition extends Model{public $timestamps=false;protected $table='clinical_cohort_definitions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/CohortMembershipSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CohortMembershipSnapshot.php	f5794b10a43f3da57c9d1e3fff386968ee07af2bcf9b80dec55f833f4f9df86

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
CohortMembershipSnapshot extends Model{public $timestamps=false;protected
$table='clinical_cohort_membership_snapshots';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/CohortVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CohortVersion.php	92cc9481609aa1ee507efc82c9c7ed038c7357908837307d498c3c07bd90d02f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
CohortVersion extends Model{public $timestamps=false;protected $table='clinical_cohort_versions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeArtifact.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeArtifact.php	7e3d7e7e2e3a8d7577c38278e2a7f96b74d60e725c906c56cb3089f242e9ebc9

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeArtifact extends Model{public $timestamps=false;protected $table='clinical_exchange_artifacts';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeDelivery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeDelivery.php	f654d329c3a128235fe7413e30f20011f883b30479d668f90d364276ead87d55

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeDelivery extends Model{public $timestamps=false;protected $table='clinical_exchange_deliveries';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeEndpoint.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeEndpoint.php	01f14317d12bbe07bce7fbb2ea19836333c31de42b946b7ad0261e4d76bd1b9c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeEndpoint extends Model{public $timestamps=false;protected $table='clinical_exchange_endpoints';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeJob.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeJob.php	44f824d44f955643cac26cc46ce277f0e2d08ad8187b0a8b395a2f283efc1a2d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeJob extends Model{public $timestamps=false;protected $table='clinical_exchange_jobs';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/InteroperabilityReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/InteroperabilityReconciliation.php	f47f903adb6788bfc39a3794c523225a0dcc3148a757fbb6e174744a63915e12

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
InteroperabilityReconciliation extends Model{public $timestamps=false;protected
$table='clinical_interoperability_reconciliations';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureDefinition.php	e0cc5099ad921c86cd11abf364d8b2f7248a026946e37f02da1dd1ef635928e5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureDefinition extends Model{public $timestamps=false;protected $table='clinical_measure_definitions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureResult.php	5422fc3645c5777b486e1ab47f41e3e24d45b8f3da4b6c95f8abad142d6fec0c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureResult extends Model{public $timestamps=false;protected $table='clinical_measure_results';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureRun.php	fa5453b1ab02c205e0d42caff7873c1bfbfb9f847d478a720ab84f634090122a

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureRun extends Model{public $timestamps=false;protected $table='clinical_measure_runs';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureVersion.php	0310365af341bd17e9e88e24e982c40c5e06edc79ece66e6d27a04493ab97dee

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureVersion extends Model{public $timestamps=false;protected $table='clinical_measure_versions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/PublicHealthSubmission.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PublicHealthSubmission.php	ff9c957a1cec6cf0c83b4b42fb85c0907e5219291b9ac6eb9efb66437edb50c3

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
PublicHealthSubmission extends Model{public $timestamps=false;protected
$table='clinical_public_health_submissions';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ResearchDatasetArtifact.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResearchDatasetArtifact.php	ee583c398cada0d4417c02bdad8ab265be9e9a1032cb5d11b35e2aeftaa4b639

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
ResearchDatasetArtifact extends Model{public $timestamps=false;protected $table='clinical_research_dataset_artifacts';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ResearchDatasetRequest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResearchDatasetRequest.php	1017c0119a2d80712ea45e8196fbf96aa77d40299dfe9d57a2f46df6cbb51a47

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
ResearchDatasetRequest extends Model{public $timestamps=false;protected
$table='clinical_research_dataset_requests';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/SubscriptionDelivery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SubscriptionDelivery.php	ceb4938c31fbc758ec60648447a42a002d376eca941e2bbe28713aa12a5f199b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
SubscriptionDelivery extends Model{public $timestamps=false;protected $table='clinical_subscription_deliveries';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/SubscriptionRegistration.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SubscriptionRegistration.php	32e5e612594aa0fbdf5b291f9439ec38b279500d6c712f5b19598fcef46883c9

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
SubscriptionRegistration extends Model{public $timestamps=false;protected $table='clinical_subscription_registrations';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/SubscriptionTopic.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Models/SubscriptionTopic.php	834de73eff3db3b69bfd3417e7f7e485cd8bdcbce5cbb84463095b9c20314d89f

CANDIDATE		
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
SubscriptionTopic extends Model{public $timestamps=false;protected $table='clinical_subscription_topics';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/TerminologyMapping.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TerminologyMapping.php	461f963b8756314254e0890fb65c6d26b0dc2516f97b6ebcfc563df073d9e3d4

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Models;use Illuminate\Database\Eloquent\Model;final class
TerminologyMapping extends Model{public $timestamps=false;protected $table='clinical_terminology_mappings';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volumes 1-11	Authoritative clinical data, privacy, release and operating governance
Main Module	Identity, consent, purpose and confidentiality decisions
FHIR layer	Profiles, CapabilityStatement, validation and mapping
Terminology	CodeSystem, ValueSet and ConceptMap services
Object storage	Encrypted artifacts and expiring delivery
Public health	Program and submission authority
Research governance	Protocol, approval and de-identification policy
Vue	Exchange jobs, subscriptions, cohorts, measures and approvals

Blocking Gaps

Approve Volume 12 as a supplemental baseline

Select supported FHIR release and implementation guides

Approve public FHIR profile catalogue and versioning

Bind authorization and consent decision services

Approve de-identification and disclosure-risk method

Bind terminology validation and mapping

Approve public-health and research governance workflows

Complete controllers, requests, resources, workers and Vue interfaces

Run conformance, privacy, tenant-isolation, bulk-scale, retry and reproducibility tests

Verification

```
composer validate --strict
```

```
composer dump-autoload --strict-psr
```

```
php artisan migrate:fresh --env=testing
```

```
php artisan migrate:rollback --step=1 --env=testing
```

```
php artisan migrate --env=testing
```

```
./vendor/bin/phpunit
```

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Clinical source-of-truth and production baseline
Volumes 1-11	Clinical, privacy, release and operational controls
HL7 FHIR R5 Subscriptions	Topic-based notification reference
HL7 FHIR Bulk Data Access	Large-dataset asynchronous exchange reference
HL7 FHIR R5 MeasureReport	Quality-measure result reference

KASHTRE CLINICAL MODULE**CODE-READY ENGINEERING DESIGN DOCUMENT**

Volume 13 Authoritative Correction Notice

This correction supersedes inherited Volume 12 exchange, cohort, measure, public-health and research entity names wherever they appear inside Volume 13. Those inherited names are retained only as historical draft evidence and shall not be implemented as the Volume 13 domain model.

Corrected Volume 13 Entity Register

Entity	Purpose
PatientPortalProjection	Authorized patient-facing clinical view
ProxyRelationship	Verified, purpose-limited and effective-dated representative authority
QuestionnaireDefinition	Versioned patient-facing questionnaire
QuestionnaireAssignment	Patient or proxy assignment and due state
PatientReportedResponse	Partial or completed structured response
CommunicationThread	Secure patient-care conversation context
PatientMessage	Versioned message and delivery evidence
RemoteMonitoringEnrollment	Approved remote monitoring program assignment
RemoteDeviceAssociation	Effective-dated device-to-patient association
RemoteMeasurement	Measured, transmitted and received values with provenance
RemoteReviewTask	Clinical review and escalation obligation
VirtualCareSession	Remote encounter continuity and outcome
EngagementReconciliation	Identity, device, delivery or mapping exception

Corrected Volume 13 Package Boundary

- Package: kashtre/clinical-engagement.
- Namespace: KashTre\Clinical\Engagement.
- Tables use the clinical_engagement, clinical_patient_message, clinical_questionnaire and clinical_remote_monitoring families.
- Patient-entered information remains distinguishable from clinician-entered information.
- A threshold breach creates review work and never an autonomous diagnosis.
- Proxy access requires current relationship, purpose, scope, consent and confidentiality clearance.

Volume 13 Required Engineering Completion

- Replace inherited interoperability source listings with engagement-specific models, migrations, actions, policies, APIs and tests.
- Add proxy expiry and revocation tests.
- Add questionnaire cardinality, type and correction tests.
- Add secure-message delivery, failure escalation and sensitive-preview tests.
- Add duplicate and out-of-order remote-measurement tests.
- Add virtual-care encounter and provenance tests.

SUPPLEMENTAL VOLUME 13**PATIENT ENGAGEMENT, REMOTE CARE AND PATIENT-REPORTED DATA****Portal Access, Proxy Relationships, Questionnaires, Messaging, Remote Monitoring and Escalation**

Version: 6.1.0 Comprehensive Draft

DRAFT FOR ENGINEERING REVIEW | PICK-AND-DROP CANDIDATE | NOT APPROVED FOR PRODUCTION

Status: Supplemental to the Phase 1-10 functional baseline

Depends on: Volumes 1 to 12

Production: Not Approved for Production

Scope Position and Governing Rule

Volume 13 is supplemental. It exposes governed representations of authoritative Clinical data without creating a second clinical source of truth. Every exchange records purpose, authorization, scope, profile version, de-identification mode, artifact hash, delivery evidence and expiry.

Patient or proxy authentication -> relationship and consent check -> questionnaire, message or device submission -> validation -> Clinical projection -> review or escalation -> acknowledgement -> follow-up and closure

Entity Register

Entity	Table	Invariant
ExchangeEndpoint	clinical_exchange_endpoints	Tenant, patient, proxy, consent and provenance preserved
ExchangeJob	clinical_exchange_jobs	Tenant, patient, proxy, consent and provenance preserved
ExchangeArtifact	clinical_exchange_artifacts	Tenant, patient, proxy, consent and provenance preserved
ExchangeDelivery	clinical_exchange_deliveries	Tenant, patient, proxy, consent and provenance preserved
SubscriptionTopic	clinical_subscription_topics	Tenant, patient, proxy, consent and provenance preserved
SubscriptionRegistration	clinical_subscription_registrations	Tenant, patient, proxy, consent and provenance preserved
SubscriptionDelivery	clinical_subscription_deliveries	Tenant, patient, proxy, consent and provenance preserved
CohortDefinition	clinical_cohort_definitions	Tenant, patient, proxy, consent and provenance preserved
CohortVersion	clinical_cohort_versions	Tenant, patient, proxy, consent and provenance preserved
CohortMembershipSnapshot	clinical_cohort_membership_snapshots	Tenant, patient, proxy, consent and provenance preserved
MeasureDefinition	clinical_measure_definitions	Tenant, patient, proxy, consent and provenance preserved
MeasureVersion	clinical_measure_versions	Tenant, patient, proxy, consent and provenance preserved
MeasureRun	clinical_measure_runs	Tenant, patient, proxy, consent and provenance preserved
MeasureResult	clinical_measure_results	Tenant, patient, proxy, consent and provenance preserved
PublicHealthSubmission	clinical_public_health_submissions	Tenant, patient, proxy, consent and provenance preserved
ResearchDatasetRequest	clinical_research_dataset_requests	Tenant, patient, proxy, consent and provenance preserved
ResearchDatasetArtifact	clinical_research_dataset_artifacts	Tenant, patient, proxy, consent and provenance preserved
TerminologyMapping	clinical_terminology_mappings	Tenant, patient, proxy, consent and provenance preserved
InteroperabilityReconciliation	clinical_interoperability_reconciliations	Tenant, patient, proxy, consent and provenance preserved

Engineering Design**Patient and proxy access**

Patient-facing views use authorized projections

Proxy authority is purpose- and time-bounded

Sensitive-content rules are enforced server-side

Patient portal never exposes Clinical tables directly

Identity or relationship ambiguity enters reconciliation

Corrections remain visible as new versions

Secure communications

Threaded patient-care communication

Patient, proxy and care-team recipient verification

Idempotent message delivery reference

Bounded delivery retry

Escalate failed urgent communication

Notification preview contains minimum necessary content

Patient-reported data

Versioned questionnaires and response assignments

Complete and partial response states

Answer validation and provenance

Patient-entered data remains distinguishable from clinician-entered data

Approved response extraction into Observations

No automatic clinical conclusion from a response

Remote monitoring

Registered device and patient association

Measured, transmitted and received times retained

Quality and plausibility flags

Duplicate and out-of-order handling

Threshold breach creates review work, not autonomous diagnosis

Virtual-care continuity

Remote-service assignment and encounter relationship required

Consent and communication preferences respected

Urgent symptoms use configured escalation instructions

Video or teleconsultation provider integration remains external

Clinical record stores encounter outcome and provenance

Failed contact attempts and follow-up retained

Permissions

Permission	Purpose
clinical.patient_portal.view	View authorized patient-facing record
clinical.proxy.manage	Manage verified proxy relationship
clinical.questionnaire.assign	Assign patient questionnaire
clinical.patient_message.respond	Respond to patient message

clinical.remote_monitoring.enrol	Enrol remote monitoring
clinical.remote_monitoring.review	Review transmitted measurement
clinical.virtual_care.document	Document virtual-care outcome
clinical.patient_reported.review	Review patient-reported data
clinical.patient_communication.escalate	Escalate urgent communication
clinical.engagement.reconcile	Resolve identity, device or delivery exception

Mandatory Tests

Unverified proxy cannot access patient content

Expired proxy authority is rejected

Patient response remains distinguishable from clinician observation

Questionnaire validation rejects invalid answer type

Duplicate device transmission is idempotent

Out-of-order measurement preserves measured time

Threshold breach creates review work once

Urgent message delivery failure escalates

Cross-tenant portal access is concealed

Notification preview excludes sensitive content

Virtual-care outcome retains encounter provenance

Correction preserves prior patient-reported version

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	058533b14c6091afb5f50b7f020a55daaf7bad6038fcbc1088989465f15709e9

KashTre Clinical Interoperability, Population Health and Research v6.1.0

Supplemental Volume 12. Pick-and-drop candidate. Bind consent/authorization, FHIR mapping, terminology, object storage, de-identification, public-health authority, research governance, subscriptions and quality-measure execution before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	3b442285ee70b14cd63b5f1953f0ebe51cbb76419ae71939d8e6fef48abc2542

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{
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  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
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      "path": "README.md",
      "sha256": "058533b14c6091afb5f50b7f020a55daaf7bad6038fcbc1088989465f15709e9",
      "bytes": 327
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    {
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      "sha256": "312e8ae53983bec7e92436b617efe1ca44419b9788ada003484ca4289426d70f",
      "bytes": 395
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    {
      "path": "config/clinical-engagement.php",
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      "bytes": 311
    },
    {
      "path": "database/migrations/2026_08_28_1100000_create_interoperability_tables.php",
      "sha256": "28b164b527d0a531075b3aec41009e2a7cbbb141efbfa58742b6b2d24f578279",
      "bytes": 5116
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    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-engagement.php",
      "sha256": "d77e8442b6a8a6fd8783eded6e65090984116f551d978d64b75383c92aa713e6",
      "bytes": 277
    },
    {

```

```

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"bytes": 1274
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"sha256": "01cdba95198ff7f50424276fdd4aec2f4ce845bb53bb831a14fb4e1869de444e",
"bytes": 216
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{
"path": "src/Contracts/DeidentificationGateway.php",
"sha256": "f56dc77a73d13224b37677009355ed77446c9576170ce24bf82a05d63d71eb9f",
"bytes": 211
},
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"sha256": "84c3d2b8d211da3cf227c07dd6f48040e551ac2a4947d5776848750e10ffd345",
"bytes": 200
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{
"path": "src/Contracts/ObjectStore.php",
"sha256": "aa8c81b228c4a25515d7657e3c139b7f205e9a2e5bb8ff05b4f5a34780a0df72",
"bytes": 225
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"bytes": 370
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"bytes": 215
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"sha256": "d5284305380c8106019d7f67988c353658a55e3e5c0f7b8dad79a2ddb189a30e",
"bytes": 234
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{
"path": "src/Enums/ExchangeStatus.php",
"sha256": "73713b6a5c3907a5fbe51998cd1c77303547c894bb5f94190f083331fbe6f835",
"bytes": 300
},

```

```

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  "bytes": 195
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  "path": "src/Enums/MeasureRunStatus.php",
  "sha256": "24c88298c37359ffb08334baaa9232065c5a909d106ea37ca04455758200839e",
  "bytes": 219
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  "path": "src/Enums/PurposeCategory.php",
  "sha256": "6f30b92e3efd3b8ff74364ad192be38ec11600e74b982cb3f171a67b5046dc8c",
  "bytes": 242
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  "path": "src/Enums/SubmissionStatus.php",
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  "bytes": 254
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  "path": "src/Enums/SubscriptionStatus.php",
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  "bytes": 219
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  "bytes": 467
},
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  "path": "src/Models/CohortMembershipSnapshot.php",
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  "bytes": 484
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  "path": "src/Models/CohortVersion.php",
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  "bytes": 461
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  "path": "src/Models/ExchangeArtifact.php",
  "sha256": "ceb976dfbddd2e78318804cbb8fe5844e6a61797cb884d5e7c90b1e4446f4354",
  "bytes": 467
},
{
  "path": "src/Models/ExchangeDelivery.php",
  "sha256": "2412cd595bac256e4c3a6d54f434f0366e1287e829f98fc2b7f8ae426acd41fa",
  "bytes": 468
},
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  "path": "src/Models/ExchangeEndpoint.php",
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  "bytes": 467
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```

```

{
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  "bytes": 457
},
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  "path": "src/Models/InteroperabilityReconciliation.php",
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  "bytes": 495
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  "bytes": 469
},
{
  "path": "src/Models/MeasureResult.php",
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  "bytes": 461
},
{
  "path": "src/Models/MeasureRun.php",
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  "bytes": 455
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{
  "path": "src/Models/MeasureVersion.php",
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  "bytes": 463
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  "path": "src/Models/PublicHealthSubmission.php",
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  "bytes": 480
},
{
  "path": "src/Models/ResearchDatasetArtifact.php",
  "sha256": "a7b296f8dcd92a40ded44a77233580ac09c3947b35a64fa6a58c80b58f4ff24c",
  "bytes": 482
},
{
  "path": "src/Models/ResearchDatasetRequest.php",
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  "bytes": 480
},
{
  "path": "src/Models/SubscriptionDelivery.php",
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  "bytes": 476
},
{
  "path": "src/Models/SubscriptionRegistration.php",
  "sha256": "d5f426dc6aeb8c9e8e223a43a963487ccf98625e2e495682a99ad65bc5ed90b5",
  "bytes": 483
},

```

```
{
  "path": "src/Models/SubscriptionTopic.php",
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  "bytes": 469
},
{
  "path": "src/Models/TerminologyMapping.php",
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  "bytes": 471
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{
  "path": "src/PatientEngagementServiceProvider.php",
  "sha256": "2882ea5d56b5f82a69be038c45e1a1767e0ffdc187330fc2595f9a7e12b18218",
  "bytes": 465
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
}
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	312e8ae53983bec7e92436b617efe1ca44419b9788ada003484ca4289426d70f

```
{
  "name": "kashtre/clinical-engagement",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\Engagement\\": "src/"
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\Engagement\\PatientEngagementServiceProvider"
          ]
        }
      }
    }
  }
}
```

config/clinical-engagement.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-engagement.php	cce96e032e6130123d86158415d4ef04de0d19aeebe19ee2e1c18959522c4e4c

```
<?php
declare(strict_types=1);return
['bulk_job_expiry_hours'=>(int)env('CLINICAL_BULK_JOB_EXPIRY_HOURS',24),'subscription_retry_limit'=>(int)env('CLINICAL_SUBSCRIPTION_RETRY_LIMIT',8),'research_export_requires_approval'=>true,'direct_identifier_export_default'=>false,'measure_reproducibility_required'=>true];
```

database/migrations/2026_08_28_1100000_create_interoperability_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_1100000_create_interoperability_tables.php	28b164b527d0a531075b3aec41009e2a7cbbb141efbfa58742b6b2d24f578279

```

<?php
declare(strict_types=1);use Illuminate\Database\Migrations\Migration;use Illuminate\Database\Schema\Blueprint;use
Illuminate\Support\Facades\Schema;return new class extends Migration{public function up():void{
Schema::create('clinical_exchange_endpoints',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('code',96);$tbl->string('endpoint_type',64);$tbl->string('base_url',500);$tbl->string('status',24);$tbl-
>json('capabilities');$tbl->unique(['tenant_id','code'],'uq_exchange_endpoint');});
Schema::create('clinical_exchange_jobs',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('requested_by_public_id');$tbl->string('purpose_code',64);$tbl->string('format',64);$tbl->json('scope');$tbl-
>string('deidentification_mode',32);$tbl->string('profile_version',96);$tbl->string('authorization_reference',240)->nullable();$tbl-
>string('status',24);$tbl->dateTime('requested_at_utc',6);$tbl->dateTime('completed_at_utc',6)->nullable();$tbl-
>dateTime('expires_at_utc',6));});
Schema::create('clinical_exchange_artifacts',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->ulid('exchange_job_public_id');$tbl->string('storage_reference',500);$tbl->string('content_type',128);$tbl-
>char('sha256',64);$tbl->unsignedBigInteger('record_count');$tbl->string('status',24);$tbl->dateTime('created_at_utc',6));});
Schema::create('clinical_subscription_registrations',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('topic_canonical',500);$tbl->string('endpoint_reference',500);$tbl->json('filters');$tbl-
>string('status',24);$tbl->dateTime('effective_from_utc',6);$tbl->dateTime('effective_to_utc',6)->nullable();});
Schema::create('clinical_subscription_deliveries',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->ulid('subscription_public_id');$tbl->string('event_reference',240);$tbl->string('status',24);$tbl-
>unsignedInteger('attempt_no');$tbl->dateTime('attempted_at_utc',6);$tbl->string('provider_reference',240)->nullable();$tbl-
>unique(['subscription_public_id','event_reference','attempt_no'],'uq_subscription_attempt');});
Schema::create('clinical_cohort_definitions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('code',96);$tbl->string('name',180);$tbl->string('status',24);$tbl->unsignedInteger('current_version')-
>nullable();$tbl->unique(['tenant_id','code'],'uq_cohort_code');});
Schema::create('clinical_cohort_versions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('cohort_definition_public_id');$tbl->unsignedInteger('version_no');$tbl->json('criteria');$tbl->string('status',24);$tbl-
>dateTime('effective_from_utc',6);$tbl->unique(['cohort_definition_public_id','version_no'],'uq_cohort_version');});
Schema::create('clinical_measure_versions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('measure_code',96);$tbl->unsignedInteger('version_no');$tbl->string('status',24);$tbl->json('logic');$tbl-
>string('implementation_version',96);$tbl->unique(['tenant_id','measure_code','version_no'],'uq_measure_version');});
Schema::create('clinical_measure_runs',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('measure_version_public_id');$tbl->string('subject_type',32);$tbl->ulid('subject_public_id');$tbl->dateTime('period_from_utc',6);$tbl-
>dateTime('period_to_utc',6);$tbl->char('input_fingerprint',64);$tbl->string('status',24);$tbl->dateTime('started_at_utc',6);$tbl-
>dateTime('completed_at_utc',6)->nullable();$tbl-
>unique(['tenant_id','measure_version_public_id','input_fingerprint'],'uq_measure_run');});
Schema::create('clinical_measure_results',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl->string('tenant_id',64);$tbl-
>ulid('measure_run_public_id');$tbl->string('result_type',32);$tbl->json('result_payload');$tbl->json('evidence');});
Schema::create('clinical_public_health_submissions',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('program_code',96);$tbl->string('status',24);$tbl->json('scope');$tbl->string('authority_reference',240);$tbl-
>dateTime('submitted_at_utc',6)->nullable();});
Schema::create('clinical_research_dataset_requests',function(Blueprint$tbl){$tbl->id();$tbl->ulid('public_id')->unique();$tbl-
>string('tenant_id',64);$tbl->string('protocol_reference',240);$tbl->string('purpose_code',64);$tbl->string('deidentification_mode',32);$tbl-
>json('scope');$tbl->string('status',24);$tbl->ulid('requested_by_public_id');$tbl->dateTime('requested_at_utc',6));});
}public function
down():void{foreach(['clinical_research_dataset_requests','clinical_public_health_submissions','clinical_measure_results','clinical_
measure_runs','clinical_measure_versions','clinical_cohort_versions','clinical_cohort_definitions','clinical_subscription_deliveries','
clinical_subscription_registrations','clinical_exchange_artifacts','clinical_exchange_jobs','clinical_exchange_endpoints']as$tbl)Schem
a::dropIfExists($tbl);};

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```
#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force
```

routes/clinical-engagement.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	routes/clinical-engagement.php	d77e8442b6a8a6fd8783eded6e65090984116f551d978d64b75383c92aa713e6

```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])-
>prefix('api/v1/clinical/engagement')->group(function(){Route::get('/jobs',fn()->response()-
>json(['error'=>['code'=>'CLN_EXCHANGE_HOST_CONTROLLER_REQUIRED'],501)]));});
```

src/Actions/CreateExchangeJob.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CreateExchangeJob.php	9417aaf0b8b4a67be742e43d818eb3f246f77d2632867e8f0e66fb2f042f529f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Actions;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Interoperability\Contracts\AuthorizationDecisionGateway;use
KashTre\Clinical\Interoperability\Data\CreateExchangeJobData;use KashTre\Clinical\Interoperability\Models\ExchangeJob;final
readonly class CreateExchangeJob{public function __construct(private PublicIdGenerator $ids,private
AuthorizationDecisionGateway $authorization){}public function execute(CreateExchangeJobData $d):ExchangeJob{$decision=$this-
>authorization->decide($d->tenantId,$d->actorId,$d->purpose,$d->scope);if(($decision['decision']??'DENY')!=='PERMIT')throw new
\DomainException('CLN_EXCHANGE_NOT_AUTHORIZED');return ExchangeJob::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$d->tenantId,'requested_by_public_id'=>$d->actorId,'purpose_code'=>$d->purpose,'format'=>$d-
>format,'scope'=>$d->scope,'deidentification_mode'=>$d->deidentificationMode,'profile_version'=>$d-
>profileVersion,'authorization_reference'=>$decision['reference']??null,'status'=>'AUTHORIZED','requested_at_utc'=>$d-
>requestedAt,'expires_at_utc'=>$d->requestedAt->modify('+'.config('clinical-engagement.bulk_job_expiry_hours').' hours'))};}
```

src/Actions/GenerateExchangeArtifact.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/GenerateExchangeArtifact.php	ef31b04f3c293768f5ff13bf52607c24475143f116f4cf82b1e257ed10bd9876

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Interoperability\Contracts\{DeidentificationGateway,FhirMapper,ObjectStore};use
KashTre\Clinical\Interoperability\Models\{ExchangeArtifact,ExchangeJob};final readonly class GenerateExchangeArtifact{public
function __construct(private PublicIdGenerator $ids,private FhirMapper $mapper,private DeidentificationGateway $deid,private
ObjectStore $store){}public function execute(string $tenantId,string $jobId,array $sourceRecords):ExchangeArtifact{return
DB::transaction(function(){use($tenantId,$jobId,$sourceRecords){$j=ExchangeJob::query()->where('tenant_id',$tenantId)-
>where('public_id',$jobId)->where('status','AUTHORIZED')->lockForUpdate()->firstOrFail();$mapped=[];foreach($sourceRecords
as$r)$mapped[]=$this->mapper->map($r['type'],$r['public_id'],$j->profile_version);$payload=$this->deid->transform($tenantId,$j-
>deidentification_mode,$mapped,$j-
```

```

>scope);$content=implode("\n",array_map(fn($x)=>json_encode($x,JSON_THROW_ON_ERROR),$payload));$sha=hash('sha256',$content);$stored=$this->store-
>putEncrypted($tenantId,"exchange/$jobId/$sha.ndjson",$content,'application/fhir+ndjson',['sha256'=>$sha]);$a=ExchangeArtifact::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'exchange_job_public_id'=>$jobId,'storage_reference'=>$stored['reference'],'content_type'=>'application/fhir+ndjson','sha256'=>$sha,'record_count'=>count($payload),'status'=>'READY','created_at_utc'=>now('UTC')));$j-
>forceFill(['status'=>'READY','completed_at_utc'=>now('UTC')])>save();return$a;});}

```

src/Actions/RunQualityMeasure.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/RunQualityMeasure.php	c8ffadfbdc9593e91305ecdcbfc45039b8ada883235cde629129bcb6020d8b88

```

<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\Interoperability\Models\{MeasureRun,MeasureResult,MeasureVersion};final readonly class
RunQualityMeasure{public function __construct(private PublicIdGenerator $ids){}public function execute(string $tenantId,string
$measureVersionId,string $subjectType,string $subjectId,\DateTimeImmutable $from,\DateTimeImmutable $to,callable
$evaluator):MeasureRun{return
DB::transaction(function()use($tenantId,$measureVersionId,$subjectType,$subjectId,$from,$to,$evaluator){$v=MeasureVersion::q
uery()->where('tenant_id',$tenantId)->where('public_id',$measureVersionId)->where('status','ACTIVE')-
>firstOrFail();$input=['measure'=>$measureVersionId,'subject'=>$subjectId,'from'=>$from->format('DATE_ATOM'),'to'=>$to-
>format('DATE_ATOM')];$hash=hash('sha256',json_encode($input,JSON_THROW_ON_ERROR));$run=MeasureRun::query()-
>firstOrCreate(['tenant_id'=>$tenantId,'measure_version_public_id'=>$measureVersionId,'input_fingerprint'=>$hash],['public_id'=>(
string)$this->ids-
>generate(),'subject_type'=>$subjectType,'subject_public_id'=>$subjectId,'period_from_utc'=>$from,'period_to_utc'=>$to,'status'=>
'RUNNING','started_at_utc'=>now('UTC')));if(!$run-
>wasRecentlyCreated)return$run;$result=$evaluator($v,$subjectType,$subjectId,$from,$to);MeasureResult::query()-
>create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$tenantId,'measure_run_public_id'=>$run-
>public_id,'result_type'=>$result['type'],'result_payload'=>$result,'evidence'=>$result['evidence']??[]);$run-
>forceFill(['status'=>'COMPLETE','completed_at_utc'=>now('UTC')])>save();return$run;});}

```

src/Contracts/AuthorizationDecisionGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/AuthorizationDecisionGateway.php	01cdba95198ff7f50424276fdd4aec2f4ce845bb53bb831a14fb4e1869de444e

```

<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface AuthorizationDecisionGateway{public
function decide(string $tenantId,string $actorId,string $purpose,array $scope):array;}

```

src/Contracts/DeidentificationGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/DeidentificationGateway.php	f56dc77a73d13224b37677009355ed77446c9576170ce24bf82a05d63d71eb9f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface DeidentificationGateway{public function
transform(string $tenantId,string $mode,array $records,array $policy):array;}
```

src/Contracts/FhirMapper.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/FhirMapper.php	84c3d2b8d211da3cf227c07dd6f48040e551ac2a4947d5776848750e10ffd345

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface FhirMapper{public function map(string
$resourceType,string $sourcePublicId,string $profileVersion):array;}
```

src/Contracts/ObjectStore.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ObjectStore.php	aa8c81b228c4a25515d7657e3c139b7f205e9a2e5bb8ff05b4f5a34780a0df72

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Contracts;interface ObjectStore{public function
putEncrypted(string $tenantId,string $path,string $content,string $contentType,array $metadata):array;}
```

src/Data/CreateExchangeJobData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/CreateExchangeJobData.php	fe7780063647fb3fe4017b69ca19b0f24b29e5549ac88c233d32d69384bb6838

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability\Data;final readonly class CreateExchangeJobData{public
function __construct(public string $tenantId,public string $actorId,public string $purpose,public string $format,public array
$scope,public string $deidentificationMode,public string $profileVersion,public \DateTimeImmutable $requestedAt){}}
```

src/Enums/CohortStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/CohortStatus.php	e492270990c76555dd1d141e03a9799928500965de77153a16aa99f5d784f76a

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum CohortStatus:string{case DRAFT='DRAFT';case
VALIDATED='VALIDATED';case APPROVED='APPROVED';case ACTIVE='ACTIVE';case RETIRED='RETIRED';}
```

src/Enums/DeidentificationMode.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DeidentificationMode.php	d5284305380c8106019d7f67988c353658a55e3e5c0f7b8dad79a2ddb189a30e

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum DeidentificationMode:string{case IDENTIFIED='IDENTIFIED';case PSEUDONYMIZED='PSEUDONYMIZED';case DEIDENTIFIED='DEIDENTIFIED';case AGGREGATED='AGGREGATED';}
```

src/Enums/ExchangeStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ExchangeStatus.php	73713b6a5c3907a5fbe51998cd1c77303547c894bb5f94190f083331fbe6f835

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum ExchangeStatus:string{case REQUESTED='REQUESTED';case AUTHORIZED='AUTHORIZED';case GENERATING='GENERATING';case READY='READY';case DELIVERED='DELIVERED';case FAILED='FAILED';case EXPIRED='EXPIRED';case REVOKED='REVOKED';}
```

src/Enums/MappingStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MappingStatus.php	1b104302edaac150f8389d9f885ce9198d47bf8f099c8bc3bee5e420b56ed0f4

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum MappingStatus:string{case DRAFT='DRAFT';case VERIFIED='VERIFIED';case ACTIVE='ACTIVE';case DEPRECATED='DEPRECATED';}
```

src/Enums/MeasureRunStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/MeasureRunStatus.php	24c88298c37359ffb08334baaa9232065c5a909d106ea37ca04455758200839e

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum MeasureRunStatus:string{case QUEUED='QUEUED';case RUNNING='RUNNING';case COMPLETE='COMPLETE';case ERROR='ERROR';case CANCELLED='CANCELLED';}
```

src/Enums/PurposeCategory.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/PurposeCategory.php	6f30b92e3efd3b8ff74364ad192be38ec11600e74b982cb3f171a67b5046dc8c

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum PurposeCategory:string{case TREATMENT='TREATMENT';case OPERATIONS='OPERATIONS';case PUBLIC_HEALTH='PUBLIC_HEALTH';case QUALITY='QUALITY';case RESEARCH='RESEARCH';}
```

src/Enums/SubmissionStatus.php

Status	Destination	SHA-256
--------	-------------	---------

PICK-AND-DROP CANDIDATE	src/Enums/SubmissionStatus.php	c31abe02a94474406c4371d5374a0e7cd7535d99ea13418ea5540c91913a97a8
-------------------------	--------------------------------	--

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum SubmissionStatus:string{case DRAFT='DRAFT';case VALIDATED='VALIDATED';case SUBMITTED='SUBMITTED';case ACCEPTED='ACCEPTED';case REJECTED='REJECTED';case CORRECTED='CORRECTED';}
```

src/Enums/SubscriptionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/SubscriptionStatus.php	3ecb1fa0c13f35860066747ff9afbb1c9a4cbd1a7089ecf511abbe71ac4cce9b

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Enums;enum SubscriptionStatus:string{case REQUESTED='REQUESTED';case ACTIVE='ACTIVE';case ERROR='ERROR';case SUSPENDED='SUSPENDED';case ENDED='ENDED';}
```

src/Models/CohortDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CohortDefinition.php	401ae71573e42f60c99f7bf9574e8115f9d44d7fe51636392860353d5edde001

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class CohortDefinition extends Model{public $timestamps=false;protected $table='clinical_cohort_definitions';protected $guarded=[];protected $casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/CohortMembershipSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CohortMembershipSnapshot.php	06d0050e9f3bbe6a013454fed38980b90acf0a89f89def93be458ecbce7de572

<?php

```
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class CohortMembershipSnapshot extends Model{public $timestamps=false;protected $table='clinical_cohort_membership_snapshots';protected $guarded=[];protected $casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','completed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/CohortVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/CohortVersion.php	eceec86bd2be07a050208518d4cf81bf89336f9c08708e41ba0d86fc068df03e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
CohortVersion extends Model{public $timestamps=false;protected $table='clinical_cohort_versions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeArtifact.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeArtifact.php	ceb976dfbddd2e78318804cbb8fe5844e6a61797cb884d5e7c90b1e4446f4354

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeArtifact extends Model{public $timestamps=false;protected $table='clinical_exchange_artifacts';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeDelivery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeDelivery.php	2412cd595bac256e4c3a6d54f434f0366e1287e829f98fc2b7f8ae426acd41fa

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeDelivery extends Model{public $timestamps=false;protected $table='clinical_exchange_deliveries';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeEndpoint.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeEndpoint.php	0b7bd5c3c5e86df1427ee104fffbca9138686a7a02cdfebab975a7c67e6c4e73

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeEndpoint extends Model{public $timestamps=false;protected $table='clinical_exchange_endpoints';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ExchangeJob.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ExchangeJob.php	b158a65390d89ca2ec8c7ee90dbfe9d96881a2a5ceaa856fac64344e41779403

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
ExchangeJob extends Model{public $timestamps=false;protected $table='clinical_exchange_jobs';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/InteroperabilityReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/InteroperabilityReconciliation.php	167d0cc8d522d1f83ecc3105fcdcd3d96df1a4189dfa9c5c79b54f4c5b26face6

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
InteroperabilityReconciliation extends Model{public $timestamps=false;protected
$table='clinical_interoperability_reconciliations';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureDefinition.php	7399444e7baf7ea81980c79063f45da4c98d43e2f498f5c83c0fef6bdd5668dc

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureDefinition extends Model{public $timestamps=false;protected $table='clinical_measure_definitions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureResult.php	3ac94cc97110d44c93514cdc4b8efe7eb060f2faf970673bf59c91810146015c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureResult extends Model{public $timestamps=false;protected $table='clinical_measure_results';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureRun.php	4826340c457a6a0a577e8481491aba6cd181f15463d6eef8d76c74d023516d2f

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureRun extends Model{public $timestamps=false;protected $table='clinical_measure_runs';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/MeasureVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/MeasureVersion.php	28d3899d0a575899935c654d072bc31cb5b3d8dabdcfffb588213d3f8fca9aef8

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
MeasureVersion extends Model{public $timestamps=false;protected $table='clinical_measure_versions';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/PublicHealthSubmission.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PublicHealthSubmission.php	3465b4092878bea641f3c7b9115d005f99017d3db4e0b4854a9d8c30bb7a210b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
PublicHealthSubmission extends Model{public $timestamps=false;protected
$table='clinical_public_health_submissions';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ResearchDatasetArtifact.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResearchDatasetArtifact.php	a7b296f8dcd92a40ded44a77233580ac09c3947b35a64fa6a58c80b58f4ff24c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
ResearchDatasetArtifact extends Model{public $timestamps=false;protected $table='clinical_research_dataset_artifacts';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','complet
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/ResearchDatasetRequest.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ResearchDatasetRequest.php	5283f1640e1ba13f1d53f050452973059da8ea53199d86cdc27c6958a217f2db

E		
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
ResearchDatasetRequest extends Model{public $timestamps=false;protected
$table='clinical_research_dataset_requests';protected $guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/SubscriptionDelivery.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SubscriptionDelivery.php	491672a7d82b4089f45f8ef3d8d7508babbc1124fed5df9209514255e6a3005e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
SubscriptionDelivery extends Model{public $timestamps=false;protected $table='clinical_subscription_deliveries';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/SubscriptionRegistration.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SubscriptionRegistration.php	d5f426dc6aeb8c9e8e223a43a963487ccf98625e2e495682a99ad65bc5ed90b5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
SubscriptionRegistration extends Model{public $timestamps=false;protected $table='clinical_subscription_registrations';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/SubscriptionTopic.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/SubscriptionTopic.php	9ddfee3f562b90caaca25e19b2ed446c08573ae43bbafdbd0b05ca4c9453b99e

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
SubscriptionTopic extends Model{public $timestamps=false;protected $table='clinical_subscription_topics';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ed_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/Models/TerminologyMapping.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/TerminologyMapping.php	43c0bb914c10ce77aca431c79c84d5dc4308ef6575af8bd489985b64b670d6f

DROP CANDIDATE	p	1
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Engagement\Models;use Illuminate\Database\Eloquent\Model;final class
TerminologyMapping extends Model{public $timestamps=false;protected $table='clinical_terminology_mappings';protected
$guarded=[];protected
$casts=['scope'=>'array','policy'=>'array','manifest'=>'array','evidence'=>'array','requested_at_utc'=>'immutable_datetime','comple
ted_at_utc'=>'immutable_datetime','expires_at_utc'=>'immutable_datetime'];}
```

src/PatientEngagementServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/PatientEngagementServiceProvider.php	2882ea5d56b5f82a69be038c45e1a1767e0ffdc187330fc2595f9a7e12b18218

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\Interoperability;use Illuminate\Support\ServiceProvider;final class
PatientEngagementServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-engagement.php','clinical-engagement');}public function boot():void{$this-
>loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-engagement.php');}}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volumes 1-12	Authoritative clinical data, privacy, release, operations and interoperability
Main Module	Identity, consent, purpose and confidentiality decisions
Patient portal	Identity, authentication and authorized record projection
Questionnaire service	Versioned definitions and response validation
Communications	Secure messaging, delivery status and preferences
Device platform	Registered device and measurement ingestion
Virtual care	Remote appointment and media-provider integration
Vue	Patient portal, proxy, questionnaires, messaging and remote-monitoring workspaces

Blocking Gaps

Approve Volume 13 as a supplemental baseline

Approve patient and proxy identity model

Approve patient-facing release and redaction policy

Bind portal authentication, consent and proxy authority

Approve questionnaire and patient-reported-data governance

Bind communications and remote-device platforms

Approve virtual-care and urgent-escalation workflows

Complete controllers, requests, resources, workers and Vue interfaces

Run proxy, privacy, tenant-isolation, device, messaging, accessibility and clinical-safety tests

Verification

composer validate --strict

composer dump-autoload --strict-psr

php artisan migrate:fresh --env=testing

php artisan migrate:rollback --step=1 --env=testing

php artisan migrate --env=testing

./vendor/bin/phpunit

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Clinical source-of-truth and production baseline
Volumes 1-11	Clinical, privacy, release and operational controls
HL7 FHIR R5 Secure communications	Topic-based notification reference
HL7 FHIR R5 Communication	Patient-care communication reference
Volumes 2, 3, 5 and 8	Observation, escalation and follow-up integration

KASHTRE CLINICAL MODULE

CODE-READY ENGINEERING DESIGN DOCUMENT

SUPPLEMENTAL VOLUME 14

CLINICAL CONTENT, CONFIGURATION, LOCALIZATION AND ACCESSIBILITY GOVERNANCE

Controlled Definitions, Validation, Promotion, Translation, Tenant Overrides, Drift Detection and Rollback

Version: 6.1.0 Comprehensive Draft

Status: Supplemental to the Phase 1-10 functional baseline

Depends on: Volumes 1 to 13

Production: Not Approved for Production

Scope Position and Governing Rule

Volume 14 is supplemental. It governs clinical configuration as controlled executable content rather than informal settings. CDEs, forms, score definitions, alert policies, questionnaires, terminology bindings, translations and tenant overrides are versioned, validated, approved, promoted and reversible.

Author content -> resolve dependencies -> validate semantics and terminology -> clinical review -> translation and accessibility review -> immutable package -> environment promotion -> drift monitoring -> rollback or supersession

Entity Register

Entity	Table	Invariant
ClinicalContentDefinition	clinical_content_definitions	Tenant, version, dependency, approval and environment provenance preserved
ClinicalContentVersion	clinical_content_versions	Tenant, version, dependency, approval and environment provenance preserved
ContentDependency	clinical_content_dependencies	Tenant, version, dependency, approval and environment provenance preserved
ContentValidationRun	clinical_content_validation_runs	Tenant, version, dependency, approval and environment provenance preserved
ContentValidationResult	clinical_content_validation_results	Tenant, version, dependency, approval and environment provenance preserved
ContentApproval	clinical_content_approvals	Tenant, version, dependency, approval and environment provenance preserved
PromotionPackage	clinical_content_promotion_packages	Tenant, version, dependency, approval and environment provenance preserved
PromotionItem	clinical_content_promotion_items	Tenant, version, dependency, approval and environment provenance preserved
PromotionRun	clinical_content_promotion_runs	Tenant, version, dependency, approval and environment provenance preserved
EnvironmentSnapshot	clinical_environment_snapshots	Tenant, version, dependency, approval and environment provenance preserved
ConfigurationDrift	clinical_configuration_drifts	Tenant, version, dependency, approval and environment provenance preserved
TranslationUnit	clinical_translation_units	Tenant, version, dependency, approval and environment provenance preserved
TranslationReview	clinical_translation_reviews	Tenant, version, dependency, approval and environment provenance preserved
AccessibilityAudit	clinical_accessibility_audits	Tenant, version, dependency, approval and environment provenance preserved
AccessibilityFinding	clinical_accessibility_findings	Tenant, version, dependency, approval and environment provenance preserved
TenantContentOverride	clinical_tenant_content_overrides	Tenant, version, dependency, approval and environment provenance preserved
RollbackRecord	clinical_content_rollbacks	Tenant, version, dependency, approval and environment provenance preserved
ContentReconciliation	clinical_content_reconciliations	Tenant, version, dependency, approval and environment provenance preserved

Engineering Design

Controlled content catalogue

- Stable content code and immutable versions
- Explicit dependency graph
- Payload hashes
- Effective dates and supersession
- No in-place production editing

Validation and approval

- Schema, semantic, terminology and reference validation
- Safety-impact and backward-compatibility evidence
- Separation of author and approver
- Failed validation blocks promotion
- Warnings require explicit disposition

Promotion and rollback

- Immutable promotion manifest
- Pre-promotion environment snapshot
- Ordered dependency-aware activation
- Idempotent package promotion
- Post-promotion verification
- Rollback restores controlled snapshot without deleting history

Localization

- Source and target locale retained
- Linguistic review separated from clinical review
- Clinical terms cannot be machine-published without review
- Fallback language explicit
- Right-to-left and pluralization behavior testable

Accessibility

- Target standard is version-pinned
- Keyboard, focus, labeling, contrast and error presentation tests
- Accessible authentication and redundant-entry checks
- Failure blocks publication for patient-facing or clinical workflow content

Tenant overrides and drift

- Override is narrow, effective-dated and attributable
- Base version remains identifiable
- Safety-critical constraints cannot be weakened locally
- Environment drift is detected by manifest comparison
- Unexpected production content enters reconciliation

Permissions

Permission	Purpose
clinical.content.create	Create controlled content version
clinical.content.validate	Run validation

clinical.content.approve	Approve content
clinical.content.translate	Create translation
clinical.content.translation_review	Approve clinical translation
clinical.content.accessibility_review	Approve accessibility evidence
clinical.content.promote	Promote approved package
clinical.content.rollback	Execute authorized rollback
clinical.content.override	Manage tenant override
clinical.content.drift.review	Resolve configuration drift

Mandatory Tests

In-place edit of active content is impossible

Missing dependency blocks validation

Failed terminology binding blocks approval

Author cannot satisfy independent approval alone

Machine translation cannot become active without clinical review

Accessibility failure blocks patient-facing publication

Promotion order respects dependencies

Repeated promotion is idempotent

Rollback restores prior manifest

Tenant override cannot weaken safety constraint

Production drift is detected

Supersession preserves historical rendering

Repository-Ready Source Listings

The ZIP is authoritative. Complete files and checksums follow.

README.md

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	README.md	b54fc51e929cf4a1873d6d7ed7627d24afa7673614654c798f7fbc0e1bb4fa20

KashTre Clinical Content, Configuration, Localization and Accessibility Governance v6.1.0

Supplemental Volume 14. Pick-and-drop candidate. Bind CDE, forms, scoring, terminology, environment promotion, translation, accessibility, tenant override and release-assurance services before VERIFIED status.

artifact-manifest.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	artifact-manifest.json	e5ca408795fce4eb1c38977e8918d48b14fbcc381bdf2b2570f350016f84beb8

```
{
  "volume": "Clinical EDD Supplemental Volume 14 Content Configuration Localization Accessibility",
  "version": "6.1.0",
  "classification": "PICK-AND-DROP CANDIDATE",
  "files": [
    {
      "path": "README.md",
      "sha256": "b54fc51e929cf4a1873d6d7ed7627d24afa7673614654c798f7fbc0e1bb4fa20",
      "bytes": 301
    },
    {
      "path": "composer.json",
      "sha256": "0bb37cb1e3b911c71d4b0b80db7b127eb46ab08f79d77c3b7b597ac11ff33289",
      "bytes": 417
    },
    {
      "path": "config/clinical-content-governance.php",
      "sha256": "e49d28aca75767045e8fa8cce16fd4caa11f7996912b1d0ee18fa59f87a6231f",
      "bytes": 231
    },
    {
      "path": "database/migrations/2026_08_28_1200000_create_content_governance_tables.php",
      "sha256": "3a081ba189e37f0c7c9b91bb7c5a03b5dc9920b7947d8bf4adc29af52c926585",
      "bytes": 4675
    },
    {
      "path": "install.sh",
      "sha256": "58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db",
      "bytes": 99
    },
    {
      "path": "routes/clinical-content-governance.php",
      "sha256": "dd85be6e9ed1032465e71d51e6d8b68dceef721724a5d882eb68bf73d4672eba",
      "bytes": 289
    }
  ]
}
```

```

"path": "src/Actions/CreateContentVersion.php",
"sha256": "8900a016225f116408e3efd36ead15f82cab2018ebc6d27ed57fb3b0802d828",
"bytes": 1179
},
{
"path": "src/Actions/PromoteContentPackage.php",
"sha256": "f5d55b8566f4ad2b65a2e68f40c72ebafb95dd7ac658dfc02916dc73811123ef",
"bytes": 1417
},
{
"path": "src/Actions/ValidateContentVersion.php",
"sha256": "60aab619231845d55654190a114c4ce6bf2e5035bdef16b234b299d916d81162",
"bytes": 1527
},
{
"path": "src/ContentGovernanceServiceProvider.php",
"sha256": "9679e8c83902ea8a3f19171f5397a1e53451ceef03385751e3f2e18d6ae167d5",
"bytes": 490
},
{
"path": "src/Contracts/AccessibilityAuditGateway.php",
"sha256": "59c6fd38c91d772cd0a7542a1196fd3197454e9e5f12911bb125a75240d1b108",
"bytes": 220
},
{
"path": "src/Contracts/ContentValidator.php",
"sha256": "2ffdada96c802592161874687269f04e94be7de3082826cfd2ba802b7d6160b8",
"bytes": 212
},
{
"path": "src/Contracts/TerminologyGateway.php",
"sha256": "1c394f0cbc9d9ed4bb0adf8aa5f4c2cfbc5f51e56f85d0ea987b35e6e1992941",
"bytes": 188
},
{
"path": "src/Data/CreateContentVersionData.php",
"sha256": "d2749136aa8b611e3bdc59fbe5911d934571210ab90879e4e29be20ba1836d26",
"bytes": 365
},
{
"path": "src/Enums/AccessibilityStatus.php",
"sha256": "f2d61c5845c2f942692a76efa535b1457dfac3e9b6b56195cb525ea6daf206bd",
"bytes": 200
},
{
"path": "src/Enums/ContentStatus.php",
"sha256": "bb0c4d83196a2ec8d9c504dc88316d035e4fffec8114499551055fa5f60fe052",
"bytes": 246
},
{
"path": "src/Enums/DriftStatus.php",
"sha256": "f62f567c49b2fb8c398d100304cb89140ea5984fb434d90001ecd74b220899b7",
"bytes": 208
},

```

```

{
  "path": "src/Enums/EnvironmentType.php",
  "sha256": "bf59992af051165eada11ee53555c0b7706fce00b050673f1982fb55dbc130f5",
  "bytes": 225
},
{
  "path": "src/Enums/OverrideStatus.php",
  "sha256": "a9c9cd64a855c331b1259d4eafe5ec94f6c1732ef17f40998e29f6336854e1cf",
  "bytes": 226
},
{
  "path": "src/Enums/PromotionStatus.php",
  "sha256": "476dd79bb6ad7b44c1a8ac8e6180354f7d9a7c98f5abeae86cb70d923ad5a9f7",
  "bytes": 297
},
{
  "path": "src/Enums/TranslationStatus.php",
  "sha256": "f31ba4f643722e105d61d9e8e6b6d1a8b0276a97369efecebb45ca33b991d064",
  "bytes": 284
},
{
  "path": "src/Enums/ValidationOutcome.php",
  "sha256": "387672667d8823bc9213b3f13123f7a806d074eea7ca7e03d47aed453b12f865",
  "bytes": 169
},
{
  "path": "src/Models/AccessibilityAudit.php",
  "sha256": "b7d764d281b1c4d3775a65053f4a5bdefb7c17743c5704efe354835248dac810",
  "bytes": 468
},
{
  "path": "src/Models/AccessibilityFinding.php",
  "sha256": "27400ba90fcc6ebfaa9655af9281bc9ed54101606b0e74558b83e1f37bb0f60b",
  "bytes": 472
},
{
  "path": "src/Models/ClinicalContentDefinition.php",
  "sha256": "d8c4e65b0c9bdca357f517ca4d0e80e53587224601c61ebbc51fe41b7c77d0a",
  "bytes": 474
},
{
  "path": "src/Models/ClinicalContentVersion.php",
  "sha256": "564e3dbc5b1c64aa34bed57080e132e26b66b353e24a06dae3ce982817cbc1e2",
  "bytes": 468
},
{
  "path": "src/Models/ConfigurationDrift.php",
  "sha256": "102000a2cbe02612dbb969758fe0c5646290a82b81e8be0327e0aa6d715f4e24",
  "bytes": 468
},
{
  "path": "src/Models/ContentApproval.php",
  "sha256": "d9456b4193d04a1f50831b761133e28ed4fcdf5afe4169fe46c8bde9cd46fd56",
  "bytes": 462
},

```

```

{
  "path": "src/Models/ContentDependency.php",
  "sha256": "809563d7dfc406940839df5ac35199dbf4a80bf0cdee4a1a4e2bc019c72ce9f3",
  "bytes": 467
},
{
  "path": "src/Models/ContentReconciliation.php",
  "sha256": "47037ba27874e73aba2a180fc1504dbe5085675f07085197c281b51e89a393ac",
  "bytes": 474
},
{
  "path": "src/Models/ContentValidationResult.php",
  "sha256": "f5e1ce30a24258ecb7f036996d26b1d6e5e15325e6b963a9a2f25f01bda09c17",
  "bytes": 479
},
{
  "path": "src/Models/ContentValidationRun.php",
  "sha256": "3964ff81b0c35aa786604cbc934974a9d99dd86f46949c782ff5e3e16cebff7c",
  "bytes": 473
},
{
  "path": "src/Models/EnvironmentSnapshot.php",
  "sha256": "88bb72d11c511fe274ba65879798cf1b846b75537207c1310d03f728af5ed8dd",
  "bytes": 470
},
{
  "path": "src/Models/PromotionItem.php",
  "sha256": "c0445519997d33bf056225c221a3bf67f127b7296d0d6b3111f99fe55c9f6441",
  "bytes": 466
},
{
  "path": "src/Models/PromotionPackage.php",
  "sha256": "79718486d4562b4b65ef3f1050d12952f7463c25392ff7e34ca52a3d152fb0ef",
  "bytes": 472
},
{
  "path": "src/Models/PromotionRun.php",
  "sha256": "9df766a5cbb04208f6890390ba5e4540bc2b8fabca3d6261c4c66f27c078b893",
  "bytes": 464
},
{
  "path": "src/Models/RollbackRecord.php",
  "sha256": "6a20e542443061d91f93a3da01ba40c0bd0fc0edd6fe8c46591cc56e444e5d6d",
  "bytes": 461
},
{
  "path": "src/Models/TenantContentOverride.php",
  "sha256": "acadbca0a314ee8bdac6e7b8b5768186ad1685c785c9d20be114500c1f3ed08f",
  "bytes": 475
},
{
  "path": "src/Models/TranslationReview.php",
  "sha256": "25dd0102e864d2f7d93b6024020638803b75a3871b39780541171b5294c2d51a",
  "bytes": 466
},

```

```
{
  "path": "src/Models/TranslationUnit.php",
  "sha256": "3f4c7fee92a63a70fae4d6ae5fbd4286c79ae19a6c1a1a758e6bf5a00c64eede",
  "bytes": 462
},
{
  "path": "src/Services/ContentValidationEngine.php",
  "sha256": "301f6c3d51ef628eb208d6490e55245c30029af5b894e2bd7a82fead9936e126",
  "bytes": 773
},
{
  "path": "verify.sh",
  "sha256": "62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2",
  "bytes": 121
}
]
```

composer.json

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	composer.json	0bb37cb1e3b911c71d4b0b80db7b127eb46ab08f79d77c3b7b597ac11ff33289

```
{
  "name": "kashtre/clinical-content-governance",
  "type": "library",
  "require": {
    "php": "^8.4",
    "illuminate/database": "^13.0",
    "illuminate/http": "^13.0",
    "illuminate/queue": "^13.0",
    "illuminate/support": "^13.0",
    "kashtre/clinical-foundation": "^6.1",
    "autoload": {
      "psr-4": {
        "KashTre\\Clinical\\ContentGovernance\\": "src/"
      },
      "extra": {
        "laravel": {
          "providers": [
            "KashTre\\Clinical\\ContentGovernance\\ContentGovernanceServiceProvider"
          ]
        }
      }
    }
  }
}
```

config/clinical-content-governance.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	config/clinical-content-governance.php	e49d28aca75767045e8fa8cce16fd4caa11f7996912b1d0ee18fa59f87a6231f

```
<?php
declare(strict_types=1);
return [
    'promotion_requires_approval' => true,
    'translation_requires_clinical_review' => true,
    'accessibility_target' => 'WCAG_2_2_AA',
    'rollback_snapshot_required' => true,
    'configuration_drift_detection' => true
];
```

database/migrations/2026_08_28_1200000_create_content_governance_tables.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	database/migrations/2026_08_28_1200000_create_content_governance_tables.php	3a081ba189e37f0c7c9b91bb7c5a03b5dc9920b7947d8bf4adc29af52c926585

```
<?php
declare(strict_types=1);
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('clinical_content_definitions', function (Blueprint $table) {
            $table->id();
            $table->ulid('public_id')->unique();
            $table->string('tenant_id', 64);
            $table->string('content_type', 96);
            $table->string('content_code', 128);
            $table->string('status', 24);
            $table->unique(['tenant_id', 'content_type', 'content_code'], 'uq_content_definition');
        });
    }
};
```

```

Schema::create('clinical_content_versions',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('definition_public_id');$t->unsignedInteger('version_no');$t->string('locale',32);$t->string('status',24);$t->json('payload');$t->char('payload_hash',64);$t->ulid('created_by_public_id');$t->dateTime('created_at_utc',6);$t->dateTime('effective_from_utc',6)->nullable();$t->dateTime('effective_to_utc',6)->nullable();$t->unique(['definition_public_id','version_no','locale'],'uq_content_version');});
Schema::create('clinical_content_dependencies',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('content_version_public_id');$t->string('dependency_type',96);$t->string('dependency_reference',240);$t->string('constraint_expression',240)->nullable();});
Schema::create('clinical_content_validation_runs',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('content_version_public_id');$t->string('overall_outcome',24);$t->char('input_fingerprint',64);$t->dateTime('started_at_utc',6);$t->dateTime('completed_at_utc',6);});
Schema::create('clinical_content_validation_results',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('validation_run_public_id');$t->string('validator_code',128);$t->string('outcome',24);$t->json('evidence');});
Schema::create('clinical_content_approvals',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('content_version_public_id');$t->string('approval_role',96);$t->string('decision',24);$t->ulid('decided_by_public_id');$t->text('comment')->nullable();$t->dateTime('decided_at_utc',6);});
Schema::create('clinical_content_promotion_packages',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->string('package_code',128);$t->string('status',24);$t->json('manifest');$t->char('manifest_hash',64);$t->ulid('created_by_public_id');$t->dateTime('created_at_utc',6);});
Schema::create('clinical_content_promotion_runs',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('promotion_package_public_id');$t->string('target_environment',64);$t->string('status',24);$t->ulid('started_by_public_id');$t->dateTime('started_at_utc',6);$t->dateTime('completed_at_utc',6)->nullable();});
Schema::create('clinical_environment_snapshots',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->string('environment_code',64);$t->string('snapshot_type',48);$t->json('manifest');$t->char('manifest_hash',64);$t->dateTime('created_at_utc',6);});
Schema::create('clinical_translation_units',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('content_version_public_id');$t->string('source_locale',32);$t->string('target_locale',32);$t->string('key',240);$t->text('source_text');$t->text('target_text')->nullable();$t->string('status',32);});
Schema::create('clinical_accessibility_audits',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('content_version_public_id');$t->string('target_standard',64);$t->string('overall_status',24);$t->json('evidence');$t->dateTime('completed_at_utc',6);});
Schema::create('clinical_tenant_content_overrides',function(Blueprint$t){$t->id();$t->ulid('public_id')->unique();$t->string('tenant_id',64);$t->ulid('base_content_version_public_id');$t->json('override_payload');$t->string('status',24);$t->dateTime('effective_from_utc',6);$t->dateTime('effective_to_utc',6)->nullable();});
}public function
down():void{foreach(['clinical_tenant_content_overrides','clinical_accessibility_audits','clinical_translation_units','clinical_environment_snapshots','clinical_content_promotion_runs','clinical_content_promotion_packages','clinical_content_approvals','clinical_content_validation_results','clinical_content_validation_runs','clinical_content_dependencies','clinical_content_versions','clinical_content_definitions']as$t)Schema::dropIfExists($t);};

```

install.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	install.sh	58225f4906f0774e97e4d15d4dad0215d5634faff4f917cb66e79616ad1571db

```

#!/usr/bin/env bash
set -euo pipefail
composer install --no-interaction
php artisan migrate --force

```

routes/clinical-content-governance.php

Status	Destination	SHA-256
PICK-AND-DROP	routes/clinical-content-	dd85be6e9ed1032465e71d51e6d8b68dceef721724a5d882eb68bf73d4672eba

CANDIDATE	governance.php	
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```
<?php
declare(strict_types=1);use Illuminate\Support\Facades\Route;Route::middleware(['api','auth:sanctum'])-
>prefix('api/v1/clinical/content-governance')->group(function(){Route::get('/catalogue',fn()->response()-
>json(['error'=>['code'=>'CLN_CONTENT_HOST_CONTROLLER_REQUIRED'],501]));});
```

src/Actions/CreateContentVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/CreateContentVersion.php	8900a016225f116408e3efd36ead15f82cabc2018ebc6d27ed57fb3b0802d828

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ContentGovernance\Data\CreateContentVersionData;use
KashTre\Clinical\ContentGovernance\Models\ClinicalContentDefinition,ClinicalContentVersion;final readonly class
CreateContentVersion{public function __construct(private PublicIdGenerator $ids){}public function
execute(CreateContentVersionData $d):ClinicalContentVersion{return
DB::transaction(function()use($d){$def=ClinicalContentDefinition::query()->firstOrCreate(['tenant_id'=>$d-
>tenantId,'content_type'=>$d->contentType,'content_code'=>$d->contentCode],['public_id'=>(string)$this->ids-
>generate(), 'status'=>'ACTIVE']);return ClinicalContentVersion::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$d->tenantId,'definition_public_id'=>$def->public_id,'version_no'=>$d->versionNo,'locale'=>$d-
>locale,'status'=>'DRAFT','payload'=>$d->payload,'payload_hash'=>hash('sha256',json_encode($d-
>payload,JSON_THROW_ON_ERROR)),'created_by_public_id'=>$d->createdByld,'created_at_utc'=>$d->createdAt]);}}
```

src/Actions/PromoteContentPackage.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Actions/PromoteContentPackage.php	f5d55b8566f4ad2b65a2e68f40c72ebafb95dd7ac658dfc02916dc73811123ef

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ContentGovernance\Models\EnvironmentSnapshot,PromotionPackage,PromotionRun;final readonly class
PromoteContentPackage{public function __construct(private PublicIdGenerator $ids){}public function execute(string
$tenantId,string $packageId,string $targetEnvironment,string $actorId,array $beforeSnapshot):PromotionRun{return
DB::transaction(function()use($tenantId,$packageId,$targetEnvironment,$actorId,$beforeSnapshot){$p=PromotionPackage::query()
->where('tenant_id',$tenantId)->where('public_id',$packageId)->where('status','APPROVED')->lockForUpdate()-
>firstOrFail();EnvironmentSnapshot::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId,'environment_code'=>$targetEnvironment,'snapshot_type'=>'PRE_PROMOTION','manifest'=>$be
foreSnapshot,'manifest_hash'=>hash('sha256',json_encode($beforeSnapshot,JSON_THROW_ON_ERROR)),'created_at_utc'=>now('
UTC')));$r=PromotionRun::query()->create(['public_id'=>(string)$this->ids-
>generate(), 'tenant_id'=>$tenantId,'promotion_package_public_id'=>$packageId,'target_environment'=>$targetEnvironment,'status'
=>'PROMOTING','started_by_public_id'=>$actorId,'started_at_utc'=>now('UTC')));$p->forceFill(['status'=>'PROMOTING'])-
>save();return$r;}}
```

src/Actions/ValidateContentVersion.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Actions/ValidateContentVersion.php	60aab619231845d55654190a114c4ce6bf2e5035bdef16b234b299d916d81162
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Actions;use Illuminate\Support\Facades\DB;use
KashTre\Clinical\Foundation\Contracts\PublicIdGenerator;use
KashTre\Clinical\ContentGovernance\Models\ClinicalContentVersion,ContentValidationResult,ContentValidationRun;use
KashTre\Clinical\ContentGovernance\Services\ContentValidationEngine;final readonly class ValidateContentVersion{public
function __construct(private PublicIdGenerator $ids,private ContentValidationEngine $engine){}public function execute(string
$tenantId,string $versionId,array $context):ContentValidationRun{return
DB::transaction(function()use($tenantId,$versionId,$context){$v=ClinicalContentVersion::query()->where('tenant_id',$tenantId)-
>where('public_id',$versionId)->firstOrFail();$r=$this->engine->evaluate($tenantId,$v->definition?->content_type??:'UNKNOWN',$v-
>payload,$context);$run=ContentValidationRun::query()->create(['public_id'=>(string)$this->ids-
>generate(),'tenant_id'=>$tenantId,'content_version_public_id'=>$versionId,'overall_outcome'=>$r['overall'],'input_fingerprint'=>$r['f
ingerprint'],'started_at_utc'=>now('UTC'),'completed_at_utc'=>now('UTC')]);foreach($r['results']as$x)ContentValidationResult::query
()->create(['public_id'=>(string)$this->ids->generate(),'tenant_id'=>$tenantId,'validation_run_public_id'=>$run-
>public_id,'validator_code'=>$x['code'],'outcome'=>$x['outcome'],'evidence'=>$x['evidence']??:[]);$v-
>forceFill(['status'=>$r['overall']=== 'FAIL'? 'DRAFT': 'REVIEW'])->save();return$run;}}}
```

src/ContentGovernanceServiceProvider.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/ContentGovernanceServiceProvider.php	9679e8c83902ea8a3f19171f5397a1e53451ceef03385751e3f2e18d6ae167d5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance;use Illuminate\Support\ServiceProvider;final class
ContentGovernanceServiceProvider extends ServiceProvider{public function register():void{$this-
>mergeConfigFrom(__DIR__.'/../config/clinical-content-governance.php','clinical-content-governance');}public function
boot():void{$this->loadMigrationsFrom(__DIR__.'/../database/migrations');$this->loadRoutesFrom(__DIR__.'/../routes/clinical-
content-governance.php');}}
```

src/Contracts/AccessibilityAuditGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/AccessibilityAuditGateway.php	59c6fd38c91d772cd0a7542a1196fd3197454e9e5f12911bb125a75240d1b108

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Contracts;interface AccessibilityAuditGateway{public
function audit(string $tenantId,string $artifactType,array $artifact,string $target):array;}
```

src/Contracts/ContentValidator.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/ContentValidator.php	2ffdada96c802592161874687269f04e94be7de3082826cfd2ba802b7d6160b8

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Contracts;interface ContentValidator{public function
validate(string $tenantId,string $contentType,array $payload,array $context):array;}
```

src/Contracts/TerminologyGateway.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Contracts/TerminologyGateway.php	1c394f0cbc9d9ed4bb0adf8aa5f4c2cfbc5f51e56f85d0ea987b35e6e1992941

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Contracts;interface TerminologyGateway{public function
validateBindings(string $tenantId,array $bindings):array;}
```

src/Data/CreateContentVersionData.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Data/CreateContentVersionData.php	d2749136aa8b611e3bdc59fbe5911d934571210ab90879e4e29be20ba1836d26

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Data;final readonly class
CreateContentVersionData{public function __construct(public string $tenantId,public string $contentType,public string
$contentCode,public int $versionNo,public array $payload,public string $locale,public string $createdById,public
\DateTimeImmutable $createdAt){}}
```

src/Enums/AccessibilityStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/AccessibilityStatus.php	f2d61c5845c2f942692a76efa535b1457dfac3e9b6b56195cb525ea6daf206bd

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum AccessibilityStatus:string{case
NOT_TESTED='NOT_TESTED';case PASS='PASS';case WARNING='WARNING';case FAIL='FAIL';}
```

src/Enums/ContentStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/ContentStatus.php	bb0c4d83196a2ec8d9c504dc88316d035e4ffec8114499551055fa5f60fe052

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum ContentStatus:string{case
DRAFT='DRAFT';case REVIEW='REVIEW';case APPROVED='APPROVED';case ACTIVE='ACTIVE';case
SUPERSEDED='SUPERSEDED';case RETIRED='RETIRED';}
```

src/Enums/DriftStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/DriftStatus.php	f62f567c49b2fb8c398d100304cb89140ea5984fb434d90001ecd74b220899b7

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum DriftStatus:string{case
MATCHED='MATCHED';case DIFFERENT='DIFFERENT';case MISSING='MISSING';case UNEXPECTED='UNEXPECTED';}
```

src/Enums/EnvironmentType.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/EnvironmentType.php	bf59992af051165eada11ee53555c0b7706fce00b050673f1982fb55dbc130f5

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum EnvironmentType:string{case
DEVELOPMENT='DEVELOPMENT';case TEST='TEST';case UAT='UAT';case STAGING='STAGING';case PRODUCTION='PRODUCTION';}
```

src/Enums/OverrideStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/OverrideStatus.php	a9c9cd64a855c331b1259d4eafe5ec94f6c1732ef17f40998e29f6336854e1cf

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum OverrideStatus:string{case
PROPOSED='PROPOSED';case APPROVED='APPROVED';case ACTIVE='ACTIVE';case EXPIRED='EXPIRED';case
REVOKED='REVOKED';}
```

src/Enums/PromotionStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/PromotionStatus.php	476dd79bb6ad7b44c1a8ac8e6180354f7d9a7c98f5abeae86cb70d923ad5a9f7

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum PromotionStatus:string{case
REQUESTED='REQUESTED';case VALIDATING='VALIDATING';case APPROVED='APPROVED';case PROMOTING='PROMOTING';case
COMPLETED='COMPLETED';case FAILED='FAILED';case ROLLED_BACK='ROLLED_BACK';}
```

src/Enums/TranslationStatus.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Enums/TranslationStatus.php	f31ba4f643722e105d61d9e8e6b6d1a8b0276a97369efecebb45ca33b991d064

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum TranslationStatus:string{case
DRAFT='DRAFT';case LINGUISTIC_REVIEW='LINGUISTIC_REVIEW';case CLINICAL_REVIEW='CLINICAL_REVIEW';case
APPROVED='APPROVED';case ACTIVE='ACTIVE';case REJECTED='REJECTED';}
```

src/Enums/ValidationOutcome.php

Status	Destination	SHA-256
PICK-AND-DROP	src/Enums/ValidationOutcome.php	387672667d8823bc9213b3f13123f7a806d074eea7ca7e03d47aed453b12f865

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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Enums;enum ValidationOutcome:string{case
PASS='PASS';case WARNING='WARNING';case FAIL='FAIL';}
```

src/Models/AccessibilityAudit.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AccessibilityAudit.php	b7d764d281b1c4d3775a65053f4a5bdefb7c17743c5704efe354835248dac810

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class AccessibilityAudit extends Model{public $timestamps=false;protected $table='clinical_accessibility_audits';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/AccessibilityFinding.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/AccessibilityFinding.php	27400ba90fcc6ebfaa9655af9281bc9ed54101606b0e74558b83e1f37bb0f60b

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class AccessibilityFinding extends Model{public $timestamps=false;protected $table='clinical_accessibility_findings';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ClinicalContentDefinition.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ClinicalContentDefinition.php	d8c4e65b0c9bdca357f517ca4d0e80e53587224601c61ebbca51fe41b7c77d0a

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ClinicalContentDefinition extends Model{public $timestamps=false;protected $table='clinical_content_definitions';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ClinicalContentVersion.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ClinicalContentVersion.php	564e3dbc5b1c64aa34bed57080e132e26b66b353e24a06dae3ce982817cbc1e2

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ClinicalContentVersion extends Model{public $timestamps=false;protected $table='clinical_content_versions';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ConfigurationDrift.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ConfigurationDrift.php	102000a2cbe02612dbb969758fe0c5646290a82b81e8be0327e0aa6d715f4e24

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ConfigurationDrift extends Model{public $timestamps=false;protected $table='clinical_configuration_drifts';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ContentApproval.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ContentApproval.php	d9456b4193d04a1f50831b761133e28ed4fcdf5afe4169fe46c8bde9cd46fd56

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ContentApproval extends Model{public $timestamps=false;protected $table='clinical_content_approvals';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ContentDependency.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ContentDependency.php	809563d7dfc406940839df5ac35199dbf4a80bf0cdee4a1a4e2bc019c72ce9f3

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ContentDependency extends Model{public $timestamps=false;protected $table='clinical_content_dependencies';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ContentReconciliation.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ContentReconciliation.php	47037ba27874e73aba2a180fc1504dbe5085675f07085197c281b51e89a393ac

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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ContentReconciliation extends Model{public $timestamps=false;protected $table='clinical_content_reconciliations';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ContentValidationResult.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ContentValidationResult.php	f5e1ce30a24258ecb7f036996d26b1d6e5e15325e6b963a9a2f25f01bda09c17

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ContentValidationResult extends Model{public $timestamps=false;protected
$table='clinical_content_validation_results';protected $guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/ContentValidationRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/ContentValidationRun.php	3964ff81b0c35aa786604cbc934974a9d99dd86f46949c782ff5e3e16cebff7c

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class ContentValidationRun extends Model{public $timestamps=false;protected
$table='clinical_content_validation_runs';protected $guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/EnvironmentSnapshot.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/EnvironmentSnapshot.php	88bb72d11c511fe274ba65879798cf1b846b75537207c1310d03f728af5ed8d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class EnvironmentSnapshot extends Model{public $timestamps=false;protected
$table='clinical_environment_snapshots';protected $guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/PromotionItem.php

Status	Destination	SHA-256
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PICK-AND-DROP CANDIDATE	src/Models/PromotionItem.php	c0445519997d33bf056225c221a3bf67f127b7296d0d6b3111f99fe55c9f6441
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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class PromotionItem extends Model{public $timestamps=false;protected $table='clinical_content_promotion_items';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/PromotionPackage.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PromotionPackage.php	79718486d4562b4b65ef3f1050d12952f7463c25392ff7e34ca52a3d152fb0ef

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class PromotionPackage extends Model{public $timestamps=false;protected
$table='clinical_content_promotion_packages';protected $guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/PromotionRun.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/PromotionRun.php	9df766a5cbb04208f6890390ba5e4540bc2b8fabca3d6261c4c66f27c078b893

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class PromotionRun extends Model{public $timestamps=false;protected $table='clinical_content_promotion_runs';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/RollbackRecord.php

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	src/Models/RollbackRecord.php	6a20e542443061d91f93a3da01ba40c0bd0fc0edd6fe8c46591cc56e444e5d6d

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class RollbackRecord extends Model{public $timestamps=false;protected $table='clinical_content_rollbacks';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/TenantContentOverride.php

Status	Destination	SHA-256
PICK-AND-	src/Models/TenantContentOverride.ph	acadbca0a314ee8bdac6e7b8b5768186ad1685c785c9d20be114500c1f3ed

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```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class TenantContentOverride extends Model{public $timestamps=false;protected
$table='clinical_tenant_content_overrides';protected $guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/TranslationReview.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDATE	src/Models/TranslationReview.php	25dd0102e864d2f7d93b6024020638803b75a3871b39780541171b5294c2d51a

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class TranslationReview extends Model{public $timestamps=false;protected $table='clinical_translation_reviews';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Models/TranslationUnit.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDATE	src/Models/TranslationUnit.php	3f4c7fee92a63a70fae4d6ae5fbd4286c79ae19a6c1a1a758e6bf5a00c64eede

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Models;use Illuminate\Database\Eloquent\Model;final
class TranslationUnit extends Model{public $timestamps=false;protected $table='clinical_translation_units';protected
$guarded=[];protected
$casts=['payload'=>'array','evidence'=>'array','dependencies'=>'array','created_at_utc'=>'immutable_datetime','effective_from_utc'
=>'immutable_datetime','effective_to_utc'=>'immutable_datetime'];}
```

src/Services/ContentValidationEngine.php

Status	Destination	SHA-256
PICK-AND- DROP CANDIDAT E	src/Services/ContentValidationEngine.p hp	301f6c3d51ef628eb208d6490e55245c30029af5b894e2bd7a82fead9936e 126

```
<?php
declare(strict_types=1);namespace KashTre\Clinical\ContentGovernance\Services;use
KashTre\Clinical\ContentGovernance\Contracts\ContentValidator;final class ContentValidationEngine{public function
__construct(private iterable $validators){}public function evaluate(string $tenantId,string $type,array $payload,array
$context):array{$results=[];$overall='PASS';foreach($this->validators as $v){if(!$v instanceof ContentValidator)continue;foreach($v-
>validate($tenantId,$type,$payload,$context)as $r){$results[]=$r;if(($r['outcome']??'FAIL')==='FAIL')$overall='FAIL';elseif(($r['outcom
e']??'PASS')==='WARNING'&&$overall==='PASS')$overall='WARNING';}}return['overall'=>$overall,'results'=>$results,'fingerprint'=>ha
sh('sha256',json_encode($results,JSON_THROW_ON_ERROR))];}
```

verify.sh

Status	Destination	SHA-256
PICK-AND-DROP CANDIDATE	verify.sh	62cb33960aa87f32609daa118dd59d51182fda0561bd5ca385efa7e33b10b8f2

```
#!/usr/bin/env bash
set -euo pipefail
composer validate --strict
composer dump-autoload --strict-psr
./vendor/bin/phpunit
```

Bindings, Gaps and Verification

Binding	Required
Volumes 2-7 and 13	CDE, forms, scores, medications, results and questionnaires
Volume 10	Release evidence and production authorization
Volume 12	FHIR profiles and terminology mappings
Main Module	Tenant and facility configuration authority
Terminology	CodeSystem, ValueSet and ConceptMap validation
Localization	Translation memory and locale rendering
Accessibility	Automated and manual audit evidence
Vue	Content catalogue, diff, approvals, promotion, drift and rollback

Blocking Gaps

Approve Volume 14 as a supplemental baseline
 Define authoritative content-type catalogue
 Approve content dependency and compatibility rules
 Bind terminology validation service
 Approve language and clinical-translation governance
 Adopt an accessibility conformance target
 Bind environment manifest and deployment services
 Complete controllers, requests, resources, workers and Vue interfaces
 Run semantic, translation, accessibility, promotion, rollback and drift tests

Verification

```

composer validate --strict
composer dump-autoload --strict-psr
php artisan migrate:fresh --env=testing
php artisan migrate:rollback --step=1 --env=testing
php artisan migrate --env=testing
./vendor/bin/phpunit
  
```

Source Register

Source	Use
KashTre_Clinical_Module_SRD_v6.1	Clinical baseline and versioned configuration authority
Volumes 1-13	Controlled Clinical content consumers
HL7 FHIR R5 ImplementationGuide	Computable implementation-rule packaging reference
W3C WCAG 2.2	Web accessibility conformance reference

KASHTRE CLINICAL MODULE**CODE-READY ENGINEERING DESIGN DOCUMENT**

Volume 15 Consolidated Engineering Expansion

The following section is normative for the DHIS2 integration candidate and expands the compact original Volume 15 section.

DHIS2 Component Architecture

- Instance Registry stores tenant, environment, base URI reference, supported DHIS2 version and capability snapshot without embedding credentials.
- Metadata Snapshot captures organisation units, data elements, category combinations, option sets, datasets, programs, stages and tracked-entity types with hash and retrieval time.
- Mapping Registry versions KashTre-to-DHIS2 mappings and prevents ad hoc production edits.
- Payload Builder creates Aggregate, Event or Tracker payloads from approved source events and reporting rules.
- Delivery Orchestrator performs validation-only or commit imports, records idempotency identity and supports synchronous or asynchronous completion.
- Import Reconciler parses summaries and object reports, links every failure to source correlation, and controls correction and resubmission.
- Drift Monitor compares active mappings with target metadata and suspends affected delivery when required metadata changes.

DHIS2 Data Objects

Entity	Purpose
Dhis2Instance	Target endpoint, environment and supported capability
Dhis2CredentialReference	Secret-manager reference only
Dhis2MetadataSnapshot	Hashed target metadata view
Dhis2OrganisationUnitMapping	Facility/client-space to organisation-unit mapping
Dhis2DataElementMapping	KashTre measure to DHIS2 data element
Dhis2CategoryMapping	Category option combination mapping
Dhis2DatasetMapping	Dataset and reporting-period contract
Dhis2ProgramMapping	Event or Tracker program mapping
Dhis2ProgramStageMapping	Program-stage and data-value mapping
Dhis2TrackedEntityMapping	Approved identifier and attribute policy
Dhis2SubmissionBatch	Aggregate, Event or Tracker submission identity
Dhis2SubmissionObject	Source-to-target object correlation
Dhis2DeliveryAttempt	Request/response and retry evidence
Dhis2ImportSummary	Overall import outcome
Dhis2ImportObjectReport	Object-level conflict or error
Dhis2MetadataDrift	Detected target metadata change
Dhis2ReconciliationCase	Correction, resubmission and closure work

DHIS2 Processing Rules

- Aggregate submissions contain no patient identifier.
- Tracker or Event disclosure requires approved public-health purpose and minimum-necessary mapping.
- The idempotency key is derived from tenant, target instance, mapping version, source public ID and source version.
- Validation-only imports create no target data.

- Asynchronous jobs retain job identity and are polled with bounded retry.
- Partial or rejected imports never become accepted silently.
- Reporting-period correction preserves earlier submission evidence.
- Metadata drift blocks only affected mappings and creates an actionable case.
- Credentials are obtained at execution time from the approved secrets platform and are never logged.

DHIS2 APIs and Workspaces Required

Area	Required candidate
API	Instance, metadata snapshot, mapping, validation, submission, retry and reconciliation controllers
Queue	Metadata refresh, payload creation, delivery, polling and reconciliation workers
Scheduler	Reporting-period generation, backlog retry and drift monitoring
Vue	Instance registry, mapping studio, submission monitor and reconciliation workspace
Observability	Delivery latency, acceptance ratio, rejection reasons, backlog age, retry count and drift alerts
Tests	Contract, privacy, tenant isolation, idempotency, metadata drift, outage, correction and resubmission suites

SUPPLEMENTAL VOLUME 15

DHIS2 INTEGRATION, PUBLIC-HEALTH REPORTING AND SURVEILLANCE

Aggregate Data Sets, Event and Tracker Programs, Metadata Mapping, Delivery and Reconciliation

Scope and Authority

KashTre remains authoritative for patient care, encounters, clinical observations and source events. DHIS2 is the destination authority for configured reporting datasets and public-health programs.

Volume 15 is supplemental to the Phase 1 to 10 baseline and depends on Volumes 1 to 14.

Integration Architecture

Authoritative KashTre event -> approved reporting rule -> metadata and organisation-unit mapping -> validation or dry run -> aggregate, Event or Tracker payload -> DHIS2 Web API -> import summary -> reconciliation and resubmission.

Aggregate Reporting

Map dataset, period, organisation unit, data element and category option combination.

Exclude patient identifiers from aggregate data.

Support correction and resubmission without deleting prior evidence.

Event and Tracker Programs

Use Event programs for standalone reportable events.

Use Tracker for approved longitudinal program records using tracked entities, enrollments, events and relationships.

Preserve source public ID, event version, occurred time, correlation ID and mapping version.

Metadata Governance

Snapshot DHIS2 metadata and identifiers.

Use effective-dated mappings for organisation units, data elements, option sets, programs and stages.

Suspend affected delivery when required target metadata drifts.

Delivery and Reconciliation

Support validation before commit.

Use idempotency keys and bounded retry.

Retain import summaries and object reports.

Create actionable reconciliation cases for rejected or partially accepted objects.

Privacy and Security

Credentials remain in the platform secrets service, not Clinical source code.

Tracker and Event disclosure requires an approved purpose and minimum-necessary mapping.

Every transmission and retry is audited.

Mandatory Acceptance Tests

Missing organisation-unit mapping blocks delivery.

Unknown data element blocks aggregate submission.

Duplicate source event does not duplicate a Tracker event.

Dry-run validation creates no target data.

Rejected object maps back to the source correlation ID.

Aggregate payload contains no patient identifier.

Cross-tenant mapping is concealed.

Metadata drift suspends affected rules.

Blocking Gaps

Confirm target DHIS2 instances and versions.

Approve datasets, Event programs and Tracker programs.

Approve Sierra Leone organisation-unit and metadata mappings.

Approve identity and pseudonymisation policy.

Bind credentials, queue, scheduler and monitoring services.

Run aggregate, Event, Tracker, outage, privacy and reconciliation tests.

Source Package Register

Artifact	Count	Status
Laravel package and configuration	2	Candidate
Contracts and actions	5	Candidate
Enums and models	19	Candidate
Migration, README and manifest	3	Candidate

Standards and Product References

DHIS2 metadata identifier schemes support stable UUIDs, codes and unique metadata attributes for mapping external systems. The current Tracker API uses /api/tracker and query endpoints for tracked entities, enrollments, events and relationships.

Repository verification is required before production approval.

Consolidated Master Registers

Master Permission Families

Family	Examples	Control
Foundation	clinical.foundation.*, clinical.audit.*	Tenant, context, audit and idempotency
Observation	clinical.observation_plan.*, clinical.observation.*	Patient, encounter, CDE, unit and provenance
Clinical workflow	clinical.order.*, clinical.medication.*, clinical.result.*, clinical.transition.*	Permission plus relationship, privilege and state
Operations and AI	clinical.operations.*, clinical.ai.*	Approved use case and human review
Interoperability	clinical.exchange.*, clinical.research.*	Purpose, authorization and minimum necessary
Engagement	clinical.patient_portal.*, clinical.proxy.*, clinical.remote_monitoring.*	Patient/proxy authority and disclosure
Content	clinical.content.*	Version, validation, approval and promotion
DHIS2	clinical.dhis2.*	Target, mapping, public-health purpose and reconciliation

Master Gap Register Seed

Gap ID	Origin	Blocking resolution
EDD-CONS-GAP-001	All	Execute all packages in the controlled KashTre repository
EDD-CONS-GAP-002	Volume 1	Approve Laravel 13, PHP 8.4 and MySQL 8.4 LTS baseline
EDD-CONS-GAP-003	Volumes 1-15	Bind Main Module identity, permissions, unit and time contracts
EDD-CONS-GAP-004	Volume 13	Replace inherited Volume 12 implementation listings
EDD-CONS-GAP-005	Volume 15	Confirm DHIS2 instances, versions, metadata and programme catalogue
EDD-CONS-GAP-006	All	Complete unified requirement-design-test-evidence traceability
EDD-CONS-GAP-007	All	Complete clean, upgrade, rollback, load, privacy and security evidence

Final Verification Gate

- All source files match controlled manifests.
- All migrations pass fresh, upgrade and rollback execution.
- All routes, queues and scheduled jobs are registered.
- All negative and cross-tenant tests pass.
- Volume 13 corrected package replaces inherited draft listings.
- Volume 15 DHIS2 contracts pass against the approved target sandbox.
- No unresolved critical or high security finding remains.
- Architecture, clinical governance, security, database, QA, operations and data-protection approvals are recorded.